

Third recirculation of SA ballot for Draft 3.4 of the
**IEC/IEEE 60802 Time-Sensitive Networking Profile for
Industrial Automation**

This is an unapproved draft prepared by the IEC/IEEE 60802 Joint Project.

NOTE – This page is not subject to ballot comments.

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TIME-SENSITIVE NETWORKING PROFILE FOR INDUSTRIAL AUTOMATION

FOREWORD

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369 This document is published as an IEC/IEEE Dual Logo standard.

370 The text of this International Standard is based on the following IEC documents:

Draft	Report on voting
65C/XX/FDIS	65C/XX/RVD

371
372 Full information on the voting for its approval can be found in the report on voting indicated in
373 the above table.

374 The language used for the development of this International Standard is English.

375 This document was drafted in accordance with the rules given in the ISO/IEC Directives, Part 2,
376 available at www.iec.ch/members_experts/refdocs. The main document types developed by IEC
377 are described in greater detail at www.iec.ch/publications/.

378 The IEC Technical Committee and IEEE Technical Committee have decided that the contents
379 of this document will remain unchanged until the stability date indicated on the IEC website
380 under webstore.iec.ch in the data related to the specific document. At this date, the document
381 will be

- 382 • reconfirmed,
- 383 • withdrawn, or
- 384 • revised.

385

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387

INTRODUCTION

This document defines time-sensitive networking profiles for industrial automation. The profile selects features, options, configurations, defaults, protocols, and procedures of bridges, end stations, and LANs to build industrial automation networks. This document also specifies YANG modules for the digital data sheet and for remote procedure calls.

The profile meets the industrial automation market objective of converging Operations Technology (OT) and Information Technology (IT) networks by defining a common, standardized network infrastructure. This objective is accomplished by taking advantage of the improvements that Time-Sensitive Networking provides to IEEE 802.1 and IEEE 802.3 standard Ethernet networks by providing guaranteed data transport with bounded low latency, low latency variation, zero congestion loss for critical traffic, and high availability.

The profile helps the convergence of industrial communication networks by referring only to international standards to build the lower layers of the communication stack and their management.

Ethernet extended with Time-Sensitive Networking technology provides the features required in the area of industrial communication networks, such as:

- meeting low latency and latency variation requirements concerning data transmission;
- efficient exchange of data records on a frequent time period;
- reliable communications with calculable downtime;
- high availability meeting application requirements;
- efficient mechanisms for bandwidth utilization of exchanges of data records, with zero congestion loss;
- improved clock synchronization mechanisms, including support of multiple gPTP domains.

TIME-SENSITIVE NETWORKING PROFILE FOR INDUSTRIAL AUTOMATION

1 Scope

This document defines time-sensitive networking profiles for industrial automation. The profiles select features, options, configurations, defaults, protocols, and procedures of bridges, end stations, and LANs to build industrial automation networks. This document also specifies YANG modules defining read-only information available online and offline as a digital data sheet. This document also specifies YANG modules for remote procedure calls and actions to address requirements arising from industrial automation networks.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes the requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO/IEC 9594-8:2020, *Information technology — Open systems interconnection — Part 8: The Directory: Public-key and attribute certificate frameworks*, available at: <https://www.iso.org/obp/ui/#iso:std:iso-iec:9594:-8:en>

IEEE Std 1588-2019, *IEEE Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems*¹

IEEE Std 1588e-2024, *Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems Amendment: MIB and YANG Data Models*

IEEE Std 802.1AB-2016, *IEEE Standard for Local and metropolitan area networks – Station and Media Access Control Connectivity Discovery*

IEEE Std 802.1ABcu-2021, *IEEE Standard for Local and metropolitan networks – Station and Media Access Control Connectivity Discovery Amendment 1: YANG Data Model*

IEEE Std 802.1AR-2018, *IEEE Standard for Local and Metropolitan Area Networks – Secure Device Identity*

IEEE Std 802.1AS-2020, *IEEE Standard for Local and Metropolitan Area Networks – Timing and Synchronization for Time-Sensitive Applications*

IEEE Std 802.1ASdm-2024, *IEEE Standard for Local and Metropolitan Area Networks: Timing and Synchronization for Time-Sensitive Applications Amendment: Hot Standby*

IEEE Std 802.1ASdn-2024, *IEEE Standard for Local and Metropolitan Area Networks: Timing and Synchronization for Time-Sensitive Applications Amendment: YANG Data Model*

IEEE Std 802.1ASdr-2024, *IEEE Standard for Local and Metropolitan Area Networks – Timing and Synchronization for Time-Sensitive Applications Amendment: Inclusive Terminology*

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- 449 IEEE Std 802.1CB-2017, *IEEE Standard for Local and metropolitan area networks – Frame*
450 *Replication and Elimination for Reliability*
- 451 IEEE Std 802.1CBcv-2021, *IEEE Standard for Local and metropolitan area networks – Frame*
452 *Replication and Elimination for Reliability – Amendment 1: Information Model, YANG Data*
453 *Model and Management Information Base Module*
- 454 IEEE Std 802.1CBdb-2021, *IEEE Standard for Local and metropolitan area networks – Frame*
455 *Replication and Elimination for Reliability – Amendment 2: Extended Stream Identification*
456 *Functions*
- 457 IEEE Std 802.1Q-2022, *IEEE Standard for Local and Metropolitan Area Network – Bridges and*
458 *Bridged Networks*
- 459 IEEE Std 802.1Qcw-2023, *IEEE Standard for Local and Metropolitan Area Networks – Bridges*
460 *and Bridged Networks, Amendment: YANG Data Models for Scheduled Traffic, Frame*
461 *Preemption, and Per-Stream Filtering and Policing*
- 462 IEEE Std 802.1Qdj-2024, *Standard for Local and Metropolitan Area Networks – Bridges and*
463 *Bridged Networks, Amendment: Configuration Enhancements for Time-Sensitive Networking*
- 464 IEEE Std 802.1Qdx-2024, *Standard for Local and Metropolitan Area Networks – Bridges and*
465 *Bridged Networks, Amendment: YANG Data Models for the Credit-Based Shaper*
- 466 IEEE Std 802.1Qdy-2025, *Standard for Local and Metropolitan Area Networks – Bridges and*
467 *Bridged Networks, Amendment: YANG Data Models for the Multiple Spanning Tree Protocol*
- 468 IEEE Std 802.3-2022, *IEEE Standard for Ethernet*
- 469 IEEE Std 802.3.2-2019, *IEEE Standard for Ethernet – YANG Data Model Definitions*
- 470 IEEE Draft Std P802.3.2a, *Standard for Ethernet – YANG Data Model Definitions*
- 471 IEEE Std 802.3cx-2023, *IEEE Standard for Ethernet Amendment 6: Media Access Control*
472 *(MAC) Service Interface and Management Parameters to Support Improved Precision Time*
473 *Protocol (PTP) Timestamping Accuracy*
- 474 IEEE Std 802.3de-2022, *Standard for Ethernet Amendment 6: Enhancements to the MAC Merge*
475 *and Time Synchronization Service Interface for Point-to-Point 10 Mb/s Single Pair Ethernet*
- 476 IETF RFC 2131, Droms, R., *Dynamic Host Configuration Protocol*, March 1997, available at
477 <https://www.rfc-editor.org/info/rfc2131>
- 478 IETF RFC 2986, Nystrom, M. and Kaliski, B., *PKCS #10: Certification Request Syntax*
479 *Specification Version 1.7*, November 2000, available at <https://www.rfc-editor.org/info/rfc2986>
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3 Terms, definitions, symbols, abbreviated terms and conventions

3.1 General

For the purposes of this document, the terms and definitions given in ITU-T G.8260, IEEE Std 802-2024, IEEE Std 802.3TM-2022, IEEE Std 802.1QTM-2022, IEEE Std 802.1ASTM-2020 and the following apply.

ISO and IEC maintain terminology databases for use in standardization at the following addresses:

- IEC Electropedia: available at <https://www.electropedia.org/>
- ISO Online browsing platform: available at <https://www.iso.org/obp>
- IEEE Standards Dictionary Online: available at <https://dictionary.ieee.org>
- ITU-T Terms and Definitions database: available at https://www.itu.int/br_tsb_terms/#/

3.2 List of terms, abbreviated terms and definitions given in various standards

For the purposes of this document, the terms and definitions given in Table 1 apply.

For ease of understanding, the most important terms used within this document are listed in Table 1 but the definitions are not repeated.

Table 1 – List of terms

Term	Source
BTCA	IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdr TM -2024
Bridge	IEEE Std 802.1Q-2022
Bridge Port	IEEE Std 802.1Q-2022
CFM	IEEE Std 802.1Q-2022
Clock	IEEE Std 802.1AS-2020
ClockTimeTransmitter	IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdr-2024
ClockTimeReceiver	IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdr-2024
ClockSource	IEEE Std 802.1AS-2020
ClockTarget	IEEE Std 802.1AS-2020
CNC	IEEE Std 802.1Q-2022
Configuration Domain	IEEE Std 802.1Qdj TM -2024
constant time error (cTE)	ITU-T G.8260

Term	Source
Customer Virtual Local Area Network (C-VLAN) component	IEEE Std 802.1Q-2022
CUC	IEEE Std 802.1Q-2022
device	IEEE Std 802.1AR™-2018
DLL	IEEE Std 802-2024
DTE	IEEE Std 802.3-2022
dynamic time error (dTE)	ITU-T G.8260
end entity (EE)	NIST Special Publication 800-57 Part 2, Revision 1
end station	IEEE Std 802-2024
Ethernet	IEEE Std 802.3-2022
FDB	IEEE Std 802.1Q-2022
FID	IEEE Std 802.1Q-2022
fingerprint	IETF RFC 7589
FQTSS	IEEE Std 802.1Q-2022
fractional frequency offset	IEEE Std 802.1AS-2020
frame	IEEE Std 802.1Q-2022
frame preemption	IEEE Std 802.1Q-2022
FRER	IEEE Std 802.1CB™-2017
gating cycle	IEEE Std 802.1Q-2022
gPTP communication path	IEEE Std 802.1AS-2020
gPTP domain	IEEE Std 802.1AS-2020
Grandmaster Clock	IEEE Std 802.1AS-2020
Grandmaster PTP Instance	IEEE Std 802.1AS-2020
IST	IEEE Std 802.1Q-2022
LAN	IEEE Std 802-2024
latency	IEEE Std 802.1Q-2022
Listener	IEEE Std 802.1Q-2022
LLDP	IEEE Std 802.1AB™-2016
LLDPDU	IEEE Std 802.1AB-2016
local clock	IEEE Std 802.1AS-2020
LocalClock	IEEE Std 802.1AS-2020
logical link	IEEE Std 802-2024
LPI	IEEE Std 802.3-2022
MAC	IEEE Std 802-2024
MMRP	IEEE Std 802.1Q-2022
MST	IEEE Std 802.1Q-2022
MVRP	IEEE Std 802.1Q-2022
NETCONF	IETF RFC 6241
PCP	IEEE Std 802.1Q-2022
PDU	IEEE Std 802.1Q-2022
PHY	IEEE Std 802.3-2022
PLS	IEEE Std 802.3-2022
Port	IEEE Std 802.1Q-2022
preciseOriginTimestamp	IEEE Std 802.1AS-2020

Term	Source
primary domain	IEEE Std 802.1ASdm™-2024
PTP End Instance	IEEE Std 802.1AS-2020
PTP Instance	IEEE Std 802.1AS-2020
PTP Link	IEEE Std 802.1AS-2020
PTP Port	IEEE Std 802.1AS-2020
PTP Relay Instance	IEEE Std 802.1AS-2020
PVID	IEEE Std 802.1Q-2022
redundancy	IEC 60050-192
residence time	IEEE Std 802.1AS-2020
secondary domain	IEEE Std 802.1ASdm-2024
signature suite	IEEE Std 802.1AR-2018
station	IEEE Std 802-2024
stream	IEEE Std 802.1Q-2022
synchronized time	IEEE Std 802.1AS-2020
Talker	IEEE Std 802.1Q-2022
time error	ITU-T G.8260
time-sensitive stream	IEEE Std 802.1Q-2022
traffic class	IEEE Std 802.1Q-2022
TLV	IEEE Std 802.3-2022
UNI	IEEE Std 802.1Q-2022
VID	IEEE Std 802.1Q-2022
VLAN	IEEE Std 802.1Q-2022
X.509	ISO/IEC 9594-8:2020
YANG	IETF RFC 6020

593

594 3.3 Terms defined in this document

595 3.3.1

596 application clock

597 clock used by the application to time events

598 Note 1 to entry: Events can be periodic or aperiodic.

599 3.3.2

600 Bridge component

601 Customer Virtual Local Area Network (C-VLAN) Bridge

602 3.3.3

603 control latency

604 time delay between the input to a sensor application and the output from an actuator application

605 Note 1 to entry: For the purposes of this document, control latency does not include latencies in the sensor,
606 actuator, or the physical system in a process.

607 3.3.4

608 deadline

609 application defined reference point that represents a time when data is required by the
610 application

3.3.5**digital data sheet**

information about the capabilities of an IA-station

3.3.6**end station component**

end station entity

3.3.7**Global Time**

synchronized time, derived from a gPTP domain, that is traceable to the PTP timescale

3.3.8**IA-controller**

industrial automation function, consisting of a comparing element and a controlling element, that performs a specified control function

Note 1 to entry: An IA-controller exchanges data with several IA-devices or other IA-controllers for the purpose of control of a system.

Note 2 to entry: The primary categories of IA-controllers are distributed control systems (DCS), programmable logic controllers (PLCs), and programmable automation controllers (PACs).

3.3.9**IA-device**

industrial automation function, consisting of sensor or actuator elements to read or write process data

Note 1 to entry: An IA-device exchanges data with an IA-controller or other IA-devices for the purpose of control of a system.

3.3.10**IA-station**

material element or assembly of one or more end station components, and zero, one or more bridge components

Note 1 to entry: IA-controllers and IA-devices are industrial automation functions of IA-stations.

3.3.11**imprinting**

<security> equipping IA-stations with an LDevID credential, corresponding trust anchor and certificate-to-name mapping instructions

3.3.12**management entity**

IA-station function responsible for configuration of Bridge components, end station components and ports

Note 1 to entry: The management entity interacts with remote management.

3.3.13**network diameter**

number of links in the longest of all the calculated shortest paths between each pair of nodes in the network

3.3.14**network provisioning**

process of defining a consistent network configuration, which is applied to all stations

3.3.15

nominal frequency

ideal frequency with zero uncertainty

Note 1 to entry: The nominal frequency of the PTP timescale is further explained in IEEE Std 1588™-2019, 7.2.1, 7.2.2, and Annex B.

3.3.16

non-proxied IA-station

IA-station which contains a ccA Bridge component or a ccA end station component

3.3.17

ppm

µHz/Hz

Note 1 to entry: The term "ppm" refers to a pure multiplier of 0,000 001 and is used in the context of this document as an SI unit term to allow readable terms conformant to various rules related to expressions.

3.3.18

proxy

IA-station whose Netconf server can be utilized by other IA-stations for the purpose of remote management

3.3.19

proxy target

IA-station which contains only ccB Bridge or end station components

3.3.20

Working Clock

synchronized time, derived from a gPTP domain, that is traceable to the PTP timescale, or to an ARB timescale that is continuous

Note 1 to entry: In general, the Working Clock is traceable to an ARB timescale; however, the Working Clock time can be correlated to a recognized timing standard.

3.4 Abbreviated terms

AEAD	Authenticated Encryption with Associated Data
AES	Advanced Encryption Standard
ARB	Arbitrary
ASCII	American Standard Code for Information Interchange
ASN	Abstract Syntax Notation
BTCA	Best timeTransmitter Clock Algorithm
CA	Certification Authority
CBC	Cipher Block Chaining
ccA	Conformance Class A
ccB	Conformance Class B
CFM	Connectivity Fault Management
CMLDS	Common Mean Link Delay Service
CMS	Cryptographic Message Syntax
CN	Common Name
CNC	Centralized Network Configuration
CRL	Certificate Revocation List
CRUDX	Create Read Update Delete eXecute
CSR	Certificate Signing Request

700	CUC	Centralized User Configuration
701	C-VLAN	Customer VLAN
702	DAC	Discretionary Access Control
703	DER	Distinguished Encoding Rules
704	DH	Diffie-Hellman
705	DHE	Diffie-Hellman Ephemeral
706	DLL	Data Link Layer
707	DMAC	Destination MAC Address
708	DNS	Domain Name Service
709	DSA	Digital Signature Algorithm
710	DTE	Data Terminal Equipment
711	EC	Elliptic Curve
712	ECC	Elliptic Curve Cryptography
713	ECDSA	Elliptic Curve Digital Signature Algorithm
714	EdDSA	Edwards-Curve Digital Signature Algorithm
715	EE	End Entity
716	FDB	Filtering Database
717	FID	Filtering Identifier
718	FQDN	Fully Qualified Domain Name
719	FQTSS	Forwarding and Queuing Enhancements for Time-Sensitive Streams
720	FRER	Frame Replication and Elimination for Reliability
721	GCM	Galois Counter Mode
722	gPTP	generalized Precision Time Protocol
723	HMAC	Keyed-Hashing for Message Authentication Code
724	HW	Hardware
725	IA	Industrial Automation
726	IDeVID	Initial Secure Device Identifier
727	IEC	International Electrotechnical Commission
728	IEEE	Institute of Electrical and Electronics Engineers
729	I-LAN	Internal Local Area Network
730	ISO	International Organization for Standardization
731	ISS	Internal Sublayer Service
732	IST	Internal Spanning Tree
733	IT	Information Technology
734	ITU	International Telecommunication Union
735	LDeVID	Locally Significant Secure Device Identifier
736	LLDP	Link Layer Discovery Protocol
737	LLDPDU	Link Layer Discovery Protocol Data Unit
738	LPI	Low Power Idle
739	LRP	Link-local Registration Protocol
740	MAC	Media Access Control
741	MD	Media-Dependent
742	MDI	Medium Dependent Interface

743	MMRP	Multiple MAC Registration Protocol
744	MST	Multiple Spanning Tree
745	MVRP	Multiple VLAN Registration Protocol
746	N/A	Not applicable
747	NACM	Network configuration Access Control Model
748	NETCONF	Network Configuration Protocol
749	NMDA	Network Management Datastore Architecture
750	NPE	Network Provisioning Entity
751	NRR	Neighbor Rate Ratio
752	OCSF	Online Certificate Status Protocol
753	OID	Object Identifier
754	OMG®	Object Management Group ²
755	OT	Operations Technology
756	OUI	Organizationally Unique Identifier
757	PCP	Priority Code Point
758	PCS	Profile Conformance Statement
759	PDU	Protocol Data Unit
760	PE	Path Entity
761	PEM	Privacy Enhanced Mail
762	PFS	Perfect Forward Secrecy
763	PHY	Physical Layer Device
764	PII	Personally Identifiable Information
765	PKCS	Public Key Cryptography Standards
766	PKI	Public Key Infrastructure
767	PKIX	Public Key Infrastructure X.509
768	PLS	Physical Signaling Sublayer
769	PPS	Pulse Per Second
770	PTP	Precision Time Protocol
771	PVID	Port VLAN Identifier
772	QoS	Quality of Service
773	RBAC	Role-Based Access Control
774	RFC	Request for Comments
775	RPC	Remote Procedure Call
776	RSA	Rivest-Shamir-Adleman
777	RAE	Resource Allocation Entity
778	SAN	Subject Alternative Name
779	SHA	Secure Hash Algorithm
780	STE	Sync Tree Entity
781	TDE	Topology Discovery Entity

² OMG®, UML® and Unified Modeling Language™ are either registered trademarks or trademarks of a product supplied by the Object Management Group, Inc. in the United States and/or other countries. This information is given for the convenience of users of this document and does not constitute an endorsement by IEEE or IEC of the product named. Equivalent products may be used if they can be shown to lead to the same results.

782	TLS	Transport Layer Security
783	TLV	Type, Length, Value
784	TOFU	Trust On First Use
785	TSN	Time-Sensitive Networking
786	TSN-IA	Time-Sensitive Networking for Industrial Automation
787	TTP	Trusted Third Party
788	UML®	Unified Modeling Language™
789	UNI	User/Network Interface
790	URL	Uniform Resource Locator
791	URN	Uniform Resource Name
792	VID	VLAN Identifier
793	VLAN	Virtual Local Area Network
794	YANG	Yet Another Next Generation data modeling language

795 3.5 Conventions

796 3.5.1 Convention for capitalizations

797 Capitalized terms are either based on the rules given in the ISO/IEC Directives Part 2 or
798 emphasize that these terms have a specific meaning throughout this document.

799 Throughout this document "bridge" is used instead of "Bridge", except when

- 800 • it occurs at the beginning of a sentence or
- 801 • it is being used as (or part of) a specific term such as "VLAN Bridge" rather than being used
802 to identify bridges (potentially of any type) in general. If "VLAN Bridge" is meant where only
803 "Bridge" is written, a change to "VLAN Bridge" would be appropriate.

804 3.5.2 Unit conventions

805 This document uses:

- 806 • Gb/s for gigabits per second,
- 807 • Mb/s for megabits per second and,
- 808 • kb/s for kilobits per second.

809 3.5.3 Conventions for YANG contents

810 YANG modules and XML instance data for YANG shown in this document use the following
811 style:

812 Text style `higher-layer-if` text style

813 Contents of a YANG module use the following style:

```
814 <ieee802-dot1q-bridge xmlns:dot1q="urn:ieee:std:802.1Q:yang:ieee802-dot1q-bridge">
815   <bridges>
816     <bridge> <!-- list -->
817       <name>functional-unit-x</name>
818     ...
```

819 3.5.4 Conventions for YANG selection and Digital Data Sheet

820 The digital data sheet expresses device capabilities and therefore, it is not required that all
821 nodes in a YANG module be included in the digital data sheet. YANG nodes in 6.4.9, marked
822 with [m], are mandatory nodes in the digital data sheet, nodes marked with [c] are conditional

mandatory if the IA-station supports the corresponding optional functionality. Nodes marked with [o] are optional nodes in the digital data sheet. These marking in no way affect whether the feature and associated YANG module is required for the IA-station. Please refer to Clause 5 for conformance criteria for the IA-station.

YANG node selections in 6.4.9 of parent nodes implicitly include all subsidiary child nodes.

4 Overview of TSN in industrial automation

4.1 Industrial application operation

Industrial network applications are based on three main types of building blocks, which can be combined in one IA-controller or provided as a combination of an IA-controller and IA-devices interconnected through a suitable communication network.

These basic building blocks are:

- IA-device Sensor subsystems, which provide input signals indicating the value of the parameter or state being monitored, such as temperature, pressure, or discrete input information;
- IA-controller subsystems, which operate on combinations of measurements and external demand settings to develop output requests, such as position corrections in a motion application;
- IA-device Actuator subsystems, which implement output requests that result in physical changes to the process or machine under control, such as a level in a storage tank, the speed of a printing press, or movement of a robot.

NOTE 1 In general, all subsystems have an internal state, based upon initial settings, and derived from execution; therefore, the application inputs are combined with the internal state to develop an updated internal state and associated outputs.

A control loop is formed when the process or machine responds to the IA-device Actuator output and produces a new measured value at the IA-device Sensor. An example of a complete loop and the data flow inside it is shown in Figure 1 and described below where an IA-controller and IA-devices are connected as end stations in the network.

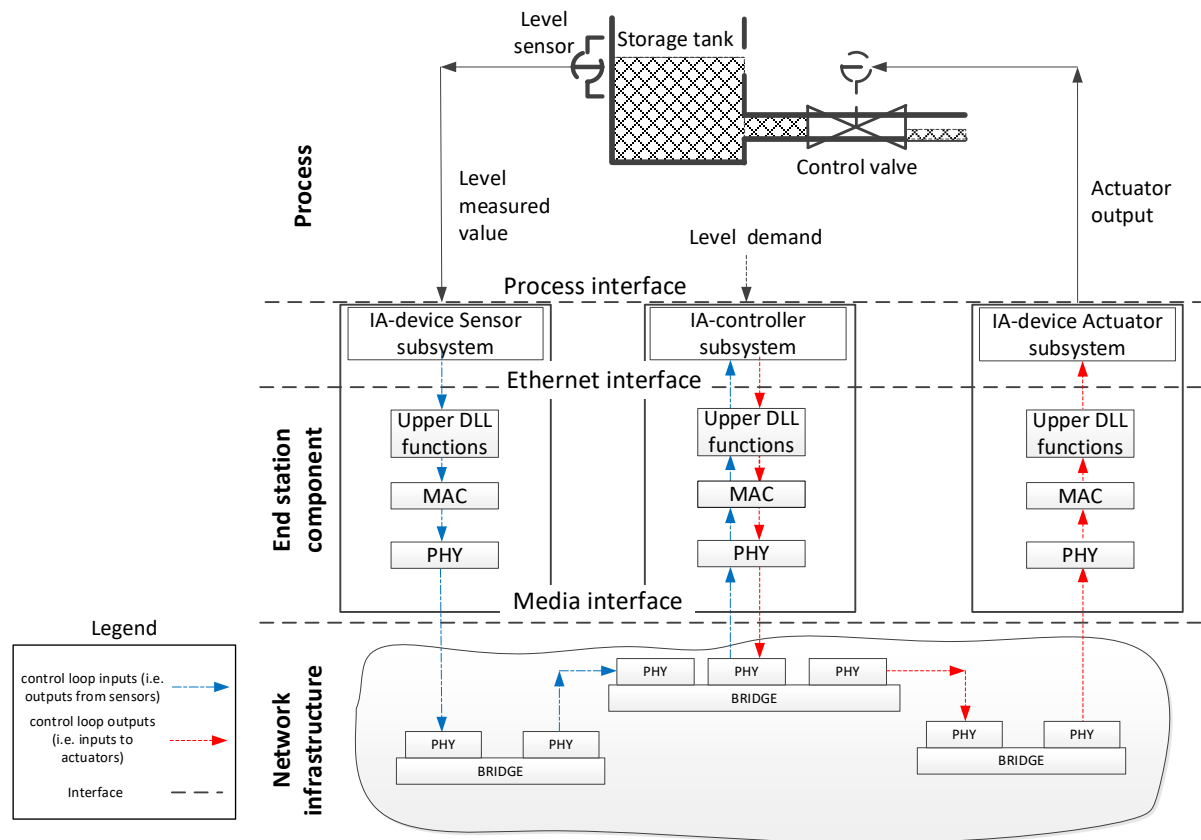


Figure 1 – Data flow in a control loop

In operation, the IA-device Sensor subsystem samples the measured value and the sampled values are transferred through the network as data packets for the IA-controller subsystem to compare with the demand value. After some computation time, the output is transferred from the IA-controller subsystem to the IA-device Actuator subsystem for implementation as a change in the external process.

This sequence repeats continuously as a regular operation using a Working Clock. The Working Clock is traceable to an ARB timescale or to the PTP timescale. Traceability to the PTP timescale is not required by all applications. For stability, the sequence repetition time (i.e., sampling time) is typically chosen to be between one fifth to one tenth of the time constant of the process response although shorter times can also be used.

NOTE 2 The actual ratio of the process response time constant to sampling time required for stability depends on the implementation.

Control latency is a critical factor in all types of control and is bounded. Components contributing to the control latency time (as shown in Figure 1) are:

- application time for sampling, computation, and processing within each IA-controller and IA-device. These are specific to the IA-device and IA-controller and known to the IA-device or IA-controller makers,

- the time for data transfer through the upper DLL functions, MAC and PHY layers within each IA-controller and IA-device. This time depends on the implementation of these components, their situation-dependent load and performance, and configuration elements related to QoS supported by these components, and
- the End Station and Bridge schedule and transfer time through the network. These are influenced by the configuration process, which allocates available bandwidth and priorities to various types of application messages.

Offline engineering of the network is possible, including the calculation of the control latency time. During system operation, management services are provided for diagnostics and checking the performance indicators of an installed network.

4.2 Industrial applications

4.2.1 General

Industrial applications can contain multiple tasks. These tasks are executed based upon time or other events. Thus, an industrial application can have multiple tasks executing on different cycles as shown in Figure 2 and Figure 3.

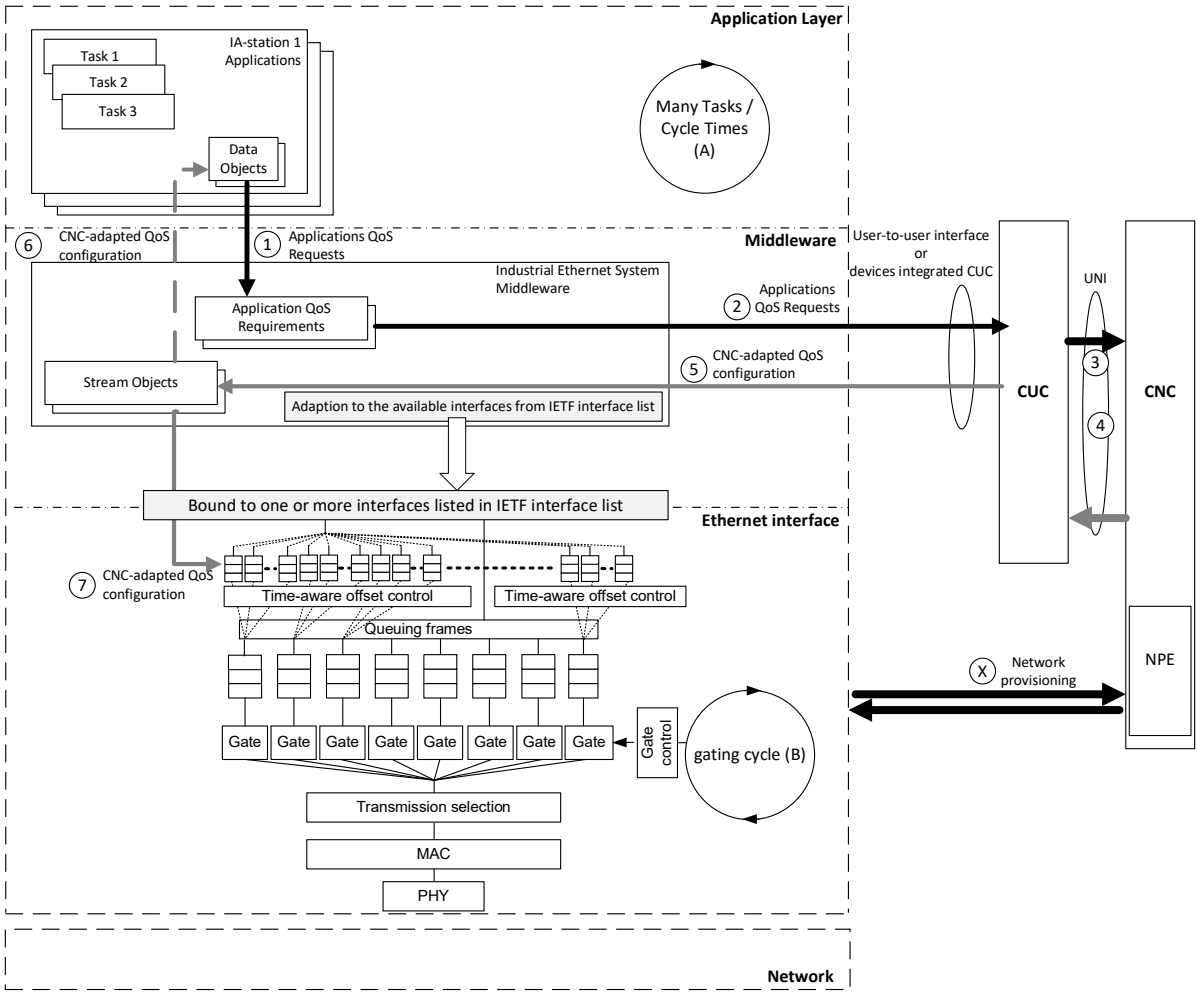
Examples of these tasks include the following.

- Background tasks, which are executed when no other task is running. There can be zero, one, or more such tasks in an industrial application.
- Main task which executes periodically. The start and execution of this task is often based upon the ARB timescale. There can be zero or one such task, in an industrial application.
- Global Time tasks. The start and execution of these tasks is often based upon Global Time (for example, at noon every day or at noon every Friday). There can be zero, one or more such tasks in an industrial application.
- Process driven tasks which are started by an event (for example, a sensor value reaches a defined point, or a process fault occurs). There can be zero, one or more such tasks in an industrial application.
- Control loop tasks which are bound to the Working Clock and started periodically. There can be zero, one or more such tasks in an industrial application.

A user defines the required automation tasks along with the data objects required as output and input for these tasks and the end station which hosts these tasks. Thus, these tasks are bound to data objects, which are exchanged between end stations per the user's definition. Many of these tasks have timing requirements, which are added as attributes to the assigned data objects. Examples of these attributes include:

- An update interval: [DataObject_Update_Interval] (time between two consecutive updates at the transmitting end station);
- A deadline: [DataObject_Deadline] (latest receive time at the end station, relative to the start of the DataObject Update Interval);
- The size of the DataObject: [DataObject_Data_Size];
- Other attributes to form a stream-list request according to IEEE Std 802.1Qdj-2024, 46.1.5.

NOTE These attributes are provided for illustration purposes. The list is not representative of all industrial applications. These are not network attributes.



Legend:

(X)

IA-stations are configured by the CNC before any CUC to CNC action happens. For example, this configuration can include the gating cycle and transmission selection. Later accesses can happen concurrently with a CUC to CNC sequence.

(1) ... (7)

Sequence of actions taken

Time-aware offset control Queues are shared between multiple middleware components.

Figure 2 – IA-station interaction with CNC – Transmit path

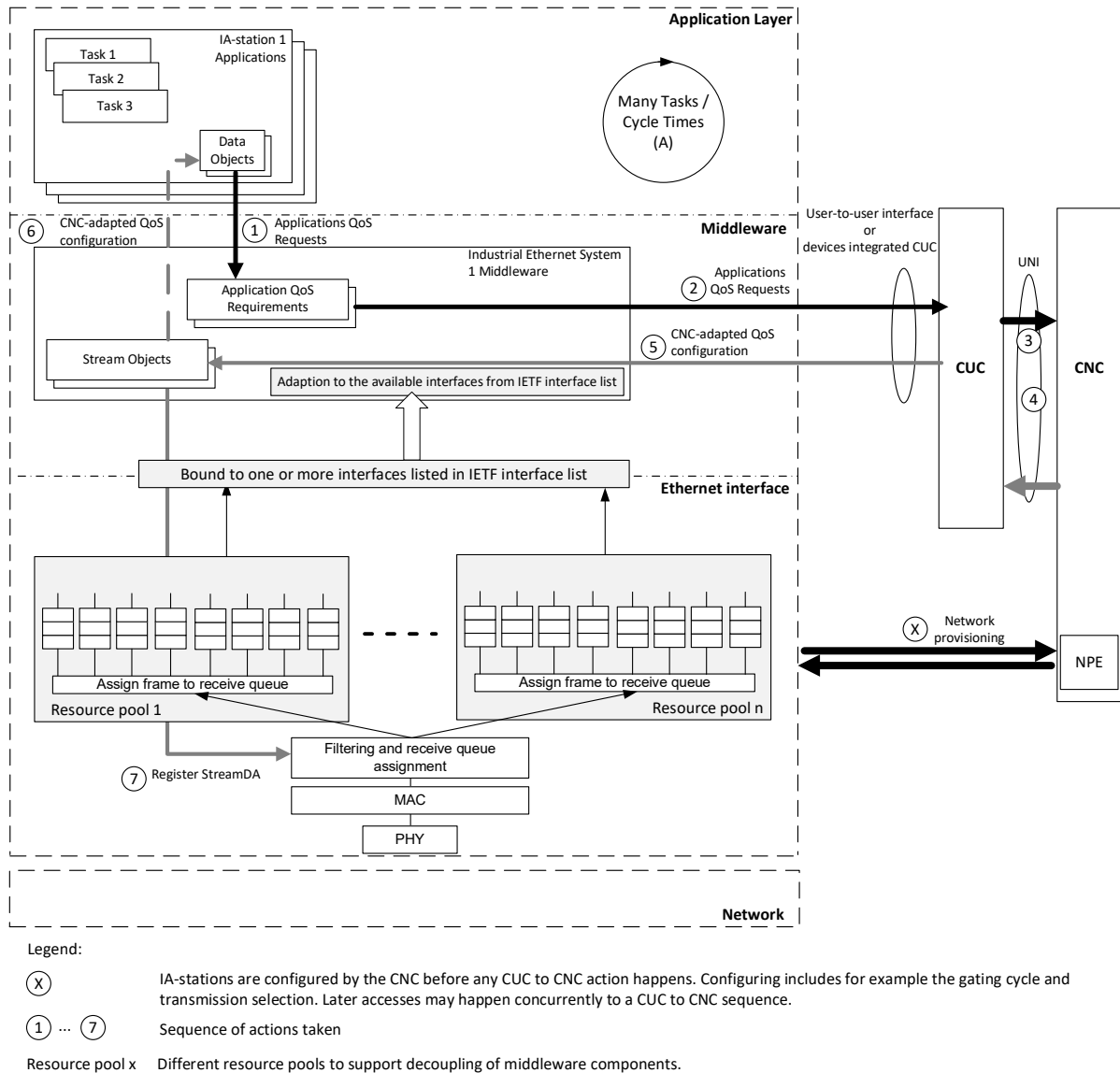


Figure 3 – IA-station interaction with CNC – Receive path

4.2.2 Control loop tasks

Multiple control loop tasks can be implemented and run in parallel in their automation devices. For example, this behavior can be implemented by using a common Working Clock, a common starting point relative to the Working Clock and a common duration for this control loop task at the involved IA-devices and IA-controllers. The data objects associated with the control loop share common values for some attributes (for example, the same values for DataObject_Update_Interval and DataObject_Deadline). Multiple control loop tasks can be implemented and run in parallel in their automation devices.

4.2.3 Start of control loop tasks

The calculation of the starting point for a control loop task is independent from the time when the device is powered up or connected to the Configuration Domain. The start of a control loop task, which is based on the Working Clock, can be calculated in the following manner: divide the Working Clock value, expressed as an integer, by the duration of the control loop task, expressed as an integer, whenever the Working Clock value increases by one. A remainder of zero provides the basis for the start of the control loop task.

NOTE The units of the Working Clock value and the units of the duration of the control loop task are the same.

Stations in the network associated with the control loop synchronize to a Working Clock using IEEE Std 802.1AS-2020.

4.3 IA-stations

An IA-station can be a simple end station acting as source or destination for control data traffic. In addition, an IA-station can be a combined functional unit that includes an end station component together with a Bridge component in one chassis. IA-stations, incorporating multiple functional units with several end station components and Bridge components within one chassis, can also be found in industrial automation. Within this kind of combined IA-station various components can be connected by internal ports and internal LANs. All components utilize a common management entity as shown in Figure 4.

Figure 4 shows an example IA-station incorporating four functional units in one chassis. Functional unit 1 and functional unit 2 each consist of a Bridge component and an end station component. The end station components are connected by internal ports via internal LANs to the Bridge components. The Bridge components include two external ports each. Functional unit 3 includes only a single end station component with one external port. Functional unit 4 includes a single end station component with two external ports.

IA-controllers and IA-devices as well as the management entity are IA-station functions acting as source of or destination for link layer data traffic. Thus, each IA-station incorporates at least one end station component where these functions can be located. Figure 4 shows that IA-station functions can either reside in a single end station component (IA-device 1, IA-controller 1, IA-device 2, IA-device 3, IA-controller 3) or in multiple end station components (IA-controller 2, management entity).

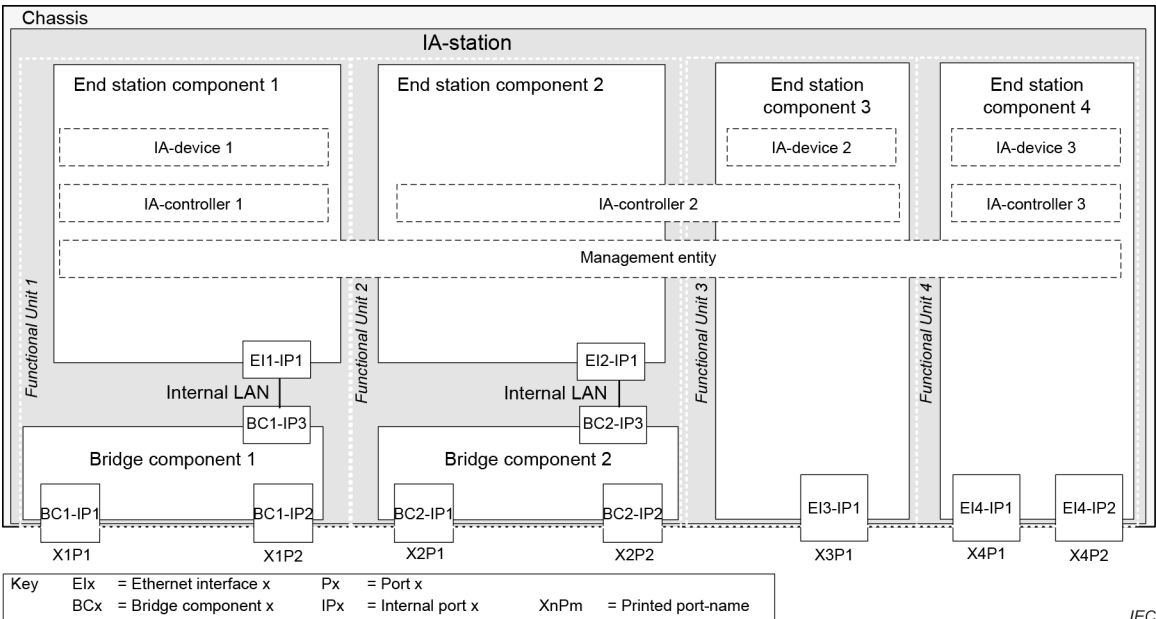


Figure 4 – IA-station example

4.4 Ethernet interface

One or more middleware components act as a layer between applications and the Ethernet interface. Figure 2 and Figure 3 show the relation between applications, middleware, Ethernet interface and the network. Various applications can run in parallel on an automation device. Data objects represent the information exchanged between applications running in different end stations. The application requirements contained in these data objects are translated by the

middleware into stream requirements for use by the CUC. This translation can be accomplished in one or both of the following ways.

- a) The user defines the data objects and translates them into stream requirements and end-station communication-configurations. A user-specific mechanism is used to configure the network components, establish paths, and the time-aware offset control.
- b) The user defines the data objects and associates them with QoS requirements for each stream (application QoS requirements). These can be forwarded as stream requirement requests by a CUC to a CNC. The CNC responds by providing a stream configuration response. The request and response are specified in IEEE Std 802.1Qdj-2024. This information is used to configure the time-aware offset control, which utilizes per-stream queues. The CUC can be integrated into the end station or can be accessed via a user-to-user protocol. The middleware uses this information for configuring Talkers and Listeners. This information is also used to add additional timing information to the data objects for application usage.

Time-aware offset control utilizes per-stream queues (see IEEE Std 802.1Q-2022, Figure 34-1) and the traffic specification of the streams, including transmission offsets, provided by the CNC to ensure the order of stream transmission.

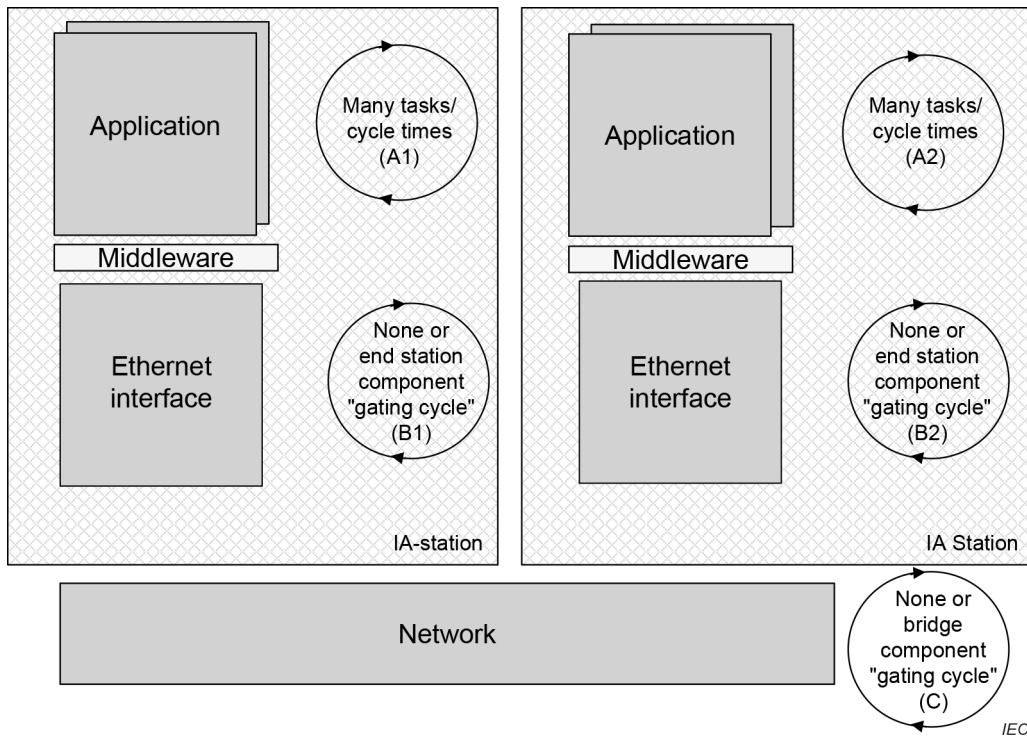


Figure 5 – Model for cycles

These automation systems, which are built from various end stations and connected via bridges, can share a common gating cycle or each station can have its own gating cycle. Alternatively, a bridge or end station can have no gating cycle (expressed as "none" in Figure 5).

4.5 Mechanisms to meet control loop latency requirements

Meeting latency requirements on a network can be accomplished using one or more combinations of the mechanisms enumerated below. The choice of a mechanism or a subset of the mechanisms listed below depends on the nature of the application(s) and the corresponding latency requirements:

- a) defining, testing, and simulating all possible application combinations and associated traffic patterns,

- b) overprovisioning the network,
- c) providing scheduled time slots for each application to transmit on the network,
- d) preempting lower priority traffic,
- e) providing scheduled time slots for certain traffic classes,
- f) time-aware offset control, and
- g) enforcing deterministic queuing delays in bridges.

NOTE This list is not comprehensive and not all mechanisms mentioned here are part of this specification. For specific mechanisms covered by this document please refer to Clause 5.

Frame preemption is specified in IEEE Std 802.1Q-2022 and IEEE Std 802.3-2022.

Reserving time on the network for certain traffic types can be done through enhancements for scheduled traffic according to IEEE Std 802.1Q-2022, 8.6.8.4. An aligned gating cycle is defined. Once a gating cycle is defined, portions of a cycle time can either be allocated to streams or classes of streams.

Multiple Talker-Listener pairs can be used for streams between end stations. Engineered time-triggered transmit can be used to coordinate transmission of all the traffic that shares a network to meet application requirements.

Creating a traffic load model in advance allows analysis of resulting traffic. It can be used to select and implement appropriate mechanisms to achieve latency requirements.

4.6 Translation between middleware and network provisioning

4.6.1 Mapping translation table

Application engineering can be done without knowledge of the network provisioning. Since the application is not aware of the network provisioning, it cannot directly map to the network configuration, for example, the use of PCP or VID as configured in the network. This problem is solved by providing a translation table, in the form of a YANG module definition, to the middleware. The IA-station's local YANG datastore contains this information.

Figure 6 shows examples of the translation models.

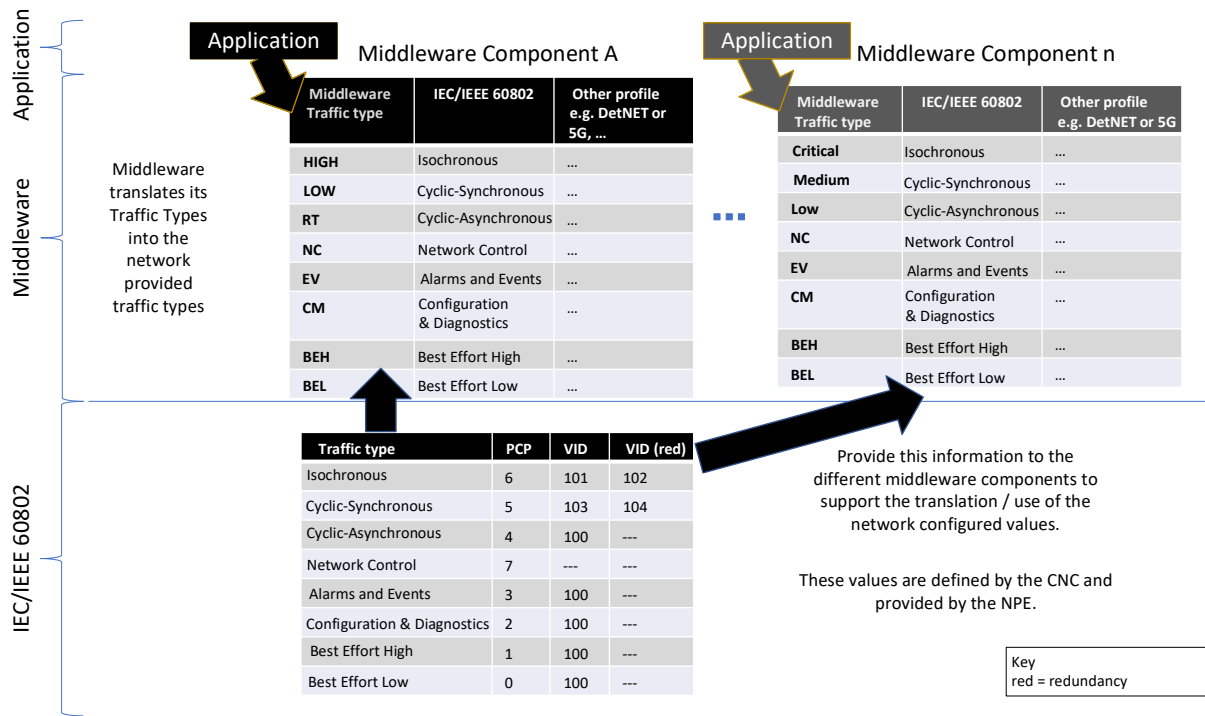


Figure 6 – Traffic type translation example

The YANG module `iecieee60802-tt-mapping.yang` is used to provide the required mapping information to all installed middleware and applications. The information in the YANG module provides the VLAN-ID, the assigned traffic types with their PCP values and redundancy information (6.4.10.8.7).

4.6.2 PTP Instances

PTP domain numbers are also configured during network provisioning. The middleware is informed about which PTP domain is assigned to which target clock by providing `descriptionDS.userDescription` names according to IEEE Std 1588-2019, 8.2.5.5, in a translation table.

`descriptionDS.userDescription` names allow the support of multiple middleware components at one IA-station using the same PTP Instances (see 6.2.13). An IA-station's local database stores this information.

Figure 7 shows examples of the translation models.

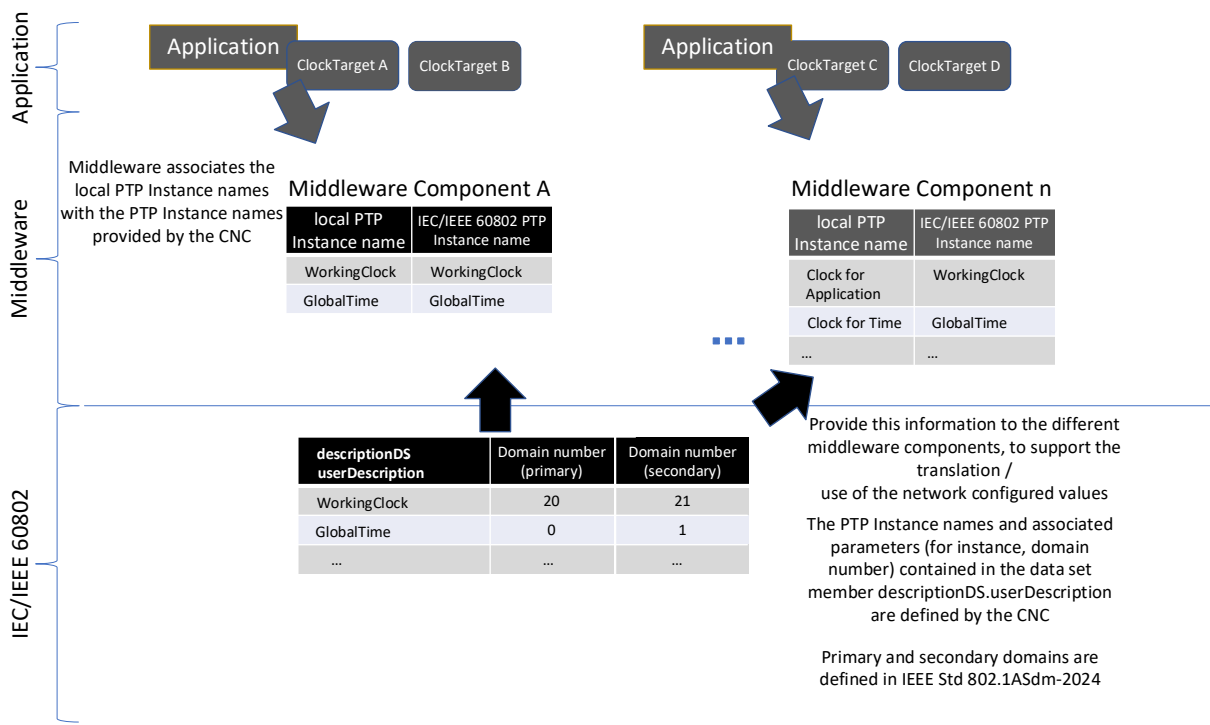


Figure 7 – PTP instance translation example

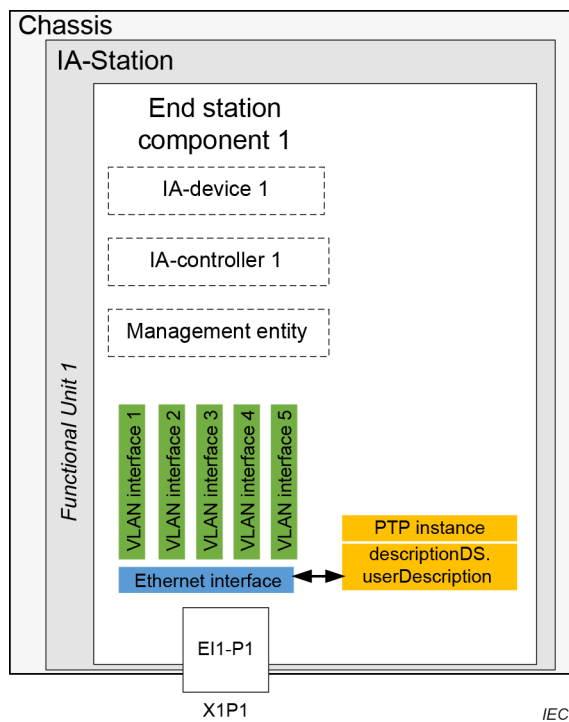


Figure 8 – descriptionDS.userDescription used for PTP instance translation

The userDescription contains the clock type (i.e., WorkingClock, GlobalTime, or both). This information is used by the middleware to align to the intended ClockTarget or ClockSource (see 6.2.13).

4.7 Industrial traffic types

4.7.1 General

Industrial automation applications make use of different traffic schemes/types for different functionalities (for example, parameterization, control, alarming). The various traffic patterns have different characteristics and thus impose different requirements on a network. To specify these traffic types, a two-step approach is used.

- a) First, define characteristics of generic traffic types (traffic type categories).
- b) Second, define instances of the generic traffic types (i.e. the traffic types).

4.7.2 Traffic type characteristics

The traffic type characteristics in Table 2 enable the identification of several distinct traffic types that are shared among sets of industrial applications.

Table 2 – Traffic type characteristics

Characteristic	Description
Cyclic	Traffic types consist of frames that can either be transmitted on a reoccurring time period (cyclic) or at no set period (acyclic). Available selections are: • required: traffic frames are transmitted cyclically, or • optional: implementation of cyclic traffic is at the discretion of the user.
Data delivery requirements	Denotes the delivery constraints for the traffic. Four options are specified. • Frame Latency: data delivery of a frame for a given Talker-Listener pair occurs within a bounded timespan. • Bandwidth: data delivery up to a certain number of frames or data size (including bursts of frames) occurs over a defined period. • Deadline: data delivery of a frame to a given Listener occurs at or before a specific point in time. • No: Denotes the case of traffic types with no special data delivery requirements
Time-triggered transmission	Talker data transmission occurs at a specific point in time based upon the Working Clock. Available selections are: • required, or • optional: implementation of time-triggered transmission is at the discretion of the user. Enhancements of scheduled traffic is only one means of achieving time-triggered transmission. Other, application-based, methods are possible
Stream-based	Indicates whether a traffic type requires registering a stream, i.e., allocating dedicated resources, for the traffic, before a given Talker can start transmitting data. • True: Data can only be transmitted after a Stream has been set up. • False: Data can be transmitted at any time.

4.7.3 Traffic type categories

4.7.3.1 General

The two-step approach described in 4.7.1 allows a clear differentiation between characteristics as seen from the "network" point of view and "application" point of view. Traffic type categories allow different IEEE 802 feature selections to achieve the goals of a specific network deployment. Four traffic type categories are identified in industrial automation systems:

- a) IA time-aware stream,
- b) IA stream,
- c) IA traffic engineered non-stream, and

d) IA non-stream.

4.7.3.2 IA time-aware stream

The characteristics of this traffic type category are shown in Table 3.

Table 3 – IA time-aware stream characteristics

Characteristics	
Cyclic	Required
Data delivery requirement	Deadline or Frame Latency
Time-triggered transmission	Required
Stream-based	True

4.7.3.3 IA stream

The characteristics of this traffic type category are shown in Table 4.

Table 4 – IA stream characteristics

Characteristics	
Cyclic	Required
Data delivery requirement	Frame Latency
Time-triggered transmission	Optional
Stream-based	True

4.7.3.4 IA traffic engineered non-stream

The characteristics of this traffic type category are shown in Table 5.

Table 5 – IA traffic engineered non-stream characteristics

Characteristics	
Cyclic	Optional
Data delivery requirement	Bandwidth
Time-triggered transmission	Optional
Stream-based	False

4.7.3.5 IA non-stream

The characteristics of this traffic type category are shown in Table 6.

Table 6 – IA non-stream characteristics

Characteristics	
Cyclic	Optional
Data delivery requirement	No
Time-triggered transmission	Optional
Stream-based	False

4.7.4 Traffic types

4.7.4.1 General

Table 7 summarizes relevant industrial automation traffic types and their associated characteristics. In an industrial automation system, other applications, such as audio or video, utilizes one of these traffic types.

Table 7 – Industrial automation traffic types summary

Traffic type name	Cyclic	Data delivery requirements	Time-triggered transmission	Stream based	Traffic type category
Isochronous	Required	Deadline	Required	True	IA time-aware stream
Cyclic-synchronous	Required	Frame Latency	Required	True	IA time-aware stream
Cyclic-asynchronous	Required	Frame Latency	Optional	True	IA stream
Alarms & events	Optional	Bandwidth	Optional	False	IA traffic engineered non-stream
Configuration & Diagnostics	Optional	Bandwidth	Optional	False	IA traffic engineered non-stream
Network Control	Optional	Bandwidth	Optional	False	IA traffic engineered non-stream
Best Effort High	Optional	No	Optional	False	IA non-stream
Best Effort Low	Optional	No	Optional	False	IA non-stream

4.7.4.2 Isochronous

Isochronous is a type of IA time-aware stream traffic. This type of traffic is transmitted cyclically using time-triggered transmission. Listeners have individual deadline requirements. Cycle times are typically in the range of microseconds to tens of milliseconds. Frame size is typically below 500 octets. Talker-Listener pairs are synchronized to the Working Clock. The network is configured by the CNC to provide zero congestion loss for this traffic type. This type of traffic is normally used in control loop tasks.

4.7.4.3 Cyclic-synchronous

Cyclic-synchronous is a type of IA time-aware stream traffic. This type of traffic is transmitted cyclically using time-triggered transmission. Talker-Listener pairs have individual latency requirements. Cycle times are typically in the range of hundreds of microseconds to hundreds of milliseconds. Frame size is unconstrained except as indicated in 5.5.1. Talker-Listener pairs are synchronized to the Working Clock. The network is configured by the CNC to provide zero congestion loss for this traffic type. This type of traffic is normally used in control loop tasks.

4.7.4.4 Cyclic-asynchronous

Cyclic-asynchronous is a type of IA stream traffic. This type of traffic is transmitted cyclically with latency requirements bounded by the interval as specified in IEEE Std 802.1Q-2022, 46.2.3.5.1. Talker-Listener pairs have individual latency requirements. Cycle times are typically in the range of milliseconds to seconds. Frame size is unconstrained except as indicated in 5.5.1. Data exchanges between Talker-Listener pairs are typically not dependent on the Working Clock. This traffic type typically tolerates limited congestion loss. The network is

configured by the CNC to handle this traffic type without loss, up to a certain number of frames or data size.

4.7.4.5 Alarms and events

Alarms and events are a type of IA traffic engineered non-stream. This type of traffic is transmitted cyclically or acyclically. This traffic expects bounded latency including time for retransmission in the range of milliseconds to hundreds of milliseconds. The source of the alarm or event typically limits the bandwidth allocated to this traffic. Frame size is unconstrained except as indicated in 5.5.1. Congestion loss can occur. Retransmission to mitigate frame loss is expected. The network is configured by the CNC to handle these frames, including bursts of frames, up to a certain number of frames or data size over a defined period.

4.7.4.6 Configuration and diagnostics

Configurations and diagnostics are a type of IA traffic engineered non-stream. This type of traffic is transmitted cyclically or acyclically. This traffic expects bounded latency, up to seconds, including time for retransmission. The source of configuration or diagnostics frames typically limits the bandwidth allocated to this traffic. Frame size is unconstrained except as indicated in 5.5.1. Congestion loss can occur. Retransmission to mitigate frame loss is expected. The network is configured by the CNC to handle these frames, including bursts of frames, up to a certain number of frames or data size over a defined period.

4.7.4.7 Network control

Network control is a type of IA traffic engineered non-stream. This type of traffic can be transmitted cyclically or acyclically. This traffic expects bounded latency including time for retransmission. Frame size is unconstrained except as indicated in 5.5.1. The network is configured by the CNC to handle these frames, including bursts of frames, up to a certain number of frames or data size over a defined period. If these limits are exceeded congestion loss can occur. Network control is comprised of services required to maintain network operation. Examples include time synchronization, loop prevention, and topology detection.

4.7.4.8 Best effort

Best effort is a type of IA non-stream. The network is configured by the CNC so that these frames do not interfere with other traffic types. These frames are forwarded when resources are available. Congestion loss resulting in frame drop can occur. It is sometimes desirable to have more than one traffic class for best effort traffic (see Table 8).

4.7.4.9 Traffic class to traffic type mapping

Table 8 provides an example for the usage of traffic classes based on the traffic type:

Table 8 – Example traffic class to traffic type mapping

Traffic class	Queue Assignment (8 queues)	Queue Assignment (4 queues)	Traffic type
7	6	2	Isochronous
6	5	1	Cyclic-Synchronous
5	4	1	Cyclic-Asynchronous
4	7	3	Network Control
3	3	0	Alarms and events
2	2	0	Configuration & Diagnostics
1	1	0	Best Effort High
0	0	0	Best Effort Low

NOTE An example of mapping of PCP and traffic type to an application is provided in Figure 6.

1138

1139 The traffic type categories definition allows different IEEE 802 feature selections to achieve
1140 specified goals. Moreover it helps in identification of the traffic protection mechanisms.
1141 Adherence to this example of a common mapping helps minimize potential conflicts between
1142 traffic types.

1143 **4.8 Security for TSN-IA**

1144 **4.8.1 General**

1145 Subclause 4.8 describes selected aspects of TSN-IA security. Protecting the management of
1146 industrial communication is the main objective of TSN-IA security. The protection of
1147 communications that use industrial traffic types is not addressed by this document.

1148 **4.8.2 Security configuration model**

1149 Security configuration is a part of system engineering and configuration. The security
1150 configuration in this document does not encompass the supply of configuration objects for
1151 middleware and application security. Security configuration settles the prerequisites for
1152 protecting the establishment and management of communications that use industrial traffic
1153 types (see 4.7). It ensures that the security features of IA-stations (including CNCs) can be
1154 used for protecting message exchanges and authorizing the resource accesses during stream
1155 establishment and management. This security configuration supplies deployment-specific
1156 configuration objects to IA-stations. They encompass:

- 1157 • instructions about cryptographic algorithms,
- 1158 • credentials and trust anchors,
- 1159 • instructions to interpret the outcome of peer entity authentication while enforcing resource
1160 access controls, and
- 1161 • access control rules and permissions.

1162 This security configuration uses NETCONF/YANG request/response exchanges, as follows.

- 1163 • The to-be-configured IA-stations act in NETCONF server role with respect to their security
1164 configuration.
- 1165 • A NETCONF client is responsible for setting-up IA-stations for security. This NETCONF
1166 client possesses information about the security relationship to be established during security
1167 configuration or about the expectations on the IA-stations in a Configuration Domain. It can
1168 be implemented as part of an interactive or automated process (for example an engineering
1169 tool, or CNC operation). As an implication, the security configuration includes options for
1170 interactive and automated setup, i.e. security configuration is done by human or non-human
1171 actors.

1172 NOTE NETCONF notifications can also be used to recognize events such as a near-term end-of-life of
1173 certificate objects, especially EE certificate objects (see IETF RFC 4210, 3.1.1).

- 1174 • The security configuration exchanges supply deployment-specific objects (trust anchors,
1175 credentials, etc.) to IA-stations and manages them. IA-stations that are in factory default
1176 state can only possess manufacturer-specific security objects (trust anchors, credentials,
1177 etc.) when booting initially. The protected NETCONF/YANG exchanges with IA-stations that
1178 are in factory default state are outlined in 4.8.3 to 4.8.6.

1179 **4.8.3 NETCONF/YANG processing**

1180 Securing NETCONF/YANG resources (for example, NETCONF sessions or managed objects)
1181 on NETCONF servers is specified by IETF RFC 6241 (NETCONF). Therefore, message
1182 exchange protection between NETCONF clients and servers as well as resource access
1183 authorization by NETCONF servers are utilized.

- IETF RFC 7589 and draft-ietf-netconf-over-tls13 (NETCONF-over-TLS) specify a solution to protect NETCONF message exchanges by TLS.
- IETF RFC 8341 (NACM) specifies three access control points, covering the request/response and notification model in NETCONF according to IETF RFC 8341, 2.1.

NETCONF servers enforce security as shown in Figure 9. The processing steps are executed upon the current configuration of the NETCONF server's YANG modules.

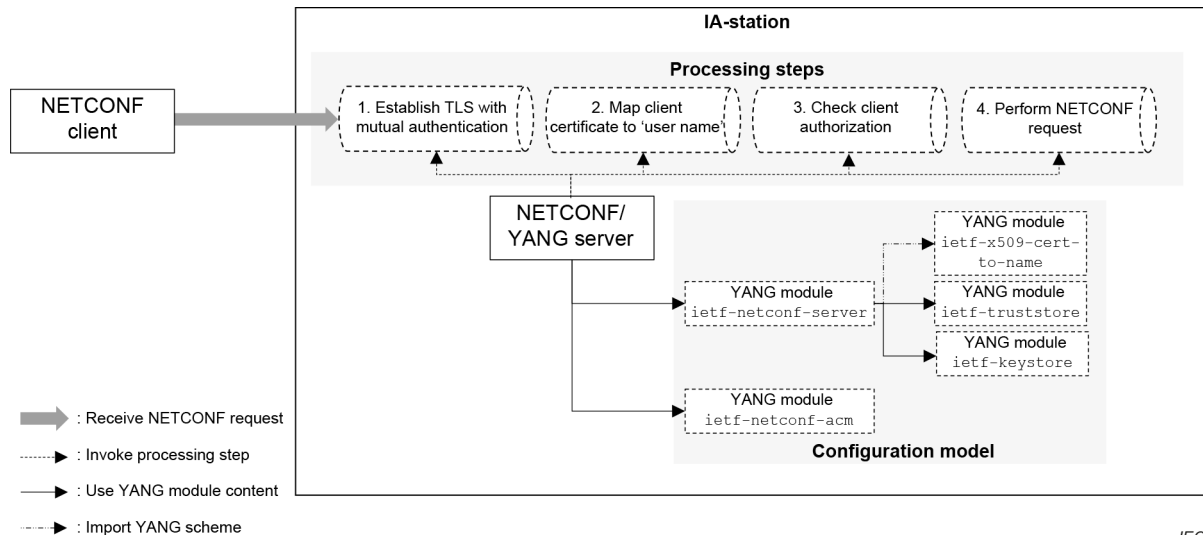


Figure 9 – NETCONF/YANG security processing steps

The processing steps on the side of NETCONF servers are as follows.

- 1) Establish a TLS connection with mutual authentication: the NETCONF server acts as TLS server and awaits connection requests of NETCONF clients (TLS clients). At the beginning of the TLS handshake, the TLS client and server negotiate the TLS protocol version to be used. During the TLS handshake the NETCONF server authenticates itself towards the NETCONF client by a credential from its ietf-keystore YANG module. In addition, the NETCONF server challenges the NETCONF client for authentication and verifies its authentication by trust anchors in its ietf-truststore YANG module according to 6.3.4. A successful mutual authentication is a prerequisite for proceeding to the next step.
- 2) Map the client certificate to a username: the NETCONF server maps the authenticated TLS client certificate to a "NETCONF username"³ by applying an ordered list of mapping instructions. These instructions are provided in its ietf-x509-cert-to-name YANG module. The applicable list item is identified by matching its configured fingerprint (according to IETF RFC 7589, Clause 7) against the certification path that was used for TLS client authentication (an end entity certificate or a CA certificate). According to the map type of the identified list item, the NETCONF server determines the "NETCONF username". This can be done by extracting information from the end entity certificate of the NETCONF client. A successful certificate-to-"NETCONF username" mapping is a prerequisite for proceeding to the next step.
- 3) Check client authorization: the NETCONF server checks if the NETCONF client has the permission to access the requested NETCONF/YANG resource based on its "NETCONF username" and the access control rules available in its ietf-netconf-acm YANG module. See 4.8.4 for more information about NETCONF/YANG access control. A successful authorization is a prerequisite for proceeding to the next step.
- 4) Perform NETCONF request: if all preceding steps succeeded, the NETCONF server performs the NETCONF request.

³ In this document and, in almost all cases, NETCONF username values do not represent human users.

1218 4.8.4 NETCONF/YANG access control

1219 NACM defines a YANG information model for describing permitted/denied access operations.
 1220 NETCONF servers are responsible for enforcing access control to their resources according to
 1221 the information in their ietf-netconf-acm YANG modules. NACM allows the description of
 1222 access-controlled resources in terms of NETCONF protocol operations, nodes in YANG
 1223 datastores or types of notification events. NACM uses character strings to represent the subject
 1224 actors, i.e. NETCONF clients. These character strings are known as "NETCONF username".
 1225 The NACM access control information of a NETCONF server is created, updated, and deleted
 1226 per IA-station. The management of this information happens along the IA-station lifecycle, for
 1227 example, manufacturing, bootstrapping, operation, maintaining, re-owning, destructing.
 1228 Moreover, the management of the NACM access control information itself is subject to NACM
 1229 access control.

1230 This document employs multiple YANG data models for fulfilling its purposes. This extends
 1231 beyond the above identified YANG modules (see 4.8.3). The NETCONF server on an IA-station
 1232 enforces access control for NETCONF/YANG resources. To meet this objective, the NETCONF
 1233 server on an IA-station is supplied with access control information for the used
 1234 NETCONF/YANG resources. NACM is employed for this purpose and profiles default access
 1235 control information for the NETCONF/YANG resources (see 6.3.2.2). This relieves other
 1236 organizations or individuals for example, manufacturers, integrators, operators, owners from
 1237 being responsible to create NACM access control information for the respective
 1238 NETCONF/YANG resources.

1239 NACM relies on character strings (known as "NETCONF username") to refer to clients. NACM
 1240 access control information as specified in this document, populates the "NETCONF username"
 1241 character strings in NACM with role names specified in 6.3.2.1.4 c). This allows to create default
 1242 NACM information without knowing actual names of individual entities. A role name can refer
 1243 to 0, 1 or more individual entities. It is the responsibility of users to assign role names to
 1244 individual entities. This happens by binding the assigned role names to the credentials of
 1245 individual entities. The current form to express this binding is a role extension in the identity
 1246 certificates of end entities defined in this document. These are NETCONF clients, i.e. these
 1247 role extensions appear in the end entity certificates of LDevID credentials for NETCONF clients.

1248 As initial step NACM maps the NETCONF username to a set of groups. The set of groups
 1249 determines the set of rules to be applied for access-controlled resources.

1250 4.8.5 Identity checking

1251 IETF RFC 7589 (NETCONF-over-TLS) specifies that NETCONF clients check the identity of
 1252 NETCONF servers (IETF RFC 7589, Clause 6) and that NETCONF servers verify the identity
 1253 of NETCONF clients (IETF RFC 7589, Clause 7).

1254 The NETCONF server identity check happens inside NETCONF clients. It matches an actual
 1255 against an expectation.

- 1256 • The actual server identity is established by the end entity certificate of the NETCONF server
 1257 (authenticated by means of TLS).
- 1258 • The expectations on server identity are established by the information that is used to
 1259 connect to the NETCONF server.

1260 IETF RFC 7589 refers to IETF RFC 6125, Clause 6, for the details of retrieving the actual and
 1261 comparing it against the expected.

1262 The NETCONF client identity check happens inside NETCONF servers. It also matches an
 1263 actual against an expectation.

- 1264 • The actual client identity is established by the end entity certificate of the NETCONF client
 1265 (authenticated by means of TLS).

- 1266 • The expectations on client identity are established by the contents of the ietf-netconf-acm
1267 and ietf-x509-cert-to-name YANG modules.

1268 The details of this check are subject to the requested NETCONF operation. IETF RFC 7589,
1269 Clause 7, specifies the mapping of an authenticated client certificate to a "NETCONF username"
1270 whose permissions are then enforced by IETF RFC 8341 (NACM). More information is provided
1271 in 4.8.3, steps 2 and 3.

1272 **4.8.6 Secure device identity**

1273 **4.8.6.1 Device identity**

1274 The term 'device' originates from IEEE Std 802.1AR-2018. It matches the term IA-station in this
1275 document.

1276 The device identity refers to a set of information items about a device that:

- 1277 • describes a device as a physical or virtual entity in a distributed system (identifier and
1278 attribute information);
- 1279 • is used by a device to describe itself as such entity (identifier and attribute information);
- 1280 • allows to interact with this device (addressing information, i.e. a specific identifier class).

1281 The targeted use case, for example application data exchanges, configuration exchanges,
1282 inventory, or ordering, determines the required amount of identity information about a device.

1283 The device identity of any single IA-station encompasses:

- 1284 • MAC addresses, IP addresses, TCP ports, DNS names;
- 1285 • ietf-hardware YANG module contents (IETF RFC 8348, Clause 3 and 7.1).

1286 **4.8.6.2 Verifiable device identity**

1287 Certain aspects of device identity are verified before relying on them during online interactions.
1288 These are examples.

- 1289 • DNS names or IP addresses are used to call the management entity of an IA-station, i.e. its
1290 NETCONF/YANG server. Their value represents the caller's expectation on the identity of
1291 their responder in network communications. Verification of the responder's identity helps
1292 defeat DNS spoofing, component impersonation and man-in-the-middle attacks. This is
1293 specified by IETF RFC 7589 and described in IETF RFC 6125, Clause 6. Passing this check
1294 is a prerequisite before NETCONF application exchanges can happen.
- 1295 • mfg-name values in instances of the ietf-hardware YANG module. These values make claims
1296 about the IA-station manufacturer. Their verification is a means to protect against
1297 counterfeiting.

1298 The verification of IA-station identity happens according to a model that is fully specified by this
1299 document. That verification can be done in a manufacturer-agnostic manner. This verification
1300 is important before supplying locally significant credentials especially LDevID to IA-stations that
1301 are in factory-default state.

1302 **4.8.6.3 Verification support mechanisms**

1303 **4.8.6.3.1 General**

1304 Subclause 4.8.6.3 considers mechanisms that support device identity verification during online
1305 interactions with IA-stations.

4.8.6.3.2 Secure transports

Sending information in plain form over a protected channel, e.g. ietf-hardware YANG module contents via NETCONF-over-TLS protects the transferred information during its transit through the network but does not vouch for the correctness of the received information, e.g. the mfg-name value.

4.8.6.3.3 Secure information

Protecting information objects by means of a cryptographic authentication code or digital signature enables verification of the authenticity and integrity of that information. These cryptographic authentication codes can use symmetric or asymmetric schemes. In case of asymmetric schemes, raw and self-signed public keys are distinguished from CA-signed public keys.

Asymmetric schemes with CA-signed public keys are preferable for the verifiable device identity use case: claimants and verifiers share a public key; the claimant possesses the corresponding private key. The establishment and storage of shared public keys uses public key certificates. For this approach self-signed CA certificates are to be established in an authentic manner. The number of self-signed CA certificates is independent from the number of verifiers (CNCs) as well as claimants (IA-stations).

4.8.6.3.4 IDevID and LDevID credentials

IDeVID and LDeVID credentials are specified by IEEE Std 802.1AR-2018. These objects are comprised of a certification path and a private key. The certification path encompasses an end entity certificate which contains verifiable device identity in a CA-signed form. The device identity verification happens after validating the certification path (IETF RFC 5280, Clause 6) and checking the proof-of-possession for the private key. The certification path validation demands trust anchors as input arguments (IETF RFC 5280, 6.1.1 input argument (d)).

Two types of credentials are distinguished by IEEE Std 802.1AR-2018:

- IDeVIDs are issued by device manufacturers. They represent an initial identity as it is known at device production-time. The initial device identity is not locally significant: it cannot contain deployment-specific information such as DNS names or IP addresses.
- LDeVIDs are issued by other actors, e.g. a device user. They represent a locally significant device identity: they can contain deployment-specific information, e.g. DNS names or IP addresses.

IEEE Std 802.1AR-2018, Clause 6, uses signature suites to describe the subject public key and the signature fields in IDeVID and LDeVID certification paths. This notion is different from TLS cipher suites.

NOTE IDeVID and LDeVID credentials also serve purposes beyond secure device identity, for instance the realization of secure transports. This facilitates the use case of NETCONF/YANG security setup from factory default state.

4.8.6.3.5 IDevID items beyond IEEE Std 802.1AR-2018

IEEE Std 802.1AR-2018 allows verification of the following identity items:

- certificate issuer (not necessarily: manufacturer) by issuer field (data type: ASN.1 Name), and
- if present: device instance by serialNumber value (data type: ASN.1 PrintableString).

NOTE 1 IEEE Std 802.1AR-2018 represents the initial device identity as an optional serialNumber attribute (OID 2.5.4.5) in the subject field of the EE certificate. This value is unique within the domain of significance of the EE certificate issuer.

NOTE 2 This verification can happen after certification path validation and the proof-of-possession checking for the private key.

The following bullet points describe options beyond IEEE Std 802.1AR-2018 for verifying the device identity of IA-stations in factory default state. It also identifies informational items for the corresponding checks:

- IA-station manufacturer check: using names that identify IA-station manufacturers, e.g. mfg-name in ietf-hardware YANG module,
- IA-station type check: using attributes that identify IA-station types, e.g. model-name, hw-revision, description in ietf-hardware YANG module, and
- IA-station instance check: using values that identify IA-station instances, e.g. serial-num in ietf-hardware YANG module.

The following model described in the bullet points applies to the verification of the initial device identity of IA-stations:

- the set of to-be-conducted checks is determined by IA-station and CNC users,
- an IA-station uses IDevID credentials to prove its device identity. The checking happens by means of online interactions in the operational network. It happens automatically and is done by CNCs. This does not depend on Configuration Domain external repositories, and
- other stakeholders, e.g. middleware/application consortia or individual manufactures are allowed to additionally express information items in IDevID credentials to reflect their device identity model. CNCs do not assess such additional information.

4.8.6.3.6 Device identity representation in IDevID and LDevID credentials

The best practices for representing verifiable device identity information in IDevID and LDevID credentials (see 6.3.3.2.2 for more information) are as follows.

- Corresponding information (actual values or references to them) appears in EE certificates:
 - IDevID EE certificates bind initial device identity items that are known by the device manufacturer at production time, e.g. mfg-name.
 - LDevID EE certificates bind locally significant device identity items that are known by other actors such as device users, e.g. DNS names or IP addresses. They can also bind initial device identity information.
 - Items that encode device naming information appear in the subjectAltName extension.
- NOTE This is specified in IETF RFC 5280, 4.2.1.6. It is further explained in IETF RFC 6125, 2.3.
- A binding can take one of following forms. Multiple forms can appear in one EE certificate:
 - by-value: the verifiable device identity information is represented by its value inside the IDevID resp. LDevID EE certificate. Examples are:
 - the product serialNumber in IDevID credentials (IEEE Std 802.1AR-2018), and
 - the hostname of the NETCONF/YANG server in LDevID credentials (IETF RFC 6125, Clause 6).
 - by-ref: the verifiable device identity information is represented by a reference inside the IDevID resp. LDevID EE certificate, not by its value.
 - The actual value can be provided by the device itself or by a device-external source.
 - If it is provided in form of an unprotected information object, then the reference object that is embedded to EE certificates includes a digest value.

5 Conformance

5.1 General

A claim of conformance to this document is a claim that the behavior of an implementation of an IA-station (see 5.5, 5.6) with its Bridge components (see 5.7, 5.8) and end station

1397 components (see 5.9, 5.10) meets the mandatory requirements of this document and may
1398 support options identified in this document.

1399 **5.2 Requirements terminology**

1400 The verbal forms for required expressions of provisions follow the conventions below.

- 1401 a) Requirements terminology is provided in the ISO/IEC Directives Part 2:2021, Clause 7. This
1402 document can be found at www.iec.ch/members_experts/refdocs.
- 1403 b) The Profile Conformance Statement (PCS) proformas (see Annex A) reflect the occurrences
1404 of the words "shall," "may," and "should" within this document.
- 1405 c) This document avoids needless repetition and apparent duplication of its formal
1406 requirements by using "is", "is not", "are", and "are not" for definitions and the logical
1407 consequences of conformant behavior. Behavior that is permitted but is neither always
1408 required nor directly controlled by an implementer or administrator, or whose conformance
1409 requirement is detailed elsewhere, is described by "can". Behavior that never occurs in a
1410 conformant implementation or system of conformant implementations is described by
1411 "cannot". The word "allow" is used as a replacement for the phrase "support the ability for",
1412 and the word "capability" means "can be configured to".

1413 **5.3 Profile conformance statement (PCS)⁴**

1414 The supplier of an implementation that is claimed to conform to this document shall provide the
1415 information necessary to identify both the supplier and the implementation and shall complete
1416 a copy of the PCS proforma provided in Annex A.

1417 **5.4 Conformance classes**

1418 This document includes conformance requirements and options that are related to an entire
1419 station, as well as conformance requirements and options that are related to single Bridge or
1420 end station components within an IA-station. Figure 10 illustrates this conformance model.

⁴ Copyright release for the PCS: Users of this document may freely reproduce the PCS contained in this document so that it can be used for its intended purpose.

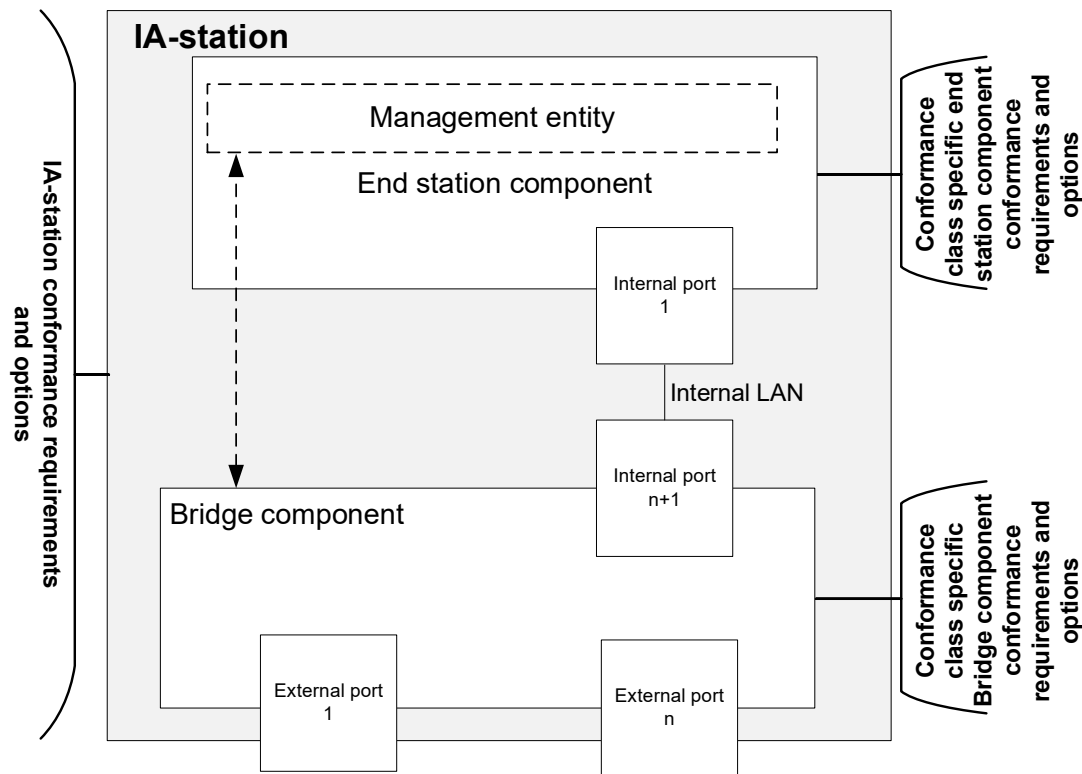


Figure 10 – IA-station conformance model

This document supports a variety of industrial use cases. In some of these use cases, support of certain TSN features can be mandatory, while in others, supporting these features can lead to non-optimal implementations. Therefore, this document defines two conformance classes that are applicable both to Bridge components and end station components. Conformance Class A (ccA) is feature-rich, i.e. tailored to use cases requiring support of many TSN-IA features. Conformance Class B (ccB) targets implementations that are more resource constrained. The details for the conformance classes are specified in 5.7 and 5.8 for Bridge components, and in 5.9 and 5.10 for end station components.

NOTE 1 It is the responsibility of the IA-station manufacturer to carefully consider the implications of mixing ccA and ccB Bridge components and end station components in a single IA-station.

NOTE 2 It is the responsibility of the user to carefully consider the implications of mixing ccA and ccB Bridge components and end station components in a single Configuration Domain.

NOTE 3 Any Bridge compliant to this document is an IA-station. Any IA-station contains a management entity (i.e. an end station component).

5.5 IA-station requirements

5.5.1 IA-station PHY and MAC requirements for external ports

External ports of IA-stations that claim conformance to this document shall meet the following requirements.

- Support the Media Access Control (MAC) service specification according to IEEE Std 802.3-2022, Clause 2.
- Support the Media Access Control (MAC) frame and packet specifications according to IEEE Std 802.3-2022, Clause 3, especially the MAC Client Data field size according to IEEE Std 802.3-2022, 3.2.7, item c).
- Support Layer Management according to IEEE Std 802.3-2022, 5.2.4.

- d) Implement at least one IEEE Std 802.3-2022 MAC that shall operate in full-duplex mode, and associated IEEE Std 802.3-2022 PHY with a data rate of at least one of the following speeds: 10 Mb/s, 100 Mb/s, 1 000 Mb/s, 2.5 Gb/s, 5 Gb/s, or 10 Gb/s together with the corresponding managed objects:
 - 1) 10BASE-T1L MAU type according to IEEE Std 802.3-2022, Clauses 22 and 146,
 - 2) 100BASE-TX or 100BASE-FX MAU types according to IEEE Std 802.3-2022 Clauses 21, 22, 24, 25, 26 and IEEE Std 802.3-2022 23A, 28A, 28B, 28C, and 28D,
 - 3) 1000BASE-T or 1000BASE-SX MAU types according to IEEE Std 802.3-2022 Clauses 28, 34, 35, 36, 37, 38, and 40, and IEEE 802.3-2022 Annexes 28A, 28B, 28C, and 28D,
 - 4) 2.5GBASE-T or 5GBASE-T MAU types according to IEEE Std 802.3-2022, Clauses 28, 125, and 126, and IEEE Std 802.3-2022 Annexes 28A, 28B, 28C, and 28D,
 - 5) 2.5GBASE-T1 or 5GBASE-T1 MAU types according to IEEE Std 802.3-2022, Clause 149,
 - 6) 10GBASE-T or 10GBASE-SR MAU types according to IEEE Std 802.3-2022, 1462 Clauses 44, 46, 49, 51, 52, 55, and IEEE Std 802.3-2022, Annexes 48A and 55A,
 - 7) 10GBASE-T1 MAU type according to IEEE Std 802.3-2022, Clause 149,
 - 8) 100BASE-T1 MAU type according to IEEE Std 802.3-2022, Clause 96, and
 - 9) 1000BASE-T1 MAU type according to IEEE Std 802.3-2022, Clause 97.
- e) Support the YANG features and nodes of the ieee802-ethernet-interface module according to 6.4.9.2.1.
- f) Implement Ethernet support for time synchronization protocols according to IEEE Std 802.3-2022, Clause 90 as amended by IEEE Std 802.3cx™-2023 and IEEE Std 802.3de™-2022 with the aTimeSyncSelectionDdmp set to 'symbol-after-sfd' according to IEEE Std 802.3cx-2023, 30.13.1.14.

Clauses and subclauses not mentioned can be implemented but are not obligatory.

5.5.2 IA-station topology discovery requirements

IA-stations for which a claim of conformance to this document is made shall support the following list of requirements.

- a) The required capabilities according to IEEE Std 802.1AB-2016, 5.3 and IEEE Std 802.1ABcu™-2021, 5.3.
- b) Topology discovery and verification according to 6.5.
- c) The YANG features and nodes of the ieee802-dot1ab-lldp module according to 6.4.9.2.2.

5.5.3 IA-station requirements for time synchronization

These requirements are related to the entire IA-station with all its PTP Instances and PTP Ports. IA-stations for which a claim of conformance to this document is made shall support the following list of requirements.

- a) PTP Instance requirements according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1AS-2020/Cor 1-2021, IEEE Std 802.1ASdr-2024, IEEE Std 802.1ASdn™-2024, and IEEE Std 802.1ASdm-2024, 5.4.1 items a) through h).

NOTE A gPTP domain in a PTP End Instance can be used for Global Time, Working Clock, or both.

- b) PTP Instance defaults for BTCA according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1AS-2020/Cor 1-2021, IEEE Std 802.1ASdr-2024, IEEE Std 802.1ASdn-2024, and IEEE Std 802.1ASdm-2024, 5.4.2a.
- c) Timing and synchronization management according to IEEE Std 802.1AS-2020, 5.4.2 items j) and k).
- d) PTP Instance requirements according to 6.2.2.
- e) PTP Protocol requirements according to 6.2.3.

- 1494 f) Clock control system requirements according to 6.2.4.
- 1495 g) Error generation limits according to 6.2.5.
- 1496 h) PTP Instance synchronization status according to 6.2.6.
- 1497 i) The transmission of the Drift_Tracking TLV according to IEEE Std 802.1AS-2020 as
1498 amended by IEEE Std 802.1AS-2020/Cor 1-2021, IEEE Std 802.1ASdr-2024, IEEE Std
1499 802.1ASdn-2024, and IEEE Std 802.1ASdm-2024, 5.4.2 item n).
- 1500 j) External port configuration capability according to IEEE Std 802.1AS-2020 as amended by
1501 IEEE Std 802.1AS-2020/Cor 1-2021, IEEE Std 802.1ASdr-2024, IEEE Std 802.1ASdn-2024,
1502 and IEEE Std 802.1ASdm-2024, 5.4.2 f).
- 1503 k) MAC-specific timing and synchronization methods for IEEE Std 802.3 full-duplex links
1504 according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1AS-2020/Cor 1-2021,
1505 IEEE Std 802.1ASdr-2024, IEEE Std 802.1ASdn-2024, and IEEE Std 802.1ASdm-2024, 5.5
1506 items a) through e) and item j), and 11.2.17.
- 1507 l) The YANG features and nodes of the:
 - 1508 i) ieee1588-ptp-tt module according to 6.4.9.2.3.1,
 - 1509 ii) ieee802-dot1as-gtp module according to 6.4.9.2.3.2, and
 - 1510 iii) ieee802-dot1as-hs module according to 6.4.9.2.3.3.
- 1511 m) The message timestamp point according to IEEE Std 802.1AS-2020, 11.3.9.
- 1512 n) The Common Mean Link Delay Service (CMLDS) according to IEEE Std 802.1AS-2020,
1513 11.2.17.
- 1514 o) The descriptionDS according to 6.2.13 and IEEE Std 1588-2019, 8.2.5.
- 1515 p) gtpCapableStateMachinesEnabled is FALSE according to IEEE Std 802.1ASdm-2024,
1516 5.4.2a and 14.8.54a.

1517 NOTE Setting gtpCapableStateMachinesEnabled to FALSE allows transmission of signaling message to be
1518 disabled by design.

1519 **5.5.4 Management requirements for non-proxied IA-stations**

1520 **5.5.4.1 Secure management exchanges**

1521 IA-stations which contain a ccA Bridge component or a ccA end station component and for
1522 which a claim of conformance to this document is made shall support the following list of
1523 requirements.

- 1524 a) NETCONF server functionality according to IETF RFC 6241 including:
 - 1525 1) candidate configuration capability as described in IETF RFC 6241, 8.3,
 - 1526 2) rollback-on-error capability as described in IETF RFC 6241, 8.5, and
 - 1527 3) validate capability as described in IETF RFC 6241, 8.6.
- 1528 b) NETCONF-over-TLS server supporting TLS version 1.2 according to 6.3.2.1.1 and 6.3.2.1.4.
- 1529 c) Secure Device Identity according to 6.3.3 and IEEE Std 802.1AR-2018, 5.3 a), using the
1530 signature suite in IEEE Std 802.1AR-2018 9.2, 5.3 d), and 5.3 i).
- 1531 d) PKIX according to 6.3.2.1.5 and IETF RFC 5280, 4.1, 4.2.1.1-3, 4.2.1.6, 6.1 and 6.2.
- 1532 e) NACM (IETF RFC 8341) supporting four different roles according to 6.3.2.1.5 c).
- 1533 f) The YANG features and nodes of the:
 - 1534 1) ietf-keystore module according to 6.4.9.2.4.1,
 - 1535 2) ietf-netconf-acm module according to 6.4.9.2.4.2 and,
 - 1536 3) ietf-truststore according to 6.4.9.2.4.3.
- 1537 g) NETCONF Event Notifications according to IETF RFC 5277 including operations according
1538 to IETF RFC 5277, Clause 2.

- 1539 h) Dynamic Subscriptions to YANG Events and Datastores over NETCONF according to
1540 6.4.7.7.
- 1541 i) NETCONF Extensions to support the Network Management Datastore Architecture (NMDA)
1542 as described in IETF RFC 8526.
- 1543 j) DHCP client according to IETF RFC 2131, 4.1, 4.2, and 4.4.
- 1544 k) At least one of the following asymmetric key pair generation methods:
1545 1) component-internal generation according to 6.3.4.3;
1546 2) component-external generation according to 6.3.4.3;
- 1547 l) Storage of at least one IDevID credential according to 6.3.3.4.2.5 and one LDevID-
1548 NETCONF credential according to 6.3.4.1.
- 1549 m) If the IA-station supports CNC functionality, that IA-station shall also support the following
1550 list of requirements.
1551 1) TSN CNC station requirements according to IEEE Std 802.1Q-2022, 5.29.
1552 2) NETCONF-over-TLS server and related client functionality 5.5.4.2.
1553 3) The common YANG modules, features, and nodes according to 6.4.9.2.
1554 4) The optional YANG modules, features, and nodes according to 6.4.9.3.
1555 5) Support the use of at least one customer VLAN Identifier for an isolation VLAN and one
1556 VLAN identifier for a default VLAN.
1557 6) Support CUC/CNC YANG modules, features and nodes according to 6.4.9.4.
- 1558 n) If the IA-station supports CUC functionality, that IA-station shall also support the following
1559 list of requirements.
1560 1) Support NETCONF-over-TLS client functionality with client related security requirements
1561 according to 5.5.4.2.
1562 2) The TSN UNI YANG module, features, and nodes according to 6.4.9.4.1.
1563 3) The ietf-netconf-client module according to 6.4.9.4.1.
- 1564 o) If the IA-station supports proxy functionality, that IA-station shall also support the
1565 iecieeee60802-proxy YANG module according to 6.4.10.8.8.
- 1566 IA-stations which contain a ccA Bridge component or a ccA end station component and for
1567 which a claim of conformance to this document is made should support internal key generation
1568 according to 6.3.4.3.2.

1569 **5.5.4.2 IA-station management YANG modules**

- 1570 IA-stations which contain a ccA Bridge component or a ccA end station component and for
1571 which a claim of conformance to this document is made shall support the YANG features and
1572 nodes for IA-station management of the:
- 1573 a) ietf-system-capabilities module according to 6.4.9.2.5.1,
- 1574 b) ietf-yang-library module according to 6.4.9.2.5.2,
- 1575 c) ietf-yang-push module according to 6.4.9.2.5.3,
- 1576 d) ietf-notification-capabilities module according to 6.4.9.2.5.4,
- 1577 e) ietf-subscribed-notifications module according to 6.4.9.2.5.5,
- 1578 f) diagnostics using YANG-Push subscriptions according to 6.4.7,
- 1579 g) ietf-netconf-monitoring module according to 6.4.9.2.5.6,
- 1580 h) ietf-system module according to 6.4.9.2.5.7,
- 1581 i) ietf-hardware module according to 6.4.9.2.5.8,
- 1582 j) ietf-interfaces module according to 6.4.9.2.5.9,
- 1583 k) ieee802-dot1q-bridge module according to 6.4.9.2.5.10,

- l) iecieee60802-ethernet-interface module according to 6.4.9.2.5.11,
- m) ietf-netconf-server according to 6.4.9.2.5.12,
- n) iecieee60802-bridge according to 6.4.9.2.5.11,
- o) ietf-subscribed-notifications according to 6.4.9.2.5.13,
- p) iecieee60802-subscribed-notifications according to 6.4.9.2.5.13, and
- q) iecieee60802-ia-station according to 6.4.9.2.5.11.
- r) ieee802-dot1q-rstp-bridge according to 6.4.9.2.5.14.

5.5.4.3 Digital data sheet

IA-stations for which a claim of conformance to this document is made shall provide a 60802 instance data file according to 6.4.8. The instance data file shall contain at least the YANG nodes of 6.4.9 that are marked with [m]. Nodes marked with [c] shall be included if the associated feature is supported.

NOTE It is the user's responsibility to ensure that the filename is unique by using a standardized mechanism (for example, GUID, URL, or ReverseDomainName).

5.6 IA-station options

5.6.1 IA-station PHY and MAC options for external ports

IA-stations for which a claim of conformance to this document is made may support the following list of requirements.

- a) Power over Ethernet (PoE) over 2 Pairs according to IEEE Std 802.3-2022, Clause 33.
- b) Power Interfaces according to IEEE Std 802.3-2022, Clause 104.
- c) Power over Ethernet according to IEEE Std 802.3-2022 Clause 145.

5.6.2 IA-station options for time synchronization

IA-stations for which a claim of conformance to this document is made may support the following list of requirements.

- a) The media-independent TimeTransmitter capability according to IEEE Std 802.1AS-2020, 5.4.2 item b) as amended by IEEE Std 802.1ASdr-2024.
- b) Grandmaster PTP Instance capability according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdr-2024 and IEEE Std 802.1ASdm-2024, 5.4.2 item c).
- c) More than one PTP port as a PTP Relay Instance according to IEEE Std 802.1AS-2020, 5.4.2 item d).
- d) Hot standby redundancy requirements according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdm-2024, 5.4.2, item m).

5.6.3 IA-station options for management

IA-stations for which a claim of conformance to this document is made may support the following list of requirements.

- a) Writable-Running capability according to IETF RFC 6241, 8.2.
- b) Confirmed Commit capability according to IETF RFC 6241, 8.4.
- c) Distinct Startup capability according to IETF RFC 6241, 8.7.
- d) URL capability according to IETF RFC 6241, 8.8.
- e) XPath capability according to IETF RFC 6241, 8.9.
- f) NETCONF-over-TLS server supporting TLS version 1.2, according to 6.3.2.1.2.
- g) NETCONF-over-TLS server supporting TLS version 1.3, according to 6.3.2.1.3.

h) PKIX according to IETF RFC 5280, 4.2.1.13, Clause 5 and 6.3.

5.6.4 Proxy target options

IA-stations which contain only ccB Bridge or end station components and for which a claim of conformance to this document is made may support the following list of requirements.

- a) Secure management exchanges according to 5.5.4.1, a) through m) including support of IA-station management YANG modules according to 5.5.4.2.
- b) Secure management exchanges using a management proxy according to 6.4.5.5 including a digital datasheet according to 5.5.4.3 including the YANG features and nodes of the iecieeee60802-proxy-target module according to 6.4.10.8.9.

5.7 Bridge component requirements

5.7.1 Common Bridge component requirements

A Bridge component implementation of any conformance class for which a claim of conformance to this document is made shall support the following list of requirements.

- a) C-VLAN component requirements according to IEEE Std 802.1Q-2022, 5.5 and 5.4 except item o) in IEEE Std 802.1Q-2022, 5.4.
- b) The use of Customer VLAN Identifiers (C-VID).
- c) FDB to contain Static and Dynamic VLAN Registration Entries for a minimum of 10 VIDs according to IEEE Std 802.1Q-2022, 8.8.

NOTE 1 An example use case for 10 VIDs: 2 VIDs for IA time-aware stream or IA stream traffic, 2 VIDs for IA time-aware stream or IA stream redundancy, 4 VIDs for IA traffic engineered non-stream or IA non-stream traffic, 1 VID for the isolation VLAN, and 1 VID for the Default VLAN (see 6.4.5.2).

- d) Translation of VIDs through support of the VID Translation Table or through support of both the VID Translation Table and Egress VID translation table on one or more Bridge Ports according to IEEE Std 802.1Q-2022, 6.9.
- e) The strict priority algorithm for transmission selection on each port for each traffic class according to IEEE Std 802.1Q-2022, 8.6.8.1.
- f) The capability to disable Priority-based flow control according to IEEE Std 802.1Q-2022, Clause 36, if it is implemented.
- g) The Priority Regeneration requirements according to IEEE Std 802.1Q-2022, 5.4.1, item o).
- h) MST according to IEEE Std 802.1Q-2022, 5.4.1.1 a) to i) and k) to o) and 6.4.2.4.
- i) TE-MSTID according to IEEE Std 802.1Q-2022, 5.5.2, 8.6, and 8.8.
- j) Spanning tree, VLAN, and TE-MSTID configuration according to 6.4.2.4.
- k) YANG features and nodes of the iecieeee60802-tt-mapping module according to 6.4.10.8.7.
- l) PSFP according to IEEE Std 802.1Q-2022, 8.6.5.2.1, with support for:
 - 1) at least 3 stream filters per port according to IEEE Std 802.1Q-2022, 8.6.5.3,
 - 2) at least one stream gate according to IEEE Std 802.1Q-2022, 8.6.5.4 with at least zero gate control list entries according to IEEE Std 802.1Q-2022, 12.31.1.4, and
 - 3) at least 3 flow meters per port according to IEEE Std 802.1Q-2022, 8.6.5.5.

NOTE 2 The flow meter parameter values Excess Information Rate (EIR) = 0, Excess burst size (EBS) = 0 and Color Mode (CM) = color_blind, as configurable by the CNC, are applicable to many industrial automation use cases. When CM = color_blind, DropOnYellow (IEEE Std 802.1Q-2022, 8.6.5.5, item h), MarkAllFramesRed (IEEE Std 802.1Q-2022, 8.6.5.1.3, item j), and MarkAllFramesRedEnable (IEEE Std 802.1Q-2022, 8.6.5.5, item i) are not used.

NOTE 3 For example, an implementation could contain one flow meter for broadcast traffic, one flow meter for multicast traffic and one flow meter for unicast traffic. This example can be realized using 49 mask and match stream identification functions per Port and 49 associated stream filter instances (one-to-one relationship) that identify and map unicast (1 stream identification function for identifying all frame with the group address bit cleared in the DA), broadcast (1 stream identification function for identifying all frames with all DA bits set) and

1674 multicast (47 stream identification function with the DA group address bit set and one other DA bit cleared for
1675 all 47 combinations).

1676 m) The YANG features and nodes for flow meter configuration according to 6.4.9.2.5.14.

1677 n) Stream identification component behaviors according to IEEE Std 802.1CB-2017, 5.3 and
1678 IEEE Std 802.1CBdbTM-2021, 5.5 d).

1679 **5.7.2 ccA Bridge component requirements**

1680 A Bridge component implementation for which a claim of conformance to ccA of this document
1681 is made shall support the following list of requirements.

1682 a) Common Bridge component requirements according to 5.7.1.

1683 b) At least 2 PTP Instances according to 5.5.3.

1684 c) Eight queues according to IEEE Std 802.1Q-2022, 8.6.6.

1685 d) Enhancements for scheduled traffic for data rates of 100 Mb/s and 1 Gb/s according to IEEE
1686 Std 802.1Q-2022, 5.4.1 items ac) and ad) including:

1687 1) TickGranularity of less than or equal to 10 ns according to IEEE Std 802.1Q-2022,
1688 8.6.9.4.16 and Table 12-32,

1689 2) support of the YANG features and nodes of the ieee802-dot1q-sched module according
1690 to 6.4.9.3.2, and

1691 3) support of the YANG features and nodes of the icieeee60802-sched-bridge module
1692 according to 6.4.9.3.3.

1693 e) Frame preemption according to IEEE Std 802.1Q-2022, 5.4.1 item ae), for data rates of
1694 100 Mb/s and 1 Gb/s, including:

1695 1) support of Interspersing Express Traffic with preemptable traffic according to IEEE
1696 Std 802.3-2022, Clause 99 as amended by IEEE Std 802.3de-2022, including support of
1697 the Additional Ethernet Capabilities for TLV in an LLDPDU to indicate supported
1698 functions of frame preemption according to IEEE Std 802.3-2022, 79.3.7, and

1699 2) support of the YANG features and nodes of the ieee802-dot1q-preemption module
1700 according to 6.4.9.3.4.

1701 **5.7.3 ccB Bridge component requirements**

1702 A Bridge component implementation for which a claim of conformance to ccB of this document
1703 is made shall support the following list of requirements.

1704 a) Common Bridge component requirements according to 5.7.1.

1705 b) At least 1 PTP Instance according to 5.5.3.

1706 c) At least four queues according to IEEE Std 802.1Q-2022, 8.6.6.

1707 **5.8 Bridge component options**

1708 **5.8.1 Common Bridge component options**

1709 A Bridge component implementation of any conformance class for which a claim of conformance
1710 to this document is made may support the operation of the credit-based shaper algorithm
1711 according to IEEE Std 802.1Q-2022, 8.6.8.2 on any Port as the transmission selection algorithm
1712 for at least 4 traffic classes including support of the YANG features and nodes of the ieee802-
1713 dot1q-cbsa-bridge module according to 6.4.9.3.5.

1714 **5.8.2 ccA Bridge component options**

1715 A Bridge component implementation for which a claim of conformance to ccA of this document
1716 is made may support the following list of requirements.

1717 a) Any or none of the common Bridge component options according to 5.8.1.

1718 b) More than 2 PTP Instances according to 5.5.3.

- c) Enhancements for scheduled traffic for data rates of 10 Mb/s, 2,5 Gb/s, 5 Gb/s, and 10 Gb/s according to IEEE Std 802.1Q-2022, 5.4.1 items ac) and ad) including:

- 1) TickGranularity of less than or equal to 10 ns according to IEEE Std 802.1Q-2022, 8.6.9.4.16 and Table 12-32,
- 2) support of the YANG features and nodes of the `ieee802-dot1q-sched` module according to 6.4.9.3.2, and
- 3) support of the YANG features and nodes of the `ieciieee60802-sched-bridge` module according to 6.4.9.3.3.

- d) Frame preemption according to IEEE Std 802.1Q-2022, 5.4.1 item ae), for data rates of 10 Mb/s, 2,5 Gb/s, 5 Gb/s, and 10 Gb/s, including:

NOTE IEEE Std 802.3de-2022, 99.1, comprises 10 Mb/s.

- 1) support of Interspersing Express Traffic with preemptable traffic according to IEEE Std 802.3-2022, Clause 99 as amended by IEEE Std 802.3de-2022, including support of the Additional Ethernet Capabilities for TLV in an LLDPDU to indicate supported functions of frame preemption according to IEEE Std 802.3-2022, 79.3.7, and
- 2) support of the YANG features and nodes of the `ieee802-dot1q-preemption` module according to 6.4.9.3.4.

5.8.3 ccB Bridge component options

A Bridge component implementation for which a claim of conformance to ccB of this document is made may support the following list of requirements.

- a) Any or none of the common Bridge component options according to 5.8.1.
- b) Up to eight queues according to IEEE Std 802.1Q-2022, 8.6.6.
- c) More than 1 PTP Instance according to 5.5.3.
- d) Enhancements for scheduled traffic according to IEEE Std 802.1Q-2022, 5.4.1, items ac) and ad) including:
 - 1) TickGranularity of less than or equal to 10 ns according to IEEE Std 802.1Q-2022, 8.6.9.4.16 and Table 12-32,
 - 2) support of the YANG features and nodes of the `ieee802-dot1q-sched` module according to 6.4.9.3.2, and
 - 3) support of the YANG features and nodes of the `ieciieee60802-sched-bridge` module according to 6.4.9.3.3.
- e) Frame preemption according to IEEE Std 802.1Q-2022, 5.4.1, item ae), including:
 - 1) support of Interspersing Express Traffic with preemptable traffic according to IEEE Std 802.3-2022, Clause 99 as amended by IEEE Std 802.3de-2022, including support of the Additional Ethernet Capabilities for TLV in an LLDPDU to indicate supported functions of frame preemption according to IEEE Std 802.3-2022, 79.3.7, and
 - 2) support of the YANG features and nodes of the `ieee802-dot1q-preemption` module according to 6.4.9.3.4.

5.9 End station component requirements

5.9.1 Common end station Component requirements

An end station component implementation of any conformance class for which a claim of conformance to this document is made shall support the following list of requirements.

- a) The use of at least one customer VID for IA traffic engineered non-stream or IA non-stream traffic.
- b) The use of an additional customer VID for IA time-aware stream traffic if that traffic type category is supported.

- c) The use of an additional customer VID for IA stream traffic if that traffic type category is supported.
- d) The use of an additional customer VID for IA time-aware stream traffic if redundancy for that traffic type category is supported.
- e) The use of an additional customer VID for IA stream traffic if redundancy for that traffic type category is supported.
- f) Participate in only a single Configuration Domain.
- g) The use of an additional customer VID for an isolation VLAN.
- h) The use of an additional customer VID for a default VLAN

5.9.2 ccA end station component requirements

An end station component implementation for which a claim of conformance to ccA of this document is made shall support the following list of requirements.

- a) Common end station component requirements according to 5.9.1.
- b) At least 2 PTP Instances according to 5.5.3.
- c) End station requirements for enhancements for scheduled traffic according to IEEE Std 802.1Q-2022, 5.25, for data rates of 100 Mb/s and 1 Gb/s including:
 - 1) TickGranularity of less than or equal to 10 ns according to IEEE Std 802.1Q-2022, 8.6.9.4.16 and Table 12-32,
 - 2) support of the YANG features and nodes of the ieee802-dot1q-sched module according to 6.4.9.3.2, and
 - 3) support of the YANG features and nodes of the icieeee60802-sched-bridge module according to 6.4.9.3.3.
- d) End station requirements for frame preemption according to IEEE Std 802.1Q-2022, 5.26, for data rates of 100 Mb/s, and 1 Gb/s, if the IA time-aware stream traffic or the IA stream traffic type categories are supported, including:
 - 1) support of Interspersing Express Traffic according to IEEE Std 802.3-2022, Clause 99 as amended by IEEE Std 802.3de-2022, including support of the Additional Ethernet Capabilities TLV in an LLDPDU to indicate supported functions of frame preemption according to IEEE Std 802.3-2022, 79.3.7 and Table 79-8, and
 - 2) support of the YANG features and nodes of the ieee802-dot1q-preemption module according to 6.4.9.3.4.

5.9.3 ccB end station component requirements

An end station component implementation for which a claim of conformance to ccB of this document is made shall support the following list of requirements:

- e) Common end station component requirements according to 5.9.1.
- f) At least 1 PTP Instance according to 5.5.3.

5.10 End station component options

5.10.1 Common end station component options

An end station component implementation of any conformance class for which a claim of conformance to this document is made may support the following list of requirements.

- a) The operation of the credit-based shaper algorithm according to IEEE Std 802.1Q-2022, 8.6.8.2 including support of the YANG features and nodes of the ieee802-dot1q-cbs module according to 6.4.9.3.5.
- b) Talker end system behaviors according to IEEE Std 802.1CB-2017, as amended by IEEE Std 802.1CBdb-2021 and IEEE Std 802.1CBcvTM-2021, 5.6, and 5.7 c) on one or more ports

and for 1 or more Compound Streams, including support of the `ieee802-dot1cb-stream-identification` and `ieee802-dot1cb-frer` YANG modules according to 6.4.9.3.6.

- c) Listener end system behaviors according to IEEE Std 802.1CB-2017, as amended by IEEE Std 802.1CBdb-2021 and IEEE Std 802.1CBcv-2021, 5.9 on one or more ports and for 1 or more Compound Streams, including support of the `ieee802-dot1cb-stream-identification` and `ieee802-dot1cb-frer` YANG modules according to 6.4.9.3.6.

5.10.2 ccA end station component options

An end station component implementation for which a claim of conformance to ccA of this document is made may support the following list of requirements.

- a) Common end station options according to 5.10.1
- b) More than 2 PTP Instances according to 5.5.3.
- c) End station requirements for enhancements for scheduled traffic according to IEEE Std 802.1Q-2022, 5.25, for data rates of 10 Mb/s, 2,5 Gb/s, 5 Gb/s, and 10 Gb/s including:
 - 1) TickGranularity of less than or equal to 10 ns according to IEEE Std 802.1Q-2022, 8.6.9.4.16 and Table 12-32,
 - 2) support of the YANG features and nodes of the `ieee802-dot1q-sched` module according to 6.4.9.3.2, and
 - 3) support of the YANG features and nodes of the `ieciieee60802-sched-bridge` module according to 6.4.9.3.3.
- d) End station requirements for frame preemption according to IEEE Std 802.1Q-2022, 5.26, for data rates of 10 Mb/s, 2,5 Gb/s, 5 Gb/s, and 10 Gb/s including:

NOTE IEEE Std 802.3de-2022, 99.1, comprises 10 Mb/s.

- 1) support of Interspersing Express Traffic according to IEEE Std 802.3-2022, Clause 99 as amended by IEEE Std 802.3de-2022, including support of the Additional Ethernet Capabilities TLV in an LLDPDU to indicate supported functions of frame preemption according to IEEE Std 802.3-2022, 79.3.7 and Table 79-8, and
- 2) support of the YANG features and nodes of the `ieee802-dot1q-preemption` module according to 6.4.9.3.4.

5.10.3 ccB end station component options

An end station component implementation for which a claim of conformance to ccB of this document is made may support the following list of requirements.

- a) Common end station component options according to 5.10.1.
- b) One or more PTP Instances according to 5.5.3.
- c) End station requirements for enhancements for scheduled traffic according to IEEE Std 802.1Q-2022, 5.25, including:
 - 1) TickGranularity of less than or equal to 10 ns according to IEEE Std 802.1Q-2022, 8.6.9.4.16 and Table 12-32,
 - 2) support of the YANG features and nodes of the `ieee802-dot1q-sched` module according to 6.4.9.3.2, and
 - 3) support of the YANG features and nodes of the `ieciieee60802-sched-bridge` module according to 6.4.9.3.3.
- d) End station requirements for frame preemption according to IEEE Std 802.1Q-2022, 5.26, including:
 - 1) support of Interspersing Express Traffic according to IEEE Std 802.3-2022, Clause 99 as amended by IEEE Std 802.3de-2022, including support of the Additional Ethernet Capabilities TLV in an LLDPDU to indicate supported functions of frame preemption according to IEEE Std 802.3-2022, 79.3.7 and Table 79-8, and

- 2) support of the YANG features and nodes of the ieee802-dot1q-preemption module according to 6.4.9.3.4.

6 Required functions for an industrial network

6.1 General

Clause 6 provides requirements specific to this document and the industrial use case.

6.2 Synchronization

6.2.1 General

An IA-station can contain more than one Grandmaster PTP Instance and PTP End Instance to support:

- a) hot-standby use cases, or
- b) Working Clock or Global Time.

For further explanation of the requirements for time synchronization, refer to Annex D.

6.2.2 PTP Instance requirements

A Grandmaster PTP Instance, a PTP Relay Instance and a PTP End Instance, and the Working Clock or Global Time clocks connected to them, shall meet the following requirements under their allowed working conditions and for their lifetime.

- a) The fractional frequency offset of the LocalClock relative to the nominal frequency shall be according to Table 9.
- b) The range of the rate of change of fractional frequency offset of the LocalClock shall be according to Table 9.
- c) During operation, the Working Clock and Global Time at Grandmaster PTP Instances and PTP End Instances shall increase monotonically, where monotonic means that for a time y that occurs after time x , the ClockTarget's timestamp of y is greater than or equal to the ClockTarget's timestamp of x .
- d) The Working Clock and Global Time at a PTP End Instance can be controlled by applying a frequency change over a period of time. The frequency applied can have a fine resolution to speed up or slow down the clock smoothly, and it has a total range of frequency adjustment.
- e) For the Global Time at a PTP End Instance, the maximum value of frequency adjustment shall be according to Table 9.
- f) For the Working Clock at a PTP End Instance, the maximum value of frequency adjustment shall be according to Table 9.

For Working Clock or Global Time, decoupled from a ClockTarget, a higher maximum value of frequency adjustment and maximum rate of change of fractional frequency offset are allowed. As soon as it is coupled (or coupled again), a) to f) apply.

Table 9 – Required values

Topic	Value
Local Clock at non-Grandmaster PTP Instance, range of fractional frequency offset relative to the nominal frequency	± 50 ppm
Local Clock at non-Grandmaster PTP Instance, range of rate of change of fractional frequency offset with respect to the nominal frequency	± 1 ppm/s
Working Clock and Global Time (acting as ClockSource) and Local Clock at Grandmaster PTP Instance, range of fractional frequency offset with respect to the nominal frequency	± 25 ppm

Topic	Value
Working Clock and Global Time (acting as ClockSource) and Local Clock at Grandmaster PTP Instance, range of rate of change of fractional frequency offset with respect to the nominal frequency (steady state, see Annex D.2.3)	± 1 ppm/s
Working Clock and Global Time (acting as ClockSource) at Grandmaster PTP Instance, range of rate of change of fractional frequency offset (transient, see Annex D.2.3)	± 3 ppm/s
Working Clock and Global Time at PTP End Instance, maximum value of frequency adjustment	± 250 ppm over any observation interval of 1 ms
NOTE 1 If the Grandmaster PTP Instance implementation is such that its Working Clock and Local Clock are the same or otherwise locked to the same frequency, the normative requirements on the Working Clock take priority over those on the Local Clock. NOTE 2 The Maximum value of frequency adjustment represents an upper bound that limits how much a PTP End Instance can change the frequency of its Working Clock or Global Time during a given period. However, the adjustment is expected to be gradual over the defined interval rather than instantaneous. NOTE 3 The example algorithms that track clock drift use up to 4 seconds of historical data and can take that length of time to respond to changes in clock drift. The example algorithm has been used for simulating cases where no fast changes in the observed frequency drift rate were observed. In most of the real-life situations, this condition can be satisfied.	

1893

1894 6.2.3 PTP protocol requirements

1895 Table 10 shows the required protocol times.

1896

Table 10 – Protocol settings

Topic	Value
Nominal time between successive Announce messages (announce interval)	1 s
Nominal time between successive Pdelay_Req messages (Pdelay_Req message transmission interval)	125 ms
Range of allowed time between successive Pdelay_Req messages	119 ms to 131 ms
Nominal time between successive Sync messages at the Grandmaster (Sync message transmission interval)	125 ms
Range of allowed time between successive Sync messages at the Grandmaster	119 ms to 131 ms
Time between reception of a Sync message and transmission of the subsequent Sync message (i.e. residence time) at a PTP Relay instance	Maximum: 15 ms Measured Mean: ≤ 5 ms
Maximum time between transmission of a Sync message and transmission of the related Follow_Up message	2,5 ms
Time between reception of a Pdelay_Req message and transmission of the subsequent Pdelay_Resp message (i.e. Pdelay turnaround time).	Maximum: 15 ms
NOTE 1 A consequence of having a single allowed value of mean sync interval is that syncLocked mode is achieved. If the timeTransmitter port sync interval is the same as that of the timeReceiver port, syncLocked mode is achieved. NOTE 2 The values contained in this table apply to both the Working Clock and Global Time.	

1897

1898 6.2.4 Clock control system requirements for PTP End Instances

1899 Table 11 shows the required clock control system characteristics at a PTP End Instance.

1900

Table 11 – Clock control system requirements

Topic	Value
Maximum Bandwidth	1,0 Hz
Minimum Bandwidth	0,9 Hz
Maximum Gain Peaking	2,2 dB
Minimum absolute value of Roll-off	20 dB/decade

Topic	Value
NOTE 1 For more information regarding the clock control system, see Annex C.	
NOTE 2 The values contained in this table apply to both the Working Clock and Global Time.	

6.2.5 Error generation limits

Table 12 shows the required limits on error generation at a Grandmaster PTP Instance. A limit on error generation for a Grandmaster PTP Instance is a limit on the amount of error it generates in the output Sync message compared to its Working Clock (acting as ClockSource) and Local Clock. See D.3.4.

Table 12 – Error generation limits for Grandmaster PTP Instance

Topic	Value
Accuracy of forwarded time information: (preciseOriginTimestamp + correctionField) in PTP timing message minus Working Clock at Grandmaster when Sync message is transmitted	Allowable range of the measured mean: - 10 ns to + 10 ns Range around the measured mean within which 90% of measurements fall: ± 7 ns Range around the measured mean within which 100% of measurements fall: ± 10 ns
Accuracy of forwarded Rate Ratio information: rateRatio field in Follow_Up information TLV minus true Rate Ratio between Working Clock at Grandmaster and Local Clock when Sync message is transmitted	Mean 0 ppm ± 0,1 ppm Standard deviation ≤ 0,1 ppm
Accuracy of syncEgressTimestamp: syncEgressTimestamp in Drift_Tracking TLV minus Local Clock when Sync message is transmitted	Range around the measured mean within which 90% of measurements fall: ± 7 ns Range around the measured mean within which 100% of measurements fall: ± 10 ns
NOTE 1 “Allowable range of the measured mean” specifies limits on constant error. “Range around the measure mean” and “Standard deviation” specify limits on dynamic error. A limit on the constant error of syncEgressTimestamp is not specified because constant error in this characteristic is not a source of time synchronization error.	
NOTE 2 This document specifies statistical normative requirements without a test duration. It is the responsibility of a test plan to determine the test duration and pass/fail criteria, taking sampling error into consideration, and striking an acceptable balance between test duration and accuracy.	

Table 13 shows the required limits on error generation at a PTP Relay Instance. A limit on error generation for a PTP Relay Instance is a limit on the amount of error it adds to the output Sync message compared to the input Sync message. These requirements are written for the case when errors due to change of fractional frequency offset of its Local Clock with respect to the nominal frequency and errors in the input Sync message are negligible with respect to the specified error generation limits. See D.3.5.

1915

Table 13 – Error generation limits for PTP Relay Instance

Topic	Value
Accuracy of residenceTime measurement (no clock drift): (preciseOriginTimestamp + correctionField) in the PTP timing message transmitted by PTP Relay Instance minus Working Clock at Grandmaster when the Sync message is transmitted while... • Working Clock (acting as ClockSource) at Grandmaster is stable. • Local Clock at upstream PTP Instance is stable • meanLinkDelay between upstream PTP Instance and PTP Relay Instance is negligible	Allowable range of the measured mean: - 2 ns to + 2 ns Range around the measured mean within which 90% of measurements fall: ± 10 ns Range around the measured mean within which 100% of measurements fall: ± 20 ns
Accuracy of Neighbor Rate Ratio measurement and correct modification of Rate Ratio (no clock drift): rateRatio field in the Follow_Up information TLV transmitted by PTP Relay Instance minus Rate Ratio from the PTP Relay Instance's Local Clock to the Working Clock at the Grandmaster while... • Working Clock (acting as ClockSource) at Grandmaster is stable. • Local Clock at upstream PTP Instance is stable	Allowable range of the measured mean: - 0,1 ppm to + 0,1 ppm Allowable measured standard deviation around the measured mean: 0,02 ppm
Accuracy of Neighbor Rate Ratio measurement and correct modification of Rate Ratio (Rate Ratio drift): rateRatio field in the Follow_Up information TLV transmitted by PTP Relay Instance minus Rate Ratio from the PTP Relay Instance's Local Clock to the Working Clock at the Grandmaster while... • WorkingClock (acting as ClockSource) at Grandmaster PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s • Local Clock at upstream PTP Instance is stable.	Allowable range of the measured mean: - 0,1 ppm to + 0,1 ppm Allowable measured standard deviation around the measured mean: 0,08 ppm

Topic	Value
Accuracy of Neighbor Rate Ratio measurement and correct modification of Rate Ratio (Rate Ratio drift and Neighbor Rate Ratio drift): rateRatio field in the Follow_Up information TLV transmitted by PTP Relay Instance minus Rate Ratio from the PTP Relay Instance's Local Clock to the Working Clock at the Grandmaster while... • WorkingClock (acting as ClockSource) at Grandmaster PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s • Local Clock at upstream PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s	Allowable range of the measured mean: - 0,1 ppm to + 0,1 ppm Allowable measured standard deviation around the measured mean: 0,08 ppm
Accuracy of Neighbor Rate Ratio Drift measurement and correct modification of Rate Ratio Drift (no clock drift) rateRatioDrift field in the Drift_Tracking TLV transmitted by PTP Relay Instance minus Rate Ratio Drift from the PTP Relay Instance's Local Clock to the Working Clock at the Grandmaster while... • WorkingClock (acting as ClockSource) atGrandmaster is stable. • Local Clock at upstream PTP Instance is stable.	Allowable range of the measured mean: - 0,1 ppm to + 0,1 ppm Allowable measured standard deviation around the measured mean: 0,02 ppm
Accuracy of Neighbor Rate Ratio Drift measurement and correct modification of Rate Ratio Drift (Rate Ratio drift) rateRatioDrift field in the Drift_Tracking TLV transmitted by PTP Relay Instance minus Rate Ratio Drift from the PTP Relay Instance's Local Clock to the Working Clock at the Grandmaster while... • WorkingClock (acting as ClockSource) at Grandmaster PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s • Local Clock at upstream PTP Instance is stable.	Allowable range of the measured mean: - 0,1 ppm to + 0,1 ppm Allowable measured standard deviation around the measured mean: 0,08 ppm

Topic	Value
Accuracy of Neighbor Rate Ratio Drift measurement and correct modification of Rate Ratio Drift (Rate Ratio drift and Neighbor Rate Ratio drift) rateRatioDrift field in the Drift_Tracking TLV transmitted by PTP Relay Instance minus Rate Ratio Drift from the PTP Relay Instance's Local Clock to the Working Clock at the Grandmaster while... • WorkingClock (acting as ClockSource) at Grandmaster PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s • Local Clock at upstream PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s	Allowable range of the measured mean: - 0,1 ppm to + 0,1 ppm Allowable measured standard deviation around the measured mean: 0,08 ppm
Accuracy of syncEgressTimestamp: syncEgressTimestamp in Drift_Tracking TLV minus Local Clock when Sync message is transmitted	Range around the measured mean within which 90% of measurements fall: ± 7 ns Range around the measured mean within which 100% of measurements fall: ± 10 ns
Accuracy of meanLinkDelay: meanLinkDelay measured by the PTP Relay Instance minus actual path delay	± 3 ns
Accuracy of Pdelay turnaround time measurement: (responseOriginTimestamp – requestReceiptTimestamp) for transmitted Pdelay_Resp and associated Pdelay_Resp_Follow_Up messages minus the actual interval between reception of the associated Pdelay_Req and transmission of the Pdelay_Resp as measured with respect to the Local Clock	Allowable range of the measured mean: - 2 ns to + 2 ns Range around the measured mean within which 90% of measurements fall: ± 10 ns Range around the measured mean within which 100% of measurements fall: ± 20 ns

Topic	Value
NOTE 1 “Allowable range of the measured mean” specifies limits on constant error. “Range around the measure mean” and “Standard deviation” specify limits on dynamic error. A limit on the constant error of syncEgressTimestamp is not specified because constant error in this characteristic is not a source of time synchronization error. NOTE 2 This document specifies statistical normative requirements without a test duration. It is the responsibility of a test plan to determine the test duration and pass/fail criteria, taking sampling error into consideration, and striking an acceptable balance between test duration and accuracy. NOTE 3 These requirements are for errors generated by the PTP Relay Instance. They assume input Sync messages with negligible errors. However, timestamps of Pdelay_Req reception and Pdelay_Resp transmission by the upstream PTP Instance should include expected timestamp errors when meanLinkDelay accuracy is assessed.	

1916

1917 Table 14 shows the required limits on error generation at a PTP End Instance. A limit on error
 1918 generation for a PTP End Instance is a limit on the amount of error it adds to its Working Clock
 1919 (acting as ClockTarget) compared to the input Sync message. These requirements are written
 1920 for the case when errors due to change of fractional frequency offset of its Local Clock with
 1921 respect to the nominal frequency and errors in the input Sync message are negligible with
 1922 respect to the specified error generation limits. See D.3.6.

1923

Table 14 – Error generation limits for PTP End Instance

Topic	Value
Accuracy of recovered clock (no clock drift): Working Clock (acting as ClockTarget) at PTP End Instance minus Working Clock (acting as Clock Source) at Grandmaster while... • WorkingClock (acting as ClockSource) at Grandmaster is stable. • Local Clock at upstream PTP Instance is stable. • meanLinkDelay between upstream PTP Instance and PTP End Instance is negligible	Allowable range of cTE: - 10 ns to + 10 ns Allowable range of dTE: - 15 ns to + 15 ns
Accuracy of recovered clock (Rate Ratio drift): Working Clock (acting as ClockTarget) at PTP End Instance minus Working Clock (acting as Clock Source) at Grandmaster while... • WorkingClock (acting as ClockSource) at Grandmaster PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s • Local Clock at upstream PTP Instance is stable. • meanLinkDelay between upstream PTP Instance and PTP Relay Instance is negligible	Allowable range of cTE: - 10 ns to + 10 ns Allowable range of dTE: - 145 ns to + 15 ns

Topic	Value
Accuracy of recovered clock (Rate Ratio drift and Neighbor Rate Ratio drift): Working Clock (acting as ClockTarget) at PTP End Instance minus Working Clock (acting as Clock Source) at Grandmaster while... • WorkingClock (acting as ClockSource) at Grandmaster PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s • Local Clock at upstream PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s • meanLinkDelay between upstream PTP Instance and PTP Relay Instance is negligible	Allowable range of cTE: - 10 ns to + 10 ns Allowable range of dTE: - 145 ns to + 15 ns
Accuracy of meanLinkDelay: meanLinkDelay measured by the PTP End Instance minus actual path delay	± 3 ns
NOTE 1 This document specifies statistical normative requirements without a test duration. It is the responsibility of a test plan to determine the test duration and pass/fail criteria, taking sampling error into consideration, and striking an acceptable balance between test duration and accuracy.	
NOTE 2 These requirements are for errors generated by the PTP End Instance. They assume input Sync messages with negligible errors. However, timestamps of Pdelay_Req reception and Pdelay_Resp transmission by the upstream PTP Instance should include expected timestamp errors when meanLinkDelay accuracy is assessed.	

1924

1925 **6.2.6 PTP Instance synchronization status**

1926 Industrial automation systems monitor the synchronization status of each PTP Instance to
1927 determine the viability of operations. This status is obtained from the isSynced global variable
1928 specified in IEEE Std 802.1ASdm-2024, 18.4.1.1.

1929 The PtpInstanceSyncStatus state machine according to IEEE Std 802.1ASdm-2024, 18.4, shall
1930 be supported by all PTP instances independent of whether hot standby is supported. Non-
1931 Grandmaster PTP Instances shall support one or more of the application interfaces of 9.3, 9.4,
1932 9.5, and 9.6 according to IEEE Std 802.1ASdm-2024. Grandmaster PTP Instances shall support
1933 the application interface of 9.2 according to IEEE Std 802.1ASdm-2024.

1934 **6.2.7 Application framework**

1935 Any step change in the time of a ClockSource or ClockTarget whose absolute value exceeds a
1936 user-defined threshold (for example 1 μs) leads to action being taken by the application or by
1937 a higher-layer entity.

1938 If the change is in Global Time, it is desirable that all consumers of that time be made aware of
1939 this change (i.e. a jump in Global Time from the value A to the value B), so that the actual time
1940 interval between the time corresponding to A and the time corresponding to B can be evaluated.

In the case of Working Clock, a time change that exceeds the user-defined threshold (for example 1 μ s) is avoided to protect assets and prevent damage. Thus, the ClockSource or ClockTarget can be decoupled (see Figure 13 and Figure 14) from the PTP-maintained clock when such a time change occurs.

In Figure 13, two ClockTargets are traceable to a reliable source of time, which should be synchronized to Global Time or Working Clock.

The status of a ClockSource, ClockTarget, ClockTimeTransmitter or ClockTimeReceiver is given by the status of the clock (see 6.2.6) as shown in Figure 11. When timestamps are provided to the application, the current ClockSource or ClockTarget state can also be provided to the application.

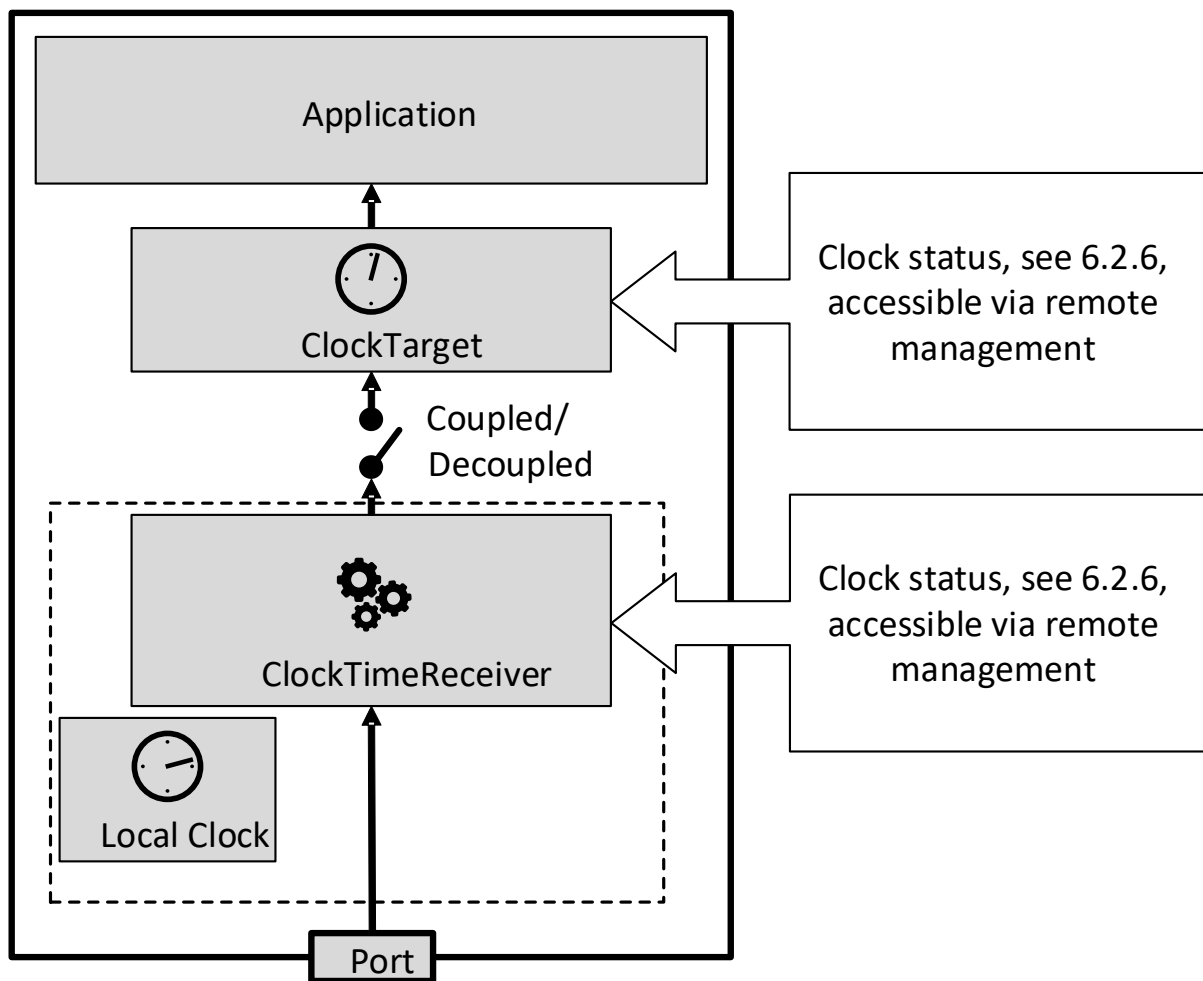


Figure 11 – Clock model

6.2.8 Working Clock domain framework

The gPTP domainNumber of a Working Clock domain is assigned by the CNC. In industrial applications, when the number of PTP Relay Instances between the Grandmaster PTP Instance and any PTP End Instance is less than or equal to 99, $\max|TE_R|$ of the synchronized time of any ClockTarget, relative to the Grandmaster ClockSource, is less than or equal to 1 μ s (see error budget A in Figure 14). Thus it is incumbent upon any PTP Instance to ensure that the requirements specified in 5.5.3, 6.2.2, 6.2.3, 6.2.4, and 6.2.5 are met.

6.2.9 Global Time domain framework

The gPTP domainNumber of a Global Time domain is assigned by the CNC. In industrial applications, when the number of PTP Relay Instances between the Grandmaster PTP Instance and any PTP End Instance is less than or equal to 99, $\max|TE_R|$ of the synchronized time of any ClockTarget, relative to the Grandmaster ClockSource, is less than or equal to 100 μ s (see error budget A in Figure 14). Thus it is incumbent upon any PTP Instance to ensure that the requirements specified in 5.5.3, 6.2.2, 6.2.3, 6.2.4, and 6.2.5 are met.

6.2.10 IA-station model for clocks

Industrial automation applications, as described in 4.1, require synchronized time that is traceable to a known source (i.e. Global Time) and a source of time synchronized to the Working Clock. Figure 12 and Figure 13 show examples of the IA-station internal model for clocks with the two PTP Instances. It is possible for the ClockSource or ClockTarget to start decoupled or become decoupled from the ClockTimeTransmitter or ClockTimeReceiver, respectively, of a PTP Instance; the ClockSource or ClockTarget runs independently of the availability of the network or a Grandmaster. For example, if the PTP Instance enters a state where isSynced is FALSE, the application might choose to decouple its clock from the PTP Instance and continue to run on its internal clock. If isSynced for the PTP Instance changes to TRUE, the application can choose to again synchronize to the PTP Instance.

Figure 13 shows the IA-station internal model for clocks, with the two PTP Instances used as ClockTimeReceiver/ClockTarget.

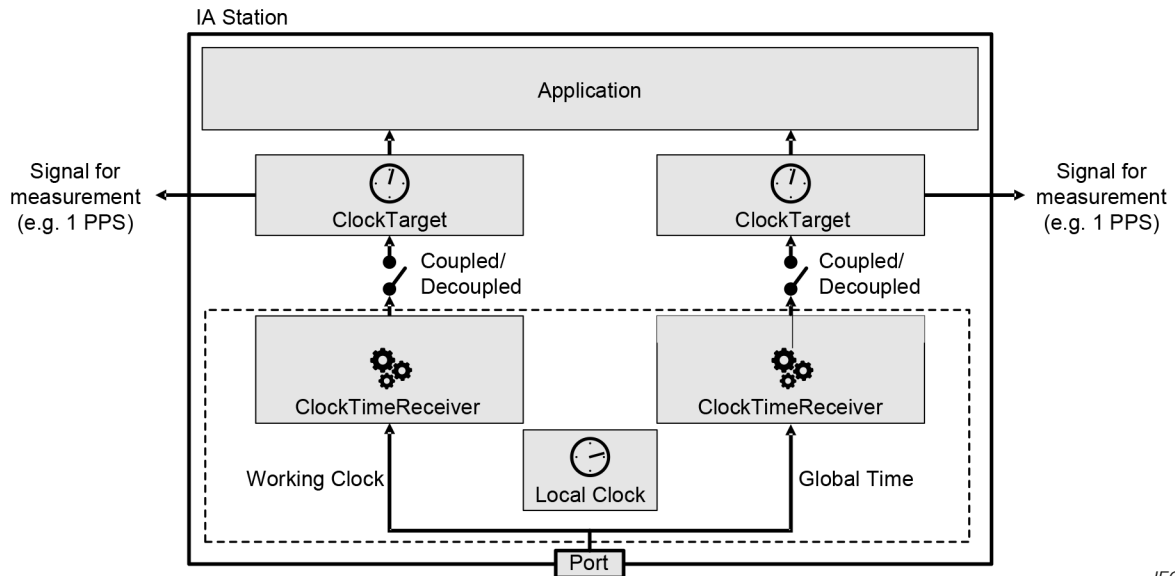


Figure 12 – Example clock usage principles for PTP End Instances

Figure 14 shows the IA-station internal model for clocks, with the two PTP Instances used as Grandmaster.

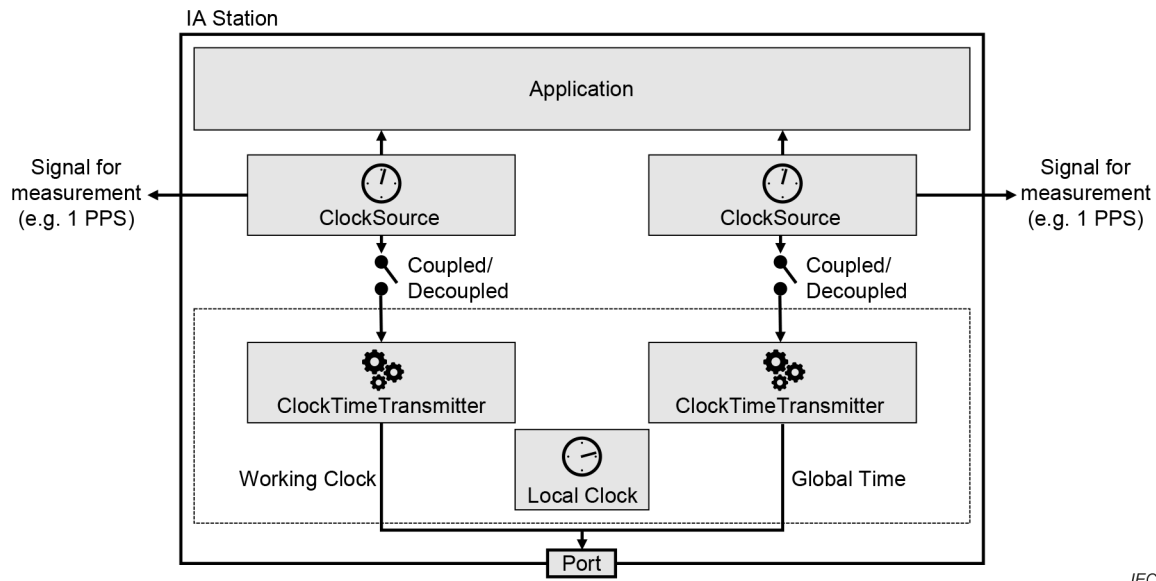


Figure 13 – Example clock usage principles for Grandmaster PTP Instances

6.2.11 Clock usage for the Ethernet interface

6.2.11.1 Time-aware offset control

Time-aware offset control (see 4.4), uses an assigned source of time and a definition when to start or to stop, which are dependent on the clock state.

The clock used is the ClockTarget or, in the case of a Grandmaster PTP Instance, the ClockSource.

IA time-aware streams are only transmitted while isSynced for the chosen ClockSource or ClockTarget is TRUE (see 6.2.6).

Thus, changes of the clock state directly influence the transmission of frames.

6.2.11.2 Gating cycle

To control the gating cycle, the gate control list uses an assigned source of time. Enabling and disabling the gate control list is dependent on the clock state.

The clock used is the ClockTarget or, in the case of a Grandmaster PTP Instance, the ClockSource.

The gating cycle is run using the chosen ClockSource or ClockTarget regardless of the value of isSynced (see 6.2.6).

6.2.12 Error model

Synchronization is transported over the entire path, from the Grandmaster PTP Instance to the PTP End Instance, through the intermediate PTP Relay Instances. All time errors, cTE and dTE, are accumulated during this process.

Time error can arise in the following processes:

- the transporting of time in PTP Instances and via PTP Links that connect PTP Instances,
- the providing of time to the Grandmaster PTP Instance, from the ClockSource entity via the ClockTimeTransmitter entity, and

c) the providing of time to a ClockTarget entity (end application) via the ClockTimeReceiver entity.

NOTE Item a) includes time error introduced in a PTP End Instance between the timeReceiver port and the ClockTimeReceiver entity, and between the ClockTimeTransmitter entity and a TimeTransmitter port.

An output synchronization signal (for example, 1 pulse per second (PPS)) synchronized to the Working Clock as shown in Figure 13 and Figure 14, at any PTP Instance, is used to measure the time error between the ClockSource of the Grandmaster and the ClockTarget of a PTP Instance that is not the Grandmaster. The additional error introduced by implementation of the output synchronization signal is in the range of –10 ns to +10 ns. Figure 14 shows the error budget principle used. These budgets do not include any deviation from the PTP timescale. Representative budgets are provided in Annex D.

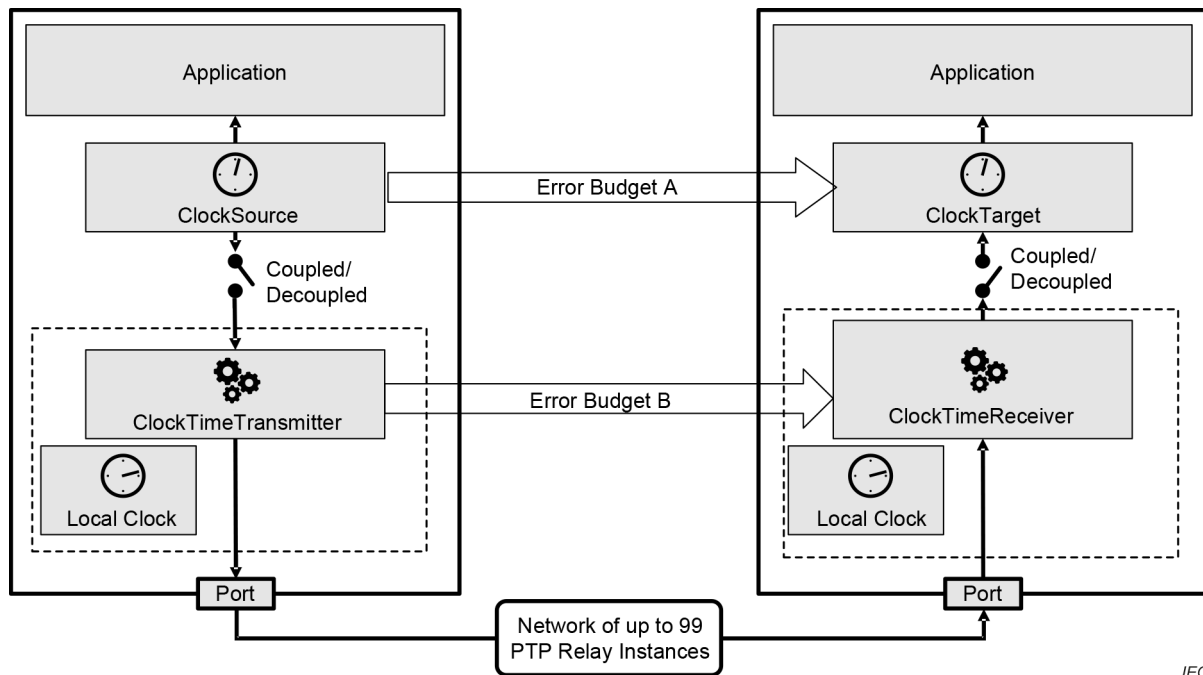


Figure 14 – Error budget scheme

Table 15 shows example values for the splitting of the available error budgets (see Figure 14).

Table 15 – Error budget

Domain	Error budget A	Error budget B
Working Clock	1 μ s	900 ns
Global Time	100 μ s	99,9 μ s

Global Time is often used for tracking events in industrial applications (i.e. sequence of events). Any usage of Global Time for time stamping of application events is allowed an error budget of 1 ms.

6.2.13 gPTP domains and PTP Instances

Any valid gPTP domain number as specified in IEEE Std 802.1AS-2020 can be used. The IEEE Std 1588-2019 attribute descriptionDS.userDescription shall be used according to Table 16 to support the translation of PTP Instances and middleware as described in 4.6.2. One gPTP domain can be used for both Working Clock and Global Time. If only one gPTP domain is used, then the requirements for the Working Clock apply (see 6.2.8).

2035

Table 16 – descriptionDS.userDescription of gPTP Domains

gPTP Domain	descriptionDS.userDescription
Working Clock (no hot standby configured)	"60802-WorkingClock"
Primary Working Clock (with configured hot standby)	"60802-Primary-WorkingClock"
Secondary Working Clock (with configured hot standby)	"60802-Secondary-WorkingClock"
Global Time (no hot standby configured)	"60802-GlobalTime"
Primary Global Time (with configured hot standby)	"60802-Primary-GlobalTime"
Secondary Global Time (with configured hot standby)	"60802-Secondary-GlobalTime"
GlobalTime and WorkingClock (no hot standby configured)	"60802-GlobalTime-WorkingClock"
Primary GlobalTime and WorkingClock (with configured hot standby)	"60802-Primary-GlobalTime-WorkingClock"
Secondary GlobalTime and WorkingClock (with hot standby configured)	"60802-Secondary-GlobalTime-WorkingClock"

2036

2037 The descriptionDS.userDescription attribute is represented in the ieee1588-ptp-tt YANG module
2038 by the `user-description` leaf in the `description-ds` container of a PTP Instance.

2039 The linking between a gPTP domain and the IETF interfaces is provided by the `underlying-`
2040 `interface`.

2041 **6.3 Security model**

2042 **6.3.1 General**

2043 Subclause 6.3 specifies the security model starting with NETCONF/YANG. It describes the
2044 security functionality, the security objects in factory default state, the imprinting of Configuration
2045 Domain-specific security objects and the secure configuration based on Configuration Domain-
2046 specific security objects.

2047 **6.3.2 Security functionality**

2048 **6.3.2.1 Message exchange protection**

2049 **6.3.2.1.1 NETCONF-over-TLSv1.2 profile**

2050 Network configuration with NETCONF/YANG is protected by NETCONF-over-TLS according to
2051 IETF RFC 7589, where the to-be-configured IA-stations act in the NETCONF server role. TLS
2052 protocol version 1.2 according to IETF RFC 5246, 6.2.3.3, 7.4.7.2 and 8.1.2 shall be supported
2053 according to the following list of requirements.

2054 a) Mutual authentication in conjunction with the IDevID and LDevID credentials according to
2055 6.3.4 and 6.3.5.

2056 b) The cipher suite TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256 according to IETF
2057 RFC 5289, 3.2 and Clause 5.

2058 NOTE IETF RFC 7589 implicitly mandates the cipher suite TLS_RSA_WITH_AES_128_CBC_SHA by referring to
2059 IETF RFC 5246. This cipher suite is not used in this document because it requires excessive asymmetric key lengths,
2060 it is not an Authenticated Encryption with Associated Data (AEAD) scheme, and it does not provide perfect forward
2061 secrecy.

2062 c) Signature algorithm ECDSA with SHA-256 and Curve P-256 according to NIST FIPS 186-5
2063 Digital Signature Standard (DSS).

2064 The TLS Certificate message from the TLS client and server contains the self-signed root
2065 certificate. This approach allows simplifying and flattening the PKI hierarchy on base of the

current TLS client certificate to NETCONF username mapping algorithm in IETF RFC 7589. Implementations shall support TLS Certificate message with at least 2 certificate objects.

NETCONF-over-SSH according to IETF RFC 6242 is not used in this document.

NOTE This document selects TLS as a secure transport for NETCONF since TLS is the better match for the case of configuration clients that rely upon unattended or automated operation. This case is dominant in industrial automation.

6.3.2.1.2 NETCONF-over-TLSv1.2 options

The following list of TLS protocol version 1.2 requirements may be supported.

- a) The cipher suites TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384 according to IETF RFC 5289, 3.2 and Clause 5, and TLS_ECDHE_ECDSA_WITH_CHACHA20_POLY1305_SHA256 according to IETF RFC 7905, Clause 2.
- b) Signature algorithms ECDSA with SHA-512 and Curve P-521 according to NIST FIPS 186-5, Ed25519 according to IETF RFC 8032, 5.1, and Ed448 according to IETF RFC 8032, 5.2.

6.3.2.1.3 NETCONF-over-TLSv1.3 profile

Network configuration with NETCONF/YANG is protected by NETCONF-over-TLS according to IETF RFC 7589 and IETF draft-ietf-netconf-over-tls13. TLS protocol version 1.3 according to IETF RFC 8446 may be used with mutual authentication for NETCONF/YANG as follows.

- a) The cipher suites TLS_AES_128_GCM_SHA256, TLS_AES_256_GCM_SHA384 and TLS_CHACHA20_POLY1305_SHA256 according to IETF RFC 8446, 9.1.
- b) One or more of the following signature schemes:
 - ecdsa_secp256r1_sha256 according to NIST FIPS 186-5 and NIST SP 800-186, 3.2.1.3, ecdsa_secp521r1_sha512 according to NIST FIPS 186-5 and NIST SP 800-186, 3.2.1.5,
 - ed25519 according to IETF RFC 8032, 5.1, and
 - ed448 according to IETF RFC 8032, 5.2.

The TLS Certificate message from the TLS client and server contains the self-signed root certificate.

6.3.2.1.4 Certificate-to-name mapping

The IETF RFC 7589 based certificate-to-name mapping procedure is as follows.

NOTE IETF RFC 7589, Clause 7, specifies that NETCONF servers map client certificates to "NETCONF usernames" and specifies a concrete mapping procedure for this purpose. This mapping is represented by the YANG module ietf-x509-cert-to-name.

The list of mapping entries has a single element containing:

- fingerprint: the fingerprint of the trust anchor for the Configuration Domain, and
- map_type: ext-60802-roles.

The map-type ext-60802-roles maps the roles provided in the id-60802-pe-roles extension (defined in 6.3.2.1.5) of the end entity certificate presented by the NETCONF client to a NETCONF username. The UTF-8 string representation of each role is added to the NETCONF username in chronological order of the enumeration values, whereas multiple roles are separated by ':' character.

2106 6.3.2.1.5 Role extension

2107 The id-60802-pe-roles extension in LDevID-NETCONF end entity certificates shall be
2108 constructed as follows:

2109 a) Extension field extnID

2110 The extnID shall provide the following OBJECT IDENTIFIER to identify the id-60802-pe-roles
2111 extension:

```
2112 id-60802 OBJECT IDENTIFIER ::= { iso (1) iso-identified-organization (3) ieee (111)
2113 standards-association-numbered-series-standards (2) lan-man-stds (802) ieee802dot1(1)
2114 ieee802dot1stds(2) 1 }
```

```
2115
2116 id-60802-pe OBJECT IDENTIFIER ::= { id-60802 1 }
```

```
2117
2118 id-60802-pe-roles OBJECT IDENTIFIER ::= { id-60802-pe 1 }
```

2119

2120 b) Extension field critical

2121 The id-60802-pe-roles extension is marked as non-critical (critical:= FALSE).

2122 c) Extension field extnValue

```
2123 60802RoleNamesSyntax ::= SEQUENCE OF 60802RoleName
```

2124

```
2125 60802RoleName ::= ENUMERATED {
2126     SecurityAdminRole (0),
2127     ConfiguratorRole (1),
2128     StreamConfiguratorRole (2),
2129     SubscriberRole (3) }
```

2130 6.3.2.2 Resource access authorization

2131 Access control to NETCONF/YANG resources shall be protected by NACM according to IETF
2132 RFC 8341.

2133 NACM specifies a YANG data model (ietf-netconf-acm) for expressing rules to control access
2134 to NETCONF/YANG resources. This document profiles NACM to deliver role-based access
2135 control.

2136 NOTE 1 NACM does not natively deliver role-based access control but can be geared by profiling.

2137 This role-based model for security resources should be applied according to the following list
2138 of requirements.

- 2139 • The global switch enable-nacm is set to true.
- 2140 • The set of NETCONF/YANG resources of an IA-station is partitioned according to the YANG
2141 modules specified in 6.4.9 with a permission-to-role assignment as listed below. An access
2142 operation is allowed through the keyword "permitted" and not allowed through the keyword
2143 "denied".

2144 NOTE 2 NACM recognizes following "access-operations": create, read, update, delete, exec and uses the term write
2145 access for the access operations "create", "delete", and "update". This document uses the terms read, write and
2146 exec access.

- 2147 • All authenticated entities (default rules): All YANG modules: read access permitted, write
2148 access denied, exec-access denied.

2149 NOTE 3 The default rules apply for YANG modules that are listed in 6.4.9 but are not listed in the rules of the
2150 individual roles.

- 2151 • Rules for StreamConfiguratorRole: YANG module ieee802-dot1q-cnc-config: write and
2152 execute operations permitted.

- 2153 • Rules for SubscriberRole:
 - 2154 – YANG module ietf-subscribed-notifications: write and execute operations permitted, and
 - 2155 – YANG module ietf-yang-push: write and execute operations permitted.
- 2156 • Rules for ConfiguratorRole: All YANG modules except those listed below, write and execute
 2157 operations permitted:
 - 2158 – YANG modules for security configuration, i.e. ietf-truststore, ietf-keystore, path to cert-
 2159 to-name nodes of ietf-netconf-server, path to tls-server-parameters nodes of ietf-
 2160 netconf-server,
 - 2161 – YANG modules for stream configuration, i.e. ieee802-dot1q-cnc-config, and
 - 2162 – YANG modules for subscription configuration, i.e. ietf-subscribed-notifications, ietf-
 2163 yang-push.
- 2164 • Rules for SecurityAdminRole:
 - 2165 – YANG module ietf-truststore, path to certificate node of IDevID trust anchor: write and
 2166 execute operations denied, and
 - 2167 – YANG module ietf-truststore (besides path to certificate node of IDevID trust anchor):
 2168 write and execute operations permitted.
 - 2169 – YANG module ietf-keystore, path to asymmetric-key node of IDevID credential: write and
 2170 execute operations denied, and
 - 2171 – YANG module ietf-keystore (besides path to asymmetric-key node of IDevID credential):
 2172 write and execute operations permitted.
 - 2173 – YANG module ietf-netconf-server (besides path to cert-to-name nodes and path to tls-
 2174 server-parameters nodes): write and execute operations denied, and
 - 2175 – YANG module ietf-netconf-server, path to cert-to-name nodes: write and execute
 2176 operations permitted.
 - 2177 – YANG module ietf-netconf-server, path to tls-server-parameters nodes: write and
 2178 execute operations permitted.

2179 In addition, the following access control should be applied for NETCONF protocol operations:

- 2180 • <lock>, <unlock>: permitted for any role specified in this document,
- 2181 • <partial-lock>, <partial-unlock>: denied (not used in this document),
- 2182 • <get> and <get-config>: mapped to a "read" access operation to the target datastore,
- 2183 • <edit-config>: permitted for any role specified in this document,
- 2184 • <copy-config>: permitted for ConfiguratorRole,
- 2185 • <delete-config>: denied (not used in this document),
- 2186 • <commit>: permitted for any role specified in this document,
- 2187 • <discard-changes>: permitted for any role specified in this document,
- 2188 • <close-session>: permitted for any role specified in this document, and
- 2189 • <kill-session>: denied (not used in this document).

2190 This document does not specify the assignment of role names to actual system entities. This is
 2191 a duty of system owners or operators.

2192 6.3.3 IDevID Profile

2193 6.3.3.1 General

2194 IA-stations shall possess IDevID credentials according to 6.3.3. CNCs shall contain trust
 2195 anchors for validating IDevID credentials.

6.3.3.2 Object Contents

6.3.3.2.1 General

The IDevID credential contents shall comply to 6.3.3.2.2, 6.3.3.2.3, and IEEE Std 802.1AR-2018, Clause 6.

6.3.3.2.2 IA-station Identity

Any IDevID EE certificate of an IA-station shall take one of the following forms:

- raw form: the IDevID EE certificate complies to IEEE Std 802.1AR-2018, Clause 8, and
- extended form: the IDevID EE certificate complies to requirements provided IEEE Std 802.1AR-2018, Clause 8. The extended form of an IDevID EE certificate shall be constructed as follows:
 - the verifiable device identity shall appear as a URN in a GeneralName of type uniformResourceIdentifier in the subjectAltName extension,
 - the URN value shall be constructed according to IETF RFC 8141 and as follows:
 - namespace identifier: ieee (see IETF RFC 8069),
 - namespace-specific string: iec-ieee-60802#verifiable-device-identity, and
 - q-component (see IETF RFC 8141, 2.3.2) to parameterize the named resource: an ampersand-separated list of keyword=value tuples with following keywords and values. These tuples can appear in any order inside the q-component:
 - the keywords: hardware-rev, serial-num, mfg-name, model-name;
 - their corresponding values from the single "chassis" component list entry in the ietf-hardware YANG module (see 6.4.9.2.5.8) that represents the management entity of the IA-station respectively from its pre-material form in percent-encoding according to IETF RFC 3986, 2.1.

NOTE 1 These are the items with the YANG property config=false from the 'component' list entry that represents the management entity of the IA-station. The config=false items firmware-rev and software-rev are excluded to avoid IDevID credential updates in case of FW or SW updates.

NOTE 2 An object looks like urn:ieee:iec-ieee-60802#verifiable-device-identity?=mfg-name=<mfg-name>&model-name=<model-name>&hardware-rev=<hardware-rev>&serial-num=<serial-num>.

NOTE 3 One IDevID EE certificate can have one subjectAltName extension which can have one or more GeneralName entries. In particular, there can be one or more GeneralName entries of type uniformResourceIdentifier. This allows other organizations (e.g. middleware and application consortia or individual manufacturers) to also represent their perception of verifiable device identity in addition to the perception of this document.

6.3.3.2.3 Signature Suites

An IDevID shall utilize the signature suite: ECDSA P-256/SHA-256 according to NIST FIPS 186-5, NIST FIPS 180-4 and NIST SP 800-186 using the algorithm identifiers according to IETF RFC 5480.

An IDevID may utilize the following signature suites:

- ECDSA P-521/SHA-512 according to NIST FIPS 186-5, NIST FIPS 180-4 and NIST SP 800-186 using the algorithm identifiers according to IETF RFC 5480,
- EdDSA instance Ed25519 according to IETF RFC 8032 using Curve25519 according to IETF RFC 7748 and using the algorithm identifiers according to IETF RFC 8410, and
- EdDSA instance Ed448 according to IETF RFC 8032 using Curve448 according to IETF RFC 7748 and using the algorithm identifiers according to IETF RFC 8410.

2240 6.3.3.3 Information Model

2241 6.3.3.3.1 General

2242 The information model for IDevID credentials and trust anchors shall comply to YANG and
2243 NMDA, in particular the YANG modules ietf-keystore and ietf-truststore, as well as subsequent
2244 subclauses of 6.3.3.3.

2245 6.3.3.3.2 Entries

2246 IDevID credentials shall be provided in form of built-in keys of an IA-station by its manufacturer.
2247 In YANG, they are modeled as config-false nodes and are represented in the 'keystore'
2248 container that is instantiated by the YANG module ietf-keystore. The private key shall use the
2249 private-key-type choice hidden-private-key, i.e. the IDevID private key is not presented in
2250 NETCONF/YANG. The details of storing and protecting IDevID private keys as well as using
2251 them for signing purposes are implementation specific.

2252 Trust anchors for IDevID credentials are CNC user-configured data objects; these objects shall
2253 be available as applied configuration (IETF RFC 8342) upon CNCs. In YANG, they are modeled
2254 as config-true nodes and are represented in the 'truststore' container that is instantiated by the
2255 YANG module ietf-truststore.

2256 NOTE IA-station built-in trust anchors for use cases such as firmware/software update are not addressed in this
2257 document.

2258 6.3.3.3.3 Entry Manifolddness

2259 An IA-station shall possess one IDevID credential with a certification path plus trust anchor
2260 information issued under the required signature suite according to 6.3.3.2.3 as part of its factory
2261 default state.

2262 If an IA-station supports an optional signature suite according to 6.3.3.2.3, it shall possess in
2263 addition one IDevID credential with a certification path plus trust anchor information issued
2264 under the optional signature suite as part of its factory default state.

2265 An IA-station can have additional IDevID credential(s) with a certification path plus trust anchor
2266 information issued under a combination of any required or any supported optional DevID
2267 signature suites.

2268 If an IA-station possesses multiple IDevID credentials, then they shall be issued by the same
2269 organization (the IA-station manufacturer). Their EE certificates shall contain the same device
2270 identity information.

2271 A CNC shall support at least one trust anchor for IDevID credentials per supported IA-station
2272 manufacturer.

2273 6.3.3.3.4 Entry Naming

2274 IDevID credentials shall be present in an 'asymmetric-key' entry that is identified as: /ietf-
2275 keystore:keystore/asymmetric-keys/asymmetric-key/name=
2276 IDevID-<SignatureSuiteName>-<CertificateSerialNumberOfEECertificate>.

2277 IDevID trust anchors shall be present in 'certificate' entries that are identified as: /ietf-
2278 truststore:truststore/certificate-bags/certificate-bag/certificate/name=
2279 IDevID-<SignatureSuiteName>-<CertificateSerialNumberOfCACertificate>.

2280 Such entries shall be present underneath a 'certificate-bag' entry that is identified as: /ietf-
2281 truststore:truststore/certificate-bags/certificate-bag/name=IDevID.

2282 6.3.3.4 Processing model**2283 6.3.3.4.1 General**

2284 The processing model for IDevID credentials and trust anchors shall comply to IEEE Std
2285 802.1AR-2018 and 6.3.3.4.

2286 6.3.3.4.2 Credentials**2287 6.3.3.4.2.1 General**

2288 IDevID credentials are used in following use cases:

- 2289 • NETCONF/YANG security setup from factory default; the number of such events scales with
2290 the number of factory resets, i.e. this use case is performed sporadically. It is conducted by
2291 CNCs and encompasses a device identity verification, and
- 2292 • device identity verification happens as a subtask during NETCONF/YANG security setup
2293 from factory default. It can also happen at the discretion of the CNC user. The details of
2294 device identity verification are also subject to given policy.

2295 In these use cases, IA-stations act in claimant role and CNCs act in verifier role:

- 2296 • IA-stations shall present the certification path of and prove private key possession for an
2297 IDevID credential, and
- 2298 • CNCs shall validate the certification path, check the proof-of-possession for the private key,
2299 and verify the obtained device identity information.

2300 6.3.3.4.2.2 Creation

2301 IA-station manufacturers select the form factor for representing verifiable device identity in
2302 IDevID credentials: raw or extended form. The details of the IDevID credential issuance process
2303 are manufacturer-specific and not addressed in this document.

2304 IA-station manufacturers are not required to offer an update feature for IDevID credentials.

2305 6.3.3.4.2.3 Distribution

2306 IA-stations shall supply IDevID credentials in form of built-in keys, see 6.3.3.3.

2307 6.3.3.4.2.4 Use

2308 Verifiers (CNCs) shall perform the following checks when they challenge claimants (IA-stations)
2309 to authenticate themselves by means of an IDevID credential.

- 2310 • IDevID certification path validation according to IETF RFC 5280, Clause 6. Whether this
2311 validation happens with or without revocation checks is at the discretion of the CNC user.
2312 It is the responsibility of the CNC user to supply a trust anchor configuration (set of trusted
2313 certificates or trusted public keys), a revocation check instruction (Boolean) and optionally,
2314 X.509 CRL objects according to IETF RFC 5280, Clause 5, to CNCs. The certification path
2315 validation is passed if and only if the IDevID EE certificate is the leaf of a valid certification
2316 path that ends with a CA certificate which is signed by a configured trust anchor and which
2317 is not revoked (if revocation check is enabled).
- 2318 • Proof-of-possession checking for the private key. The proof-of-possession check is passed
2319 if and only if the IA-station possesses the private key which matches the public key in the
2320 IDevID EE certificate.

- It is the responsibility of the CNC user to establish and supply to CNCs: a device identity verification policy which determines the verifiable device identity subset that shall be checked by the CNC for the IA-stations in a Configuration Domain. This is a subset of {hardware-rev, serial-num, mfg-name, model-name}. The empty subset ("no-identity-check") as well as the whole set are allowed. The device identity verification for an IA-station instance shall behave according to the following list of requirements.
 - If this subset is empty, then the device identity check is passed. If the user chooses not to verify identity, information about the devices is considered unreliable. Tracking the unverified status of such devices is the responsibility of the user. It is the responsibility of the user to establish policies for the use of such devices.
 - If this subset is non-empty, then the CNC performs the following expected vs. actual check for each verifiable device identity item in this subset:
 - The check for any item in this subset is passed if the expected value (from ietf-hardware YANG module) matches the actual value (from the verifiable device identity URN value for this document in the subjectAltName extension of the IDevID EE certificate). This check fails if the IDevID has raw form.
 - The device identity check is passed if it is passed for all items in the subset.

IDevIDs in raw form (without verifiable device identity URN) can be used if the device identity verification setting option "no-identity-check" is employed. This allows to perform the NETCONF/YANG security setup from factory default for IA-stations with IDevID credentials in raw form. From CNC perspective these IA-stations remain anonymous.

NOTE This document does not specify a mechanism for device identity verification for IDevIDs in raw form. Whether and how device identity checks for such IA-stations are done in an offline mode is at the discretion of CNC users.

6.3.3.4.2.5 Storage

Credentials shall be stored persistently upon an IA-station. The details for implementing this persistent storage are IA-station manufacturer-specific and not addressed in this document.

IA-stations shall support storage of at least one IDevID credential.

6.3.3.4.2.6 Revocation

It is the responsibility of IA-station manufacturers to report revocation for the IDevID credentials issued by them in form of X.509 CRL objects according to IETF RFC 5280, Clause 5. These objects are made available in a form that allows relying parties, i.e. CNC users to retrieve them at their own discretion.

CNC users decide whether they support IDevID certification path validation with or without revocation:

- if revocation checks are disabled, then certificate path validation shall be performed according to IETF RFC 5280, 6.1, or
- if revocation checks are enabled, then certificate path validation shall be performed according to IETF RFC 5280, 6.1 and 6.3.

NOTE It is the responsibility of CNC users to obtain up-to-date X.509 CRL objects from manufactures and make them locally available for verifiers.

6.3.3.4.3 Trust anchors

6.3.3.4.3.1 General

Trust anchors are input arguments for certification path validation according to IETF RFC 5280, 6.1.1, input argument (d). Relying parties decide about these input arguments in a discretionary fashion, i.e. these objects are not created and distributed as literal trust anchor objects but in a pre-material form of, for example, self-signed certificate objects.

2367 NOTE The digital signature in self-signed certificates do not vouch for authenticity of this object: Actor X can issue
 2368 self-signed certificates featuring the name of actor A that cannot be distinguished from self-signed certificates issued
 2369 by A. The mechanisms to verify the authenticity of self-signed certificates are not addressed in this document.

2370 The trust anchors for use cases where IA-stations act in claimant role are determined by CNC
 2371 users.

2372 **6.3.3.4.3.2 Creation**

2373 The details of the issuance and update processes for trust anchors for validation of IDevID
 2374 credentials are not addressed by this document.

2375 **6.3.3.4.3.3 Distribution**

2376 With respect to use cases where IA-stations act in claimant role, e.g. NETCONF/YANG security
 2377 setup and device identity verification, the following model applies.

- 2378 • Issuers (IA-station manufacturers) create and distribute trust anchors. Issuers also provide
 2379 out-of-band means that allow relying parties to check the authenticity of these objects, and
- 2380 • Relying parties (CNC users) check the authenticity of trust anchors and decide about their
 2381 acceptance as trust anchors for certification path validation in a discretionary manner and
 2382 configure their verifiers (CNCs) accordingly.

2383 The details of distribution and validation of trust anchors are not addressed by this document.

2384 **6.3.3.4.3.4 Use**

2385 Trust anchors for IDevID credentials are used for certification path validation according to IETF
 2386 RFC 5280, 6.1.1 d). This concerns CNCs with respect to the use cases NETCONF/YANG
 2387 security setup from factory default, device identity verification.

2388 **6.3.3.4.3.5 Storage**

2389 Trust anchors for IDevID credentials shall be stored persistently upon CNCs. The details for
 2390 implementing this persistent storage are not addressed in this document.

2391 **6.3.3.4.3.6 Revocation**

2392 IA-station manufacturers are not required to support an authority revocation feature for IDevID
 2393 credential certification authorities.

2394 **6.3.4 Security setup based on IDevID**

2395 **6.3.4.1 General**

2396 IA-stations in factory default state shall conduct a security setup sequence for the Configuration
 2397 Domain. This sequence consists of the following steps, each step is described in 6.3.4.

- 2398 • imprintTrustAnchor: imprint of a Configuration Domain specific trust anchor to an IA-station
 2399 that allows to validate LDevID-NETCONF certificates presented by communication partners.
- 2400 • imprintCredential: imprint of a Configuration Domain specific credential to an IA-station, i.e.
 2401 a private key and the corresponding X.509 v3 end entity certificate according to ISO/IEC
 2402 9594-8 as profiled in IETF RFC 5280, Clause 4, (plus intermediate CA certificates, if
 2403 applicable) plus self-signed root CA certificate that serves as own LDevID credential.
- 2404 • imprintCertToNameMapping: imprint a Configuration Domain specific certificate-to-name
 2405 mapping to an IA-station.

2406 Credentials shall be stored persistently upon an IA-station. The details for implementing this
 2407 persistent storage are IA-station manufacturer-specific and not addressed in this document.

2408 IA-stations shall support storage of at least one LDevID-NETCONF credential.

2409 **6.3.4.2 imprintTrustAnchor**

2410 IA-stations in factory default state shall support the imprinting of a single Configuration Domain
2411 specific trust anchor according to IETF RFC 6024 via NETCONF-over-TLS according to a
2412 procedure called "provisional accept of client certificate", which uses an IDevID credential on
2413 NETCONF and TLS server side (IA-station) and a LDevID credential on NETCONF and TLS
2414 client side (for example, a CNC) and operates as follows at the NETCONF and TLS server.

- 2415 a) Challenge the client for TLS client authentication according to IETF RFC 7589 by sending
2416 a CertificateRequest message with an empty certificate_authorities entry.
- 2417 b) Perform certification path validation according to IETF RFC 5280, Clause 6, for the contents
2418 of the client's Certificate message. This certification path validation fails due to a missing
2419 trust anchor for the LDevID credential.
- 2420 c) Provisionally accept the failing certification path validation when the reason is "no matching
2421 trust anchor" (and only this reason) and proceed with the TLS exchange.
- 2422 d) Expect the client to send a trust anchor for LDevID over the provisionally accepted TLS
2423 session (no other object type).
- 2424 e) If the trust anchor in the NETCONF application payload was accepted, then redo the priorly
2425 failing certification path validation using this trust anchor, see step b).
- 2426 f) If this certification path revalidation is successful, then keep the TLS session alive and send
2427 an <rpc-reply> with success. The client then is expected to perform the NETCONF
2428 exchanges for imprintCredential (described in 6.3.4.3) and for imprintCertToNameMapping
2429 (described in 6.3.4.4) via the already established TLS session.
- 2430 g) If this certification path revalidation is not successful, then terminate the TLS session. The
2431 usual NETCONF/YANG hygiene applies. This is expected to remove the entry in the ietf-
2432 truststore that was created in step d).

2433 NOTE This "provisional accept of client certificate" is a mirrored version of the "provisional accept of server cert"
2434 in IETF RFC 8995.

2435 The "provisional accept of client cert" in factory default state shall skip the certificate-to-name
2436 mapping and shall use the NACM recovery session, i.e. skip permission checking. In this model
2437 all authenticated clients are accepted as authorized for doing the first imprinting of the LDevID
2438 credential and the corresponding trust anchor. Only contextual checks such as "once only when
2439 being in factory default state" are feasible. This model is also known as "trust on first use"
2440 (TOFU) and, e.g. also allows to read contents of the ietf-hardware module by the client for an
2441 extended identity check.

2442 The imprinting NETCONF client checks the actual server identity that is stated by the IA-station
2443 on TLS level.

2444 The NETCONF client checks the IDevID end entity certificate presented by the NETCONF
2445 server on TLS level for existence of subjectAltName extension with GeneralName entries of
2446 type uniformResourceIdentifier. If an entry contains the namespace identifier and the
2447 namespace-specific string as defined in 6.3.3.2.2, the presented server certificate is in
2448 extended form, otherwise it is in raw form.

2449 In case that the server certificate is in raw form, the following matching can be done:

- 2450 • Match a list of accepted (or blocked) manufacturers against the issuer or subject field entries
2451 of the certificate.
- 2452 • Match a list of accepted (or blocked) product instances against the product serial number
2453 from the subject field per accepted manufacturer.
- 2454 • Match the end entity certificate object as a whole against a list of pinned certificates.

2455 In case that the server certificate is in extended form, the following additional matching can be
 2456 done: match the q-components included in the verifiable device identity according to
 2457 6.3.3.2.2 against those that can be read out from the corresponding leaves of the YANG module
 2458 ietf-hardware or against reference values obtained by a method not addressed in this document.

2459 Details of how the matching happens depend on the implementation of the client that performs
 2460 this imprinting.

2461 The LDevID-NETCONF trust anchor certificate shall be imprinted using the truststore container
 2462 of the ietf-truststore module with:

- 2463 • /ts:truststore/ts:certificate-bags/ts:certificate-bag/ts:name = IEC60802,
- 2464 • /ts:truststore/ts:certificate-bags/ts:certificate-bag/[ts:name=IEC60802]/,
 - 2465 – ts:certificate/ts:name = IEC60802-LDevID,
 - 2466 – ts:certificate/ts:cert-data containing the IEC60802-LDevID trust anchor certificate data
 - 2467 object of type trust-anchor-cert-cms according to IETF RFC 9640, i.e., enveloped in
 - 2468 Base64-encoded CMS SignedData in degenerated form "certs-only" (no signature
 - 2469 value), and
 - 2470 – The imprintTrustAnchor step shall use the NETCONF operation <edit-config> according
 - 2471 to IETF RFC 6241 for the truststore container. The NETCONF operation <commit> is not
 - 2472 yet applied, but rather after successful completion of all security setup sequence steps.

2473 6.3.4.3 imprintCredential

2474 6.3.4.3.1 General

2475 The LDevID-NETCONF end entity certificate shall be provided as X.509 v3 public key certificate
 2476 according to ISO/IEC 9594-8 as profiled in IETF RFC 5280, Clause 4, with the following criteria:

- 2477 • contains the FQDN of the NETCONF server in its subjectAltName extension according to
- 2478 IETF RFC 7589, Clause 6, and IETF RFC 6125, 2.2 and B.7;
- 2479 • contains a public key and is signed by a signature suite according to 5.5.4.2 or 5.6.3;
- 2480 • contains a digitalSignature in its keyUsage extension;
- 2481 • has a finite validity period.

2482 NOTE The actual length of the validity period is at the discretion of the user of the Configuration Domain.

2483 Depending on the key generation capabilities, different steps are applied to this keystore
 2484 container.

2485 6.3.4.3.2 Internal key generation

2486 For IA-station with internal key generation capabilities, two NETCONF exchanges are
 2487 performed. Processing steps for the first NETCONF exchange shall be applied as follows at the
 2488 NETCONF server.

2489 a) Receive and process the NETCONF request message with action <generate-csr> and input
 2490 values as follows:

- 2491 • /ks:keystore/ks:asymmetric-keys/ks:asymmetric-key/[ks:name=LDevID_NETCONF]/ks:
- 2492 generate-csr/ks:input/ks:csr-format containing derived identity p10-csr-format according
- 2493 to IETF RFC 9640, and
- 2494 • /ks:keystore/ks:asymmetric-keys/ks:asymmetric-key/[ks:name=LDevID_NETCONF]/ks:
- 2495 generate-csr/ks:input/ks:csr-info containing a Base64-encoded PKCS#10
- 2496 CertificationRequestInfo according to IETF RFC 2986, Clause 4.

- 2497 b) Base64-decode the <csr-info> value and parse it as a PKCS#10 CertificationRequestInfo
2498 object.
- 2499 c) Extract the algorithm information from the child element SubjectPublicKeyInfo of
2500 CertificationRequestInfo and randomly generate a key pair for the specified algorithm.
- 2501 d) Internally store the private key together with its metadata for example, algorithm information,
2502 <name> value in a secure manner.
- 2503 e) Put the public key into the (parsed) PKCS#10 CertificationRequestInfo.
- 2504 f) Serialize the PKCS#10 CertificationRequestInfo (including the public key).
- 2505 g) Use the private key to create signature value for the (serialized) PKCS#10
2506 CertificationRequestInfo (including the public key).
- 2507 h) Create a NETCONF reply message with /ks:keystore/ks:asymmetric-keys/ks:asymmetric-
2508 key/[ks:name=LDevID-NETCONF]/ks:generate-csr/ks:output/ks:p10-csr containing the data
2509 object of the previous step.

2510 In the second NETCONF exchange, the LDevID-NETCONF end entity certificate (plus
2511 intermediate CA certificates) shall be imprinted using the keystore container of the ietf-keystore
2512 module with:

- 2513 • /ks:keystore/ks:asymmetric-keys/ks:asymmetric-key/ks:name = LDevID-NETCONF,
- 2514 • /ks:keystore/ks:asymmetric-keys/ks:asymmetric-key/[ks:name=LDevID-NETCONF]/,
2515 – ks:certificates/ks:certificate/ks:name = LDevID-NETCONF, and
2516 – ks:certificates/ks:certificate/ks:cert-data containing the certificate chain LDevID-
2517 NETCONF end entity certificate (plus intermediate CA certificates, if applicable) plus
2518 self-signed root CA certificate as data object of type end-entity-cert-cms according to
2519 IETF RFC 9640

2520 The imprintCredential step shall use the NETCONF operation <edit-config> according to IETF
2521 RFC 6241 for the keystore container. The NETCONF operation <commit> is not yet applied, but
2522 rather after successful completion of all security setup sequence steps.

2523 **6.3.4.3.3 External key generation**

2524 External key generation can be used for IA-stations without internal key generation capability.
2525 For external key generation, one NETCONF exchange is performed.

2526 The LDevID-NETCONF private key and end entity certificate (plus intermediate CA certificates)
2527 shall be imprinted using the keystore container of the ietf-keystore module with:

- 2528 • /ks:keystore/ks:asymmetric-keys/ks:asymmetric-key/ks:name = LDevID-NETCONF,
- 2529 • /ks:keystore/ks:asymmetric-keys/ks:asymmetric-key/[ks:name=LDevID-NETCONF]/,
2530 – ks:certificates/ks:certificate/ks:name = LDevID-NETCONF,
2531 – ks:certificates/ks:certificate/ks:public-key-format describing the encoding of the public
2532 key of the selected cryptographic algorithm according to IETF RFC 9640,
2533 – ks:certificates/ks:certificate/ks:public-key containing the public key value in the selected
2534 public-key-format,
2535 – ks:certificates/ks:certificate/ks:private-key-format describing the encoding of the private
2536 key of the selected cryptographic algorithm according to IETF RFC 9640,
2537 – ks:certificates/ks:certificate/ks:cleartext-private-key containing the private key value in
2538 the selected private-key-format,

2539 NOTE The private key is confidentiality protected by NETCONF-over-TLS even if the option <cleartext-private-key>
2540 is used.

- 2541 – ks:certificates/ks:certificate/ks:name = LDevID-NETCONF, and
- 2542 – ks:certificates/ks:certificate/ks:cert-data containing the certificate chain LDevID-
2543 NETCONF end entity certificate (plus intermediate CA certificates, if applicable) plus

2544 self-signed root CA certificate as data object of type end-entity-cert-cms according to
2545 IETF RFC 9640.

2546 The imprintCredential step shall use the NETCONF operation <edit-config> according to IETF
2547 RFC 6241 for the keystore container. The NETCONF operation <commit> is not yet applied, but
2548 rather after successful completion of all security setup sequence steps.

2549 External key generation can introduce security vulnerabilities during the generation and loading
2550 process. Ensuring those processes are secure is the responsibility of the user and not
2551 addressed in this document.

2552 **6.3.4.4 imprintCertToNameMapping**

2553 The Configuration Domain specific certificate-to-name mapping is imprinted in the ietf-netconf-
2554 server YANG module under the following node.

- 2555 • /ncs:netconf-server/ncs:listen/ncs:endpoint/ncs:tls/ncs:netconf-server-
2556 parameters/ncs:client-identity-mappings/ncs:cert-to-name, with the following leaves:
 - 2557 – id = 1,
 - 2558 – fingerprint = Configuration Domain specific fingerprint of the LDevID-NETCONF trust
2559 anchor using the hash algorithm sha256 according to IETF RFC 7589, Clause 7, and
 - 2560 – map-type = ext-60802-roles.

2561 The application of this map-type is described in 6.3.5, steps e) and f).

2562 The imprintCertToNameMapping step uses the NETCONF operation <edit-config> according to
2563 IETF RFC 6241 for the certificate-to-name mapping. Afterwards the NETCONF operation
2564 <commit> is applied to finalize the security setup sequence steps and to leave the factory
2565 default state.

2566 **6.3.5 Secure configuration based on LDevID-NETCONF**

2567 Configuration by NETCONF/YANG is protected by NETCONF-over-TLS as described in 6.3.2.1
2568 and NACM as described in 6.3.2.2. The NETCONF/YANG servers and clients shall use LDevID
2569 credentials for authentication.

2570 The procedure called "provisional accept of client certificate" as described in 6.3.4.2 is not
2571 applied anymore if the IA-station has left the factory default state. Instead, after successful
2572 establishment of a TLS session according to IETF RFC 7589 and IETF draft-ietf-netconf-over-
2573 tls13, the NETCONF server shall perform a certificate-to-name mapping and authorization
2574 check as follows.

- 2575 a) Compare the fingerprint of the trust anchor of the NETCONF client's certification path with
2576 the fingerprint contained in cert-to-name list entries of the x509c2n container for equal
2577 values.
- 2578 b) If no cert-name list entry match is found, then terminate the TLS session.
- 2579 c) If a cert-to-name list entry match is found, then verify if the map-type is equal to ext-60802-
2580 roles.
- 2581 d) If the map-type does not match, then terminate the TLS session.
- 2582 e) If the map-type value matches, then extract the role values from the id-60802-pe-roles
2583 certificate extension of the NETCONF client's TLS-authenticated end entity certificate. The
2584 output is a list of string values from the enumeration of specified role names according to
2585 this document.
- 2586 f) The list of role name string values is provided as input to NACM for permission checking.
2587 The access to the requested resource is checked according to the rules configured in the
2588 nacm container of the ietf-netconf-acm YANG module.

2589 The NETCONF client checks if the expected identity to address the NETCONF server (IP
2590 address or DNS name) matches to the actual server identity that is stated by the IA-station on
2591 TLS level. This shall be done by comparing the expected identity with the subjectAltName
2592 extension of the TLS authenticated LDevID-NETCONF end entity certificate of the NETCONF
2593 server.

2594 6.4 Management

2595 6.4.1 General

2596 Subclause 6.4 describes a model for configuration, deployment, and management of an
2597 industrial automation network.

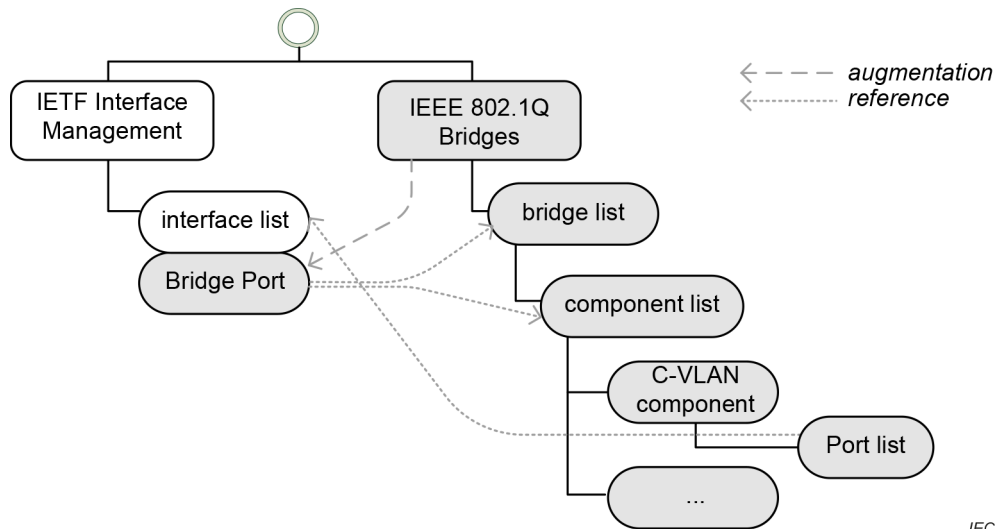
2598 6.4.2 IA-station management model

2599 6.4.2.1 General

2600 The management model of IA-stations covers simple end station IA-stations as well as
2601 combined IA-stations as described in 4.3. The IA-station management model is applied for
2602 topology discovery, network provisioning and stream establishment.

2603 6.4.2.2 IEEE 802.1Q management model

2604 In industrial automation both Bridge and end station components make use of IEEE 802.1Q
2605 specified functionality (for example, traffic classes, gate control). Thus, the IEEE 802.1Q
2606 management model is the basic management model to be applied to all IA-stations. Figure 15
2607 shows the implementation of the IEEE Std 802.1Q Bridge model in YANG as specified in IEEE
2608 Std 802.1Q-2022, Clause 48. The IETF Interface Management YANG data model is specified
2609 in IETF RFC 8343.



IEC

2610
2611 **Figure 15 – Generic IEEE 802.1Q YANG Bridge management model**

2612 The IEEE 802.1Q Bridge model is organized as a bridge list where each bridge includes an
2613 underlying component list (for example, C-VLAN components). Each component has a Port list
2614 attached with references to the representation of the ports in the IETF interface list. The
2615 managed data of the ports is defined as Bridge Port augmentation to the IETF interface model.
2616 Each Bridge Port includes a reference to its bridge and component instances in the IEEE
2617 802.1Q Bridge model.

2618 The YANG data model for an IEEE 802.1Q Bridge is applied to IA-stations as follows.

- 2619 • Each functional unit of an IA-station is modeled as bridge entry in the bridge list.

- Each Bridge and end station component of an IA-station is modeled as C-VLAN component.
- IA-station components belonging to the same functional unit are added to the component list of this functional unit's bridge entry.
- Each IA-station external or internal port is modeled as Bridge Port.

IA-station ports belonging to the same component are added to the Port list of the related component list entry.

Further YANG data models which are relevant for IA-stations are described in 6.4.9.

6.4.2.3 Internal LAN connection model

The modeling of internal connections between C-VLAN components within an IA-station is aligned to IEEE Std 802.1Q-2022, 17.3.2.2, which includes an I-LAN interface. As shown in Figure 16, the I-LAN interface is modeled as an ilan IETF interface object (see IETF RFC 7224) together with appropriate `higher-layer-if` and `lower-layer-if` reference objects to describe the internal connection.

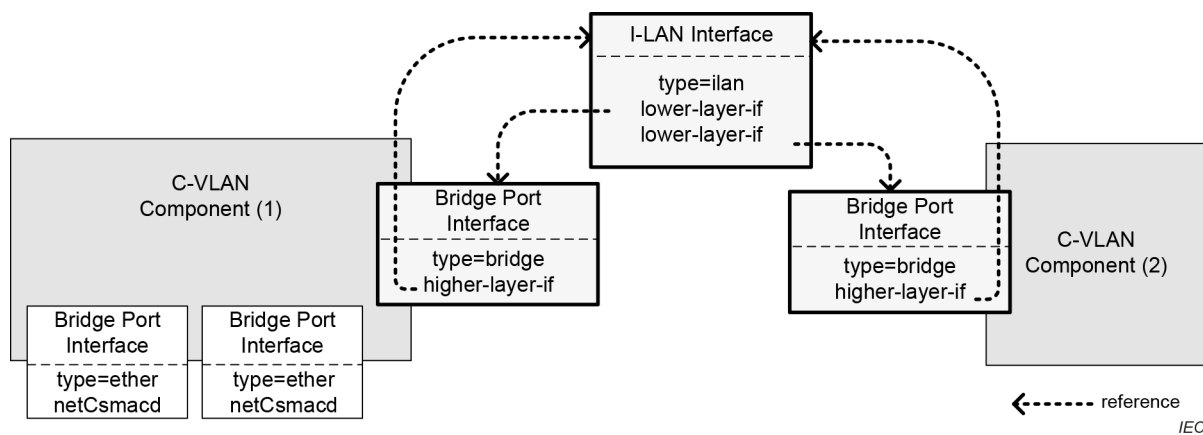


Figure 16 – Internal LAN connection management model

This internal LAN connection model comprises three configuration steps.

- The internal Ports of the C-VLAN components are modelled as IETF interfaces of type bridge with Bridge Port augmentation.
- An additional I-LAN interface of type ilan as described in IETF RFC 7224 is created.
- The I-LAN interface references the internal Bridge Port interfaces of the connected C-VLAN components as lower-layer-if, and the internal Bridge Port interfaces of the connected C-VLAN components reference the I-LAN interface as higher-layer-if.

Figure 17 shows the application of this model to the example IA-station of Figure 16.

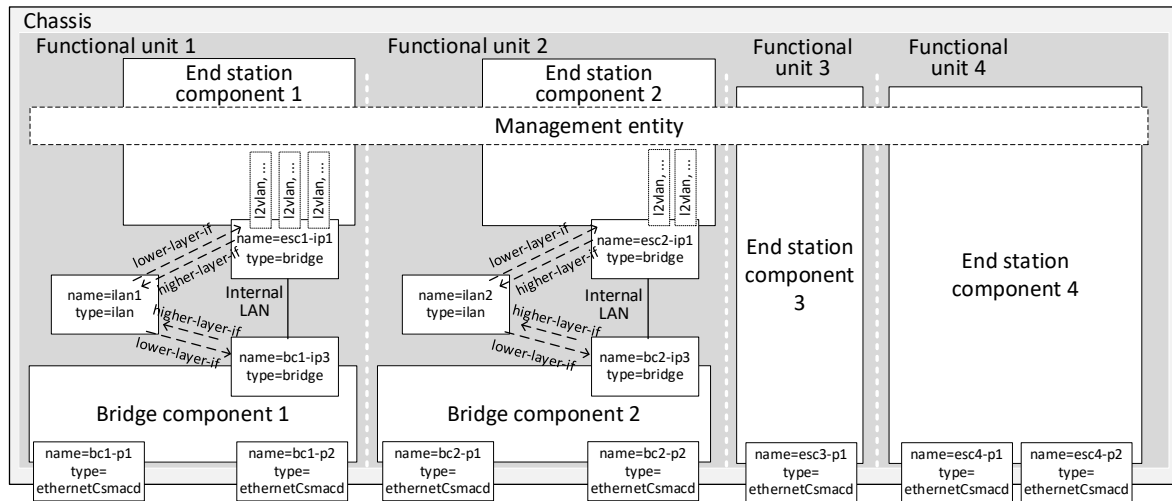


Figure 17 – IA-station example with IETF interfaces

NOTE Figure 17 represents an abstract model and is not intended to imply a particular implementation or partitioning.

Late binding of IA-station applications to the configured VLANs and priorities is performed using the `ieciieee60802-tt-mapping` YANG module in 6.4.10.8.7.

6.4.2.4 Spanning Tree, VLAN and TE-MSTID configuration

C-VLAN Bridge components of IA-stations shall support:

- the Common and Internal Spanning Tree (CIST) calculated by the Multiple Spanning Tree Algorithm and Protocol (MSTP), and
- the Traffic Engineering Multiple Spanning Tree Instance Identifier (TE-MSTID) as specified in IEEE Std 802.1Q-2022, 5.5.2.

The MSTP configuration is either default or accomplished by IA-station specific means.

CNCs configure VLANs in the `vlan` list in the `bridge-vlan` container of the `ieee802-dot1q-bridge` YANG module. Ports are assigned to a `vlan` as `static-filtering-entries` in a `filtering-database`.

NOTE `vlan`, in lowercase, refers to a YANG element.

VLANs are assigned to filtering databases in the `vid-to-fid` list of the `bridge-vlan` container. The filtering databases, and in consequence the VLANs, are by default assigned to the MSTP calculated Internal Spanning Tree and can be assigned to the TE-MSTID by management. IA-time-aware streams and IA-streams are assigned to the TE-MSTID.

TE-MSTID assignment is accomplished via the `bridge-mst` container of the `ieee802-dot1q-bridge` YANG module.

The following example shows the VID/FID/MSTID configuration of an IA-station's C-VLAN Bridge component, which supports three VLANs in three Forwarding Databases (VID 100 in FID 1, VID 101 in FID 2 and VID 102 in FID 3). FID 2 and FID 3 – and in consequence VID 101 and VID 102 – are assigned to the TE-MSTID. FID 1 – and in consequence VID 100 – is not assigned to a MSTID and thus, is implicitly assigned to the Internal Spanning Tree (IST).

Figure 18 shows the representation of this example configuration in the MST configuration table.

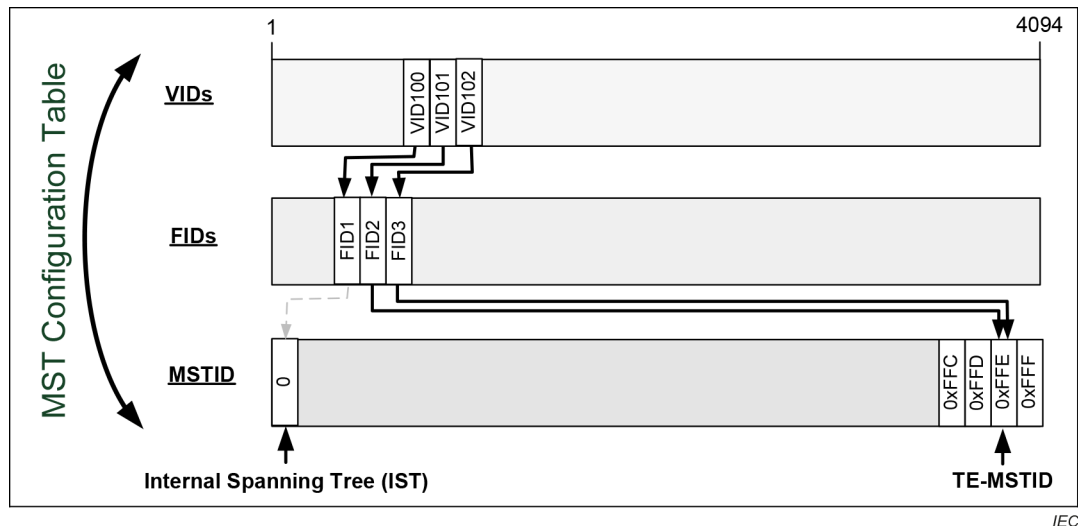


Figure 18 – VID/FID/MSTID example

The YANG-based configuration of this example is shown as YANG instance data snippet of the ieee802-dot1q-bridge YANG module. Herein the MST configuration table is included in component "bridge-component-x", which is part of bridge "functional-unit-x". It is the responsibility of the user to ensure that traffic type names are consistent with those provided in the YANG module iecieee60802-tt-mapping (see 6.4.10.8.7).

```

<ieee802-dot1q-bridge xmlns:dot1q="urn:ieee:std:802.1Q:yang:ieee802-dot1q-bridge">
  <bridges>
    <bridge> <!-- list -->
      <name>functional-unit-x</name>
      ...
      <component> <!-- list -->
        <name>bridge-component-x</name>
        ...
        <bridge-vlan>
          <version>2</version> <!-- MST supported -->
          ...
          <vlan>
            <vid>100</vid>
            <name>60802-best-effort-hl-100</name> <!-- best effort high and
low -->
          </vlan>
          <vlan>
            <vid>101</vid>
            <name>60802-isochronous-101</name> <!-- isochronous -->
          </vlan>
          <vlan>
            <vid>102</vid>
            <name>60802-isochronous-102R</name> <!-- isochronous -->
          </vlan>
          ...
          <vid-to-fid>
            <vid>100</vid>
            <fid>1</fid>
          </vid-to-fid>
          <vid-to-fid>
            <vid>101</vid>
            <fid>2</fid>
          </vid-to-fid>
          <vid-to-fid>
            <vid>102</vid>
            <fid>3</fid>
          </vid-to-fid>
        </bridge-vlan>
      ...
    </bridge>
  </bridges>

```

```

2718         <bridge-mst>
2719             ...
2720             <fid-to-mstid> <!-- list -->
2721                 <!-- fid 1 is implicitly assigned to mstid 0 -->
2722                 <fid>2</fid>
2723                 <mstid>4094</mstid> <!-- TE-MSTID -->
2724             </fid-to-mstid>
2725             <fid-to-mstid> <!-- list -->
2726                 <fid>3</fid>
2727                 <mstid>4094</mstid> <!-- TE-MSTID -->
2728             </fid-to-mstid>
2729         </bridge-mst>
2730         ...
2731     </component>
2732 </bridge>
2733 </bridges>
2734 </ieee802-dot1q-bridge>

```

2735 6.4.3 Discovery of IA-station internal structure

2736 LLDP provides information about the external connectivity of IA-stations. To identify the internal
2737 structure of complex IA-stations (see 4.3) the IEEE 802.1Q management model (see 6.4.2.2)
2738 and the IETF Interface management model are applied.

- 2739 • The functional units of an IA-station are represented as bridge entries in the bridge-list.
- 2740 • The components of a functional unit are represented as component entries in the associated
2741 bridge entry's component-list.
- 2742 • Internal LAN connections between components of a functional unit are identified by I-LAN
2743 entries in the IETF interface list (6.4.2.3).

2744 6.4.4 Network engineering model

2745 To understand the requirements for network configuration, deployment and management, an
2746 engineering model covering industrial use cases is required. The "fully centralized model"
2747 described in IEEE Std 802.1Q-2022, 46.1.3.3 includes two functional entities: the CUC and the
2748 CNC. The relationship between user and network configuration is described in IEEE Std
2749 802.1Q-2022, Clause 46. This document further elaborates this relationship to address use
2750 cases for industrial automation. A conceptual block diagram of a CNC is shown in Figure 19,
2751 which adds further details to the CNC specified in IEEE Std 802.1Q-2022 to serve the industrial
2752 automation use case. The following functional entities are introduced.

2753 a) The Topology Discovery Entity (TDE)

2754 The topology discovery entity is responsible for the topology discovery (i.e. Bridge
2755 component and end station component discovery). The TDE also performs a topology
2756 verification in cases where an expected topology is provided by the engineering tool. The
2757 resulting topology information is used by the CNC. The TDE detects added or removed IA-
2758 stations, including internal structure and connectivity. Thus, the CNC becomes aware of
2759 them. Overall, the TDE discovers and maintains an inventory of the devices, including their
2760 capabilities and the topology they form.

2761 b) The Path Entity (PE)

2762 The PE computes, establishes and maintains the forwarding paths for the IA time-aware
2763 stream and IA stream traffic type categories according to 4.7.3.

2764 c) The Sync Tree Entity (STE)

2765 The STE computes, establishes and maintains the sync trees. For example, for Working
2766 Clock and Global Time.

2767 d) The Resource Allocation Entity (RAE)

2768 The RAE is responsible for the allocation of the resources that are necessary for all traffic
2769 type categories, according to 4.7.3, to meet their requirements via their forwarding paths.
2770 For example, frame buffers at egress ports and FDB entries.

2771 e) The Network Provisioning Entity (NPE)

2772 The NPE applies a network policy provided by the Engineering Tool to the IA-stations within
 2773 the Configuration Domain. It uses the information discovered by the TDE to create a network
 2774 configuration based upon this policy which is then applied to all IA-stations. The CNC uses
 2775 the chosen network configuration together with the discovered IA-stations and their
 2776 capabilities as input for its stream calculation and deployment.

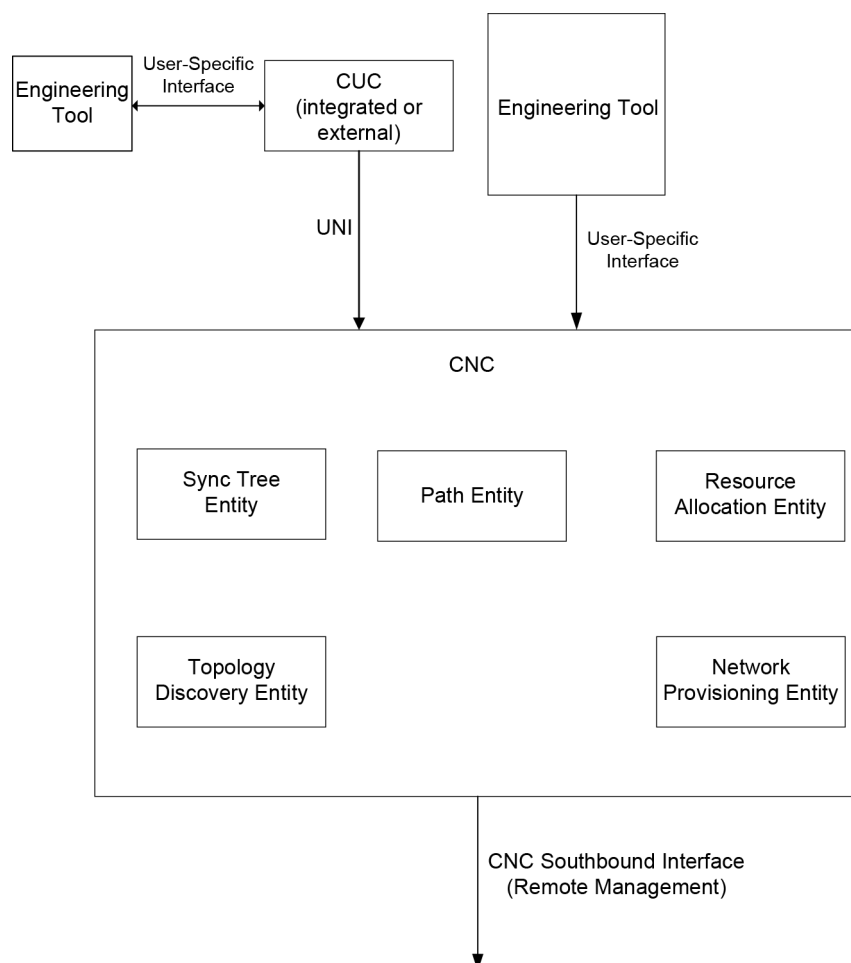
2777 A CNC includes these functional entities. The implementation of these functional entities and
 2778 the CNC can vary. The means of communication among these functional entities is
 2779 implementation dependent.

2780 If there are multiple CNCs in one Configuration Domain, then, by some means not addressed
 2781 by this document, only a single CNC is in charge at any time in the given Configuration Domain.

2782 The CNC can be in a dedicated station or integrated into any IA-controller or IA-device.
 2783 Generally, its engineering tool interface is user-specific and can only work with the compatible
 2784 engineering tools. The definition of this interface is not addressed in this document.

2785 The CUC can be in a dedicated station or integrated into any IA-controller or IA-device.
 2786 Generally, the CUC is user-specific. In industrial automation use cases, an IA-controller
 2787 integrated CUC is very likely.

2788 For stream establishment, the UNI of the CNC component is exposed.



IEC

Figure 19 – Structure and interfaces of a CNC

2791 Figure 20 shows an example of the structure of an IA-station which the CNC can discover and
 2792 manage.

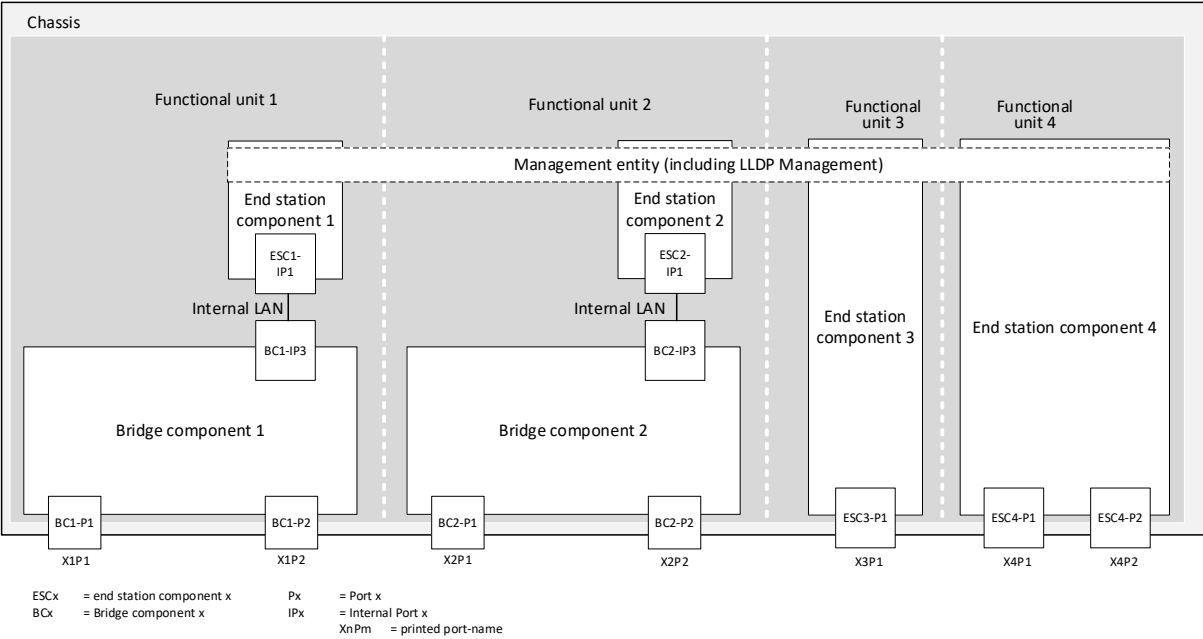


Figure 20 – IA-station structure example

Figure 21 shows the interaction of IA-stations with the CNC.

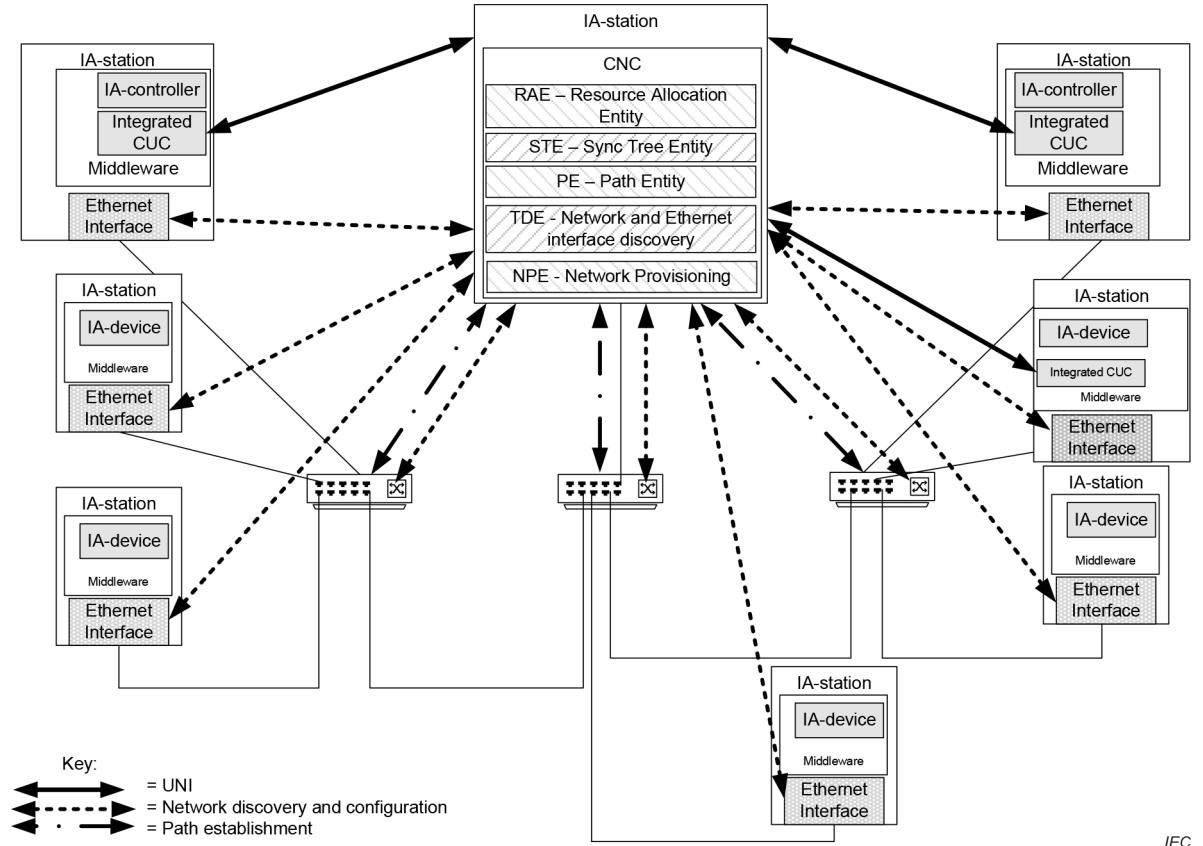


Figure 21 – CNC interaction

6.4.5 Operation

6.4.5.1 General

A representative model for network configuration is shown in Figure 22. This diagram maintains the traditional role of the IA-controller and the IA-device in an industrial automation network. IA-devices and IA-controllers require configuration from engineering tools (refer to engineering tools A, B, D, and E). These tools and associated interfaces are not addressed by this document. In this example, engineering tool C communicates directly with the CNC to provide traffic requirements for the network. The protocols that the engineering tool uses for communication with end stations are specific to the user application.

The UNI is the interface to the CNC which is serviced by NETCONF over TLS. The UNI service recognizes that industrial automation communications are typically connection oriented. There is a communication initiator, typically in an IA-controller, which is responsible for establishing those connections, determining what data is of interest and providing the required update rate. So, while an application/middleware of an IA-station (for example a Drive) understands what information it can produce and the maximum rate at which that information can be provided, until an IA-controller establishes a connection with that device, it does not know where that information goes and what update rate is required to close the control loop. The IA-controller gets this information from its engineering tool. There can be multiple IA-controllers in each Configuration Domain. The CNC uses the topology, the device capabilities, the device configuration, and the traffic specifications from the user to calculate a path for each Talker-Listener pair. The UNI then provides stream identification (VLAN, DMAC, etc.) to the middleware.

The operational management model, see Figure 22, reflects the model used in industrial automation. Figure 22 shows an active CNC managing multiple IA-stations. Each station can wholly incorporate a CUC and interact with the CNC directly.

Security requirements (see 6.3) are an important consideration for these networks and are integrated into the design, configuration, and deployment of any management model.

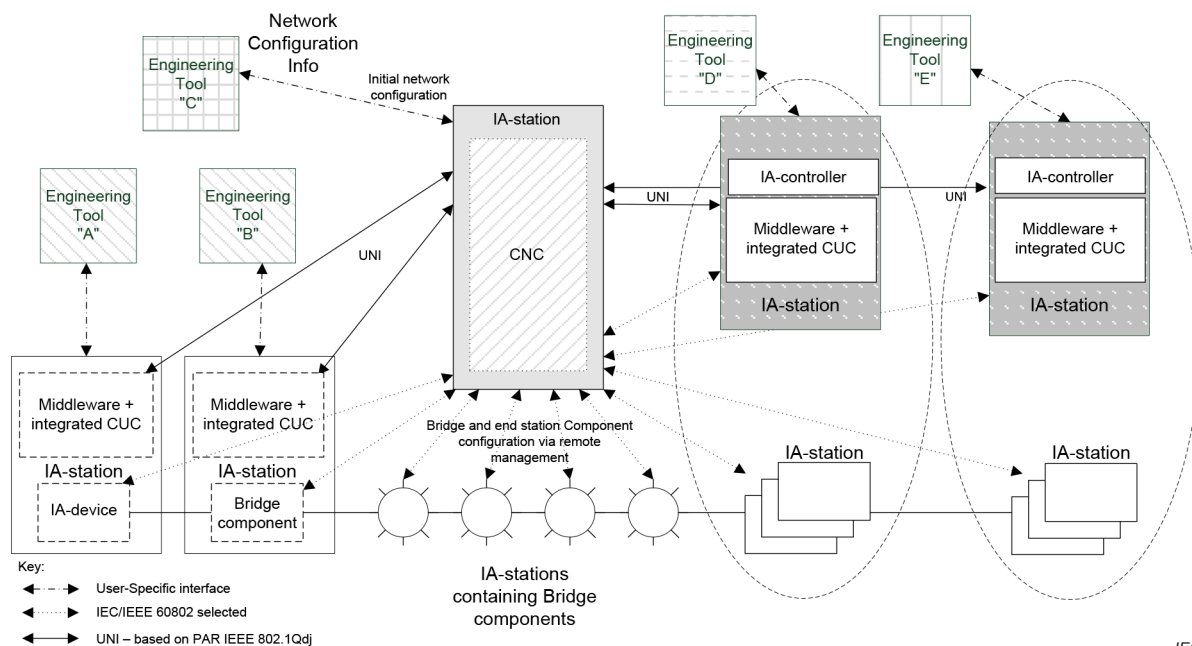


Figure 22 – Operational management model

Figure 23 shows the steps that are typically performed in the scope of the CUC-CNC interaction.

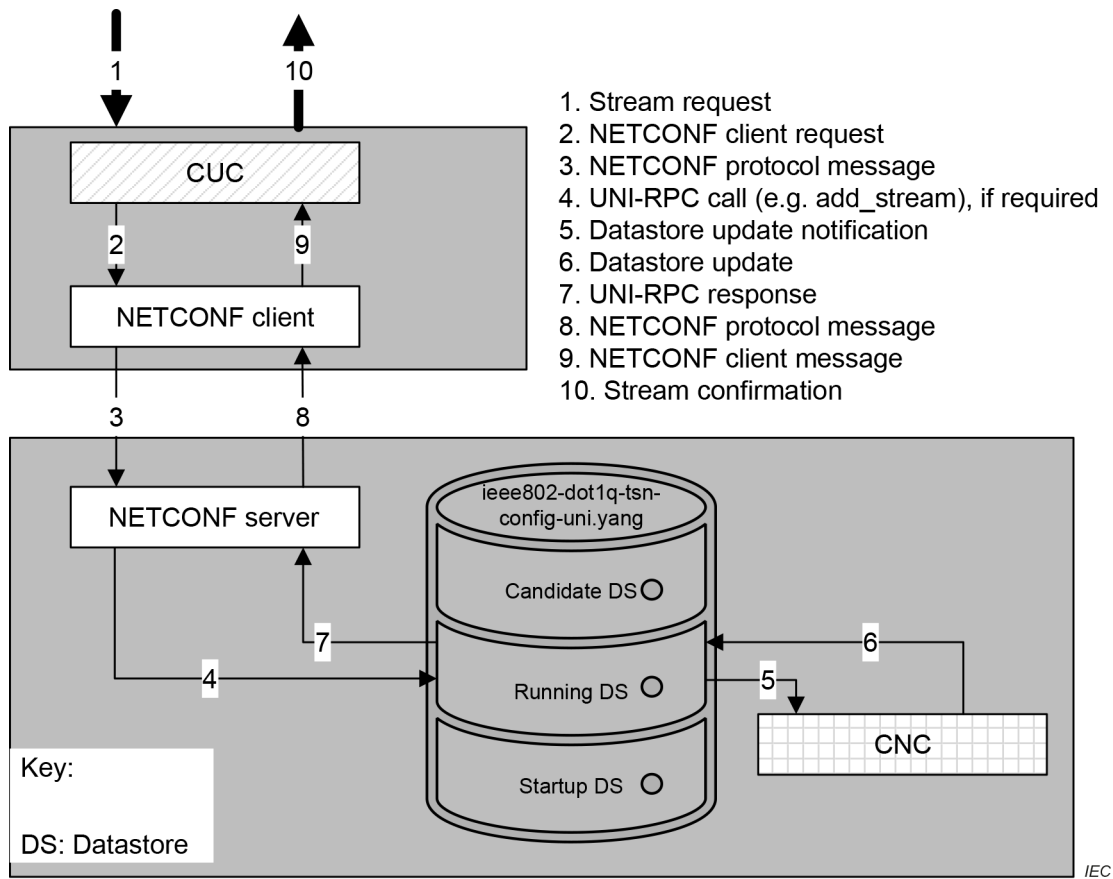


Figure 23 – UNI service model

After the computation of the paths and the scheduling or shaping configuration have been done, the CNC configures the IA-stations via NETCONF client. The typical steps that are performed in this process are shown in Figure 24 below.

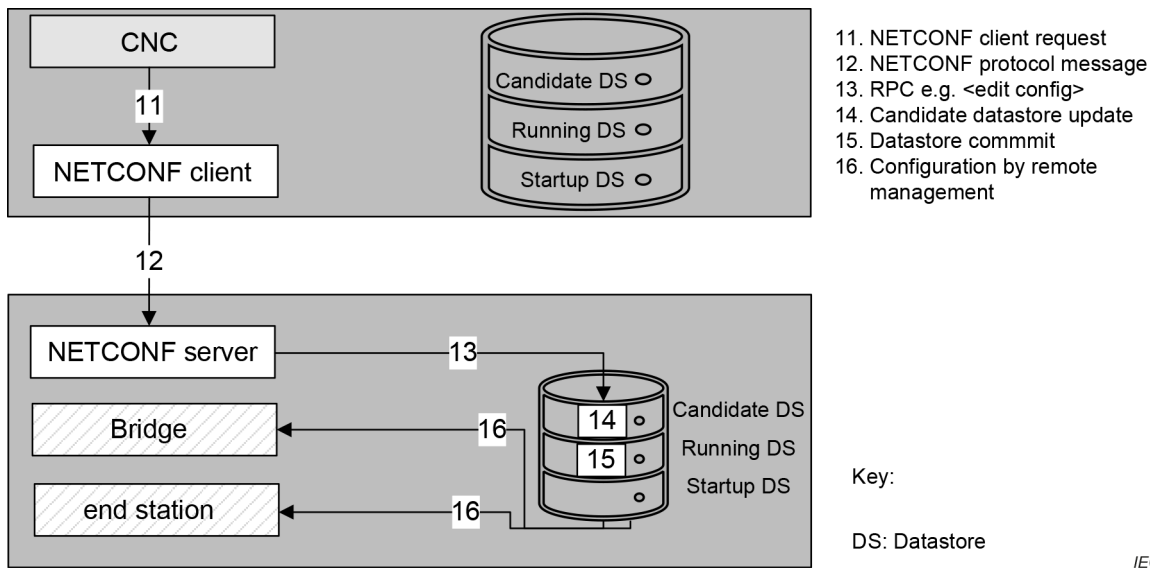


Figure 24 – CNC southbound

Instances of NETCONF servers and clients within a Configuration Domain are shown in Figure 25. IA-stations that contain a CNC or CUC entity contain both a NETCONF server and a NETCONF client. The CUC utilizes a NETCONF client for the UNI. The CNC utilizes a

NETCONF server to accommodate the UNI as well as remote network management of the end stations and bridges that are contained in the same chassis as the CNC entity. The CNC utilizes a NETCONF client for the southbound interface of the CNC, i.e. for the remote management of the bridges and end stations in the scope of stream configuration. All IA-stations have a NETCONF server to make remote management possible. The NETCONF server used by the CNC serves multiple NETCONF Clients (CUCs) within a single Configuration Domain whose requests clients can occur simultaneously.

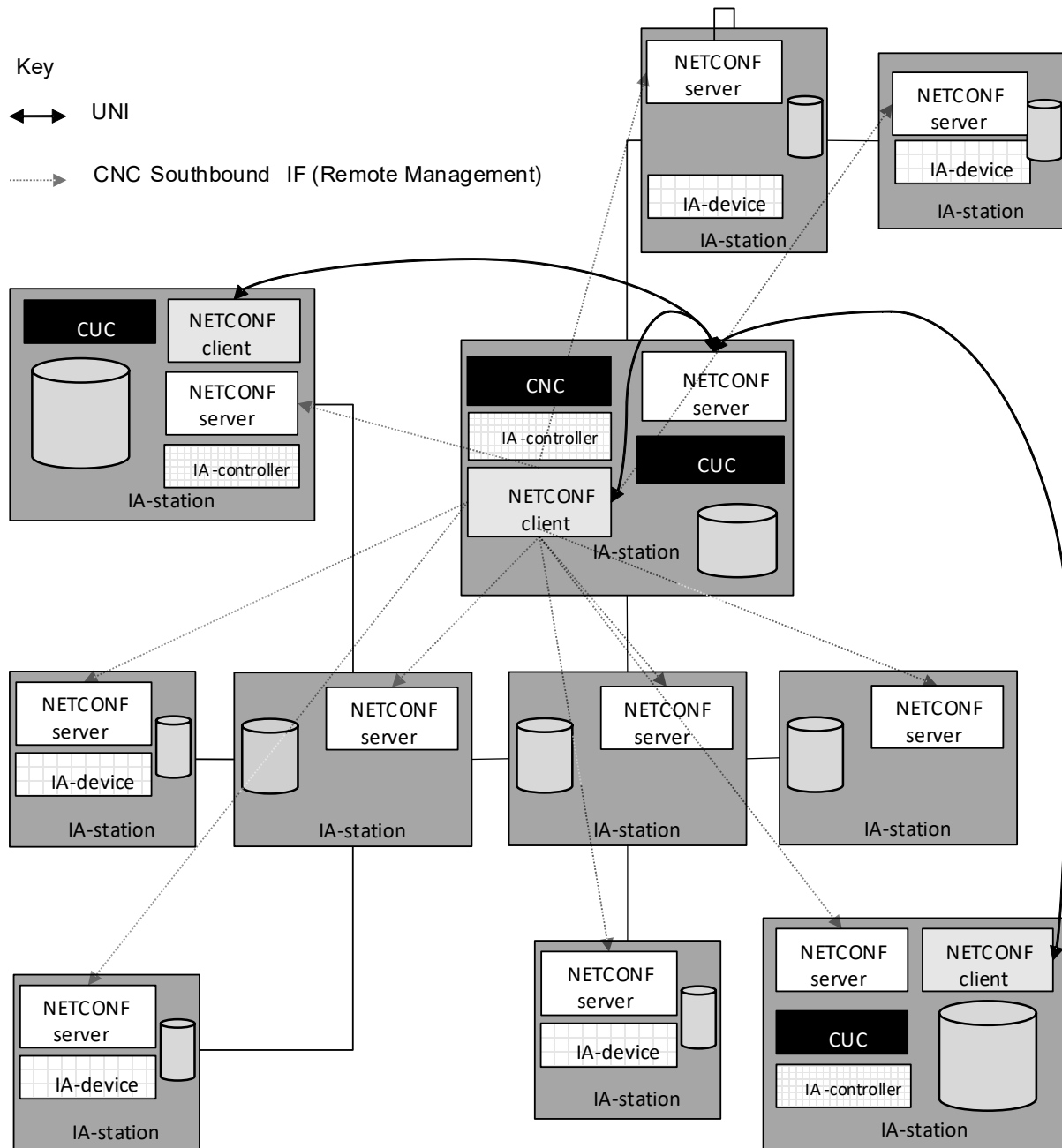


Figure 25 – NETCONF usage in a Configuration Domain

6.4.5.2 Domain port states

A CNC manages available network resources and assigns them to the IA-stations. Management of the network resources is only possible if the CNC owns these resources. Thus, no connected station is allowed to make use of network resources that are not granted by the CNC. The security configuration of a connected station allows remote access for the CNC.

Protection of the network resources is done by managing the ports (see Figure 26) at the boundary of the Configuration Domain. The state of any newly connected station is unknown. The CNC is responsible for determining if the newly connected station is added to the Configuration Domain and configuring the IA-station appropriately.

This port state model avoids any assumptions about configuration of added stations or network portions.

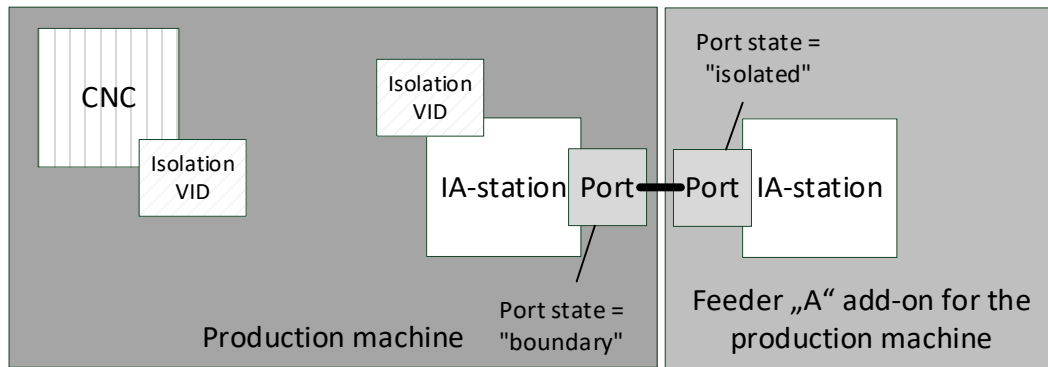


Figure 26 – Boundary port model

Ports of an IA-station that is a member of a Configuration Domain have different states:

- Isolated – a station connected via this port can only exchange information with a CNC. The CNC is responsible for establishing an isolation VID and for onboarding the station. In the isolated state:
 - the port gets to or remains in isolated state in case of a link down event, e.g. when nothing is connected, or no link is established;
 - the port gets to or remains in isolated state in case of a link up event;
 - the port stays in isolated state as long as the neighbor is unknown, not able to enter Boundary state.
- Boundary – a station connected via this port is not part of the Configuration Domain, but is allowed to access devices inside the Configuration Domain and to pass traffic through the Configuration Domain
- Inside – a station connected via this port is part of the Configuration Domain

The determination of whether a given port of an IA-station remains in the Isolated state or transitions to the Boundary or Inside state is performed by the CNC using remote management. A port acts as a domain boundary if it is in the Isolated or Boundary state.

For example, a port could be configured as follows:

- Isolated state:
 - port is IST boundary,
 - port is not part of a sync tree,
 - port uses VLAN stripping for egress,
 - port uses VLAN assignment and priority regeneration to assign all traffic to an isolated VLAN, and
 - port uses an ingress rate limiter to control the amount of traffic for the Configuration Domain.
- Boundary state:
 - port is part of IST,

- 2887 – port is part of a sync tree,
- 2888 – port uses VLAN stripping for egress,
- 2889 – port uses VLAN assignment and priority regeneration to assign all traffic to a default
- 2890 VLAN, and
- 2891 – port uses an ingress rate limiter to control the amount of traffic for the Configuration
- 2892 Domain.
- 2893 • Inside state:
 - 2894 – port is part of IST,
 - 2895 – port is part of a sync tree, and
 - 2896 – port is part of the active topology for stream and non-stream traffic.

2897 An example workflow includes the following steps executed by the CNC:

- 2898 a) Topology discovery:
 - 2899 1) Case A: Link down / Port not connected:
 - 2900 i) set port to isolated state;
 - 2901 ii) configure a NETCONF subscription "on data change" to the port state leaf
 - 2902 2) Case B: Neighbor is not a Configuration Domain member:
 - 2903 i) set port to boundary state;
 - 2904 ii) configure a NETCONF subscription "on data change" to the port state leaf.
 - 2905 3) Case C: Neighbor is not a Configuration Domain member – but part of expected topology:
 - 2906 i) set port to boundary state;
 - 2907 ii) configure the neighbor station as Configuration Domain member;
 - 2908 iii) set port to inside state.
- 2909 b) NETCONF subscription trigger:
 - 2910 Issued to the CNC upon change of subscribed YANG data.

2911 **6.4.5.3 Engineered network**

2912 For an offline engineered (based on the available digital data sheets of the used IA-stations)
 2913 centralized approach with fixed topology, fixed stations and fixed paths, the user provides traffic
 2914 requirements, path information, topology information and expected network configuration to the
 2915 CNC. The CNC then uses the TDE, RAE and the NPE to perform the calculation of paths,
 2916 resources, and stream schedules necessary to meet the specified traffic requirements and
 2917 deploys the result of these calculations via remote management. The CNC also provides the
 2918 relevant results to the CUC via the UNI. The CUC then configures the end stations using the
 2919 user-to-user interface (see Figure 3).

2920 The workflow for this example consists of the following steps:

- 2921 a) The user determines:
 - 2922 1) the expected network topology,
 - 2923 2) the expected stations and its capabilities, value ranges and quantities,
 - 2924 3) the expected paths and resources,
 - 2925 4) the required streams, and
 - 2926 5) the requirements for IA non-stream traffic.

2927 This step focuses on network capabilities including the Ethernet interface of the end stations.
 2928 For example, if the end station is a sensor, the user considers the Ethernet interface capabilities
 2929 of the sensor as they apply to the physical world.

- 2930 b) Engineering Tool provides this information to the CNC via a user-specific interface.

2931 Although the communication between the CNC and any Engineering Tool is user-specific, the
2932 CNC obtains all necessary information via the integrated TDE and NPE.

2933 c) The CNC uses the TDE to discover the topology and checks it against the expected
2934 topology. The NPE is used to configure the IA-stations of the Configuration Domain.

2935 d) The CNC uses STE and NPE to setup, validate, and monitor synchronization configuration
2936 in the Configuration Domain.

2937 e) The CNC uses the information from engineering item a), steps 1 to 5, above to respond to
2938 requests from middleware (with integrated CUC) using UNI. These requests are handled
2939 using the already established communication paths received from the user.

2940 If the CNC is not required after commissioning, then the CNC can be removed after setting up
2941 the IA-stations. That requires that all IA-stations have a persistent storage for the data provided
2942 by the CNC.

2943 **6.4.5.4 Dynamic topology**

2944 **6.4.5.4.1 General**

2945 For a centralized approach with a dynamic topology and dynamic paths, the user provides the
2946 network policy to the CNC. The TDE performs topology discovery including IA-station
2947 capabilities (YANG representation of the digital data sheet) and the NPE performs network
2948 configuration for the CNC. IA-stations then provide traffic requirements via the middleware to
2949 the CNC via the UNI. The CNC then uses the TDE, RAE, and NPE to perform the calculation of
2950 paths, resources, and stream schedules necessary to meet the specified traffic requirements
2951 and deploys the result of these calculations via remote management. The CNC also provides
2952 the relevant results to the CUC via the UNI. The CUC then configures the end stations using
2953 the user-to-user interface (see Figure 3).

2954 The workflow for this example consists of the following steps.

2955 a) The user determines the network policy and provides it to the CNC.

2956 b) The TDE continuously discovers the physical network topology and station capabilities of
2957 each station using remote management.

2958 c) The NPE uses the information gathered in steps a) to b) to configure the IA-stations in the
2959 Configuration Domain.

2960 d) The CNC uses STE and NPE to setup, validate and monitor time synchronization
2961 configuration in the Configuration Domain.

2962 The CNC uses the information from steps a) to d) to respond to requests from middleware using
2963 UNI. The CNC establishes streams in the bridges via a remote management protocol.

2964 **6.4.5.4.2 Adding an IA-station**

2965 Each IA-station added to the Configuration Domain is discovered by the TDE and receives the
2966 initial network configuration from the NPE. After this, the middleware can request stream
2967 establishment.

2968 When an IA-station is added to the network, it is isolated until the CNC determines that its traffic
2969 requirements can be accommodated without disrupting other traffic (see 6.4.5.2).

2970 **6.4.5.4.3 Removing an IA-station**

2971 Each IA-station removed from the Configuration Domain is discovered by the TDE. A
2972 neighboring station can receive an updated network configuration by the NPE. After this, the
2973 removed IA-station is no longer part of the Configuration Domain.

6.4.5.4.4 Replacing an IA-station

In the simplest case, replacing an IA-station is simply the sequence of removing an IA-station (6.4.5.4.3) and adding an IA-station (6.4.5.4.2). In more complex cases, the user can take other precautions or actions following deployment.

6.4.5.5 Management proxies

6.4.5.5.1 General

Implementing NETCONF server functionality can be burdensome for resource constrained IA-stations. IA-stations which contain only ccB Bridge or end station components may utilize the NETCONF server of another IA-station as a management proxy. IA-stations supporting proxy functionality (proxies), maintain a list of the types of proxy functionality supported as well as the IA-stations for which they act as proxy (proxy targets). Proxy types are described in E.1.3. The information is maintained in the YANG module `ieciieee60802-proxy` as described in 6.4.10.8.8.

6.4.5.5.2 Proxy operation

When onboarding a new proxy target, a different process is followed. Proxies communicate with proxy targets via the isolation VLAN. When a new IA station is connected, the following sequence is followed.

- a) The IA-station to which the proxy target is connected (neighbor), which is already part of the configuration domain, reports a change in link status to the CNC.
- b) The CNC reads the stored LLDP information related to the proxy target from the neighbor via NETCONF. The neighbor can be a proxied device in which case the LLDP information about the proxy target would be retrieved using the neighbor's proxy.
- c) Each proxy target identifies itself using the 60802 device info TLV described in Annex E.
- d) Based on this TLV data, the CNC determines whether the connected station:
 - natively speaks NETCONF/YANG (no information regarding proxy operation is included), or
 - requires a proxy, and if so, which one.
- e) The proxy onboards the new device via a user-specific protocol. This user-specific protocol is not described in this document. Ensuring the security of this user-specific mechanism is not described in this document. If security equivalent to the secure management exchanges as described in 5.5.4.1 is not implemented in the user-specific protocol, security of the network can be compromised.
- f) The proxy communicates the result of the Authentication to the CNC. For instance, the CNC can utilize dynamic subscriptions and the on-change feature of the `ietf-yang-push` module to automatically be notified when the new proxy target is added to the proxy's list of proxy targets. Based upon the results of the authentication, the CNC sets the boundary port state appropriately
- g) The CNC communicates with the appropriate proxy and configures the device accordingly. The proxy handles the NETCONF parameterization and translates it into the required organization-defined or vendor-specific protocol and configures the proxy target.

6.4.5.6 Engineered network extended by dynamic topology

The engineered and dynamic topology workflows can be used together. For instance, modular machines, robot tool changers or more general plug & produce can add or remove modules. The basic machine is handled as an engineered network. Additional modules or removed modules are handled dynamically.

6.4.6 Engineered time-synchronization spanning tree

6.4.6.1 General

Engineered time-synchronization spanning tree (sync tree) for a given gPTP domain refers to the usage of external port configuration instead of BTCA for the construction of a desired sync tree with the Grandmaster PTP Instance as the root (see IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdr-2024 and IEEE Std 802.1ASdm-2024, 10.3.1). The Grandmaster PTP Instance can reside in a dedicated grandmaster-capable IA-station.

One of the advantages of engineered sync trees is to enable a planned, deterministic, and stable configuration of the IEEE Std 802.1AS-2020 sync tree for a given gPTP domain. For example, this approach prevents sync tree changes in case of IA-station addition or removal from the network. Hot standby (see IEEE Std 802.1ASdm-2024) is a use case of an engineered sync tree.

6.4.6.2 Sync tree requirements

If an engineered synchronization spanning tree is used, the sync tree requirements for all participating PTP Instances in a gPTP domain are specified in 5.5.3, h).

6.4.6.3 STE phases

6.4.6.3.1 General

The STE follows the logical sequence described in 6.4.6.3 if an engineered sync tree is utilized in a gPTP domain. Each STE phase describes an externally observable behavior of the participating PTP Instances in a gPTP domain.

6.4.6.3.2 Discovery phase

In discovery phase, STE utilizes the topology discovered by the TDE to verify the capabilities and status of participating IA-stations via a diagnostics entity (see 6.4.7.1) by reading the following managed objects.

- The status of oper-status parameter is up (see IETF RFC 8343) for all participating Ethernet links.
- The status of isMeasuringDelay (see IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdm-2024, 14.16.4) is TRUE for all PTP Ports.
- The status of asCapable (see IEEE Std 802.1AS-2020, 14.8.7) is TRUE for all PTP Ports.
- The status of asCapableAcrossDomains (see IEEE Std 802.1AS-2020, 14.16.5) is TRUE for all LinkPorts.
- The status of gmCapable (see IEEE Std 802.1AS-2020, 14.2.7) is TRUE, only applicable to the Grandmaster PTP Instance.

STE should use the information collected via managed objects and the discovered topology to verify the constraints on the gPTP domain, for example:

- Verify that the number of PTP Relay Instances (hops) between the Grandmaster PTP Instance and any given timeReceiver PTP End Instance is within the limit prescribed by for example, CNC.

6.4.6.3.3 Provisioning phase

In provisioning phase, STE should apply the desired configuration to all participating PTP Instances, for example:

- externalPortConfigurationEnabled is set to true,

- 3061 • the desiredState of all PTP ports of the Grandmaster PTP Instance is set to
3062 TimeTransmitterPort,
- 3063 • the desiredState of exactly one PTP port of all the other PTP Instances is set to
3064 TimeReceiverPort,
- 3065 • the desiredState of remaining PTP ports that are part of sync tree in non-Grandmaster PTP
3066 Relay Instances is set to TimeTransmitterPort, and
- 3067 • the desiredState of all other PTP ports is set to PassivePort.

3068 Then STE should validate, for example, the syncLocked (see IEEE Std 802.1AS-2020, 14.8.52)
3069 parameter is TRUE for all PTP ports of PTP Relay Instances that are in TimeTransmitterPort
3070 state.

3071 **6.4.6.4 Adding an IA-station**

3072 Each IA-station added to the gPTP domain is discovered by STE via TDE. It is the responsibility
3073 of the CNC to on-board this newly added station. IA-stations can receive an updated gPTP
3074 configuration via STE.

3075 A newly installed IA-station can disrupt the operation of a gPTP domain. The extent of disruption
3076 is dependent on the location of the IA-station in the gPTP domain and the type of PTP Instance
3077 running on that IA-station. For example, if PTP Instances are arranged in a daisy-chain
3078 formation and an IA-station with a non-Grandmaster Relay Instance is installed in the middle of
3079 a daisy-chain, then this change can disrupt the operation of downstream Instances.

3080 **6.4.6.5 Removing an IA-station**

3081 The removal of a station from the gPTP domain is detected by STE via TDE. IA-stations can
3082 receive an updated gPTP configuration via STE.

3083 **6.4.6.6 Replacing an IA-station**

3084 An IA-station replacement follows the sequence of removing an IA-station according to 6.4.6.5
3085 and adding an IA-station according to 6.4.6.4.

3086 **6.4.7 Diagnostics**

3087 **6.4.7.1 General**

3088 Diagnosis for an IA-station is done by monitoring YANG representation of the IA-station's local
3089 database.

3090 A vendor can implement an observer in a diagnostics entity, which could reside in the CNC.
3091 This diagnostics entity uses the information provided by remote management to define the
3092 monitored objects and set up fitting notifications.

3093 **6.4.7.2 Observer model**

3094 A diagnostic entity can select any objects described via YANG and observe them via NETCONF.
3095 The NETCONF binding is specified in IETF RFC 8640, and the subscription model in IETF RFC
3096 8641. NETCONF messages can be pipelined, i.e. a client can invoke multiple RPCs without
3097 having to wait for RPC result messages first. RPC messages are specified in IETF RFC 6241,
3098 and notification messages are specified in IETF RFC 5277. To reduce the load on the diagnostic
3099 entity when many stations are providing notifications, the diagnostic objects can be monitored
3100 and notifications can be retrieved from individual IA-stations.

3101 Figure 27 shows the model of a diagnostic observer.

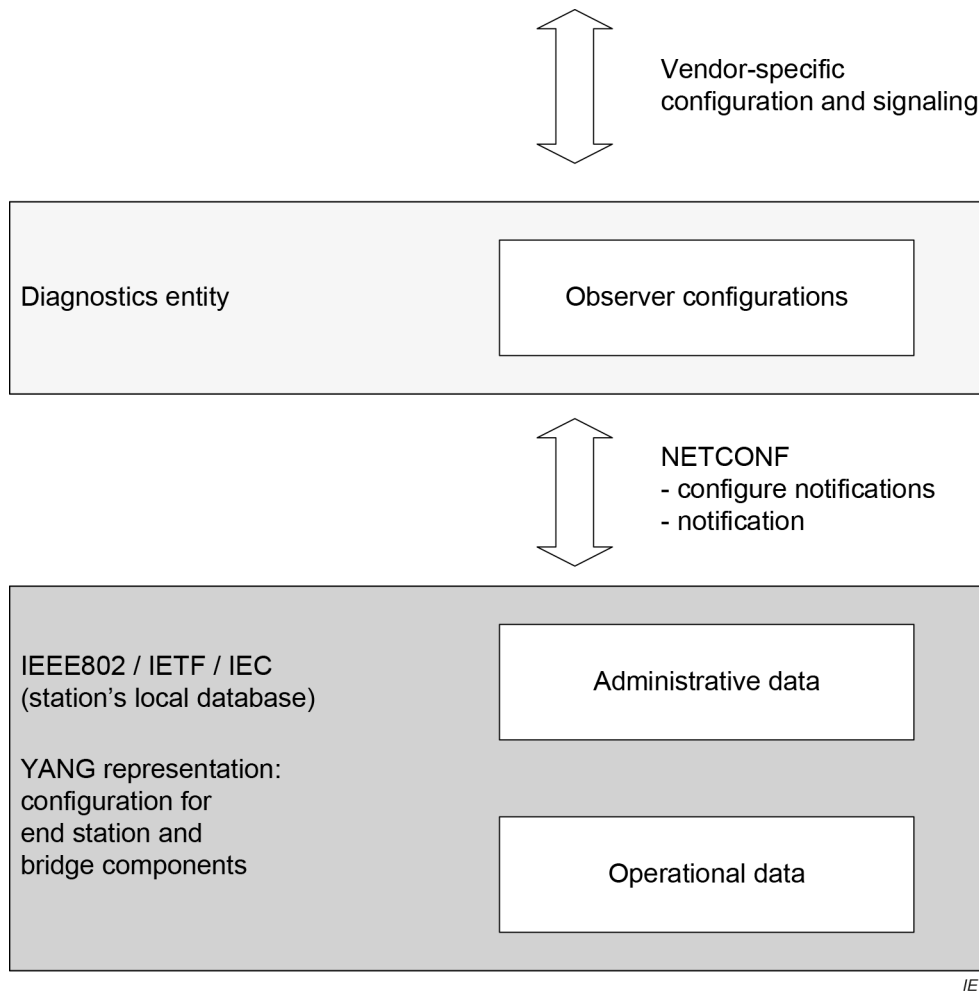


Figure 27 – Observer model

6.4.7.3 Usage of YANG Push

For diagnostics, an IA-station shall support YANG-Push subscriptions according to IETF RFC 8641 (YANG Push) and IETF RFC 8639 (Subscribed Notifications).

IA-stations shall support the "subtree" selection filter as specified in IETF RFC 8641, 3.6

6.4.7.4 Mandatory RPCs

An IA-station shall support following RPCs as specified in IETF RFC 8641:

- a) establish-subscription,
- b) modify-subscription,
- c) delete-subscription, and
- d) kill-subscription.

6.4.7.5 Mandatory notifications

An IA-station shall support following notifications as specified in IETF RFC 8641:

- a) subscription-resumed,
- b) subscription-modified,
- c) subscription-terminated,

- 3119 d) subscription-suspended,
- 3120 e) push-update, and
- 3121 f) push-change-update.

3122 **6.4.7.6 Mandatory diagnostics data nodes**

3123 An IA-station shall provide the following data nodes for NETCONF subscriptions.

- 3124 a) Change of link-status per Ethernet port:

3125 /ietf-interfaces/interface/oper-status

- 3126 b) Change of MAU-type per Ethernet port:

3127 /ieee802-ethernet-lldp/lldp/port/ operational-mau-type

- 3128 c) Change of sync-status:

- 3129 1) per PTP Instance:

3130 for every application-clock: /ieciieee60802-
3131 bridge/bridges/bridge/component/clock/is-synced

- 3132 2) per hot standby Instance:

3133 /dotlas-hs/ptp/common-services/hss/hot-standby-system-
3134 list/hot-standby-system-ds/hot-standby-system-state

- 3135 d) Data to be provided as periodic time-aligned subscriptions:

- 3136 1) Dropped frames statistic counters per Ethernet interface

- 3137 – /ietf-interfaces/interface/statistics/in-octets
- 3138 – /ietf-interfaces/interface/statistics/in-discards
- 3139 – /ietf-interfaces/interface/statistics/in-errors
- 3140 – /ietf-interfaces/interface/statistics/out-octets
- 3141 – /ietf-interfaces/interface/statistics/out-discards
- 3142 – /ietf-interfaces/interface/statistics/out-errors

- 3143 2) VLAN specific counters per Ethernet Interface and VLAN ID

- 3144 – /ieee802-dot1q-bridge/interfaces/interface/bridge-
3145 port/statistics/octets-rx
- 3146 – /ieee802-dot1q-bridge/interfaces/interface/bridge-
3147 port/statistics/octets-tx
- 3148 – /ieee802-dot1q-bridge/interfaces/interface/bridge-
3149 port/statistics/forward-outbound
- 3150 – /ieee802-dot1q-bridge/interfaces/interface/bridge-
3151 port/statistics/discard-inbound

3152 **6.4.7.7 Usage of NETCONF notifications**

3153 IA-stations shall implement the binding of a stream of events according to IETF RFC 8640
3154 (NETCONF Notifications) using the "encode-xml" feature and the "NETCONF" event stream of
3155 IETF RFC 8639.

3156 An IA-station shall support dynamic subscriptions according to 6.4.7.6, IETF RFC 8640 Clauses
3157 5, 6 and 7. The number of dynamic subscriptions shall be reported.

6.4.8 Data sheet

6.4.8.1 General

Data sheets containing the capabilities, value ranges and quantities of IA-stations allow a user to select appropriate IA-stations and enable users to configure a system using online and offline engineering. See Annex B for quantities in a representative Configuration Domain.

Online data sheets are modeled using YANG. YANG modeling is used for the offline data sheet to keep the offline (6.4.5.3) and online (6.4.5.4) format the same.

6.4.8.2 Digital data sheet of an IA-station

Both engineering models, offline via an engineering tool and online with plug & produce by the CNC, require information about the capabilities of an IA-station, for example, states, configurations, or supported features. An example depicting the creation of a digital data sheet is provided in Figure 28.

This data is extracted from the implemented YANG modules, which are available in the local database of the IA-station.

The data from the implemented YANG modules is also available offline in the form of a digital data sheet of an IA-station as a digital data sheet file.

The digital data sheet of an IA-station provides a collection of all instantiated data nodes of all YANG modules that are required by this document (see 6.4.9). A manufacturer may reduce the instance data set by removing statistical config-false YANG nodes from the digital data sheet.

The digital data sheet does not contain any additional information that is not modeled by the YANG modules that exist in the local database of the IA-station.

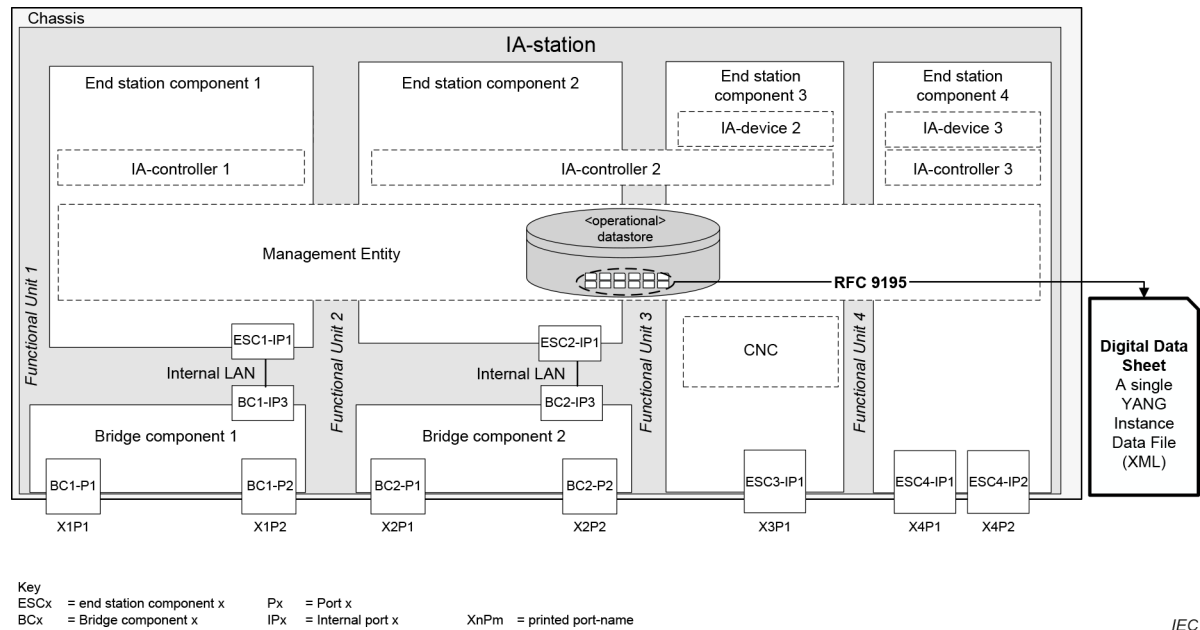
The data sheet contains a single instance data set. It carries complete configuration and state data of each YANG module that is present in the local database of the IA-station.

The identity of the datastore with which the instance data set is associated and the format of the YANG instance data set shall be reported according to IETF RFC 9195. The file format is based on the XML encoding. It is created by applying the respective XML encoding rules for the YANG structure of all YANG modules included in the digital data sheet.

A user uses the information from the digital data sheet to understand the quantities and capabilities of an IA-station, which is required for successful offline engineering of the network.

The features of a CNC are used for offline and online engineering or diagnostics. For this purpose, YANG modules are used that allow structured access to the local database of the CNC according to 6.4.9.2.5.11.

Any IA-station can include a CNC entity in which case the collection of YANG modules of such IA-station includes all CNC specific YANG modules for example, the ieee802-dot1q-cnc-config YANG module. Since all IA-stations meet the requirements from 5.5.4, the CNC related YANG instance data is automatically included in the digital data sheet of the IA-station that hosts the CNC as described in 6.4.9.2.



IEC

Figure 28 – Creation of the digital data sheet of an IA-station

6.4.9 YANG representation of managed objects and nodes

6.4.9.1 General

All managed objects shall be represented in the YANG 1.1 format as described in IETF RFC 7950. The markings (i.e. [m], [o], [c]) indicate whether the node is included in the digital data sheet (see 3.5.4). These markings are independent of the conformance criteria for an IA-station (see 5.2).

6.4.9.2 Common YANG modules, features, and nodes

6.4.9.2.1 IEEE standard for Ethernet

IA-stations shall support the `ieee802-ethernet-interface` YANG module according to IEEE Std 802.3.2™-2019 as modified by IEEE Draft Std P802.3.2a with the following nodes:

- [o] `/ieee802-ethernet-interface/interfaces/interface/ethernet/duplex`
- [o] `/ieee802-ethernet-interface/interfaces/interface/ethernet/speed`
- [o] `/ieee802-ethernet-interface/interfaces/interface/ethernet/flow-control/pause/direction` (if the feature "ethernet-pause" is supported)
- [o] `ieee802-ethernet-lldp/lldp/port/operational-mau-type`
- [o] `ieee802-ethernet-lldp/lldp/port/preemption-supporated`
- [o] `ieee802-ethernet-lldp/lldp/port/preemption-enabled`
- [o] `ieee802-ethernet-lldp/lldp/port/preemption-active`
- [o] `ieee802-ethernet-lldp/lldp/port/additional-fragment-size`
- [o] `ieee802-ethernet-lldp/lldp/port/local-max-frame-size`

3217 **6.4.9.2.2 Station and media access control connectivity discovery**

3218 IA-stations shall support the following nodes from the ieee802-dot1ab-lldp YANG module
3219 according to IEEE Std 802.1ABcu-2021 with values and value ranges according to 6.5.

- 3220 [o] /ieee802-dot1ab-lldp/lldp/message-fast-tx
- 3221 [o] /ieee802-dot1ab-lldp/lldp/message-tx-hold-multiplier
- 3222 [o] /ieee802-dot1ab-lldp/lldp/message-tx-interval
- 3223 [o] /ieee802-dot1ab-lldp/lldp/reinit-delay
- 3224 [o] /ieee802-dot1ab-lldp/lldp/tx-credit-max
- 3225 [o] /ieee802-dot1ab-lldp/lldp/tx-fast-init
- 3226 [o] /ieee802-dot1ab-lldp/lldp/notification-interval
- 3227 [o] /ieee802-dot1ab-lldp/lldp/remote-statistics
- 3228 [o] /ieee802-dot1ab-lldp/lldp/remote-statistics/last-change-time
- 3229 [o] /ieee802-dot1ab-lldp/lldp/remote-statistics/remote-inserts
- 3230 [o] /ieee802-dot1ab-lldp/lldp/remote-statistics/remote-deletes
- 3231 [o] /ieee802-dot1ab-lldp/lldp/remote-statistics/remote-drops
- 3232 [o] /ieee802-dot1ab-lldp/lldp/remote-statistics/remote-ageouts
- 3233 [m] /ieee802-dot1ab-lldp/lldp/local-system-data
- 3234 [o] /ieee802-dot1ab-lldp/lldp/local-system-data/chassis-id-subtype
- 3235 [o] /ieee802-dot1ab-lldp/lldp/local-system-data/chassis-id
- 3236 [o] /ieee802-dot1ab-lldp/lldp/local-system-data/system-name
- 3237 [o] /ieee802-dot1ab-lldp/lldp/local-system-data/system-description
- 3238 [m] /ieee802-dot1ab-lldp/lldp/local-system-data/system-capabilities-
3239 supported
- 3240 [o] /ieee802-dot1ab-lldp/lldp/local-system-data/system-capabilities-
3241 enabled
- 3242 [o] /ieee802-dot1ab-lldp/lldp/port
- 3243 [o] /ieee802-dot1ab-lldp/lldp/port/name
- 3244 [o] /ieee802-dot1ab-lldp/lldp/port/dest-mac-address
- 3245 [o] /ieee802-dot1ab-lldp/lldp/port/admin-status

3246 [o] /ieee802-dot1ab-lldp/lldp/port/notification-enable

3247 [o] /ieee802-dot1ab-lldp/lldp/port/tlvs-tx-enable

3248 [o] /ieee802-dot1ab-lldp/lldp/port/message-fast-tx

3249 [o] /ieee802-dot1ab-lldp/lldp/port/message-tx-hold-multiplier

3250 [o] /ieee802-dot1ab-lldp/lldp/port/message-tx-interval

3251 [o] /ieee802-dot1ab-lldp/lldp/port/reinit-delay

3252 [o] /ieee802-dot1ab-lldp/lldp/port/tx-credit-max

3253 [o] /ieee802-dot1ab-lldp/lldp/port/tx-fast-init

3254 [o] /ieee802-dot1ab-lldp/lldp/port/notification-interval

3255 [o] /ieee802-dot1ab-lldp/lldp/port/management-address-tx-port

3256 [o] /ieee802-dot1ab-lldp/lldp/port/port-id-subtype

3257 [o] /ieee802-dot1ab-lldp/lldp/port/port-id

3258 [o] /ieee802-dot1ab-lldp/lldp/port/port-desc

3259 [o] /ieee802-dot1ab-lldp/lldp/port/remote-systems-data

3260 **6.4.9.2.3 Synchronization**

3261 **6.4.9.2.3.1 Timesync**

3262 IA-stations shall support the ieee1588-ptp-tt YANG module according to IEEE Std 1588e™-
3263 2024 with the following features:

- 3264 • cmls (Common Mean Link Delay Service), and
- 3265 • external-port-config.

3266 IA-stations shall support the ieee1588-ptp-tt YANG module according to IEEE Std 1588e-2024
3267 with the following nodes:

3268 [o] /ieee1588-ptp-tt/ptp/instances/instance/instance-index

3269 [o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/clock-identity

3270 [o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/number-ports

3271 [o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/priority1

3272 [o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/domain-number

3273 [o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/time-receiver-
3274 only

3275 [o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/sdo-id

3276 [o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/instance-enable

3277 [o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/external-port-
3278 config-enable

3279 [o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/instance-type

3280 [o] /ieee1588-ptp-tt/ptp/instances/instance/description-ds/user-
3281 description

3282 [o] /ieee1588-ptp-tt/ptp/instances/ports/port/port-index

3283 [o] /ieee1588-ptp-tt/ptp/instances/ports/port/underlying-interface

3284 [o] /ieee1588-ptp-tt/ptp/instances/ports/port/port-ds/port-state

3285 [o] /ieee1588-ptp-tt/ptp/instances/ports/port/port-ds/delay-mechanism

3286 [o] /ieee1588-ptp-tt/ptp/instances/ports/port/port-ds/port-enable

3287 [o] /ieee1588-ptp-tt/ptp/instances/ports/port/external-port-config-
3288 port-ds/desired-state

3289 [o] /ieee1588-ptp-tt/ptp/common-services/cmlDs/default-ds/clock-
3290 identity

3291 [o] /ieee1588-ptp-tt/ptp/common-services/cmlDs/default-ds/number-link-
3292 ports

3293 [o] /ieee1588-ptp-tt/ptp/common-services/cmlDs/ports/port/port-index

3294 [o] /ieee1588-ptp-tt/ptp/common-services/cmlDs/ports/port/underlying-
3295 interface

3296 [o] /ieee1588-ptp-tt/ptp/common-services/cmlDs/ports/port/link-port-
3297 ds/port-identity/clock-identity

3298 [o] /ieee1588-ptp-tt/ptp/common-services/cmlDs/ports/port/link-port-
3299 ds/port-identity/port-number

3300 [o] /ieee1588-ptp-tt/ptp/common-services/cmlDs/ports/port/link-port-
3301 ds/domain-number

3302 [o] /ieee1588-ptp-tt/ptp/common-services/cmlDs/ports/port/link-port-
3303 ds/service-measurement-valid

3304 [o] /ieee1588-ptp-tt/ptp/common-services/cmlDs/ports/port/link-port-
3305 ds/mean-link-delay

3306 [o] /ieee1588-ptp-tt/ptp/common-services/cmlDs/ports/port/link-port-
3307 ds/scaled-neighbor-rate-ratio

3308 [o] /ieee1588-ptp-tt/ptp/common-services/cmlDs/ports/port/link-port-
3309 ds/log-min-pdelay-req-interval

3310 [m] /ieee1588-ptp-tt/ptp/common-services/cmlsds/ports/port/link-port-
3311 ds/version-number

3312 [m] /ieee1588-ptp-tt/ptp/common-services/cmlsds/ports/port/link-port-
3313 ds/minor-version-number

3314 [o] /ieee1588-ptp-tt/ptp/common-services/cmlsds/ports/port/link-port-
3315 ds/delay-asymmetry

3316 **6.4.9.2.3.2 Timesync (ieee802-dot1as-gptp)**

3317 IA-stations shall support the ieee802-dot1as-gptp YANG module according to EEE Std
3318 802.1ASdn-2024™ with the following nodes:

3319 [o] /ieee802-dot1as-gptp/ptp/instances/instance/default-ds/gm-capable

3320 [o] /ieee802-dot1as-gptp/ptp/instances/instance/default-ds/current-
3321 utc-offset-valid

3322 [o] /ieee802-dot1as-gptp/ptp/instances/instance/default-ds/ptp-
3323 timescale

3324 [o] /ieee802-dot1as-gptp/ptp/instances/ports/port/port-ds/sync-
3325 receipt-timeout

3326 [o] /ieee802-dot1as-gptp/ptp/instances/ports/port/port-ds/current-one-
3327 step-tx-oper

3328 [o] /ieee802-dot1as-gptp/ptp/instances/ports/port/port-ds/use-mgt-one-
3329 step-tx-oper

3330 [o] /ieee802-dot1as-gptp/ptp/instances/ports/port/port-ds/mgt-one-
3331 step-tx-oper

3332 [o] /ieee802-dot1as-gptp/ptp/instances/ports/port/port-ds/sync-locked

3333 [o] /ieee802-dot1as-gptp/ptp/common-services/cmlsds/ports/port/link-
3334 port-ds/cmlsds-link-port-enabled

3335 [o] /ieee802-dot1as-gptp/ptp/common-services/cmlsds/ports/port/link-
3336 port-ds/is-measuring-delay

3337 [o] /ieee802-dot1as-gptp/ptp/common-services/cmlsds/ports/port/link-
3338 port-ds/as-capable-across-domains

3339 [o] /ieee802-dot1as-gptp/ptp/common-services/cmlsds/ports/port/link-
3340 port-ds/mean-link-delay-thresh

3341 [o] /ieee802-dot1as-gptp/ptp/common-services/cmlsds/ports/port/link-
3342 port-ds/current-log-pdelay-req-interval

3343 [o] /ieee802-dot1as-gptp/ptp/common-services/cmlsds/ports/port/link-
3344 port-ds/use-mgt-log-pdelay-req-interval

3345 [o] /ieee802-dot1as-gptp/ptp/common-services/cmlsds/ports/port/link-
3346 port-ds/mgt-log-pdelay-req-interval

3347 [o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
3348 port-ds/current-compute-rate-ratio

3349 [o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
3350 port-ds/use-mgt-compute-rate-ratio

3351 [o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
3352 port-ds/mgt-compute-rate-ratio

3353 [o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
3354 port-ds/current-compute-mean-link-delay

3355 [o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
3356 port-ds/use-mgt-compute-mean-link-delay

3357 [o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
3358 port-ds/mgt-compute-mean-link-delay

3359 [o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
3360 port-ds/allowed-lost-responses

3361 [o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
3362 port-ds/allowed-faults

3363 **6.4.9.2.3.3 Timesync (ieee802-dot1as-hs)**

3364 IA-stations shall support the ieee802-dot1as-hs YANG module according to IEEE Std
3365 802.1ASdm-2024 with the following nodes:

3366 [o] /ieee802-dot1as-hs/ptp/instances/instance/parent-ds/gmPresent

3367 [o] /ieee802-dot1as-hs/ptp/instances/instance/ptp-instance-ds/is-
3368 synced

3369 [o] /ieee802-dot1as-hs/ptp/instances/instance/ptp-instance-ds/nrr-
3370 pdelay

3371 [o] /ieee802-dot1as-hs/ptp/instances/instance/ptp-instance-ds/nrr-sync

3372 [o] /ieee802-dot1as-hs/ptp/instances/instance/ptp-instance-ds/nrr-
3373 comp-method

3374 [o] /ieee802-dot1as-hs/ptp/instances/instance/ptp-instance-ds/as-
3375 capable-across-domains

3376 [o] /ieee802-dot1as-hs/ptp/instances/instance/ptp-instance-ds/gptp-
3377 capable-state-machines-enabled

3378 **6.4.9.2.4 Security configuration modules**

3379 **6.4.9.2.4.1 YANG module for a keystore**

3380 IA-stations shall support the ietf-keystore YANG module according to IETF RFC 9642 with the
3381 following features:

- 3382 • central-truststore-supported, and
- 3383 • asymmetric-keys.

3384 IA-stations shall support the ietf-keystore YANG module according to IETF RFC 9642 with the
3385 following nodes:

- 3386 [o] /ietf-keystore/keystore/asymmetric-keys/asymmetric-key/name
- 3387 [o] /ietf-keystore/keystore/asymmetric-keys/asymmetric-key/public-key-
3388 format
- 3389 [o] /ietf-keystore/keystore/asymmetric-keys/asymmetric-key/public-key
- 3390 [o] /ietf-keystore/keystore/asymmetric-keys/asymmetric-key/private-
3391 key-format
- 3392 [o] /ietf-keystore/keystore/asymmetric-keys/asymmetric-key/hidden-
3393 private-key
- 3394 [o] /ietf-keystore/certificates/certificate/name
- 3395 [o] /ietf-keystore/certificates/certificate/cert-data
- 3396 [o] /ietf-keystore/certificates/certificate/expiration-date
- 3397 [o] /ietf-keystore/certificates/certificate/csr-info
- 3398 [o] /ietf-keystore/certificates/certificate/certificate-signing-
3399 request

3400 **6.4.9.2.4.2 Network configuration access control**

3401 IA-stations shall support the ietf-netconf-acm YANG module according to IETF RFC 8341 with
3402 the following nodes:

- 3403 [o] /ietf-netconf-acm/nacm/enable-nacm
- 3404 [o] /ietf-netconf-acm/nacm/read-default
- 3405 [o] /ietf-netconf-acm/nacm/write-default
- 3406 [o] /ietf-netconf-acm/nacm/exec-default
- 3407 [o] /ietf-netconf-acm/nacm/enable-external-groups
- 3408 [o] /ietf-netconf-acm/nacm/groups
- 3409 [o] /ietf-netconf-acm/nacm/rule-list

3410 **6.4.9.2.4.3 A YANG data module for a truststore**

3411 IA-stations shall support the ietf-truststore YANG module according to IETF RFC 9641 with the
3412 following features:

- 3413 • central-keystore-supported, and
- 3414 • certificates.

3415 IA-stations shall support the ietf-truststore YANG module according to IETF RFC 9641 with the
3416 following nodes:

3417 [o] /ietf-truststore/truststore/certificate-bags/certificate-bag/name

3418 [o] /ietf-truststore/truststore/certificate-bags/certificate-
3419 bag/certificate/name

3420 [o] /ietf-truststore/truststore/certificate-bags/certificate-
3421 bag/certificate/cert-data

3422 [o] /ietf-truststore/truststore/certificate-bags/certificate-
3423 bag/certificate/expiration-date

3424 **6.4.9.2.5 IA-station management**

3425 **6.4.9.2.5.1 System capabilities**

3426 IA-stations shall support the ietf-system-capabilities YANG modules according to IETF RFC
3427 9196 with the following nodes:

3428 [m] /ietf-system-capabilities/system-capabilities/datastore-
3429 capabilities/datastore

3430 [m] /ietf-system-capabilities/system-capabilities/datastore-
3431 capabilities/per-node-capabilities

3432 [m] /ietf-system-capabilities/system-capabilities/datastore-
3433 capabilities/on-change-supported

3434 **6.4.9.2.5.2 YANG library**

3435 IA-stations shall support the ietf-yang-library YANG module according to IETF RFC 8525 with
3436 the following nodes:

3437 [m] /ietf-yang-library/yang-library/module-set

3438 [m] /ietf-yang-library/yang-library/schema

3439 [m] /ietf-yang-library/yang-library/datastore

3440 [m] /ietf-yang-library/yang-library/content-id

3441 **6.4.9.2.5.3 YANG push**

3442 IA-stations shall support the ietf-yang-push YANG module according to IETF RFC 8641, 4.1,
3443 with the on-change feature.

3444 IA-stations shall support the ietf-yang-push YANG module according to IETF RFC 8641, 4.1,
3445 with the following nodes:

3446 [o] /ietf-subscribed-notifications/filters/selection-filter

3447 [o] /ietf-subscribed-
3448 notifications/subscriptions/subscription/target/datastore

3449 [o] /ietf-subscribed-notifications/subscriptions/subscription/update-
3450 trigger

3451 **6.4.9.2.5.4 YANG notification capabilities**

3452 IA-stations shall support the ietf-notification-capabilities YANG module according to IETF RFC
3453 9196 with the following nodes:

3454 [m] /ietf-notification-capabilities/system-capabilities/subscription-
3455 capabilities

3456 [m] /ietf-notification-capabilities/system-capabilities/datastore-
3457 capabilities/per-node-capabilities/subscription-capabilities

3458 **6.4.9.2.5.5 YANG notifications**

3459 IA-stations shall support the ietf-subscribed-notifications YANG module according to IETF RFC
3460 8639 with the following features:

- 3461 • Configured,
- 3462 • encode-xml, and
- 3463 • subtree.

3464 IA-stations shall support the ietf-subscribed-notifications YANG module according to IETF RFC
3465 8639 with the following nodes:

3466 [o] /ietf-subscribed-notifications/streams/stream/name

3467 [o] /ietf-subscribed-notifications/streams/stream/description

3468 [o] /ietf-subscribed-notifications/streams/stream/replay-support

3469 [o] /ietf-subscribed-notifications/streams/stream/replay-log-creation-
3470 time

3471 [o] /ietf-subscribed-notifications/streams/stream/replay-log-aged-time

3472 [o] /ietf-subscribed-notifications/filters/stream-filter/name

3473 [o] /ietf-subscribed-notifications/filters/stream-filter/filter-spec

3474 [o] /ietf-subscribed-notifications/subscriptions/subscription/id

3475 [o] /ietf-subscribed-notifications/subscriptions/subscription/target

3476 [o] /ietf-subscribed-notifications/subscriptions/subscription/stop-
3477 time

3478 [o] /ietf-subscribed-notifications/subscriptions/subscription/dscp

3479 [o] /ietf-subscribed-
3480 notifications/subscriptions/subscription/weighting

3481 [o] /ietf-subscribed-
3482 notifications/subscriptions/subscription/dependency

3483 [o] /ietf-subscribed-
3484 notifications/subscriptions/subscription/transport

3485 [o] /ietf-subscribed-notifications/subscriptions/subscription/encoding

3486 [o] /ietf-subscribed-notifications/subscriptions/subscription/purpose

3487 [o] /ietf-subscribed-
3488 notifications/subscriptions/subscription/notification-message-origin

3489 [o] /ietf-subscribed-
3490 notifications/subscriptions/subscription/configured-subscription-state

3491 [o] /ietf-subscribed-
3492 notifications/subscriptions/subscription/receivers

3493 **6.4.9.2.5.6 NETCONF monitoring**

3494 IA-stations shall support the ietf-netconf-monitoring YANG module according to IETF RFC 6022
3495 with the following nodes:

3496 [m] /ietf-netconf-monitoring/netconf-state/capabilities

3497 [m] /ietf-netconf-monitoring/netconf-state/datastores

3498 [m] /ietf-netconf-monitoring/netconf-state/schemas

3499 **6.4.9.2.5.7 System management**

3500 IA-stations shall support the ietf-system YANG module according to IETF RFC 7317 with the
3501 following nodes:

3502 [o] /ietf-system/system/contact

3503 [o] /ietf-system/system/hostname

3504 [o] /ietf-system/system/location

3505 **6.4.9.2.5.8 Hardware management**

3506 IA-stations shall support the ietf-hardware YANG module according to IETF RFC 8348 with the
3507 following nodes:

3508 [m] /ietf-hardware/hardware/component/name

3509 [m] /ietf-hardware/hardware/component/class

3510 [m] /ietf-hardware/hardware/component/description

3511 [m] /ietf-hardware/hardware/component/hardware-rev

3512 [m] /ietf-hardware/hardware/component/software-rev

3513 [o] /ietf-hardware/hardware/component/serial-num

3514 [m] /ietf-hardware/hardware/component/mfg-name

3515 [m] /ietf-hardware/hardware/component/model-name

3516 An IA-station shall provide exactly one /ietf-hardware/component with class "chassis" and may
3517 provide further components with other classes.

3518 **6.4.9.2.5.9 Interface management**

3519 IA-stations shall support the ietf-interfaces YANG module according to IETF RFC 8343 with the
3520 following nodes:

3521 [m] /ietf-interfaces/interfaces/interface/name

3522 [m] /ietf-interfaces/interfaces/interface/description

3523 [m] /ietf-interfaces/interfaces/interface/type

3524 [o] /ietf-interfaces/interfaces/interface/enabled

3525 [o] /ietf-interfaces/interfaces/interface/oper-status

3526 [o] /ietf-interfaces/interfaces/interface/phys-address

3527 [o] /ietf-interfaces/interfaces/interface/higher-layer-if

3528 [o] /ietf-interfaces/interfaces/interface/lower-layer-if

3529 [o] /ietf-interfaces/interfaces/interface/speed

3530 [o] /ietf-interfaces/interfaces/interface/statistics/discontinuity-
3531 time

3532 [o] /ietf-interfaces/interfaces/interface/statistics/in-octets

3533 [o] /ietf-interfaces/interfaces/interface/statistics/in-discards

3534 [o] /ietf-interfaces/interfaces/interface/statistics/in-errors

3535 [o] /ietf-interfaces/interfaces/interface/statistics/out-octets

3536 [o] /ietf-interfaces/interfaces/interface/statistics/out-discards

3537 [o] /ietf-interfaces/interfaces/interface/statistics/out-errors

3538 **6.4.9.2.5.10 Bridge and end station component management**

3539 **6.4.9.2.5.10.1 General**

3540 IA-stations shall support the ieee802-dot1q-bridge YANG module according to
3541 IEEE Std 802.1Q-2022, Clause 48, as amended by IEEE Std 802.1Qcw™-2023 with the
3542 following feature: ingress-filtering.

3543 IA-stations shall support the ieee802-dot1q-bridge YANG module according to
3544 IEEE Std 802.1Q-2022, Clause 48, as amended by IEEE Std 802.1Qcw-2023 with the following
3545 nodes. A distinction is made between nodes that shall be supported by bridge and end station
3546 components, or by bridge components only.

3547 **6.4.9.2.5.10.2 Bridge nodes to be supported by bridge and end station components**

- 3548 [m] /ieee802-dot1q-bridge/bridges/bridge/name
- 3549 [o] /ieee802-dot1q-bridge/bridges/bridge/address
- 3550 [m] /ieee802-dot1q-bridge/bridges/bridge/bridge-type
- 3551 [m] /ieee802-dot1q-bridge/bridges/bridge/ports
- 3552 [m] /ieee802-dot1q-bridge/bridges/bridge/components
- 3553 [m] /ieee802-dot1q-bridge/bridges/bridge/component/name
- 3554 [o] /ieee802-dot1q-bridge/bridges/bridge/component/id
- 3555 [m] /ieee802-dot1q-bridge/bridges/bridge/component/type
- 3556 [o] /ieee802-dot1q-bridge/bridges/bridge/component/traffic-class-
3557 enabled
- 3558 [m] /ieee802-dot1q-bridge/bridges/bridge/component/ports
- 3559 [o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-port
- 3560 [m] /ieee802-dot1q-bridge/bridges/bridge/component/capabilities
- 3561 [m] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
3562 database/size
- 3563 [o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
3564 database/static-vlan-registration-entries
- 3565 [o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
3566 database/vlan-registration-entry

3567 **6.4.9.2.5.10.3 Filtering-database nodes to be supported by bridge components**

- 3568 [o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
3569 database/aging-time
- 3570 [o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
3571 database/static-entries
- 3572 [o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
3573 database/dynamic-entries
- 3574 [o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
3575 database/dynamic-vlan-registration-entries
- 3576 [o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
3577 database/mac-address-registration-entries
- 3578 [o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
3579 database/filtering-entry

3580 6.4.9.2.5.10.4 Permanent-database nodes to be supported by bridge components

3581 [m] /ieee802-dot1q-bridge/bridges/bridge/component/permanent-
3582 database/size

3583 [o] /ieee802-dot1q-bridge/bridges/bridge/component/permanent-
3584 database/static-entries

3585 [o] /ieee802-dot1q-bridge/bridges/bridge/component/permanent-
3586 database/static-vlan-registration-entries

3587 [o] /ieee802-dot1q-bridge/bridges/bridge/component/permanent-
3588 database/filtering-entry

3589 6.4.9.2.5.10.5 Bridge-vlan nodes to be supported by bridge and end station components

3590 [m] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/version

3591 [m] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/max-
3592 vids

3593 [o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-
3594 vlan/override-default-pvid

3595 [o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/vlan

3596 6.4.9.2.5.10.6 Bridge-vlan nodes to be supported by bridge components

3597 [m] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/max-
3598 msti

3599 [o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/vid-to-
3600 fid-allocation

3601 [o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/fid-to-
3602 vid-allocation

3603 [o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/vid-to-
3604 fid

3605 6.4.9.2.5.10.7 Bridge-mst nodes to be supported by bridge components

3606 [o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-mst

3607 6.4.9.2.5.10.8 Bridge-port nodes to be supported by bridge and end station components

3608 [m] /ietf-interfaces/interfaces/interface/bridge-port/bridge-name

3609 [m] /ietf-interfaces/interfaces/interface/bridge-port/component-name

3610 [m] /ietf-interfaces/interfaces/interface/bridge-port/port-type

3611 [o] /ietf-interfaces/interfaces/interface/bridge-port/pvid

3612 [o] /ietf-interfaces/interfaces/interface/bridge-port/default-priority

3613 [m] /ietf-interfaces/interfaces/interface/bridge-port/traffic-class

3614 [o] /ietf-interfaces/interfaces/interface/bridge-port/statistics
 3615 [m] /ietf-interfaces/interfaces/interface/bridge-port/capabilities
 3616 [m] /ietf-interfaces/interfaces/interface/bridge-port/type-capabilities
 3617 [o] /ietf-interfaces/interfaces/interface/bridge-port/transmission-
 3618 selection-algorithm-table

3619 **6.4.9.2.5.10.9 Bridge-port nodes to be supported by bridge component ports**

3620 [o] /ietf-interfaces/interfaces/interface/bridge-port/priority-
 3621 regeneration
 3622 [o] /ietf-interfaces/interfaces/interface/bridge-port/acceptable-frame
 3623 [o] /ietf-interfaces/interfaces/interface/bridge-port/enable-ingress-
 3624 filtering
 3625 [o] /ietf-interfaces/interfaces/interface/bridge-port/enable-vid-
 3626 translation-table
 3627 [o] /ietf-interfaces/interfaces/interface/bridge-port/vid-translations
 3628 [o] /ietf-interfaces/interfaces/interface/bridge-port/enable-egress-
 3629 vid-translation-table
 3630 [o] /ietf-interfaces/interfaces/interface/bridge-port/egress-vid-
 3631 translations

3632 **6.4.9.2.5.11 IEC/IEEE 60802 YANG modules**

3633 IA-stations shall support the `ieciieee60802-ethernet-interface` YANG module according to this
 3634 document with the following nodes:

3635 [m] /ieciieee60802-ethernet-
 3636 interface/interfaces/interface/ethernet/supported-mau-types/mau-type
 3637 [m] /ieciieee60802-ethernet-
 3638 interface/interfaces/interface/ethernet/supported-mau-
 3639 types/preemption-supported

3640 IA-stations shall support the `ieciieee60802-bridge` YANG module according to this document
 3641 with the following nodes:

3642 [m] /ieciieee60802-bridge/interfaces/interface/bridge-port/max-burst-
 3643 params
 3644 [m] /ieciieee60802-bridge/interfaces/interface/bridge-port/committed-
 3645 data-rates
 3646 [m] /ieciieee60802-bridge/interfaces/interface/bridge-
 3647 port/transmission-selection-algorithm
 3648 [m] /ieciieee60802/interfaces/interface/bridge-port/supported-resource-
 3649 pools

3650 [m] /iecieee60802-bridge/bridges/bridge/component/frer-supported

3651 [m] /iecieee60802-bridge/bridges/bridge/component/max-redundant-
3652 streams

3653 [m] /iecieee60802-bridge/bridges/bridge/component/max-fids

3654 [m] /iecieee60802-bridge/bridges/bridge/component/max-fdb-entries

3655 [m] /iecieee60802-bridge/bridges/bridge/component/delay-variance

3656 [m] /iecieee60802-bridge/bridges/bridge/component/max-ptp-instances

3657 [m] /iecieee60802-bridge/bridges/bridge/component/max-hot-standby-
3658 systems

3659 [m] /iecieee60802-bridge/bridges/bridge/component/clock

3660 [m] /iecieee60802/interfaces/interface/bridge-port/phy-latencies/mau-
3661 type

3662 [m] /iecieee60802/interfaces/interface/bridge-port/phy-
3663 latencies/latency

3664 [m] /iecieee60802/interfaces/interface/bridge-port/phy-
3665 latencies/latency-variation

3666 **IA-stations shall support the iecieee60802-ia-station YANG module according to this document**
3667 **with the following nodes:**

3668 [m] /iecieee60802-ia-station/ia-station-capabilities/capability-lldp

3669 [m] /iecieee60802-ia-station/ia-station-capabilities/capability-timesync

3670 [m] /iecieee60802-ia-station/ia-station-capabilities/capability-keystore

3671 [m] /iecieee60802-ia-station/ia-station-capabilities/capability-truststore

3672 [m] /iecieee60802-ia-station/ia-station-capabilities/capability-nacm

3673 [m] /iecieee60802-ia-station/ia-station-capabilities/capability-yang-library

3674 [m] /iecieee60802-ia-station/ia-station-capabilities/capability-yang-push

3675 [m] /iecieee60802-ia-station/ia-station-capabilities/capability-yang-
3676 notifications

3677 [m] /iecieee60802-ia-station/ia-station-capabilities/capability-netconf-
3678 monitoring

3679 [m] /iecieee60802-ia-station/ia-station-capabilities/capability-netconf-
3680 client

3681 [m] /iecieee60802-ia-station/ia-station-capabilities/capability-cnc-config

3682 [m] /ieciieee60802-ia-station/ia-station-capabilities/capability-sched-
3683 traffic

3684 [m] /ieciieee60802-ia-station/ia-station-capabilities/capability-frame-
3685 preemption

3686 IA-stations shall support the ieciieee60802-subscribed-notifications YANG module according to
3687 this document with the following nodes:

3688 [m] /ieciieee60802-subscribed-notifications/subscriptions/max-
3689 subscriptions

3690 [m] /ieciieee60802-subscribed-notifications/subscriptions/max-on-
3691 change-subscription-leaves

3692 [m] /ieciieee60802-subscribed-notifications/subscriptions/max-periodic-
3693 subscription-leaves

3694 [m] /ieciieee60802-subscribed-notifications/subscriptions/max-periodic-
3695 subscription-interval

3696 IA-stations shall support the ieciieee60802-proxy YANG module according to this document with
3697 the following nodes:

3698 [c] /ieciieee60802-proxy/proxy/max-num-proxy-targets

3699 [c] /ieciieee60802-proxy/proxy/proxy-types

3700 [o] /ieciieee60802-proxy/proxy/proxy-targets

3701 IA-stations shall support the ieciieee60802-proxy-target YANG module according to this
3702 document with the following nodes:

3703 [c] /ieciieee60802-proxy-target/proxy-target/required-proxy-type

3704 **6.4.9.2.5.12 NETCONF server**

3705 IA-stations shall support the ietf-netconf-server YANG module according to draft-ietf-netconf-
3706 netconf-client-server, 3.1.1, with the following features:

- 3707 • tls-call-home, and
- 3708 • central-netconf-server-supported.

3709 IA-stations shall support the ietf-netconf-server YANG module according to draft-ietf-netconf-
3710 netconf-client-server, 3.3, with the following nodes:

3711 [o] /ietf-netconf-server/netconf-server/listen/idle-timeout

3712 [o] /ietf-netconf-server/netconf-server/listen/endpoint/name

3713 [o] /ietf-netconf-server/netconf-
3714 server/listen/endpoint/transport/tls/netconf-server-parameters

3715 [o] /ietf-netconf-server/netconf-
3716 server/listen/endpoint/transport/tls/tls-server-parameters

3717 [o] /ietf-netconf-server/netconf-server/call-home/netconf-client/name

3718 [o] /ietf-netconf-server/netconf-server/call-home/netconf-
3719 client/endpoints/endpoint/name

3720 [o] /ietf-netconf-server/netconf-server/call-home/netconf-
3721 client/endpoints/endpoint/transport/tls/netconf-server-parameters

3722 [o] /ietf-netconf-server/netconf-server/call-home/netconf-
3723 client/endpoints/endpoint/transport/tls/tls-server-parameters

3724 **6.4.9.2.5.13 Flow Meter Management**

3725 IA-stations which incorporate a bridge component shall support the ieee802-dot1q-stream-
3726 filters-gates YANG module according to IEEE Std 802.1Qcz-2023 as amended by IEEE Std
3727 802.1Qcw-2023 with the following nodes:

3728 [o] /ieee802-dot1q-bridge/bridges/bridge/component/stream-
3729 filters/stream-filter-instance-table/stream-filter-instance-id

3730 [o] /ieee802-dot1q-bridge/bridges/bridge/component/stream-
3731 filters/stream-filter-instance-table/stream-handle

3732 [o] /ieee802-dot1q-bridge/bridges/bridge/component/stream-
3733 filters/stream-filter-instance-table/flow-meter-ref

3734 [o] /ieee802-dot1q-bridge/bridges/bridge/component/stream-
3735 filters/stream-filter-instance-table/flow-meter-enable

3736 [m] /ieee802-dot1q-bridge/bridges/bridge/component/stream-filters/max-
3737 stream-filter-instances

3738 [o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/stream-
3739 gates/stream-gate-instance-table

3740 [m] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/stream-
3741 gates/max-stream-gate-instances

3742 [m] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/stream-
3743 gates/supported-list-max

3744 IA-stations which incorporate a bridge component shall support the ieee802-dot1cb-stream-
3745 identification YANG module according to IEEE Std 802.1CBcv-2021 as amended by IEEE Std
3746 802.1CBdb-2021 with the following nodes:

3747 [o] /ieee802-dot1cb-stream-identification/stream-identity/index

3748 [o] /ieee802-dot1cb-stream-identification/stream-identity/handle

3749 [o] /ieee802-dot1cb-stream-identification/stream-identity/out-
3750 facing/input-port

3751 [o] /ieee802-dot1cb-stream-identification/stream-
3752 identity/parameters/mask-and-match-stream-identification/destination-
3753 mac-mask

3754 [o] /ieee802-dot1cb-stream-identification/stream-
3755 identity/parameters/mask-and-match-stream-identification/destination-
3756 mac-match

3757 NOTE For example, an implementation could contain per out-facing/input-port one mask and match stream
3758 identification for broadcast traffic, one mask and match stream identification for multicast traffic and one mask and
3759 match stream identification for unicast traffic.

3760 IA-stations which incorporate a bridge component shall support the ieee802-dot1q-psfp-bridge
3761 YANG module according to IEEE Std 802.1Qcw-2023 with the following nodes:

3762 [o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
3763 meters/flow-meter-instance-table/flow-meter-instance-id

3764 [o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
3765 meters/flow-meter-instance-table/committed-information-rate

3766 [o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
3767 meters/flow-meter-instance-table/committed-burst-size

3768 [o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
3769 meters/flow-meter-instance-table/excess-information-rate

3770 [o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
3771 meters/flow-meter-instance-table/excess-burst-size

3772 [o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
3773 meters/flow-meter-instance-table/coupling-flag

3774 [o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
3775 meters/flow-meter-instance-table/color-mode

3776 [o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
3777 meters/flow-meter-instance-table/drop-on-yellow

3778 [m] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
3779 meters/max-flow-meter-instances

3780 **6.4.9.2.5.14 RSTP/MSTP**

3781 IA-stations shall support the ieee802-dot1q-rstp-bridge YANG module according to IEEE Std
3782 802.1Qdy™-2025 with the following nodes.

3783 [o] /ieee802-dot1q-rstp-bridge/bridges/bridge/rstp

3784 [o] /ieee802-dot1q-rstp-bridge/interfaces/interface/bridge-port/rstp

3785 IA-stations shall support the ieee802-dot1q-mstp-bridge YANG module according to IEEE Std
3786 802.1Qdy-2025 with the following nodes.

3787 [o] /ieee802-dot1q-mstp-bridge/bridges/bridge/component/bridge-
3788 mst/bridge-mstp

3789 [o] /ieee802-dot1q-mstp-bridge/interfaces/interface/bridge-port/port-
3790 mstp

3791 **6.4.9.3 Optional YANG data models, features, and nodes**

3792 **6.4.9.3.1 General**

3793 The following YANG modules, features and nodes shall be supported by IA-stations if the
3794 functionality they describe is included.

3795 **6.4.9.3.2 Scheduled traffic**

3796 IA-stations supporting the enhancements for scheduled traffic shall support the ieee802-dot1q-
3797 sched-bridge YANG module according to IEEE Std 802.1Qcw-2023 with the following feature:
3798 `scheduled-traffic`.

3799 IA-stations supporting the enhancements for scheduled traffic shall support the ieee802-dot1q-
3800 sched-bridge YANG module according to IEEE Std 802.1Qcw-2023 with the following nodes:

3801 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3802 `parameter-table/queue-max-sdu-table`

3803 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3804 `parameter-table/gate-enabled`

3805 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3806 `parameter-table/admin-gate-states`

3807 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3808 `parameter-table/oper-gate-states`

3809 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3810 `parameter-table/admin-control-list`

3811 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3812 `parameter-table/oper-control-list`

3813 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3814 `parameter-table/admin-cycle-time`

3815 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3816 `parameter-table/oper-cycle-time`

3817 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3818 `parameter-table/admin-cycle-time-extension`

3819 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3820 `parameter-table/oper-cycle-time-extension`

3821 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3822 `parameter-table/admin-base-time`

3823 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3824 `parameter-table/oper-base-time`

3825 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3826 `parameter-table/config-change`

3827 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3828 `parameter-table/config-change-time`

3829 [o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
3830 parameter-table/tick-granularity

3831 [o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
3832 parameter-table/current-time

3833 [o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
3834 parameter-table/config-pending

3835 [o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
3836 parameter-table/config-change-error

3837 [c] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
3838 parameter-table/supported-list-max

3839 [c] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
3840 parameter-table/supported-cycle-max

3841 [c] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
3842 parameter-table/supported-interval-max

3843 **6.4.9.3.3 IEC/IEEE 60802 YANG modules**

3844 IA-stations that support enhancements for scheduled traffic shall support the iecieee60802-
3845 sched-bridge YANG module according to this document with the following nodes:

3846 [c] /ieciieee60802-sched-bridge/interfaces/interface/bridge-port/gate-
3847 parameter-table/min-gating-times

3848 **6.4.9.3.4 Frame preemption**

3849 IA-stations supporting frame preemption according to IEEE Std 802.1Q-2022, 5.4.1, ae), shall
3850 support the ieee802-dot1q-preemption-bridge YANG module according to IEEE Std
3851 802.1Qcw-2023 with the following feature: frame-preemption.

3852 IA-stations supporting frame preemption according to IEEE Std 802.1Q-2022, 5.4.1, ae), shall
3853 support the ieee802-dot1q-preemption-bridge YANG module according to IEEE Std
3854 802.1Qcw-2023 with the following nodes:

3855 [o] ieee802-dot1q-preemption-bridge/interfaces/interface/bridge-
3856 port/frame-preemption-parameters/frame-preemption-status-table

3857 [o] ieee802-dot1q-preemption-bridge/interfaces/interface/bridge-
3858 port/frame-preemption-parameters/preemption-active

3859 **6.4.9.3.5 Credit-based shaper**

3860 IA-stations supporting the credit-based shaper according to IEEE Std 802.1Q-2022, 8.6.8.2,
3861 shall support the ieee802-dot1q-cbsa-bridge YANG module according to IEEE Std 802.1Qdx™-
3862 2024 with the following nodes.

3863 [c] /ieee802-dot1q-cbsa-bridge/interfaces/interface/bridge-
3864 port/cbsa/cbsa-parameter-table/traffic-class

3865 [c] /ieee802-dot1q-cbsa-bridge/interfaces/interface/bridge-
3866 port/cbsa/cbsa-parameter-table/admin-idle-slope

3867 [c] /ieee802-dot1q-cbsa-bridge/interfaces/interface/bridge-
3868 port/cbsa/cbsa-parameter-table/oper-idle-slope

3869 **6.4.9.3.6 FRER**

3870 IA-stations supporting FRER according to 5.10.1 item b) or item c), shall support the
3871 ieee802-dot1cb-stream-identification and ieee802-dot1cb-frer YANG modules according to
3872 IEEE Std 802.1CBcv-2021 with the following nodes:

3873 [o] /ieee802-dot1cb-stream-identification/stream-identity/index

3874 [o] /ieee802-dot1cb-stream-identification/stream-identity/handle

3875 [o] /ieee802-dot1cb-stream-identification /stream-identity/out-
3876 facing/input-port

3877 [o] /ieee802-dot1cb-stream-identification /stream-identity/out-
3878 facing/output-port

3879 [o] /ieee802-dot1cb-stream-identification /stream-
3880 identity/parameters/null-stream-identification

3881 [o] /ieee802-dot1cb-frer/frer/sequence-generation/index

3882 [o] /ieee802-dot1cb-frer/frer/sequence-generation/stream

3883 [o] /ieee802-dot1cb-frer/frer/sequence-generation/direction-out-facing

3884 [o] /ieee802-dot1cb-frer/frer/sequence-recovery/index

3885 [o] /ieee802-dot1cb-frer/frer/sequence-recovery/stream

3886 [o] /ieee802-dot1cb-frer/frer/sequence-recovery/port

3887 [o] /ieee802-dot1cb-frer/frer/sequence-recovery/direction-out-facing

3888 [o] /ieee802-dot1cb-frer/frer/sequence-recovery/algorithm/vector

3889 [o] /ieee802-dot1cb-frer/frer/sequence-identification/port

3890 [o] /ieee802-dot1cb-frer/frer/sequence-identification/direction-out-
3891 facing

3892 [o] /ieee802-dot1cb-frer/frer/sequence-identification/stream

3893 [o] /ieee802-dot1cb-frer/frer/sequence-identification/encapsulation/r-
3894 tag

3895 [o] /ieee802-dot1cb-frer/frer/stream-split

3896 **6.4.9.4 CUC/CNC YANG**

3897 **6.4.9.4.1 NETCONF Client**

3898 IA-stations with CNC or CUC functionality shall support the ietf-netconf-client YANG module
3899 according to draft-ietf-netconf-netconf-client-server, 2.1.1, with the following features:

- 3900 • `tls-initiate`,
- 3901 • `tls-listen`, and
- 3902 • `central-netconf-client-supported`.

3903 **6.4.9.4.2 YANG module for CNC-Config**

3904 IA-stations with CNC or CUC functionality shall support the ieee802-dot1q-cnc-config YANG
3905 module according to IEEE Std 802.1Qdj-2024 with the node:
3906 `[o] /ieee802-dot1q-cnc-config/cnc-config`

3907 **6.4.10 YANG Data Model**

3908 **6.4.10.1 General**

3909 Subclause 6.4.10 specifies the YANG data model for IA-stations. YANG (IETF RFC 7950) is a
3910 data modelling language used to model configuration data and state data for remote network
3911 management protocols. The selected YANG-based remote network management protocol is
3912 NETCONF (IETF RFC 6241). A YANG module specifies the organization and rules for the
3913 management data, and a mapping from YANG to the specific encoding enables the data to be
3914 understood correctly by both client (e.g. network manager) and server (e.g. IA-stations).

3915 **6.4.10.2 YANG framework**

3916 The core of the YANG module for IEC/IEEE 60802 IA-stations consists of YANG "augment"
3917 statements, used to add members to the tree of existing YANG modules plus one new module
3918 for IEC/IEEE 60802 specific objects.

3919 **6.4.10.3 IEC/IEEE 60802 Specific managed objects**

3920 **6.4.10.3.1 General**

3921 Subclause 6.4.10.3 defines the set of managed objects, and their functionality, that provides
3922 additional information about an IA-station that is required by a CNC to calculate network
3923 configurations.

3924 IEC/IEEE 60802 specific managed objects are specified:

- 3925 • per Ethernet interface, i.e. external port, in 6.4.10.3.2,
- 3926 • per end station component internal or external port in 6.4.10.3.3 and 6.4.10.3.4,
- 3927 • per bridge component internal or external port in 6.4.10.3.4,
- 3928 • per end station component in 6.4.10.3.5 and 6.4.10.3.7,
- 3929 • per bridge component in 6.4.10.3.6 and 6.4.10.3.7, and
- 3930 • per IA-station in 6.4.10.3.8.

3931 IEC/IEEE 60802 specific managed objects for CNC entities are specified in 6.4.10.3.9.

6.4.10.3.2 IEC/IEEE 60802 managed objects per Ethernet interface**6.4.10.3.2.1 supportedMauTypes**

The list of supported MAU Types including the data:

a) mauType

The value is the supported MAU Type derived from the list position of the corresponding dot3MauType as listed in IETF RFC 4836, Clause 5.

b) preemptionSupported

The Boolean value indicates if preemption is supported by the MAU Type.

NOTE The operational MAU Type of an Ethernet interface is provided as leaf operational-mau-type of the ieee802-ethernet-lldp YANG module. The operational MAU Type is included in the supportedMauTypes list.

6.4.10.3.3 IEC/IEEE 60802 managed objects per end station component port**6.4.10.3.3.1 worstCasePacketGap**

The value is the worst case maximum inter-packet gap between consecutive frames in a traffic burst expressed in bit-times.

NOTE Minimum interPacketGap is defined in IEEE Std 802.3-2022, 1.4.362. The worst-case-packet-gap is never less than the minimum interPacketGap.

6.4.10.3.3.2 maxBurstFrames

The value is the maximum number of frames that can be sent with minimal inter packet gap.

6.4.10.3.3.3 maxBurstBytes

The value is the maximum number of octets that can be sent with minimal inter packet gap.

6.4.10.3.3.4 committedDataRates

The list of committed data rates per traffic class and supported line speed including the data:

a) committedInformationRate

The value is the bandwidth limit in kbit/s.

b) committedBurstSize

The value is the burst size limit in octets.

6.4.10.3.4 IEC/IEEE 60802 managed objects per bridge or end station component port**6.4.10.3.4.1 transmissionSelectionAlgorithm**

The list of supported transmission selection algorithms according to IEEE Std 802.1Q-2022, 8.6.8, per traffic class.

6.4.10.3.4.2 supportedResourcePools

The list of supported buffer resource pools including the data:

a) resourcePoolName

The value is the name of a resource pool.

b) coveredTimeInterval

The value specifies the covered buffering time given as rational number of seconds for the highest supported link speed.

c) resourcePoolTrafficClasses

3970 The list of the traffic classes to be served by the resource pool.

3971 **6.4.10.3.4.3 minGatingTimes**

3972 The list of minimum gating times per supported line speed including the data:

3973 a) minCycleTime

3974 The value is the minimum value supported by this port of the AdminCycleTime and
3975 OperCycleTime parameters given as rational number of seconds.

3976 b) minIntervalTime

3977 The value is the minimum value supported by this port of the TimeIntervalValue parameter
3978 in nanoseconds.

3979 **6.4.10.3.5 IEC/IEEE 60802 managed objects per end station component**

3980 **6.4.10.3.5.1 frerSupported**

3981 The value indicates if FRER is supported.

3982 **6.4.10.3.5.2 maxRedundantStreams**

3983 The value is the maximum number of supported redundant streams.

3984 **6.4.10.3.6 IEC/IEEE 60802 managed objects per bridge component**

3985 **6.4.10.3.6.1 delayVariance**

3986 The value indicates variance in delay depending upon the use of a singleValue or
3987 multipleValues (see 6.4.10.3.6.2).

3988 **6.4.10.3.6.2 delayTimes**

3989 The list of minimum and maximum frame length independent and frame length dependent delay
3990 time values of frames as they pass through a bridge component. These values are given:

- 3991 • per supported MAU Type pair and traffic class, if delayVariance is singleValue, or
- 3992 • per port pair with supported MAU Types and traffic class, if delayVariance is multipleValues.

3993 The list includes the data:

3994 a) independentDelayMin

3995 The value is the minimum delay portion that is independent of frame length according to
3996 IEEE Std 802.1Q-2022, 12.32.1.1.

3997 b) independentDelayMax

3998 The value is the maximum delay portion that is independent of frame length according to
3999 IEEE Std 802.1Q-2022, 12.32.1.1.

4000 c) dependentDelayMin

4001 The value is the minimum delay portion that is dependent on frame length according to IEEE
4002 Std 802.1Q-2022, 12.32.1.2.

4003 d) dependentDelayMax

4004 The value is the maximum delay portion that is dependent on frame length according to
4005 IEEE Std 802.1Q-2022, 12.32.1.2.

4006 **6.4.10.3.6.3 latency**

4007 The PHY latency in ns.

4008 **6.4.10.3.6.4 latencyVariation**

4009 The PHY latency variation in ns.

4010 **6.4.10.3.7 IEC/IEEE 60802 managed objects per bridge or end station component**

4011 **6.4.10.3.7.1 maxFids**

4012 The value is the maximum number of supported FIDs.

4013 **6.4.10.3.7.2 maxFdbEntries**

4014 The list of the maximum number of static (6.4.10.3.7.3) and dynamic (6.4.10.3.7.4) FDB entries
4015 per FDB.

4016 **6.4.10.3.7.3 maxStaticFdbEntries**

4017 The value is the maximum number of static FDB entries.

4018 **6.4.10.3.7.4 maxDynamicFdbEntries**

4019 The value is the maximum number of dynamic FDB entries.

4020 **6.4.10.3.7.5 maxPtpInstances**

4021 The value is the maximum number of supported PTP Instances.

4022 **6.4.10.3.7.6 maxHotStandbySystems**

4023 The value is the maximum number of supported HotStandbySystem entities (see IEEE Std
4024 802.1ASdm-2024).

4025 **6.4.10.3.7.7 clockList**

4026 The list of supported application clock entities including the data:

4027 a) clockIdentity

4028 The clock identity of the application clock.

4029 b) clockTarget

4030 The Boolean value indicates if the application clock is a clock target (TRUE) or clock source
4031 (FALSE).

4032 c) arbSupported

4033 The Boolean value indicates if the application clock supports the ARB timescale.

4034 d) ptpSupported

4035 The Boolean value indicates if the application clock supports the PTP timescale.

4036 e) hotStandbySupported

4037 The Boolean value indicates if the application clock supports hot standby.

4038 f) attachedPtpInstance

4039 The value is a reference to the PTP or hot standby Instance, that is attached to the
4040 application clock.

4041 g) isSynced

4042 The Boolean value indicates if the application clock is either synchronized to the attached
4043 PTP Instance (TRUE) or to an internal/external ClockSource (FALSE).

4044 **6.4.10.3.8 IEC/IEEE 60802 managed objects per IA-station**

4045 **6.4.10.3.8.1 maxSubscriptions**

4046 The value is the maximum number of supported NETCONF Server subscriptions.

4047 **6.4.10.3.8.2 maxOnChangeSubscriptionLeaves**

4048 The value is the maximum number of supported leaves for NETCONF Server on-change
4049 subscriptions according to IETF RFC 8641.

4050 **6.4.10.3.8.3 maxPeriodicSubscriptionLeaves**

4051 The value is the maximum number of supported leaves for NETCONF Server periodic
4052 subscriptions according to IETF RFC 8641.

4053 **6.4.10.3.8.4 minPeriodicSubscriptionInterval**

4054 The value is the minimum periodic subscription interval in centiseconds (0,01 seconds) for
4055 NETCONF Server periodic subscriptions according to IETF RFC 8641.

4056 **6.4.10.3.8.5 capabilityLLDP**

4057 This Boolean value indicates if LLDP is supported.

4058 **6.4.10.3.8.6 capabilityTimesync**

4059 This Boolean value indicates if Timesync is supported.

4060 **6.4.10.3.8.7 capabilityKeystore**

4061 This Boolean value indicates if Keystore is supported.

4062 **6.4.10.3.8.8 capabilityNACM**

4063 This Boolean value indicates if NACM is supported.

4064 **6.4.10.3.8.9 capabilityTruststore**

4065 This Boolean value indicates if Truststore is supported.

4066 **6.4.10.3.8.10 capabilityYangLibrary**

4067 This Boolean value indicates if YANG library is supported.

4068 **6.4.10.3.8.11 capabilityYangPush**

4069 This Boolean value indicates if Yang Push is supported.

4070 **6.4.10.3.8.12 capabilityYangNotifications**

4071 This Boolean value indicates if YANG notifications is supported.

4072 **6.4.10.3.8.13 capabilityNetconfMonitoring**

4073 This Boolean value indicates if NETCONF Monitoring is supported.

4074 **6.4.10.3.8.14 capabilityNetconfClient**

4075 This Boolean value indicates if NETCONF client is supported.

4076 **6.4.10.3.8.15 capabilityCncConfig**

4077 This Boolean value indicates if CNC configuration is supported.

4078 **6.4.10.3.8.16 capabilitySchedTraffic**

4079 This Boolean value indicates if scheduled traffic is supported.

4080 **6.4.10.3.8.17 capabilityFramePreemption**

4081 This Boolean value indicates if frame preemption is supported.

4082 **6.4.10.3.9 IEC/IEEE 60802 managed objects per proxy station**

4083 **6.4.10.3.9.1 maxNumProxyTargets**

4084 The value indicates how many proxy targets can be handled by this proxy.

4085 **6.4.10.3.9.2 proxyType**

4086 OID of proxy targets handled by this proxy.

4087 **6.4.10.3.9.3 device**

4088 The identification of the device.

4089 **6.4.10.3.9.4 deviceType**

4090 The type of the device.

4091 **6.4.10.3.9.5 proxyAddr**

4092 The ip-address that is used by the CNC to configure the proxy target.

4093 **6.4.10.3.9.6 port**

4094 The port-number that is used by the CNC to configure the proxy target.

4095 **6.4.10.3.9.7 deviceAddr**

4096 The ip-address that is used by the proxy to address the proxy target.

4097 **6.4.10.3.9.8 macAddr**

4098 The mac-address that is used by the proxy to address the proxy target.

4099 **6.4.10.3.10 IEC/IEEE 60802 managed objects per proxy target**

4100 **6.4.10.3.10.1 proxyType**

4101 OID of proxy that this proxy target requires.

4102 **6.4.10.3.11 IEC/IEEE 60802 managed objects for CNC entities**

4103 **6.4.10.3.11.1 maxConfigurationDomains**

4104 The value is the maximum number of supported Configuration Domains.

4105 **6.4.10.3.11.2 maxCUCs**

4106 The value is the maximum number of supported CUC entities.

4107 **6.4.10.3.11.3 maxIAstations**

4108 The value is the maximum number of supported IA-stations.

4109 **6.4.10.3.11.4 maxNetworkDiameter**

4110 The value is the maximum supported network diameter.

4111 **6.4.10.3.11.5 maxStreams**

4112 The value is the maximum number of supported streams.

4113 **6.4.10.3.11.6 maxNumSeamlessTrees**

4114 The value is the maximum number of trees supported for seamless redundancy of a stream.

4115 **6.4.10.3.11.7 hotStandbySupported**

4116 The Boolean value indicates if hot standby is supported.

4117 **6.4.10.4 RPCs and actions specific to this document**

4118 **6.4.10.4.1 RPC ia-factory-reset**

4119 **6.4.10.4.1.1 General**

4120 In contrast to the original factory-reset RPC in IETF RFC 8808, this RPC puts the device into a
4121 state where a subsequent configuration by a CNC component results in a functioning IA-station
4122 according to this document. Depending on the factory default configuration, after being reset,
4123 the device can become unreachable on the network.

4124 **6.4.10.4.1.2 Input**

4125 There is no input.

4126 **6.4.10.4.1.3 Output**

4127 There is no output.

4128 **6.4.10.4.2 Action add-streams**

4129 **6.4.10.4.2.1 General**

4130 This Action requests a CNC to add a list of streams.

4131 **6.4.10.4.2.2 Input**

4132 The RPC requires the following input parameters.

- 4133 a) Cuid – The ID of the CUC for which the streams are to be added.
- 4134 b) StreamId – The Stream ID is a unique identifier of a Stream request and corresponding
4135 configuration.
- 4136 c) Container Talker – The Talker container contains:
 - 4137 – Talker's behavior for Stream (how/when transmitted),
 - 4138 – Talker's requirements from the network, and

- 4139 – TSN capabilities of the Talker's interface(s).
- 4140 d) List Listener – Each Listener list entry contains:
- 4141 – Listener's requirements from the network, and
- 4142 – TSN capabilities of the Listener's interface(s).

4143 **6.4.10.4.2.3 Output**

4144 Result – Status information indicating if Stream addition has been successful.

4145 **6.4.10.4.3 Action remove-listener**

4146 **6.4.10.4.3.1 General**

4147 This Action removes listeners from a stream.

4148 **6.4.10.4.3.2 Input**

4149 List Listener – A list of indices of listeners to be removed from a stream.

4150 **6.4.10.4.3.3 Output**

4151 Result – Status information indicating if Stream addition has been successful.

4152 **6.4.10.5 IEC/IEEE 60802 YANG data models**

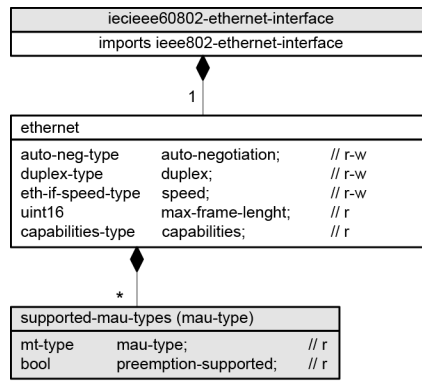
4153 A representation similar to UML® is used to provide an overview of the hierarchy of the
4154 IEC/IEEE 60802 YANG data model.

4155 A UML-like representation of the management model is provided in Figure 29, Figure 30,
4156 Figure 31, Figure 32, Figure 33, Figure 34, Figure 35, Figure 36, and Figure 37. The purpose
4157 of a UML-like diagram is to express the model design in a concise manner. The structure of the
4158 UML-like representation shows the name of the object followed by a list of properties for the
4159 object. The properties indicate their type and accessibility. It should be noted that UML-like
4160 representation is meant to express simplified semantics for the properties. It is not meant to
4161 provide the specific datatype used to encode the object in either MIB or YANG.

4162 NOTE – OMG® UML® 2.5 conventions together with C++ language constructs are used as a representation to convey
4163 model structure and relationships.

4164 For all UML® figures, data that is imported from original modules is shown in white, and data
4165 that is augmented according to IEC/IEEE 60802 is shown in grey.

4166 Figure 29, Figure 30, Figure 31, Figure 32, Figure 33, Figure 34, Figure 35, Figure 36, and
4167 Figure 37 provide an overview of the IEC/IEEE 60802 augmentations.



4168

IEC

4169

Figure 29 – Module iecieee60802-ethernet-interface

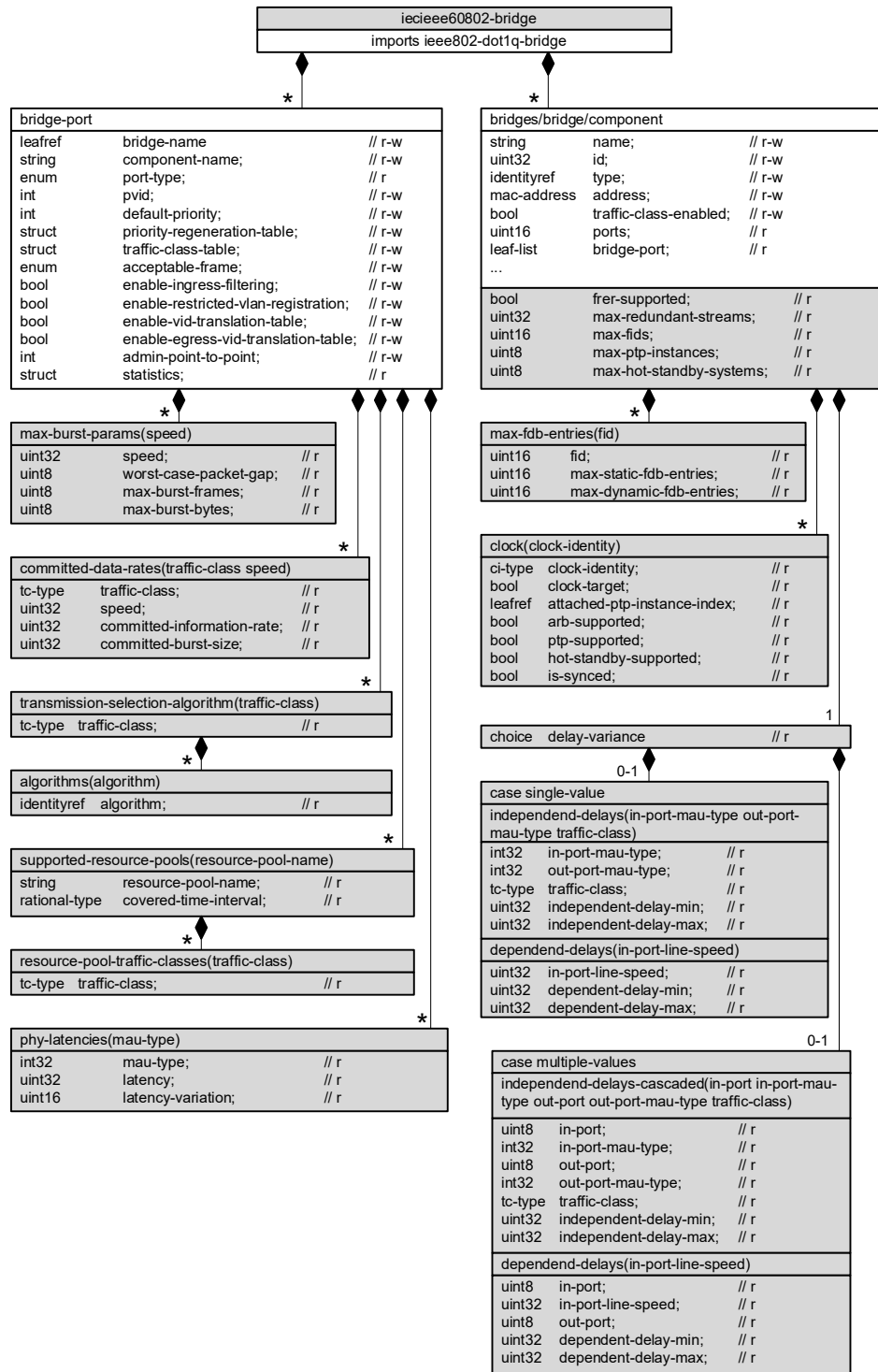
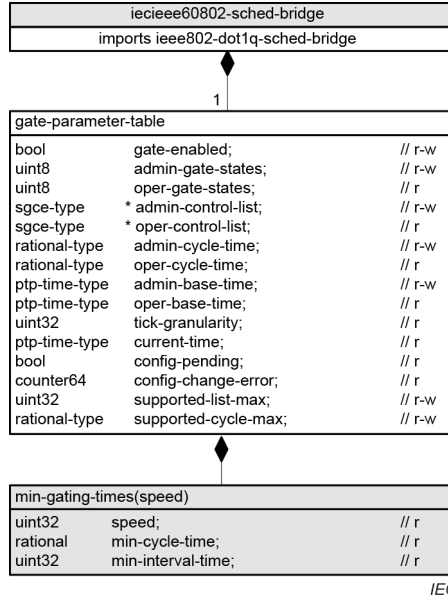
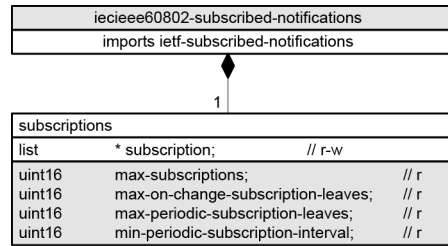


Figure 30 – Module iecieee60802-bridge



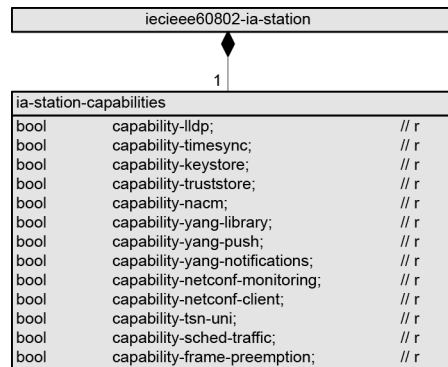
IEC

Figure 31 – Module icieeee60802-dot1-sched-bridge



IEC

Figure 32 – Module icieeee60802-subscribed-notifications



IEC

Figure 33 – Module icieeee60802-ia-station

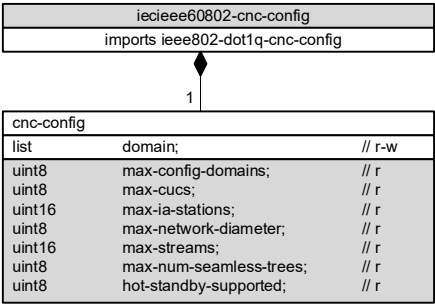


Figure 34 – Module icieeee60802-cnc-config

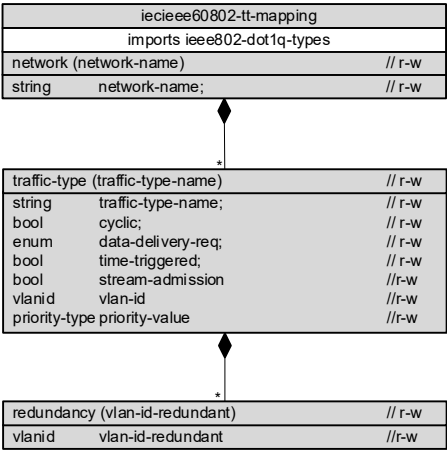


Figure 35 – Module icieeee60802-tt-mapping

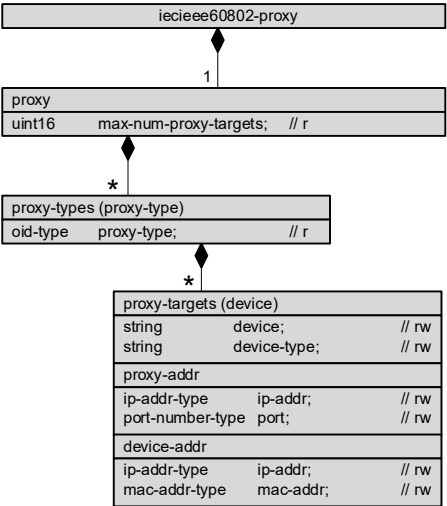


Figure 36 – Module icieeee60802-proxy

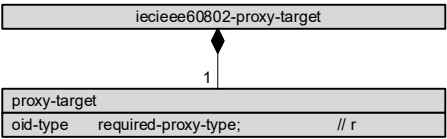


Figure 37 – icieeee60802-proxy-target

6.4.10.6 Structure of IEC/IEEE 60802 YANG data models

The YANG data models specified by this standard use the YANG modules summarized in Table 17.

In the YANG module definitions, if any discrepancy between the "description" text and the corresponding definition in any other part of this standard occurs, the definitions outside Clause 6 take precedence.

Table 17 – Summary of the YANG modules

Module	Description
ieee802-ethernet-interface	This module contains YANG definitions for configuring IEEE Std 802.3 Ethernet Interfaces.
ietf-interfaces	This module contains a collection of YANG definitions for managing network interfaces.
ieciieee60802-ethernet-interface	This module augments ieee802-ethernet-interface.
ieee802-types	This module contains a collection of generally useful derived data types for IEEE YANG data models.
ieee802-dot1q-bridge	This module describes the bridge configuration model for IEEE 802.1Q Bridges.
ieee802-dot1q-types	This module contains common types used within dot1q-bridge modules.
ieciieee60802-bridge	This module augments ieee802-dot1q-bridge.
ieee802-dot1q-sched-bridge	This module provides for management of IEEE Std 802.1Q Bridges that support Scheduled Traffic Enhancements.
ieciieee60802-dot1q-sched-bridge	This module augments ieee802-dot1q-sched-bridge.
ieee802-dot1cb-frer	This module provides management objects that control the frame replication and elimination from IEEE Std 802.1CB-2017.
ieee1588-ptp-tt	This module defines a data model for the configuration and state of IEEE Std 1588 clocks.
ietf-netconf-acm	This module provides management for the Network Configuration Access Control Model.
ieee802-dot1q-cnc-config	This module provides the Time-Sensitive Networking (TSN) User/Network Interface (UNI) for the exchange of information between CUC and CNC that are required to configure TSN Streams in a TSN network.
ieciieee60802-cnc-config	This module augments ieee802-dot1q-cnc-config.
ieciieee60802-ia-station	This module provides read-only information about the capabilities and RPCs for IEC/IEEE 60802 IA-stations.
ietf-subscribed-notifications	This module defines a YANG data model for subscribing to event records and receiving matching content in notification messages.
ieciieee60802-subscribed-notifications	This module augments ietf-subscribed-notifications.
ieciieee60802-tt-mapping	This module provides information on characteristics of traffic classes configured in a network to allow applications to map application traffic classes to them
ieciieee60802-proxy.yang	This module contains configuration and capability information for IEC/IEEE 60802 proxy stations
ieciieee60802-proxy-target.yang	This module contains capability information for IEC/IEEE 60802 proxy targets

6.4.10.7 YANG schema tree definitions

6.4.10.7.1 General

The schema tree is provided as an overview of the YANG modules. The symbols and their meaning are specified in YANG Tree Diagrams (IETF RFC 8340).

6.4.10.7.2 Module iecieee60802-ethernet-interface

```

4198 module: iecieee60802-ethernet-interface
4199
4200 augment /if:interfaces/if:interface/eth-if:ethernet:
4201     +-ro supported-mau-types* [mau-type]
4202     +-ro mau-type                uint32
4203     +-ro preemption-supported?   boolean
4204
4205

```

6.4.10.7.3 Module iecieee60802-bridge

```

4206 module: iecieee60802-bridge
4207
4208 augment /if:interfaces/if:interface/bridge:bridge-port:
4209     +-ro max-burst-params* [speed]
4210     | +-ro speed                uint32
4211     | +-ro worst-case-packet-gap? uint8
4212     | +-ro max-burst-frames?     uint8
4213     | +-ro max-burst-bytes?     uint8
4214     +-ro committed-data-rates* [traffic-class speed]
4215     | +-ro traffic-class        dot1q-types:traffic-class-type
4216     | +-ro speed                uint32
4217     | +-ro committed-information-rate? uint32
4218     | +-ro committed-burst-size?  uint32
4219     +-ro transmission-selection-algorithm* [traffic-class]
4220     | +-ro traffic-class        dot1q-types:traffic-class-type
4221     | +-ro algorithms* [algorithm]
4222     | | +-ro algorithm         identityref
4223     +-ro supported-resource-pools* [resource-pool-name]
4224     | +-ro resource-pool-name   string
4225     | +-ro covered-time-interval
4226     | | +-ro numerator?        uint32
4227     | | +-ro denominator?      uint32
4228     | +-ro resource-pool-traffic-classes* [traffic-class]
4229     | | +-ro traffic-class      dot1q-types:traffic-class-type
4230     +-ro phy-latencies* [mau-type]
4231     | +-ro mau-type            int32
4232     | +-ro latency?            uint32
4233     | +-ro latency-variation?  uint16
4234     augment /bridge:bridges/bridge:bridge/bridge:component:
4235     | +-ro frer-supported?      boolean
4236     | +-ro max-redundant-streams? uint32
4237     | +-ro max-fids?            uint16
4238     | +-ro max-fdb-entries* [fid]
4239     | | +-ro fid                uint16
4240     | | +-ro max-static-fdb-entries? uint16
4241     | | +-ro max-dynamic-fdb-entries? uint16
4242     | +-ro (delay-variance)?
4243     | | +-: (single-value)
4244     | | | +-ro independent-delays* [in-port-mau-type out-port-mau-type
4245     | | | traffic-class]
4246     | | | | +-ro in-port-mau-type    int32
4247     | | | | +-ro out-port-mau-type   int32
4248     | | | | +-ro traffic-class       dot1q-types:traffic-class-type
4249     | | | | +-ro independent-delay-min? uint32
4250     | | | | +-ro independent-delay-max? uint32
4251     | | | +-ro dependent-delays* [in-port-line-speed]
4252     | | | | +-ro in-port-line-speed  uint32
4253     | | | | +-ro dependent-delay-min? uint32
4254     | | | | +-ro dependent-delay-max? uint32
4255     | | | +-: (multiple-values)
4256     | | | | +-ro independent-delays-cascaded* [in-port in-port-mau-type out-
4257     | | | | port out-port-mau-type traffic-class]
4258

```

```

4259 | | +-ro in-port uint8
4260 | | +-ro in-port-mau-type int32
4261 | | +-ro out-port uint8
4262 | | +-ro out-port-mau-type int32
4263 | | +-ro traffic-class dot1q-types:traffic-class-type
4264 | | +-ro independent-delay-min? uint32
4265 | | +-ro independent-delay-max? uint32
4266 | +-ro dependent-delays-cascaded* [in-port in-port-line-speed out-
4267 port]
4268 | +-ro in-port uint8
4269 | +-ro in-port-line-speed uint32
4270 | +-ro out-port uint8
4271 | +-ro dependent-delay-min? uint32
4272 | +-ro dependent-delay-max? uint32
4273 +-ro max-ptp-instances? uint8
4274 +-ro max-hot-standby-systems? uint8
4275 +-ro clock* [clock-identity]
4276 | +-ro clock-identity ptp-tt:clock-identity
4277 | +-ro clock-target? boolean
4278 | +-ro attached-ptp-instance-index? -> /ptp-
4279 tt:ptp/instances/instance/instance-index
4280 | +-ro arb-supported? boolean
4281 | +-ro ptp-supported? boolean
4282 | +-ro hot-standby-supported? boolean
4283 | +-ro is-synced? boolean

```

6.4.10.7.4 Module iecieee60802-sched-bridge

```

4285 module: iecieee60802-sched-bridge
4286
4287 augment /if:interfaces/if:interface/bridge:bridge-port/sched-bridge:gate-
4288 parameter-table:
4289 | +-ro min-gating-times* [speed]
4290 | +-ro speed uint32
4291 | +-ro min-cycle-time
4292 | | +-u ieee802:rational-grouping
4293 | +-ro min-interval-time? uint32

```

6.4.10.7.5 Module iecieee60802-cnc-config

```

4295 module: iecieee60802-cnc-config
4296
4297 augment /dot1q-cnc-config:cnc-config:
4298 | +-ro max-config-domains? uint8
4299 | +-ro max-cucs? uint8
4300 | +-ro max-ia-stations? uint16
4301 | +-ro max-network-diameter? uint8
4302 | +-ro max-streams? uint16
4303 | +-ro max-num-seamless-trees? uint8
4304 | +-ro hot-standby-supported? uint8
4305 +-x add_streams
4306 | +-w input
4307 | | +-w cuc-id? string
4308 | | +-w stream-list* [stream-id]
4309 | | +-w stream-id tsn-types:stream-id-type
4310 | | +-w talker
4311 | | | +-w stream-rank
4312 | | | | +-w rank? uint8
4313 | | | +-w end-station-interfaces* [mac-address interface-name]
4314 | | | | +-w mac-address string
4315 | | | | +-w interface-name string
4316 | | | +-w data-frame-specification* [index]
4317 | | | | +-w index uint8
4318 | | | +-w (field)?

```

```

4319         |         |         |         +-:(ieee802-mac-addresses)
4320         |         |         |         |   +--w ieee802-mac-addresses
4321         |         |         |         |       +--w destination-mac-address?   string
4322         |         |         |         |       +--w source-mac-address?       string
4323         |         |         |         +-:(ieee802-vlan-tag)
4324         |         |         |         |   +--w ieee802-vlan-tag
4325         |         |         |         |       +--w priority-code-point?   uint8
4326         |         |         |         |       +--w vlan-id?               uint16
4327         |         |         |         +-:(ipv4-tuple)
4328         |         |         |         |   +--w ipv4-tuple
4329         |         |         |         |       +--w source-ip-address?       inet:ipv4-
4330 address    |         |         |         |
4331         |         |         |         |       +--w destination-ip-address?   inet:ipv4-
4332 address    |         |         |         |
4333         |         |         |         |       +--w dscp?                   uint8
4334         |         |         |         |       +--w protocol?               uint16
4335         |         |         |         |       +--w source-port?             uint16
4336         |         |         |         |       +--w destination-port?        uint16
4337         |         |         |         +-:(ipv6-tuple)
4338         |         |         |         |   +--w ipv6-tuple
4339         |         |         |         |       +--w source-ip-address?       inet:ipv6-
4340 address    |         |         |         |
4341         |         |         |         |       +--w destination-ip-address?   inet:ipv6-
4342 address    |         |         |         |
4343         |         |         |         |       +--w dscp?                   uint8
4344         |         |         |         |       +--w protocol?               uint16
4345         |         |         |         |       +--w source-port?             uint16
4346         |         |         |         |       +--w destination-port?        uint16
4347         |         |         |         +--w traffic-specification
4348         |         |         |         |   +--w interval
4349         |         |         |         | |   +--w numerator?   uint32
4350         |         |         |         | |   +--w denominator? uint32
4351         |         |         |         | |   +--w max-frames-per-interval?   uint16
4352         |         |         |         | |   +--w max-frame-size?             uint16
4353         |         |         |         | |   +--w transmission-selection?     uint8
4354         |         |         |         | |   +--w time-aware!
4355         |         |         |         | |   +--w earliest-transmit-offset?   uint32
4356         |         |         |         | |   +--w latest-transmit-offset?     uint32
4357         |         |         |         | |   +--w jitter?                     uint32
4358         |         |         |         +--w user-to-network-requirements
4359         |         |         |         |   +--w num-seamless-trees?   uint8
4360         |         |         |         |   +--w max-latency?         uint32
4361         |         |         |         +--w interface-capabilities
4362         |         |         |         |   +--w vlan-tag-capable?     boolean
4363         |         |         |         |   +--w cb-stream-iden-type-list* uint32
4364         |         |         |         |   +--w cb-sequence-type-list*  uint32
4365         |         |         |         +--w listener* [index]
4366         |         |         |         |   +--w index                 uint32
4367         |         |         |         |   +--w end-station-interfaces* [mac-address interface-name]
4368         |         |         |         | |   +--w mac-address         string
4369         |         |         |         | |   +--w interface-name      string
4370         |         |         |         +--w user-to-network-requirements
4371         |         |         |         |   +--w num-seamless-trees?   uint8
4372         |         |         |         |   +--w max-latency?         uint32
4373         |         |         |         +--w interface-capabilities
4374         |         |         |         |   +--w vlan-tag-capable?     boolean
4375         |         |         |         |   +--w cb-stream-iden-type-list* uint32
4376         |         |         |         |   +--w cb-sequence-type-list*  uint32
4377         +-rw output
4378         +-rw result?   boolean
4379 augment /dot1q-cnc-config:cnc-config/dot1q-cnc-config:domain/dot1q-cnc-
4380 config:cuc/dot1q-cnc-config:stream:
4381         +--x remove_listener

```

```

4382         +--w input
4383         |   +--w listener* [index]
4384         |       +--w index      uint32
4385         +--rw output
4386         +--rw result?   boolean

```

4387 **6.4.10.7.6 Module iecieeee60802-ia-station**

```

4388 module: iecieeee60802-ia-station
4389   +--ro ia-station-capabilities
4390     +--ro capability-lldp?          boolean
4391     +--ro capability-timesync?      boolean
4392     +--ro capability-keystore?      boolean
4393     +--ro capability-truststore?    boolean
4394     +--ro capability-nacm?          boolean
4395     +--ro capability-yang-library?  boolean
4396     +--ro capability-yang-push?     boolean
4397     +--ro capability-yang-notifications? boolean
4398     +--ro capability-netconf-monitoring? boolean
4399     +--ro capability-netconf-client? boolean
4400     +--ro capability-cnc-config?    boolean
4401     +--ro capability-sched-traffic? boolean
4402     +--ro capability-frame-preemption? boolean
4403
4404   rpcs:
4405     +--x ia-factory-reset

```

4406 **6.4.10.7.7 Module iecieeee60802-subscribed-notifications**

```

4407 module: iecieeee60802-subscribed-notifications
4408
4409   augment /sn:subscriptions:
4410     +--ro max-subscriptions?          uint16
4411     +--ro max-on-change-subscription-leaves? uint16
4412     +--ro max-periodic-subscription-leaves? uint16
4413     +--ro min-periodic-subscription-interval? uint16

```

4414 **6.4.10.7.8 Module iecieeee-tt-mapping**

```

4415 module: iecieeee60802-tt-mapping
4416   +--rw tt-mapping
4417     +--rw network* [network-name]
4418       +--rw network-name      string
4419       +--rw traffic-type* [traffic-type-name]
4420         +--rw traffic-type-name string
4421         +--rw cyclic?          boolean
4422         +--rw data-delivery-req? enumeration
4423         +--rw time-triggered?   boolean
4424         +--rw stream-based?     boolean
4425         +--rw vlan-id?          dot1q-types:vlanid
4426         +--rw redundancy* [vlan-id-redundant]
4427         |   +--rw vlan-id-redundant dot1q-types:vlanid
4428         +--rw priority-value?    dot1q-types:priority-type

```

4429 **6.4.10.7.9 Module iecieeee60802-proxy**

```

4430 module: iecieeee60802-proxy
4431   +--rw proxy!
4432     +--ro max-num-proxy-targets?  uint16
4433     +--ro proxy-types* [proxy-type]
4434       +--ro proxy-type           string
4435       +--ro proxy-targets* [device]
4436         +--ro device             string
4437         +--ro device-type?       string
4438         +--ro proxy-addr
4439         |   +--ro ip-addr?       inet:ip-address

```



```

4440         |  +--ro port?          inet:port-number
4441         +--ro device-addr
4442         +--ro ip-addr?         inet:ip-address
4443         +--ro mac-addr?       ieee:mac-address

```

4444 6.4.10.7.10 Module `ieciieee60802-proxy-target`

```

4445 module: iecieee60802-proxy-target
4446   +--rw proxy-target!
4447   +--ro required-proxy-type?  string

```

4448 6.4.10.8 YANG modules^{5,6}

4449 6.4.10.8.1 Module `ieciieee60802-ethernet-interface`

```

4450 module iecieee60802-ethernet-interface {
4451   yang-version 1.1;
4452   namespace
4453     "urn:ieee:std:60802:yang:ieciieee60802-ethernet-interface";
4454   prefix ia-eth-if;
4455
4456   import ieee802-ethernet-interface {
4457     prefix eth-if;
4458   }
4459   import ietf-interfaces {
4460     prefix if;
4461   }
4462
4463   organization
4464     "IEEE 802.1 Working Group and IEC subcommittee 65C:
4465     Industrial networks, of IEC technical committee 65:
4466     Industrial-process measurement, control and automation";
4467   contact
4468     "WG-URL: http://ieee802.org/1/
4469     WG-EMail: stds-802-1-1@ieee.org
4470
4471     Contact: IEEE 802.1 Working Group Chair
4472             Postal: C/O IEEE 802.1 Working Group
4473             IEEE Standards Association
4474             445 Hoes Lane
4475             Piscataway, NJ 08854
4476             USA
4477
4478     E-mail: stds-802-1-chairs@ieee.org";
4479   description
4480     "Management objects that provide information about IEC/IEEE 60802
4481     IA-Stations as specified in IEC/IEEE 60802.
4482
4483     Copyright (C) IEC/IEEE (2025).
4484     This version of this YANG module is part of IEC/IEEE 60802;
4485     see the standard itself for full legal notices.";
4486
4487   revision 2025-03-12 {
4488     description "Published as part of IEC/IEEE 60802-2025.
4489     The following reference statement identifies each referenced
4490     IEEE Standard as updated by applicable amendments.";
4491     reference
4492       "IEC/IEEE 60802 TSN profile for industrial automation:

```

⁵ Copyright release for YANG: Users of this document may freely reproduce the YANG modules contained in this document so that they can be used for their intended purpose.

⁶ An ASCII version of each YANG module defined in this document is attached to the PDF of this document and can also be obtained from the IEEE 802.Website at <https://1.ieee802.org/yang-modules/>.

```

4493     IEC/IEEE 60802-2025.
4494     IEEE Std 802.1Q Bridges and Bridged Networks:
4495     IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
4496     IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
4497     IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2025.";
4498 }
4499
4500 augment "/if:interfaces/if:interface/eth-if:ethernet" {
4501     description
4502         "Augment IEEE Std 802.3 ethernet.";
4503     list supported-mau-types {
4504         key "mau-type";
4505         config false;
4506         description
4507             "Contains a list of supported MAU parameters.";
4508         leaf mau-type {
4509             type int32;
4510             config false;
4511             description
4512                 "The value is the supported MAU Type derived from the list
4513                 position of the corresponding dot3MauType as listed in
4514                 Clause 5 of IETF RFC 4836.";
4515             reference
4516                 "Item a) in 6.4.10.3.2.1 of IEC/IEEE 60802";
4517         }
4518         leaf preemption-supported {
4519             type boolean;
4520             config false;
4521             description
4522                 "The Boolean value indicates if preemption is supported by
4523                 the MAU Type.";
4524             reference
4525                 "Item b) in 6.4.10.3.2.1 of IEC/IEEE 60802";
4526         }
4527     }
4528 }
4529 }
4530

```

4531 **6.4.10.8.2 Module iecieee60802-bridge**

```

4532 module iecieee60802-bridge {
4533     yang-version 1.1;
4534     namespace "urn:ieee:std:60802:yang:ieciieee60802-bridge";
4535     prefix ia-bridge;
4536
4537     import ieee802-types {
4538         prefix ieee802;
4539     }
4540     import ieee802-dot1q-bridge {
4541         prefix bridge;
4542     }
4543     import ietf-interfaces {
4544         prefix if;
4545     }
4546     import ieee802-dot1q-types {
4547         prefix dot1q-types;
4548     }
4549     import ieee1588-ptp-tt {
4550         prefix ptp-tt;
4551     }
4552
4553     organization
4554         "IEEE 802.1 Working Group and IEC subcommittee 65C:

```

```
4555     Industrial networks, of IEC technical committee 65:
4556     Industrial-process measurement, control and automation";
4557     contact
4558     "WG-URL: http://ieee802.org/1/
4559     WG-EMail: stds-802-1-1@ieee.org
4560
4561     Contact: IEEE 802.1 Working Group Chair
4562             Postal: C/O IEEE 802.1 Working Group
4563             IEEE Standards Association
4564             445 Hoes Lane
4565             Piscataway, NJ 08854
4566             USA
4567
4568     E-mail: stds-802-1-chairs@ieee.org";
4569     description
4570     "Management objects that provide information about
4571     IEC/IEEE 60802 IA-Stations as specified in IEC/IEEE 60802.
4572
4573     Copyright (C) IEC/IEEE (2025).
4574     This version of this YANG module is part of IEC/IEEE 60802;
4575     see the standard itself for full legal notices.";
4576
4577     revision 2025-03-12 {
4578         description
4579         "Published as part of IEC/IEEE 60802-2025.
4580         The following reference statement identifies each referenced
4581         IEEE Standard as updated by applicable amendments.";
4582         reference
4583         "IEC/IEEE 60802 TSN profile for industrial automation:
4584         IEC/IEEE 60802-2025.
4585         IEEE Std 802.1Q Bridges and Bridged Networks:
4586         IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
4587         IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
4588         IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2025.";
4589     }
4590
4591     augment "/if:interfaces/if:interface/bridge:bridge-port" {
4592         description
4593         "Augment IEEE Std 802.1 bridge.";
4594         list max-burst-params {
4595             key "speed";
4596             config false;
4597             description
4598             "The list of maximum burst parameters per supported line
4599             speed.";
4600             leaf speed {
4601                 type uint32;
4602                 description
4603                 "This value is the line speed in Mbps.";
4604             }
4605             leaf worst-case-packet-gap {
4606                 type uint8;
4607                 config false;
4608                 description
4609                 "The value is the worst case maximum inter-packet gap
4610                 between consecutive frames in a traffic burst expressed
4611                 in bit-times.";
4612                 reference
4613                 "Item a) in 6.4.10.3.3.1 of IEC/IEEE 60802";
4614             }
4615             leaf max-burst-frames {
4616                 type uint8;
4617                 config false;
```

```

4618         description
4619             "The value is the maximum number of frames that can be sent
4620             with minimal inter packet gap.";
4621         reference
4622             "Item b) in 6.4.10.3.3.1 of IEC/IEEE 60802";
4623     }
4624     leaf max-burst-bytes {
4625         type uint8;
4626         config false;
4627         description
4628             "The value is the maximum number of octets that can be sent
4629             with minimal inter packet gap.";
4630         reference
4631             "Item c) in 6.4.10.3.3.1 of IEC/IEEE 60802";
4632     }
4633 }
4634 list committed-data-rates {
4635     key "traffic-class speed";
4636     config false;
4637     description
4638         "The list of committed data rates per traffic class and
4639         supported line speed.";
4640     leaf traffic-class {
4641         type dot1q-types:traffic-class-type;
4642         description
4643             "The traffic class of the entry (0..7).";
4644         reference
4645             "8.6.6 of IEEE Std 802.1Q";
4646     }
4647     leaf speed {
4648         type uint32;
4649         description
4650             "This value is the line speed in Mbps.";
4651     }
4652     leaf committed-information-rate {
4653         type uint32;
4654         config false;
4655         description
4656             "The value is the bandwidth limit in kbit/s.";
4657         reference
4658             "Item a) in 6.4.10.3.3.2 of IEC/IEEE 60802";
4659     }
4660     leaf committed-burst-size {
4661         type uint32;
4662         config false;
4663         description
4664             "The value is the burst size limit in bytes.";
4665         reference
4666             "Item b) in 6.4.10.3.3.2 of IEC/IEEE 60802";
4667     }
4668 }
4669 list transmission-selection-algorithm {
4670     key "traffic-class";
4671     config false;
4672     description
4673         "The list of supported transmission selection algorithms
4674         according to 8.6.8 of IEEE Std 802.1Q per traffic class.";
4675     leaf traffic-class {
4676         type dot1q-types:traffic-class-type;
4677         config false;
4678         description
4679             "Traffic class. (0..7)";
4680         reference

```

```
4681         "8.6.6 of IEEE Std 802.1Q";
4682     }
4683     list algorithms {
4684         key "algorithm";
4685         config false;
4686         description
4687             "The list of supported transmission selection algorithms
4688             according to 8.6.8 of IEEE Std 802.1Q for this traffic
4689             class.";
4690         leaf algorithm {
4691             type identityref {
4692                 base dot1q-types:transmission-selection-algorithm;
4693             }
4694             config false;
4695             description
4696                 "Transmission selection algorithm";
4697             reference
4698                 "8.6.8 of IEEE Std 802.1Q";
4699         }
4700     }
4701 }
4702 list supported-resource-pools {
4703     key "resource-pool-name";
4704     config false;
4705     description
4706         "The list of supported buffer resource pools.";
4707     leaf resource-pool-name {
4708         type string;
4709         config false;
4710         description
4711             "The value is the name of a resource pool.";
4712         reference
4713             "Item a) in 6.4.10.3.4.2 of IEC/IEEE 60802";
4714     }
4715     container covered-time-interval {
4716         config false;
4717         uses ieee802:rational-grouping;
4718         description
4719             "The value is the covered buffering time given as rational
4720             number of seconds for the highest supported link speed.";
4721         reference
4722             "Item b) in 6.4.10.3.4.2 of IEC/IEEE 60802";
4723     }
4724     list resource-pool-traffic-classes {
4725         key "traffic-class";
4726         config false;
4727         description
4728             "The list of the traffic classes to be served by the
4729             resource pool.";
4730         reference
4731             "Item c) in 6.4.10.3.4.2 of IEC/IEEE 60802";
4732         leaf traffic-class {
4733             type dot1q-types:traffic-class-type;
4734             description
4735                 "The traffic class of the entry.";
4736             reference
4737                 "8.6.6 of IEEE Std 802.1Q";
4738         }
4739     }
4740 }
4741 list phy-latencies {
4742     key "mau-type";
4743     config false;
```

```

4744     description
4745         "Latency information for each MAU type";
4746     leaf mau-type {
4747         type int32;
4748         config false;
4749         description
4750             "The MAU type";
4751     }
4752     leaf latency {
4753         type uint32;
4754         config false;
4755         description
4756             "The PHY latency in ns";
4757         reference
4758             "6.4.10.3.6.3 of IEC/IEEE 60802";
4759     }
4760     leaf latency-variation {
4761         type uint16;
4762         config false;
4763         description
4764             "The PHY latency variation in ns";
4765         reference
4766             "6.4.10.3.6.4 of IEC/IEEE 60802";
4767     }
4768 }
4769 }
4770
4771 augment "/bridge:bridges/bridge:bridge/bridge:component" {
4772     description
4773         "Augment IEEE Std 802.1 bridge component.";
4774     leaf frer-supported {
4775         type boolean;
4776         config false;
4777         description
4778             "The Boolean value indicates if FRER is supported.";
4779         reference
4780             "6.4.10.3.5.1 of IEC/IEEE 60802";
4781     }
4782     leaf max-redundant-streams {
4783         type uint32;
4784         config false;
4785         description
4786             "The value is the maximum number of supported redundant
4787             streams.";
4788         reference
4789             "6.4.10.3.5.2 of IEC/IEEE 60802";
4790     }
4791     leaf max-fids {
4792         type uint16;
4793         config false;
4794         description
4795             "The value is the maximum number of supported FIDs.";
4796         reference
4797             "6.4.10.3.7.1 of IEC/IEEE 60802";
4798     }
4799     list max-fdb-entries {
4800         key "fid";
4801         config false;
4802         description
4803             "The list of the maximum number of static and dynamic
4804             FDB entries per FID.";
4805         reference
4806             "6.4.10.3.7.2 of IEC/IEEE 60802";

```

```
4807     leaf fid {
4808         type uint16;
4809         config false;
4810         description
4811             "The FID number";
4812     }
4813     leaf max-static-fdb-entries {
4814         type uint16;
4815         config false;
4816         description
4817             "The value is the maximum number of static FDB
4818             entries.";
4819         reference
4820             "6.4.10.3.7.3 of IEC/IEEE 60802";
4821     }
4822     leaf max-dynamic-fdb-entries {
4823         type uint16;
4824         config false;
4825         description
4826             "The value is the maximum number of dynamic FDB entries.";
4827         reference
4828             "6.4.10.3.7.4 of IEC/IEEE 60802";
4829     }
4830 }
4831 choice delay-variance {
4832     config false;
4833     description
4834         "The value indicates variance in delay depending upon the use
4835         of a singleValue or multipleValues.";
4836     reference
4837         "6.4.10.3.6.1 of IEC/IEEE 60802";
4838     case single-value {
4839         list independent-delays {
4840             key "in-port-mau-type out-port-mau-type traffic-class";
4841             config false;
4842             description
4843                 "The list of minimum and maximum frame length
4844                 independent delay time values of frames as they pass
4845                 through a bridge component.";
4846             reference
4847                 "6.4.10.3.6.2 of IEC/IEEE 60802";
4848             leaf in-port-mau-type {
4849                 type int32;
4850                 config false;
4851                 description
4852                     "The MAU type of the input port";
4853             }
4854             leaf out-port-mau-type {
4855                 type int32;
4856                 config false;
4857                 description
4858                     "The MAU type of the input port";
4859             }
4860             leaf traffic-class {
4861                 type dot1q-types:traffic-class-type;
4862                 config false;
4863                 description
4864                     "The traffic class of the entry.";
4865                 reference
4866                     "8.6.6 of IEEE Std 802.1Q";
4867             }
4868             leaf independent-delay-min {
4869                 type uint32;
```

```

4870         config false;
4871         description
4872             "The value is the minimum delay portion that is
4873             independent of frame length according to 12.32.1.1.
4874             of IEEE 802.1Q";
4875         reference
4876             "Item a) in 6.4.10.3.6.2 of IEC/IEEE 60802";
4877     }
4878     leaf independent-delay-max {
4879         type uint32;
4880         config false;
4881         description
4882             "The value is the maximum delay portion that is
4883             independent of frame length according to 12.32.1.1.
4884             of IEEE 802.1Q";
4885         reference
4886             "Item b) in 6.4.10.3.6.2 of IEC/IEEE 60802";
4887     }
4888 }
4889 list dependent-delays {
4890     key "in-port-line-speed";
4891     config false;
4892     description
4893         "The list of minimum and maximum frame length dependent
4894         delay time values of frames as they pass through a
4895         bridge component";
4896     reference
4897         "6.4.10.3.6.2 of IEC/IEEE 60802";
4898     leaf in-port-line-speed {
4899         type uint32;
4900         config false;
4901         description
4902             "This value is the line speed in Mbps.";
4903     }
4904     leaf dependent-delay-min {
4905         type uint32;
4906         config false;
4907         description
4908             "The value is the minimum delay portion that is
4909             dependent on frame length according to 12.32.1.2.
4910             of IEEE 802.1Q";
4911         reference
4912             "Item c) in 6.4.10.3.6.2 of IEC/IEEE 60802";
4913     }
4914     leaf dependent-delay-max {
4915         type uint32;
4916         config false;
4917         description
4918             "The value is the maximum delay portion that is
4919             dependent on frame length according to 12.32.1.2.
4920             of IEEE 802.1Q";
4921         reference
4922             "Item d) in 6.4.10.3.6.2 of IEC/IEEE 60802";
4923     }
4924 }
4925 }
4926 case multiple-values {
4927     list independent-delays-cascaded {
4928         key "in-port in-port-mau-type out-port out-port-mau-type
4929             traffic-class";
4930         config false;
4931         description
4932             "The list of minimum and maximum frame length

```



```
4933         independent delay time values of frames as they pass
4934         through a bridge component.";
4935     reference
4936         "6.4.10.3.6.2 of IEC/IEEE 60802";
4937     leaf in-port {
4938         type uint8;
4939         config false;
4940         description
4941             "The port number of the input port";
4942     }
4943     leaf in-port-mau-type {
4944         type int32;
4945         config false;
4946         description
4947             "The MAU type of the input port";
4948     }
4949     leaf out-port {
4950         type uint8;
4951         config false;
4952         description
4953             "The port number of the output port";
4954     }
4955     leaf out-port-mau-type {
4956         type int32;
4957         config false;
4958         description
4959             "The MAU type of the input port";
4960     }
4961     leaf traffic-class {
4962         type dot1q-types:traffic-class-type;
4963         config false;
4964         description
4965             "The traffic class of the entry.";
4966         reference
4967             "8.6.6 of IEEE Std 802.1Q";
4968     }
4969     leaf independent-delay-min {
4970         type uint32;
4971         config false;
4972         description
4973             "The value is the minimum delay portion that is
4974             independent of frame length according to 12.32.1.1.
4975             of IEEE 802.1Q";
4976         reference
4977             "Item a) in 6.4.10.3.6.2 of IEC/IEEE 60802";
4978     }
4979     leaf independent-delay-max {
4980         type uint32;
4981         config false;
4982         description
4983             "The value is the maximum delay portion that is
4984             independent of frame length according to 12.32.1.1.
4985             of IEEE 802.1Q";
4986         reference
4987             "Item b) in 6.4.10.3.6.2 of IEC/IEEE 60802";
4988     }
4989 }
4990 list dependent-delays-cascaded {
4991     key "in-port in-port-line-speed out-port";
4992     config false;
4993     description
4994         "The list of minimum and maximum frame length dependent
4995         delay time values of frames as they pass through a
```

```

4996         bridge component";
4997     reference
4998         "6.4.10.3.6.2 of IEC/IEEE 60802";
4999     leaf in-port {
5000         type uint8;
5001         config false;
5002         description
5003             "The port number of the input port";
5004     }
5005     leaf in-port-line-speed {
5006         type uint32;
5007         config false;
5008         description
5009             "This value is the line speed in Mbps.";
5010     }
5011     leaf out-port {
5012         type uint8;
5013         config false;
5014         description
5015             "The port number of the output port";
5016     }
5017     leaf dependent-delay-min {
5018         type uint32;
5019         config false;
5020         description
5021             "The value is the minimum delay portion that is
5022             dependent on frame length according to 12.32.1.2.
5023             of IEEE 802.1Q";
5024         reference
5025             "Item c) in 6.4.10.3.6.2 of IEC/IEEE 60802";
5026     }
5027     leaf dependent-delay-max {
5028         type uint32;
5029         config false;
5030         description
5031             "The value is the maximum delay portion that is
5032             dependent on frame length according to 12.32.1.2.
5033             of IEEE 802.1Q";
5034         reference
5035             "Item d) in 6.4.10.3.6.2 of IEC/IEEE 60802";
5036     }
5037 }
5038 }
5039 }
5040 leaf max-ptp-instances {
5041     type uint8;
5042     config false;
5043     description
5044         "The value is the maximum number of supported PTP
5045         Instances.";
5046     reference
5047         "6.4.10.3.7.5 of IEC/IEEE 60802";
5048 }
5049 leaf max-hot-standby-systems {
5050     type uint8;
5051     config false;
5052     description
5053         "The value is the maximum number of supported
5054         HotStandbySystem entities";
5055     reference
5056         "6.4.10.3.7.6 of IEC/IEEE 60802";
5057 }
5058 list clock {

```

```
5059     key "clock-identity";
5060     config false;
5061     description
5062         "The list of supported application clock entities.";
5063     reference
5064         "6.4.10.3.7.7 of IEC/IEEE 60802";
5065     leaf clock-identity {
5066         type ptp-tt:clock-identity;
5067         config false;
5068         description
5069             "The clock identity of the application clock.";
5070         reference
5071             "Item a) in 6.4.10.3.7.7 of IEC/IEEE 60802";
5072     }
5073     leaf clock-target {
5074         type boolean;
5075         config false;
5076         description
5077             "The Boolean value indicates if the application clock is a
5078             clock target (TRUE) or clock source (FALSE).";
5079         reference
5080             "Item b) in 6.4.10.3.7.7 of IEC/IEEE 60802";
5081     }
5082     leaf attached-ptp-instance-index {
5083         type leafref {
5084             path "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-
5085 tt:instance-index";
5086         }
5087         config false;
5088         description
5089             "The value is a reference to the index of the PTP or hot
5090             standby Instance, that is attached to the application
5091             clock.";
5092         reference
5093             "Item f) in 6.4.10.3.7.7 of IEC/IEEE 60802";
5094     }
5095     leaf arb-supported {
5096         type boolean;
5097         config false;
5098         description
5099             "The Boolean value indicates if the application clock
5100             supports the ARB timescale.";
5101         reference
5102             "Item c) in 6.4.10.3.7.7 of IEC/IEEE 60802";
5103     }
5104     leaf ptp-supported {
5105         type boolean;
5106         config false;
5107         description
5108             "The Boolean value indicates if the application clock
5109             supports the PTP timescale.";
5110         reference
5111             "Item d) in 6.4.10.3.7.7 of IEC/IEEE 60802";
5112     }
5113     leaf hot-standby-supported {
5114         type boolean;
5115         config false;
5116         description
5117             "The Boolean value indicates if the application clock
5118             supports the hot standby.";
5119         reference
5120             "Item e) in 6.4.10.3.7.7 of IEC/IEEE 60802";
5121     }
```

```

5122     leaf is-synced {
5123         type boolean;
5124         config false;
5125         description
5126             "The Boolean value indicates if the application clock is
5127             either synchronized to the attached PTP Instance (TRUE)
5128             or to an internal/external ClockSource (FALSE).";
5129         reference
5130             "Item g) in 6.4.10.3.7.7 of IEC/IEEE 60802";
5131     }
5132 }
5133 }
5134 }

```

5135 **6.4.10.8.3 Module iecieee60802-sched-bridge**

```

5136 module iecieee60802-sched-bridge {
5137     yang-version 1.1;
5138     namespace "urn:ieee:std:60802:yang:ieciieee60802-sched-bridge";
5139     prefix ia-sched-bridge;
5140
5141     import ieee802-types {
5142         prefix ieee802;
5143     }
5144     import ieee802-dot1q-bridge {
5145         prefix bridge;
5146     }
5147     import ieee802-dot1q-sched-bridge {
5148         prefix sched-bridge;
5149     }
5150     import ietf-interfaces {
5151         prefix if;
5152     }
5153
5154     organization
5155         "IEEE 802.1 Working Group and IEC subcommittee 65C:
5156         Industrial networks, of IEC technical committee 65:
5157         Industrial-process measurement, control and automation";
5158     contact
5159         "WG-URL: http://ieee802.org/1/
5160         WG-EMail: stds-802-1-1@ieee.org
5161
5162         Contact: IEEE 802.1 Working Group Chair
5163         Postal: C/O IEEE 802.1 Working Group
5164         IEEE Standards Association
5165         445 Hoes Lane
5166         Piscataway, NJ 08854
5167         USA
5168
5169         E-mail: stds-802-1-chairs@ieee.org";
5170     description
5171         "Management objects that provide information about IEC/IEEE 60802
5172         IA-Stations as specified in IEC/IEEE 60802.
5173
5174         Copyright (C) IEC/IEEE (2025).
5175         This version of this YANG module is part of IEC/IEEE 60802;
5176         see the standard itself for full legal notices.";
5177
5178     revision 2025-03-12 {
5179         description "Published as part of IEC/IEEE 60802-2025.
5180         The following reference statement identifies each referenced
5181         IEEE Standard as updated by applicable amendments.";
5182         reference
5183             "IEC/IEEE 60802 TSN profile for industrial automation:

```

```

5184     IEC/IEEE 60802-2025.
5185     IEEE Std 802.1Q Bridges and Bridged Networks:
5186     IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
5187     IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
5188     IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2025.";
5189 }
5190
5191 augment "/if:interfaces/if:interface/bridge:bridge-port/sched-
5192 bridge:gate-parameter-table" {
5193     description
5194         "Augment IEEE Std 802.1 bridge/gate-parameter-table.";
5195     list min-gating-times {
5196         key "speed";
5197         config false;
5198         description
5199             "The list of minimum gating times per supported line speed.";
5200         reference
5201             "6.4.10.3.4.3 of IEC/IEEE 60802";
5202         leaf speed {
5203             type uint32;
5204             config false;
5205             description
5206                 "This value is the line speed in Mbps.";
5207         }
5208         container min-cycle-time {
5209             uses ieee802:rational-grouping;
5210             description
5211                 "The value is the minimum value supported by this port of
5212                 the AdminCycleTime and OperCycleTime parameters given as
5213                 rational number of seconds.";
5214             reference
5215                 "Item a) in 6.4.10.3.4.3 of IEC/IEEE 60802";
5216         }
5217         leaf min-interval-time {
5218             type uint32;
5219             description
5220                 "The value is the minimum value supported by this port of
5221                 the TimeIntervalValue parameter in nanoseconds.";
5222             reference
5223                 "Item b) in 6.4.10.3.4.3 of IEC/IEEE 60802";
5224         }
5225     }
5226 }

```

6.4.10.8.4 Module iecieee60802-cnc-config

```

5228 module iecieee60802-cnc-config {
5229     yang-version 1.1;
5230     namespace "urn:ieee:std:60802:yang:ieciieee60802-cnc-config";
5231     prefix ia-cnc-config;
5232
5233     import ieee802-dot1q-cnc-config {
5234         prefix dot1q-cnc-config;
5235     }
5236     import ieee802-dot1q-tsn-types {
5237         prefix tsn-types;
5238     }
5239
5240     organization
5241         "IEEE 802.1 Working Group and IEC subcommittee 65C:
5242         Industrial networks, of IEC technical committee 65:
5243         Industrial-process measurement, control and automation";
5244     contact
5245         "WG-URL: http://ieee802.org/1/

```

```

5246     WG-EMail: stds-802-1-1@ieee.org
5247
5248     Contact: IEEE 802.1 Working Group Chair
5249             Postal: C/O IEEE 802.1 Working Group
5250             IEEE Standards Association
5251             445 Hoes Lane
5252             Piscataway, NJ 08854
5253             USA
5254
5255     E-mail: stds-802-1-chairs@ieee.org";
5256 description
5257     "Management objects that provide information about IEC/IEEE 60802
5258     IA-Stations as specified in IEC/IEEE 60802.
5259
5260     Copyright (C) IEC/IEEE (2025).
5261     This version of this YANG module is part of IEC/IEEE 60802;
5262     see the standard itself for full legal notices.";
5263
5264 revision 2025-01-10 {
5265     description "Published as part of IEC/IEEE 60802-2025.
5266     The following reference statement identifies each referenced
5267     IEEE Standard as updated by applicable amendments.";
5268     reference
5269         "IEC/IEEE 60802 TSN profile for industrial automation:
5270         IEC/IEEE 60802-2025.
5271         IEEE Std 802.1Q Bridges and Bridged Networks:
5272         IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
5273         IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
5274         IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2025.";
5275 }
5276
5277 augment "/dot1q-cnc-config:cnc-config" {
5278     description
5279         "Augment main container in cnc-config.";
5280     leaf max-config-domains {
5281         type uint8;
5282         config false;
5283         description
5284             "The value is the maximum number of supported configuration
5285             domains.";
5286         reference
5287             "6.4.10.3.9.1 of IEC/IEEE 60802";
5288     }
5289     leaf max-cucs {
5290         type uint8;
5291         config false;
5292         description
5293             "The value is the maximum number of supported CUC entities.";
5294         reference
5295             "6.4.10.3.9.2 of IEC/IEEE 60802";
5296     }
5297     leaf max-ia-stations {
5298         type uint16;
5299         config false;
5300         description
5301             "The value is the maximum number of supported IA-stations.";
5302         reference
5303             "6.4.10.3.9.3 of IEC/IEEE 60802";
5304     }
5305     leaf max-network-diameter {
5306         type uint8;
5307         config false;
5308         description

```

```

5309         "The value is the maximum supported network diameter.";
5310     reference
5311         "6.4.10.3.9.4 of IEC/IEEE 60802";
5312 }
5313 leaf max-streams {
5314     type uint16;
5315     config false;
5316     description
5317         "The value is the maximum number of supported streams.";
5318     reference
5319         "6.4.10.3.9.5 of IEC/IEEE 60802";
5320 }
5321 leaf max-num-seamless-trees {
5322     type uint8;
5323     config false;
5324     description
5325         "The value is the maximum number of trees supported for
5326         seamless redundancy of a stream.";
5327     reference
5328         "6.4.10.3.9.6 of IEC/IEEE 60802";
5329 }
5330 leaf hot-standby-supported {
5331     type uint8;
5332     config false;
5333     description
5334         "The Boolean value indicates if PTP hot standby is
5335         supported.";
5336     reference
5337         "6.4.10.3.9.7 of IEC/IEEE 60802";
5338 }
5339 action add_streams {
5340     description
5341         "This Action requests a CNC to add a list of streams.";
5342     input {
5343         leaf cuc-id {
5344             type string;
5345             description
5346                 "The CUC ID where the streams are to be added";
5347         }
5348         list stream-list {
5349             key "stream-id";
5350             description
5351                 "List of Streams that should be added.";
5352             leaf stream-id {
5353                 type tsn-types:stream-id-type;
5354                 description
5355                     "The Stream ID is a unique identifier of a Stream
5356                     request and corresponding configuration. It is used to
5357                     associate a CUC's Stream request with a CNC's
5358                     corresponding response.";
5359             }
5360             container talker {
5361                 description
5362                     "The Talker container contains: - Talker's behavior for
5363                     Stream (how/when transmitted) - Talker's requirements
5364                     from the network - TSN capabilities of the Talker's
5365                     interface(s).";
5366                 uses tsn-types:group-talker;
5367             }
5368             list listener {
5369                 key "index";
5370                 description
5371                     "Each Listener list entry contains: - Listener's

```

```

5372         requirements from the network - TSN capabilities of
5373         the Listener's interface(s).";
5374     leaf index {
5375         type uint32;
5376         description
5377             "This index is provided in order to provide a unique
5378             key per list entry.";
5379     }
5380     uses tsn-types:group-listener;
5381 }
5382 }
5383 }
5384 output {
5385     leaf result {
5386         type boolean;
5387         description
5388             "Returns status information indicating if Stream addition
5389             has been successful.";
5390     }
5391 }
5392 }
5393 }
5394
5395 augment "/dot1q-cnc-config:cnc-config/dot1q-cnc-config:domain/dot1q-cnc-
5396 config:cuc/dot1q-cnc-config:stream" {
5397     description
5398         "Augment stream list in tsc-config-uni.";
5399     action remove_listener {
5400         description
5401             "This Action removes listeners from a stream.";
5402         input {
5403             list listener {
5404                 key "index";
5405                 description
5406                     "Each Listener list entry contains: - Listener's
5407                     requirements from the network - TSN capabilities of the
5408                     Listener's interface(s).";
5409                 leaf index {
5410                     type uint32;
5411                     description
5412                         "This index is provided in order to provide a unique
5413                         key per list entry.";
5414                 }
5415             }
5416         }
5417         output {
5418             leaf result {
5419                 type boolean;
5420                 description
5421                     "Returns status information indicating if listene removal
5422                     has been successful.";
5423             }
5424         }
5425     }
5426 }
5427 }

```

6.4.10.8.5 Module iecieee60802-ia-station

```

5429 module iecieee60802-ia-station {
5430     yang-version 1.1;
5431     namespace "urn:ieee:std:60802:yang:ieciieee60802-ia-station";
5432     prefix ias;
5433 }

```



```
5434 import ietf-datastores {
5435     prefix ds;
5436     reference
5437         "IETF RFC 8342: Network Management Datastore Architecture
5438         (NMDA)";
5439 }
5440 import ietf-netconf-acm {
5441     prefix nacm;
5442     reference
5443         "IETF RFC 8341: Network Configuration Access Control Model";
5444 }
5445
5446 organization
5447     "IEEE 802.1 Working Group and IEC subcommittee 65C:
5448     Industrial networks, of IEC technical committee 65:
5449     Industrial-process measurement, control and automation";
5450 contact
5451     "WG-URL: http://ieee802.org/1/
5452     WG-EMail: stds-802-1-1@ieee.org
5453
5454     Contact: IEEE 802.1 Working Group Chair
5455             Postal: C/O IEEE 802.1 Working Group
5456             IEEE Standards Association
5457             445 Hoes Lane
5458             Piscataway, NJ 08854
5459             USA
5460
5461     E-mail: stds-802-1-chairs@ieee.org";
5462 description
5463     "Capability information and reset to factory defaults
5464     functionality for IEC/IEEE 60802 IA-Stations as specified in
5465     IEC/IEEE 60802.
5466
5467     Copyright (C) IEC/IEEE (2025).
5468     This version of this YANG module is part of IEC/IEEE 60802;
5469     see the standard itself for full legal notices.";
5470
5471 revision 2025-03-12 {
5472     description "Published as part of IEC/IEEE 60802-2025.
5473     The following reference statement identifies each referenced
5474     IEEE Standard as updated by applicable amendments.";
5475     reference
5476         "IEC/IEEE 60802 TSN profile for industrial automation:
5477         IEC/IEEE 60802-2025.
5478         IEEE Std 802.1Q Bridges and Bridged Networks:
5479         IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
5480         IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
5481         IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2025.";
5482 }
5483
5484 feature ia-factory-default-datastore {
5485     description
5486         "Indicates that the factory default configuration is
5487         available as a datastore.";
5488 }
5489
5490 identity ia-factory-default {
5491     if-feature "ia-factory-default-datastore";
5492     base ds:datastore;
5493     description
5494         "This read-only datastore contains the factory default
5495         configuration for the device that will be used to replace
5496         the contents of the read-write conventional configuration
```

```

5497         datastores during a 'ia-factory-reset' RPC operation.";
5498     }
5499
5500     container ia-station-capabilities {
5501         config false;
5502         description
5503             "This container provides read only information about an
5504             ia-station's capabilities.";
5505         reference
5506             "IEC/IEEE 60802 - YANG Data Model";
5507         leaf capability-lldp {
5508             type boolean;
5509             config false;
5510             description
5511                 "The value is true if the device supports LLDP.";
5512             reference
5513                 "6.4.10.3.8.5 of IEC/IEEE 60802";
5514         }
5515         leaf capability-timesync {
5516             type boolean;
5517             config false;
5518             description
5519                 "The value is true if the device supports Timesync.";
5520             reference
5521                 "6.4.10.3.8.6 of IEC/IEEE 60802";
5522         }
5523         leaf capability-keystore {
5524             type boolean;
5525             config false;
5526             description
5527                 "The value is true if the device supports Keystore.";
5528             reference
5529                 "6.4.10.3.8.7 of IEC/IEEE 60802";
5530         }
5531         leaf capability-truststore {
5532             type boolean;
5533             config false;
5534             description
5535                 "The value is true if the device supports Truststore.";
5536             reference
5537                 "6.4.10.3.8.9 of IEC/IEEE 60802";
5538         }
5539         leaf capability-nacm {
5540             type boolean;
5541             config false;
5542             description
5543                 "The value is true if the device supports NACM.";
5544             reference
5545                 "6.4.10.3.8.8 of IEC/IEEE 60802";
5546         }
5547         leaf capability-yang-library {
5548             type boolean;
5549             config false;
5550             description
5551                 "The value is true if the device supports YANG library.";
5552             reference
5553                 "6.4.10.3.8.10 of IEC/IEEE 60802";
5554         }
5555         leaf capability-yang-push {
5556             type boolean;
5557             config false;
5558             description
5559                 "The value is true if the device supports YANG push.";

```

```
5560         reference
5561             "6.4.10.3.8.11 of IEC/IEEE 60802";
5562     }
5563     leaf capability-yang-notifications {
5564         type boolean;
5565         config false;
5566         description
5567             "The value is true if the device supports YANG
5568             notifications.";
5569         reference
5570             "6.4.10.3.8.12 of IEC/IEEE 60802";
5571     }
5572     leaf capability-netconf-monitoring {
5573         type boolean;
5574         config false;
5575         description
5576             "The value is true if the device supports NETCONF
5577             monitoring.";
5578         reference
5579             "6.4.10.3.8.13 of IEC/IEEE 60802";
5580     }
5581     leaf capability-netconf-client {
5582         type boolean;
5583         config false;
5584         description
5585             "The value is true if the device supports NETCONF client.";
5586         reference
5587             "6.4.10.3.8.14 of IEC/IEEE 60802";
5588     }
5589     leaf capability-cnc-config {
5590         type boolean;
5591         config false;
5592         description
5593             "The value is true if the device supports the CNC configuration.";
5594         reference
5595             "6.4.10.3.8.15 of IEC/IEEE 60802";
5596     }
5597     leaf capability-sched-traffic {
5598         type boolean;
5599         config false;
5600         description
5601             "The value is true if the device supports scheduled
5602             traffic.";
5603         reference
5604             "6.4.10.3.8.16 of IEC/IEEE 60802";
5605     }
5606     leaf capability-frame-preemption {
5607         type boolean;
5608         config false;
5609         description
5610             "The value is true if the device supports frame preemption.";
5611         reference
5612             "6.4.10.3.8.17 of IEC/IEEE 60802";
5613     }
5614 }
5615
5616 rpc ia-factory-reset {
5617     nacm:default-deny-all;
5618     description
5619         "The server resets all datastores to their factory
5620         default contents and any nonvolatile storage back to
5621         factory condition, deleting all dynamically
5622         generated files, including those containing keys,
```

certificates, logs, and other temporary files.

Depending on the factory default configuration, after being reset, the device may become unreachable on the network.

In contrast to the original factory-reset RPC in IETF RFC 8808, this RPC puts the device into a state where a subsequent configuration by a CNC component results in a functioning 60802 IA-station";

```
}
}
```

6.4.10.8.6 Module `iecieeee60802-subscribed-notifications`

```
module iecieeee60802-subscribed-notifications {
  yang-version 1.1;
  namespace
    "urn:ieee:std:60802:yang:iecieeee60802-subscribed-notifications";
  prefix ia-sn;

  import ietf-subscribed-notifications {
    prefix sn;
  }

  organization
    "IEEE 802.1 Working Group and IEC subcommittee 65C:
    Industrial networks, of IEC technical committee 65:
    Industrial-process measurement, control and automation";
  contact
    "WG-URL: http://ieee802.org/1/
    WG-EMail: stds-802-1-1@ieee.org

    Contact: IEEE 802.1 Working Group Chair
    Postal: C/O IEEE 802.1 Working Group
    IEEE Standards Association
    445 Hoes Lane
    Piscataway, NJ 08854
    USA

    E-mail: stds-802-1-chairs@ieee.org";
  description
    "Management objects that provide information about IEC/IEEE 60802
    IA-Stations as specified in IEC/IEEE 60802.

    Copyright (C) IEC/IEEE (2025).
    This version of this YANG module is part of IEC/IEEE 60802;
    see the standard itself for full legal notices.";
  revision 2024-02-19 {
    description "Published as part of IEC/IEEE 60802-2025.
    The following reference statement identifies each referenced
    IEEE Standard as updated by applicable amendments.";
    reference
      "IEC/IEEE 60802 TSN profile for industrial automation:
      IEC/IEEE 60802-2025.
      IEEE Std 802.1Q Bridges and Bridged Networks:
      IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
      IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
      IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2025.";
  }

  augment "/sn:subscriptions" {
```

```
5684     description
5685         "Augment subscriptions in ietf-subscribed-notifications.";
5686     leaf max-subscriptions {
5687         type uint16;
5688         config false;
5689         description
5690             "The value is the maximum number of supported NETCONF Server
5691             subscriptions.";
5692         reference
5693             "6.4.10.3.8.1 of IEC/IEEE 60802";
5694     }
5695     leaf max-on-change-subscription-leaves {
5696         type uint16;
5697         config false;
5698         description
5699             "The value is the maximum number of supported leaves for
5700             NETCONF Server on-change subscriptions according to IETF
5701             RFC 8641.";
5702         reference
5703             "6.4.10.3.8.2 of IEC/IEEE 60802";
5704     }
5705     leaf max-periodic-subscription-leaves {
5706         type uint16;
5707         config false;
5708         description
5709             "The value is the maximum number of supported leaves for
5710             NETCONF Server periodic subscriptions according to IETF
5711             RFC 8641.";
5712         reference
5713             "6.4.10.3.8.3 of IEC/IEEE 60802";
5714     }
5715     leaf min-periodic-subscription-interval {
5716         type uint16;
5717         config false;
5718         description
5719             "The value is the minimum periodic subscription interval in
5720             centiseconds (0.01 seconds) for NETCONF Server periodic
5721             subscriptions according to IETF RFC 8641.";
5722         reference
5723             "6.4.10.3.8.4 of IEC/IEEE 60802";
5724     }
5725 }
5726 }
```

5727 **6.4.10.8.7 Module iecieee60802-tt-mapping**

```
5728 module iecieee60802-tt-mapping {
5729     yang-version "1.1";
5730     namespace "urn:ieee:std:60802:yang:iecieee60802-tt-mapping";
5731     prefix tt-mapping;
5732
5733     import ieee802-dot1q-types {
5734         prefix dot1q-types;
5735     }
5736
5737     organization
5738         "IEEE 802.1 Working Group and IEC subcommittee 65C:
5739         Industrial networks, of IEC technical committee 65:
5740         Industrial-process measurement, control and automation";
5741     contact
5742         "WG-URL: http://ieee802.org/1/
5743         WG-EMail: stds-802-1-1@ieee.org
5744
```

```

5745     Contact: IEEE 802.1 Working Group Chair
5746     Postal: C/O IEEE 802.1 Working Group
5747             IEEE Standards Association
5748             445 Hoes Lane
5749             Piscataway, NJ 08854
5750             USA
5751
5752     E-mail: stds-802-1-chairs@ieee.org";
5753     description
5754     "This module provides information on characteristics of traffic classes
5755     configured in a network to allow applications to map application
5756 traffic
5757     classes to them.
5758
5759     Copyright (C) IEC/IEEE (2025).
5760
5761     This version of this YANG module is part of IEC/IEEE 60802;
5762     see the standard itself for full legal notices.";
5763     revision 2024-12-13 {
5764         description
5765         "Published as part of IEC/IEEE 60802-2025.
5766
5767         The following reference statement identifies each referenced
5768         IEEE Standard as updated by applicable amendments.";
5769         reference
5770         "IEC/IEEE 60802 TSN profile for industrial automation:
5771         IEC/IEEE 60802-2025.
5772         IEEE Std 802.1Q Bridges and Bridged Networks:
5773         IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
5774         IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
5775         IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2025.";
5776     }
5777
5778     container tt-mapping {
5779         description
5780         "Top-level container for the traffic class module.";
5781         list network {
5782             key "network-name";
5783             description
5784             "A list of networks, each containing one or more sets of traffic
5785 class
5786             characteristics.";
5787             leaf network-name {
5788                 type string;
5789                 description
5790                 "The network-name uniquely identifies a network.";
5791             }
5792             list traffic-type {
5793                 key "traffic-type-name";
5794                 description
5795                 "A list of traffic types in the network, each with its associated
5796                 traffic type characteristics.";
5797                 leaf traffic-type-name {
5798                     type string;
5799                     description
5800                     "The traffic-type-name uniquely identifies the traffic type.";
5801                 }
5802                 leaf cyclic {
5803                     type boolean;
5804                     description
5805                     "Indicates if cyclic transmission is mandatory (TRUE) or
5806 optional (FALSE)
5807                     for this traffic type.";

```

```
5808         reference
5809         "4.7.2 of IEC/IEEE 60802-2025";
5810     }
5811     leaf data-delivery-req {
5812         type enumeration {
5813             enum frame-latency {
5814                 value 0;
5815                 description
5816                 "Data delivery of a frame for a given Talker-Listener pair
5817                  occurs within a bounded timespan.";
5818             }
5819             enum flow-latency {
5820                 value 1;
5821                 description
5822                 "Data delivery up to a certain number of frames or data
5823 size          (including bursts of frames) occurring over a defined
5824                 period.";
5825             }
5826             enum deadline {
5827                 value 2;
5828                 description
5829                 "Data delivery of a frame to a given Listener occurs at or
5830                 before a specific point in time.";
5831             }
5832             enum no {
5833                 value 3;
5834                 description
5835                 "No special data delivery requirements.";
5836             }
5837         }
5838     }
5839     description
5840     "The data-delivery-req provides information on the data
5841     delivery requirements for this traffic type.";
5842     reference
5843     "4.7.2 of IEC/IEEE 60802-2025";
5844 }
5845 leaf time-triggered {
5846     type boolean;
5847     description
5848     "Indicates if it is required (TRUE) or optional (FALSE) that
5849     transmission occurs at a specific point in time based upon
5850     the Working Clock.";
5851     reference
5852     "4.7.2 of IEC/IEEE 60802-2025";
5853 }
5854 leaf stream-based {
5855     type boolean;
5856     default true;
5857     description
5858     "Indicates if traffic of this traffic type requires
5859     registering a stream, i.e., allocating dedicated
5860     resources, for the traffic, before a given Talker
5861     can start transmitting data (TRUE), or if registering
5862     a stream is not required (FALSE).";
5863     reference
5864     "4.7.2 of IEC/IEEE 60802-2025";
5865 }
5866 leaf vlan-id {
5867     type dot1q-types:vlanid;
5868     description
5869     "The VID this traffic type is mapped to.";
5870 }
```

```

5871     list redundancy {
5872         key "vlan-id-redundant";
5873         description
5874             "A list of VIDs this traffic type is mapped to for redundant
5875             transmission. There is one entry in the list per redundant
5876             transmission ";
5877         leaf vlan-id-redundant {
5878             type dot1q-types:vlanid;
5879             description
5880                 "The VID this traffic type is mapped for a redundant
5881 transmission.";
5882         }
5883     }
5884     leaf priority-value {
5885         type dot1q-types:priority-type;
5886         description
5887             "The PCP this traffic type is mapped to.";
5888     }
5889 }
5890 }
5891 }
5892 }

```

5893 **6.4.10.8.8 Module iecieee60802-proxy**

```

5894 module iecieee60802-proxy {
5895     yang-version 1.1;
5896     namespace "urn:ieee:std:60802:yang:ieciieee60802-proxy";
5897     prefix pr;
5898
5899     import ieee802-types {
5900         prefix ieee;
5901     }
5902     import ietf-inet-types {
5903         prefix inet;
5904     }
5905
5906     organization
5907         "IEEE 802.1 Working Group and IEC subcommittee 65C:
5908         Industrial networks, of IEC technical committee 65:
5909         Industrial-process measurement, control and automation";
5910     contact
5911         "WG-URL: http://ieee802.org/1/
5912         WG-EMail: stds-802-1-1@ieee.org
5913
5914         Contact: IEEE 802.1 Working Group Chair
5915                 Postal: C/O IEEE 802.1 Working Group
5916                 IEEE Standards Association
5917                 445 Hoes Lane
5918                 Piscataway, NJ 08854
5919                 USA
5920
5921         E-mail: stds-802-1-chairs@ieee.org";
5922     description
5923         "Configuration and capability information for IEC/IEEE 60802 proxy
5924         targets as specified in IEC/IEEE 60802.
5925
5926         Copyright (C) IEC/IEEE (2025).
5927         This version of this YANG module is part of IEC/IEEE 60802;
5928         see the standard itself for full legal notices.";
5929
5930     revision 2025-03-12 {
5931         description
5932             "Published as part of IEC/IEEE 60802-2025.

```



```
5933     The following reference statement identifies each referenced
5934     IEEE Standard as updated by applicable amendments.";
5935 reference
5936     "IEC/IEEE 60802 TSN profile for industrial automation:
5937     IEC/IEEE 60802-2025.
5938     IEEE Std 802.1Q Bridges and Bridged Networks:
5939     IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
5940     IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
5941     IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2025.";
5942 }
5943
5944 container proxy {
5945     presence "The presence of this container indicates that this IA station
5946     provides proxy functionality for proxy targets";
5947     description
5948         "This container provides information about a
5949         60802 IA station with proxy functionality.";
5950     reference
5951         "IEC/IEEE 60802 - YANG Data Model";
5952     leaf max-num-proxy-targets {
5953         type uint16;
5954         config false;
5955         description
5956             "The value indicates how many proxy targets can be handled by this
5957 proxy component.";
5958         reference
5959             "6.4.10.3.9.1 of IEC/IEEE 60802";
5960     }
5961     list proxy-types {
5962         key "proxy-type";
5963         config false;
5964         description
5965             "Supported proxy types";
5966         leaf proxy-type {
5967             type string {
5968                 pattern '[0-9A-F]{2}(-[0-9A-F]{2}){2}';
5969             }
5970             config false;
5971             description
5972                 "OID of proxy targets handled by this proxy.";
5973             reference
5974                 "6.4.10.3.9.2 of IEC/IEEE 60802";
5975         }
5976         list proxy-targets {
5977             key "device";
5978             description
5979                 "List of proxy targets that are connected to this proxy";
5980             leaf device {
5981                 type string;
5982                 description
5983                     "The identification of the device.";
5984                 reference
5985                     "6.4.10.3.9.3 of IEC/IEEE 60802";
5986             }
5987             leaf device-type {
5988                 type string;
5989                 description
5990                     "The type of the device.";
5991                 reference
5992                     "6.4.10.3.9.4 of IEC/IEEE 60802";
5993             }
5994         }
5995     }
5996     container proxy-addr {
5997         description
```

```

5996         "This container provides the address information
5997         that is used by the CNC to configure the proxy target.";
5998     leaf ip-addr {
5999         type inet:ip-address;
6000         description
6001             "The ip-address that is used by the CNC to configure
6002             the proxy target.";
6003         reference
6004             "6.4.10.3.9.5 of IEC/IEEE 60802";
6005     }
6006     leaf port {
6007         type inet:port-number;
6008         description
6009             "The port-number that is used by the CNC to configure
6010             the proxy target.";
6011         reference
6012             "6.4.10.3.9.6 of IEC/IEEE 60802";
6013     }
6014 }
6015 container device-addr {
6016     description
6017         "This container provides the address information
6018         that is used by the proxy to address the proxy target.";
6019     leaf ip-addr {
6020         type inet:ip-address;
6021         description
6022             "The ip-address that is used by the proxy to address
6023             the proxy target.";
6024         reference
6025             "6.4.10.3.9.7 of IEC/IEEE 60802";
6026     }
6027     leaf mac-addr {
6028         type ieee:mac-address;
6029         description
6030             "The mac-address that is used by the proxy to address
6031             the proxy target.";
6032         reference
6033             "6.4.10.3.9.8 of IEC/IEEE 60802";
6034     }
6035 }
6036 }
6037 }
6038 }
6039 }

```

6.4.10.8.9 Module iecieee60802-proxy-target

```

6041 module iecieee60802-proxy-target {
6042     yang-version 1.1;
6043     namespace "urn:ieee:std:60802:yang:ieciieee60802-proxy-target";
6044     prefix cd;
6045
6046     organization
6047         "IEEE 802.1 Working Group and IEC subcommittee 65C:
6048         Industrial networks, of IEC technical committee 65:
6049         Industrial-process measurement, control and automation";
6050     contact
6051         "WG-URL: http://ieee802.org/1/
6052         WG-EMail: stds-802-1-1@ieee.org
6053
6054         Contact: IEEE 802.1 Working Group Chair
6055         Postal: C/O IEEE 802.1 Working Group
6056         IEEE Standards Association

```

```
6057         445 Hoes Lane
6058         Piscataway, NJ 08854
6059         USA
6060
6061         E-mail: stds-802-1-chairs@ieee.org";
6062     description
6063         "Capability information for IEC/IEEE 60802 proxy targets as
6064         specified in IEC/IEEE 60802.
6065
6066         Copyright (C) IEC/IEEE (2025).
6067         This version of this YANG module is part of IEC/IEEE 60802;
6068         see the standard itself for full legal notices.";
6069
6070     revision 2025-01-08 {
6071         description
6072             "Published as part of IEC/IEEE 60802-2025.
6073             The following reference statement identifies each referenced
6074             IEEE Standard as updated by applicable amendments.";
6075         reference
6076             "IEC/IEEE 60802 TSN profile for industrial automation:
6077             IEC/IEEE 60802-2025.
6078             IEEE Std 802.1Q Bridges and Bridged Networks:
6079             IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
6080             IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
6081             IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2025.";
6082     }
6083
6084     container proxy-target {
6085         presence "The presence of this container indicates that an IA station
6086             is a proxy target that requires a proxy device";
6087         description
6088             "This container provides information about a
6089             60802 proxy target device.";
6090         reference
6091             "IEC/IEEE 60802 - YANG Data Model";
6092         leaf required-proxy-type {
6093             type string {
6094                 pattern '[0-9A-F]{2}(-[0-9A-F]{2}){2}';
6095             }
6096             config false;
6097             description
6098                 "OID of proxy that this proxy target requires.";
6099             reference
6100                 "6.4.10.3.10.1 of IEC/IEEE 60802";
6101         }
6102     }
6103 }
```

6104 6.5 Topology discovery and verification

6105 6.5.1 Topology discovery and verification requirements

6106 Electrical engineering of machines with multiple IA-stations includes the definition of the
6107 machine internal network topology (i.e. the engineered topology).

6108 The machine internal network topology includes type specific data of IA-stations (for example
6109 model name or manufacturer name) as well as instance specific data (for example IP addresses
6110 or DNS names).

The electrical engineering data of the network topology is used:

- during commissioning so that machine planning and installation are identical;
- by the TDE during operation to verify that the actual topology of the Configuration Domain matches the engineered topology;
- by maintenance staff during repair to easily identify failed IA-stations, ports, or links to be replaced.

Repair and replacement of an IA-station do not require verification of the updated engineered topology so that the TDE does not produce a verification error.

IA-stations are necessarily pre-configured when they are repaired or replaced. IA-stations report type and instance data as described in 6.5.3.

6.5.2 Topology discovery overview

6.5.2.1 General

LLDP enables the discovery of IA-stations, their external ports, and their external connectivity. A Topology Discovery Entity can query LLDP data by remote management to derive the physical network topology.

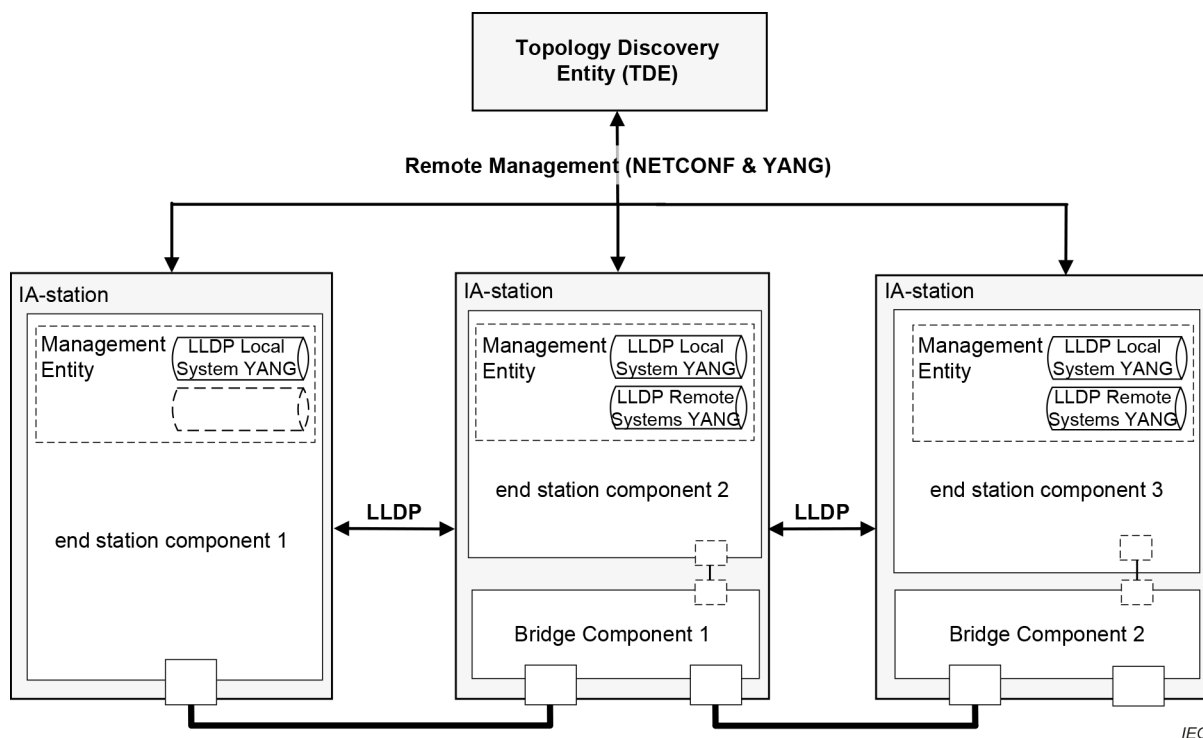


Figure 38 – Usage example of LLDP

Figure 38 illustrates a network showing the LLDP agent implementations in an IA-station consisting of a single end station component and two IA-stations with end station and Bridge components (see 4.3). The LLDP protocol is used to convey neighborhood information among peers, and NETCONF is used between the TDE and the IA-stations to query this neighborhood information from the IA-stations. This information allows the TDE to discover IA-stations and the physical network topology.

NOTE A Topology Discovery Entity (TDE) can be run from anywhere in the network with reachability to the to-be-discovered devices.

IA-stations announce themselves via LLDP to support discovery by the TDE. Announcements contain the management address (see 6.5.2.4.6) and system capabilities (see 6.5.2.4.5) for the discovery operation. The announced system capabilities information enables the TDE to identify IA-stations with multiple end station and Bridge components. The TDE can use the definitions in 6.4.3 for the discovery of the internal structure of such IA-stations.

To allow for operational behavior and exchanged information, IA-stations support the local system YANG (see 6.4.9.2.2). IA-stations that include a Bridge component additionally support the processing of received LLDP messages and support the remote systems YANG (see 6.4.9.2.2).

6.5.2.2 LLDP operational control parameters

LLDP defines several operational parameters that control the protocol behavior (see IEEE Std 802.1AB-2016, 10.5.1). These parameter definitions apply to all external ports of an IA-station.

NOTE According to IEEE Std 802.1AB-2016, 9.1.1 c), changes to the local system that impact information exchanged via LLDP immediately trigger the transmission of an LLDPDU to communicate the local changes as quickly as possible to any neighboring systems.

An IA-station shall support LLDP transmit mode (adminStatus enabledTxOnly) on an external end station component port and may support transmit and receive mode (adminStatus enabledRxBTx) on that port (see IEEE Std 802.1AB-2016, 10.5.1).

An IA-station shall support LLDP transmit and receive mode (adminStatus enabledRxBTx) on an external Bridge component port (see IEEE Std 802.1AB-2016, 10.5.1).

6.5.2.3 LLDPDU transmission, reception, and addressing

The destination address to be used for LLDPDU transmission (dest-mac-address) shall be the nearest bridge group MAC address, i.e., 01-80-C2-00-00-0E, on all ports to limit the scope of LLDPDU propagation to a single physical link (see IEEE Std 802.1AB-2016, 7.1 a).

NOTE IEEE Std 802.1AB-2016 defines LLDPDUs to be transmitted untagged, i.e. frames do not carry priority information for traffic class selection. At the same time, IEEE Std 802.1AB-2016 neither specifies a well-defined device-internal priority nor management capabilities for the configuration of the traffic class to be used for the transmission of LLDPDUs. It is the user's responsibility to prevent LLDPDUs from interfering with the transmission of time-critical control data.

6.5.2.4 LLDP TLV selection

6.5.2.4.1 General

An IA-station transmitting LLDPDUs shall include the LLDP TLVs selected in 6.5.2.4 and may include additional TLVs (tlvs-tx-enable). An IA-station receiving LLDPDUs shall process LLDPDUs.

Each LLDPDU shall contain the following LLDP TLVs specified in IEEE Std 802.1AB-2016, 8.5:

- exactly one Chassis ID TLV according to 6.5.2.4.2,
- exactly one Port ID TLV according to 6.5.2.4.3,
- exactly one Time To Live TLV according to 6.5.2.4.4,
- exactly one System Capabilities TLV according to 6.5.2.4.5, and
- one or more Management Address TLVs according to 6.5.2.4.6.

NOTE The concatenation of the Chassis ID and Port ID fields enables the recipient of an LLDPDU to identify the sending LLDP agent/port.

In addition, each LLDPDU shall contain the 60802 device info TLV described in Annex E.

6179 **6.5.2.4.2 Chassis ID TLV**

6180 The Chassis ID field shall contain the same value for all transmitted LLDPDUs independent
6181 from the transmitting port of the IA-station, i.e. be a non-volatile identifier which is unique within
6182 the context of the administrative domain.

6183 The Chassis ID subtype field (chassis-id-subtype) should contain subtype 4, indicating that the
6184 Chassis ID field (chassis-id) contains a MAC address to achieve the Chassis ID's desired
6185 uniqueness. For IA-stations with multiple unique MAC addresses, any one of the IA-station's
6186 MAC addresses may be used and shall be the same for all external ports of that IA-station.

6187 **6.5.2.4.3 Port ID TLV**

6188 The Port ID field shall contain the same value for all transmitted LLDPDUs for a given external
6189 port, i.e. be a non-volatile, IA-station-unique identifier of the LLDPDU-transmitting port.

6190 The Port ID subtype field (port-id-subtype) should contain subtype 5, indicating that the Port ID
6191 field contains the port interface name (name) according to IETF RFC 8343.

6192 IA-stations should restrict the system-defined port ID to read-only access and a maximum name
6193 length of 255 characters. The names should match the port names printed on the chassis.

6194 **6.5.2.4.4 Time To Live TLV**

6195 The Time To Live value shall be set according to IEEE Std 802.1AB-2016, 8.5.4.

6196 **6.5.2.4.5 System capabilities TLV**

6197 An IA-station consisting of a single end station component shall set the system capabilities and
6198 enabled capabilities fields (system-capabilities-supported, system-capabilities-enabled) to
6199 Station Only (i.e. bit 8 set to 1) for all transmitted LLDPDUs.

6200 An IA-station consisting of at least one end station component and at least one Bridge
6201 component shall set the system capabilities and enabled capabilities fields to Station Only (i.e.
6202 bit 8 set to 1) and C-VLAN component (i.e. bit 9 set to 1) for all transmitted LLDPDUs.

6203 NOTE The combination of the Station Only and C-VLAN component flags is used as a marker indicating to the TDE
6204 that the internal structure of the IA-station consists of multiple components. This is a deliberate deviation from IEEE
6205 Std 802.1AB-2016, Table 8-4, which states in a footnote: "The Station Only capability is intended for devices that
6206 implement only an end station capability, and for which none of the other capabilities in the table apply. Bit 8 therefore
6207 is not set in conjunction with any other bits."

6208 **6.5.2.4.6 Management address TLV**

6209 An IA-station shall announce at least one IPv4 address by which its Management entity (see
6210 4.3) can be reached (management-address-tx-port).

6211 **6.5.2.5 LLDP remote systems data**

6212 An IA-station supporting the remote systems YANG shall be able to store information from at
6213 least one neighbor per external port.

6214 Receiving LLDPDUs from more neighbors than supported on a given port shall result in the last
6215 one received being saved to the remote systems YANG as described in
6216 IEEE Std 802.1AB-2016, 9.2.7.7.5.

6.5.3 Topology verification overview

Topology verification checks discovered topologies against engineered topologies. Topology verification data includes for every IA-station:

- model name,
- manufacturer name, and
- management address.

Topology verification data includes for every external port of an IA-station:

- port name, and
- remote connection (i.e. management address and port name of connected IA-station).

To support topology verification IA-stations shall support LLDP YANG data as specified in 6.4.9.2.2 and Hardware Management YANG data as specified in 6.4.9.2.5.8.

IA-station hardware instance specific data like MAC addresses or serial numbers are not considered for topology verification. This kind of data changes after a repair and replacement operation and thus, induces a topology verification error.

6.6 CNC

6.6.1 General

Subclause 6.6 describes stream destination MAC address handling at the CNC.

6.6.2 Stream destination MAC address range

A CNC manages the destination MAC address for requested streams. This destination MAC address together with the VID identifies the path used for these streams. Thus, a stream destination MAC address is unique together with the VID in a Configuration Domain.

Figure 39 shows the possible selections of a CNC for a contiguous address range. The CNC selects an OUI and an offset of the address range for the stream destination MAC addresses.

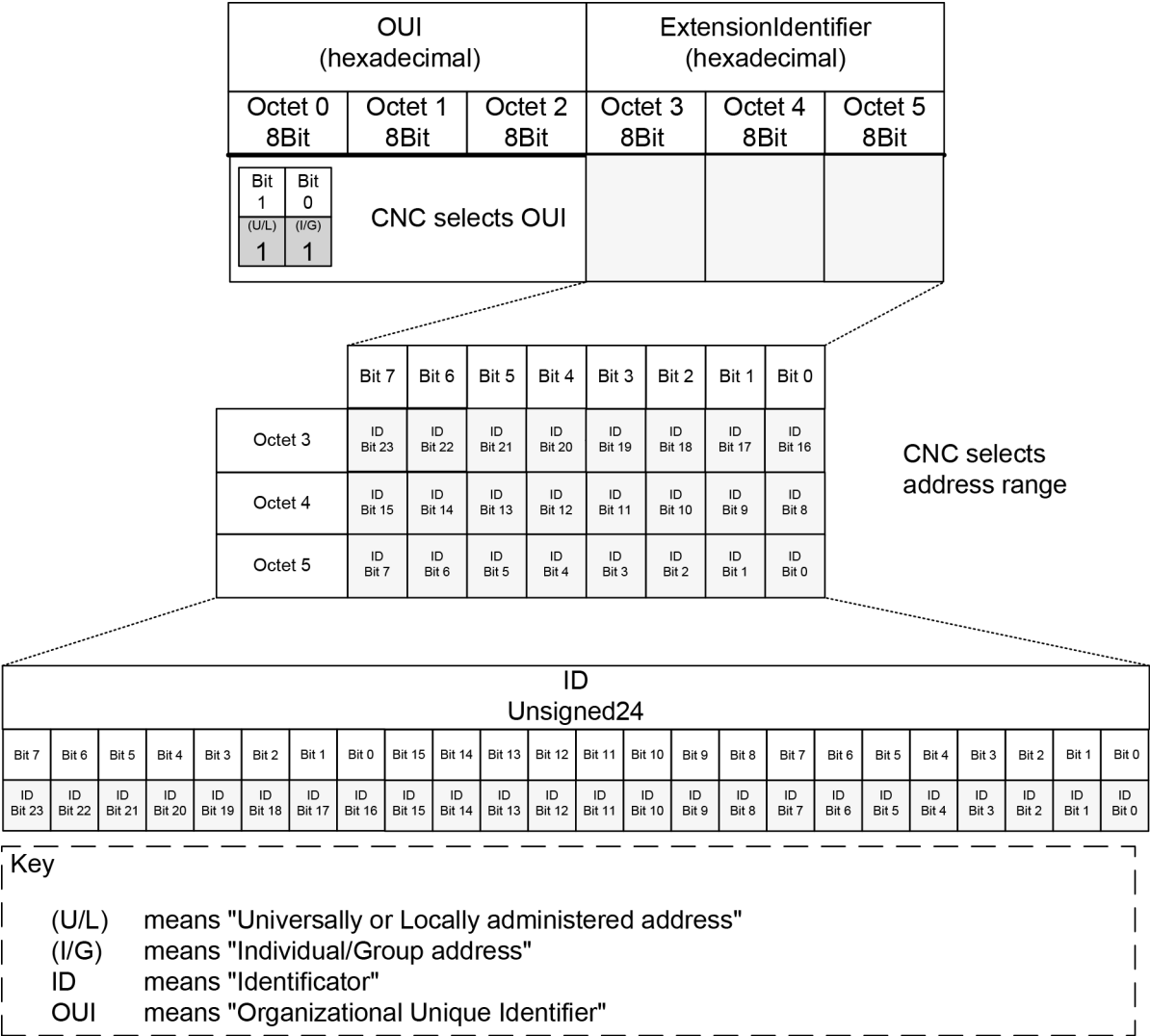
An address range of 2 048 stream destination MAC addresses allows together with a VID the usage of 2 048 streams. Each additional VID used for streams allows an additional 2 048 streams.

EXAMPLE

CNC selected OUI:= 00-80-C2

CNC selected address range:= 0.2047

CNC selected VID:= 101



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IEC

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Figure 39 – Stream destination MAC Address

Annex A (normative)

PCS proforma – Time-sensitive networking profile for industrial automation

A.1 General⁷

The supplier of an implementation that is claimed to conform to the profile specified in this document shall complete the corresponding Profile Conformance Statement (PCS) proforma, which is presented in a tabular format based on the format used for Protocol Implementation Conformance Statement (PICS) proformas.

The tables do not contain an exhaustive list of all requirements that are stated in the referenced standards; for example, if a row in a table asks whether the implementation is conformant to Standard X, and the answer "Yes" is chosen, then it is assumed that it is possible, for that implementation, to fill out the PCS proforma specified in Standard X to show that the implementation is conformant; however, the tables in this document only further refine those elements of conformance to Standard X where particular answers are required for the profiles specified here.

A completed PCS proforma is the PCS for the implementation in question. The PCS is a statement of which capabilities and options of the protocol have been implemented. The PCS can have several uses, including use by the following.

- a) Protocol implementer, as a checklist to reduce the risk of failure to conform to the document through oversight.
 - b) Supplier and acquirer, or potential acquirer, of the implementation, as a detailed indication of the capabilities of the implementation, stated relative to the common basis for understanding provided by the standard PCS proforma.
 - c) User, or potential user, of the implementation, as a basis for initially checking the possibility of interworking with another implementation.
- NOTE While interworking can never be guaranteed, failure to interwork can often be predicted from incompatible PCS.
- d) Protocol tester, as the basis for selecting appropriate tests against which to assess the claim for conformance of the implementation.
 - e) The user, to verify whether the IA-station, as described by the PCS, fulfills use-case requirements.

A.2 Abbreviated terms and special symbols

A.2.1 Status symbols

M	mandatory
O	optional
O.n	optional, but support of at least one of the group of options labeled by the same numeral n is required
X	prohibited
pred	conditional-item symbol, including predicate identification (see A.3.4)
¬	logical negation, applied to a conditional item's predicate

⁷ Copyright release for the PCS: Users of this document may freely reproduce the PCS contained in this document so that they can be used for their intended purpose.

6291 **A.2.2 General abbreviated terms**

6292 N/A not applicable

6293 PCS Profile Conformance Statement

6294 **A.3 Instructions for completing the PCS proforma**

6295 **A.3.1 General structure of the PCS proforma**

6296 The first part of the PCS proforma, implementation identification and protocol summary, is to
6297 be completed as indicated with the information necessary to identify fully both the supplier and
6298 the implementation.

6299 The main part of the PCS proforma is a fixed-format questionnaire, divided into several
6300 subclauses, each containing a number of individual items. Answers to the questionnaire items
6301 are to be provided in the rightmost column, either by simply marking an answer to indicate a
6302 restricted choice (usually Yes or No) or by entering a value or a set or range of values. There
6303 are some items where two or more choices from a set of possible answers can apply; all relevant
6304 choices are to be marked. Each item is identified by an item reference in the first column. The
6305 second column contains the question to be answered; the third column records the status of
6306 the item – whether support is mandatory, optional, or conditional (see also A.3.4). The fourth
6307 column contains the reference or references to the material that specifies the item in the main
6308 body of this document, and the fifth column provides the space for the answers.

6309 The PCS indicates support of one of the conformance classes, ccA or ccB, per bridge and end-
6310 station component, specified in this profile.

6311 A single IA-station can incorporate the functionality of one or more of the functions listed in this
6312 PCS. For example, an IA-station could have both an end station component and a Bridge
6313 component.

6314 A supplier can also provide (or be required to provide) further information, categorized as either
6315 additional information (see A.3.2) or exception information (see A.3.3). When present, each
6316 kind of further information is to be provided in a further subclause of items labeled Ai or Xi,
6317 respectively, for cross-referencing purposes, where (i) is any unambiguous identification for the
6318 item (for example, simply a numeral). There are no other restrictions on its format and
6319 presentation.

6320 A completed PCS proforma, including any Additional Information and Exception Information, is
6321 the Protocol Implementation Conformation Statement for the implementation in question.

6322 NOTE Where an implementation is capable of being configured in more than one way, a single PCS can be used
6323 to describe all such configurations. However, the supplier has the choice of providing more than one PCS, each
6324 covering some subset of the implementation's configuration capabilities, in case that makes for easier and clearer
6325 presentation of the information.

6326 **A.3.2 Additional information**

6327 Items of Additional Information allow a supplier to provide further information intended to assist
6328 the interpretation of the PCS. It is not intended or expected that a large quantity be supplied,
6329 and a PCS can be considered complete without any such information. Examples can be an
6330 outline of the ways in which a (single) implementation can be set up to operate in a variety of
6331 environments and configurations, or information about aspects of the implementation that are
6332 outside the scope of this document but that have a bearing on the answers to some items.

6333 References to items of Additional Information can be entered next to any answer in the
6334 questionnaire and can be included in items of Exception Information.

A.3.3 Exception information

It can occasionally happen that a supplier wishes to answer an item with mandatory status (after any conditions have been applied) in a way that conflicts with the indicated requirement. No preprinted answer will be found in the Support column for this item. Instead, the supplier shall write the missing answer into the Support column, together with an Xi reference to an item of Exception Information and shall provide the appropriate rationale in the Exception item itself.

An implementation for which an Exception item is required in this way does not conform to this document.

NOTE A possible reason for the situation described previously is that a defect in this document has been reported, a correction for which is expected to change the requirement not met by the implementation.

A.3.4 Conditional status

A.3.4.1 Conditional items

The PCS proforma contains a number of conditional items. These are items for which both the applicability of the item itself, and its status if it does apply (mandatory or optional) are dependent on whether certain other items are supported.

Where a group of items is subject to the same condition for applicability, a separate preliminary question about the condition appears at the head of the group, with an instruction to skip to a later point in the questionnaire if the "Not Applicable" (N/A) answer is selected. Otherwise, individual conditional items are indicated by a conditional symbol in the Status column.

A conditional symbol is of the form "pred: S" where pred is a predicate as described in A.3.4.2, and S is a status symbol, M or O.

If the value of the predicate is true (see A.3.4.2), the conditional item is applicable, and its status is indicated by the status symbol following the predicate: the answer column is to be marked in the usual way. If the value of the predicate is false, the "Not Applicable" (N/A) answer is to be marked.

A.3.4.2 Predicates

A predicate is one of the following:

- a) an item-reference for an item in the PCS proforma: the value of the predicate is true if the item is marked as supported and is false otherwise,
- b) a predicate-name, for a predicate defined as a Boolean expression constructed by combining item-references using the Boolean operator OR: the value of the predicate is true if one or more of the items is marked as supported, or
- c) the logical negation symbol "¬" prefixed to an item-reference or predicate-name: the value of the predicate is true if the value of the predicate formed by omitting the "¬" symbol is false, and vice versa.

Each item whose reference is used in a predicate or predicate definition, or in a preliminary question for grouped conditional items, is indicated by an asterisk in the Item column.

A.4 Common requirements

A.4.1 Instructions

One instance of Clause A.4 shall be filled out per IA-station.

A.4.2 Implementation identification

The entire PCS proforma is a form that shall be filled out by a supplier according to Table A.1.

Table A.1 – Implementation identification template

Supplier	
Contact point for queries about the PCS	
Implementation Name(s) and Version(s)	
Other information necessary for full identification, for example, name(s) and version(s) of machines and operating system names	

Only the first three items are required for all implementations; other information can be completed as appropriate in meeting the requirement for full identification. The terms "Name" and "Version" should be interpreted appropriately to correspond with a supplier's terminology (for example, Type, Series, Model).

A.4.3 Profile summary

Table A.2 shows the profile summary template.

Table A.2 – Profile summary template

Identification of profile specification	IEC/IEEE 60802 – Time-Sensitive Networking profile for industrial automation			
Identification of amendments (Amd) and corrigenda (Corr) to the PCS proforma that have been completed as part of the PCS	Amd	:	Corr	:
	Amd	:	Corr	:
Have any Exception items been required? (See A.3.3: the answer "Yes" means that the implementation does not conform to IEC/IEEE 60802)	No	[]	Yes	[]
Date of Statement				

A.4.4 Implementation summary

The form in Table A.3 is used to indicate the type of system that the PCS describes.

Table A.3 – Implementation type

Item	Feature	Status	References	Support
BC-CCA-N	State the number of Conformance Class A bridge components implemented by the IA-station.	O	5.7.2, 5.8.2	Number _____
BC-CCB-N	State the number of Conformance Class B bridge components implemented by the IA-station.	O	5.7.3, 5.8.3	Number _____
ESC-CCA-N	State the number of Conformance Class A end station components implemented by the IA-station.	O.1	5.9.2, 5.10.2	Number _____
ESC-CCB-N	State the number of Conformance Class B end station components implemented by the IA-station.	O.1	5.9.3, 5.10.3	Number _____
CNC	Does the IA-station include a CNC?	O	5.5.4.1 m)	Yes [] No []
CUC	Does the IA-station include a CUC?	O	5.5.4.1 n)	Yes [] No []
PRX	Does the IA-station support proxy functionality?	O	5.5.4.1 o)	Yes [] No []

A.5 IA-station requirements and options

A.5.1 Instructions

One instance of Clause A.5 shall be filled out for an IA-station.

A.5.2 IA-station requirements

The form in Table A.4 is used to indicate the IA-station requirements.

Table A.4 – IA-station requirements

Item	Feature	Status	References	Support
IASTA-1	Does the IA-station support PHY and MAC requirements for external ports?	M	5.5.1	Yes []
IASTA-2	Does the IA-station support topology discovery requirements?	M	5.5.2	Yes []
IASTA-3	Does the IA-station support requirements for time synchronization?	M	5.5.3	Yes []
IASTA-4	Does the IA-station provide a digital data sheet?	M	5.5.4.3	Yes []

A.5.3 Management Requirements for non-proxied IA-stations

The form in Table A.5 is used to indicate management requirements for IA-stations containing conformance class A components

Table A.5 – Additional requirements for non-proxied IA-stations

Item	Feature	Status	References	Support
IASTA-5	Does the IA-station support requirements for Secure management exchanges?	M	5.5.4.1 a) through l)	Yes []
IASTA-6	Number of of Dynamic Subscriptions to YANG Events and Datastores over NETCONF	M	5.5.4.1 h)	Number _____
IASTA-7	Does the IA-station support management YANG modules?	M	5.5.4.2	Yes []

A.5.4 IA-station PHY and MAC options for external ports

The form in Table A.6 is used to indicate PHY and MAC options for external ports.

Table A.6 – IA-station PHY and MAC options

Item	Feature	Status	References	Support
DOT3-1	Does the IA-station support PoE over 2 pairs?	O	5.6.1 a)	Yes [] No [] N/A []
DOT3-2	Does the IA-station support Power Interfaces?	O	5.6.1 b)	Yes [] No [] N/A []
DOT3-3	Does the IA-station support PoE?	O	5.6.1 c)	Yes [] No [] N/A []

A.5.5 IA-station options for time synchronization

The form in Table A.7 is used to indicate options for time synchronization.

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Table A.7 – IA-station time synchronization options

Item	Feature	Status	References	Support
PTP-1	Does the IA-station support media-independent TimeTransmitter capability according to IEEE Std 802.1AS-2020, 5.4.2 item b) as amended by IEEE Std 802.1ASdr-2024?	O	5.6.2 a)	Yes [] No []
PTP-2	Does the IA-station support Grandmaster PTP Instance capability Grandmaster PTP Instance capability according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdr-2024 and IEEE Std 802.1ASdm-2024, 5.4.2 item c)?	O	5.6.2 b)	Yes [] No []
PTP-3	Does the IA-station support more than one PTP port as a PTP Relay Instance according to IEEE Std 802.1AS-2020, 5.4.2 item d)?	O	5.6.2 c)	Yes [] No []
PTP-4	Does the IA-station support hot standby redundancy requirements according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdm-2024, 5.4.2, item m)?	O	5.6.2 d)	Yes [] No []

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6410 **A.5.6 IA-station secure management exchange options**

6411 The form in Table A.8 is used to indicate options for secure management exchange.

6412

Table A.8 – IA-station secure management exchange options

Item	Feature	Status	References	Support
SECMGMT-5	Does the IA-station support Writable-Running capability?	O	5.6.3 a)	Yes [] No []
SECMGMT-6	Does the IA-station support Confirmed Commit capability?	O	5.6.3 b)	Yes [] No []
SECMGMT-7	Does the IA-station support Distinct Startup capability?	O	5.6.3 c)	Yes [] No []
SECMGMT-8	Does the IA-station support URL capability?	O	5.6.3 d)	Yes [] No []
SECMGMT-9	Does the IA-station support XPath capability?	O	5.6.3 e)	Yes [] No []
SECMGMT-10	NETCONF-over-TLS server supporting TLS version 1.2?	O	5.6.3 f)	Yes [] No []
SECMGMT-11	Does the IA-station support NETCONF-over-TLS server supporting TLS version 1.3?	O	5.6.3 g)	Yes [] No []
SECMGMT-12	Does the IA-station support PKIX?	O	5.6.3 h)	Yes [] No []

6413

6414 **A.5.7 Proxy target requirements**

6415 The form in Table A.9 is used to indicate requirements for proxy targets.

6416

Table A.9 – Proxy target requirements

Item	Feature	Status	References	Support
PT-1	Does the IA-station support Secure management exchanges?	O	5.6.4 a)	Yes [] No []
PT-1	Does the IA-station support Secure management exchanges using a management proxy?	O	5.6.4 b)	Yes [] No []

6417

6418 **A.6 Bridge component**

6419 **A.6.1 Bridge component requirements**

6420 The form in Table A.10 is used to indicate bridge component requirements.

Table A.10 – Bridge component requirements

Item	Feature	Status	References	Support
BC-1	Does the bridge component support the common bridge component requirements?	M	5.7.1	Yes []
BC-2	Does the bridge component support ccA bridge component requirements?	O.2	5.7.2	Yes [] No []
BC-3	Does the bridge component support ccB bridge component requirements?	O.2	5.7.3	Yes [] No []

A.6.2 Instructions

One instance of Clause A.6 shall be filled out per bridge component implemented by an IA-station.

A.6.3 Bridge Common bridge component options

The form in Table A.11 is used to indicate bridge component options.

Table A.11 – Common bridge component options

Item	Feature	Status	References	Support
BC-4	Does the bridge component support the operation of the credit-based shaper algorithm?	O	5.8.1	Yes [] No []

A.6.4 Conformance class A bridge component options

The form in Table A.12 is used to indicate options for bridge components conforming to conformance class A.

Table A.12 – Conformance class A bridge component options

Item	Feature	Status	References	Support
CCA-BC-1	Does the bridge component support any of the common bridge component options?	O	5.8.2 a)	Yes [] No [] N/A []
CCA-BC-2	Does the bridge component support more than 2 PTP instances?	O	5.8.2 b)	Yes [] No [] N/A []
CCA-BC-3	State the number of PTP instances supported by the bridge component.	CCA-BC-2:M	5.8.2 b)	Number ____
CCA-BC-4	Does the bridge component support enhancements for scheduled traffic for the 10 Mb/s, 2,5 Gb/s, 5 Gb/s, or 10 Gb/s data rates?	O	5.8.2 c)	Yes [] No [] N/A []
CCA-BC-5	Does the bridge component support frame preemption for the 10 Mb/s, 2,5 Gb/s, 5 Gb/s, or 10 Gb/s data rates?	O	5.8.2 d)	Yes [] No [] N/A []

A.6.5 Conformance class B bridge component options

The form in Table A.13 is used to indicate options for bridge components conforming to conformance class B.

Table A.13 – Conformance class B bridge component options

Item	Feature	Status	References	Support
CCB-BC-1	Does the bridge component support any of the common bridge component options?	O	5.8.3 a)	Yes [] No [] N/A []
CCB-BC-2	Does the bridge component support more than 4 but not more than 8 egress queues?	O	5.8.3 b)	Yes [] No [] N/A []

CCB-BC-3	State the number of egress queues supported by the bridge component.	CCB-BC-2:M	5.8.3 b)	Number ____
CCB-BC-4	Does the bridge component support more than 1 PTP Instance?	O	5.8.3 c)	Yes [] No [] N/A []
CCB-BC-5	State the number of PTP Instances supported by the bridge component.	CCB-BC-4:M	5.8.3 c)	Number ____
CCB-BC-6	Does the bridge component support enhancements for scheduled traffic?	O	5.8.3 d)	Yes [] No [] N/A []
CCB-BC-7	Does the bridge component support frame preemption?	O	5.8.3 e)	Yes [] No [] N/A []

6439

6440 **A.7 End station component**

6441 **A.7.1 Instructions**

6442 One instance of Clause A.7 shall be filled out per end station component implemented by an
6443 IA-station.

6444 **A.7.2 Common end station component requirements**

6445 The form in Table A.14 is used to indicate common requirements for end station components.

6446 **Table A.14 – Common end station component requirements**

Item	Feature	Status	References	Support
ESC-1	Does the end station component support the common end station component requirements?	M	5.9.1	Yes []
ESC-2	Does the end station component support the ccA end station component requirements?	O.3	5.9.2	Yes [] No []
ESC-3	Does the end station component support the ccB end station component requirements?	O.3	5.9.3	Yes [] No []

6447

6448 **A.7.3 Common end station component options**

6449 The form in Table A.15 is used to indicate options for end station components.

6450 **Table A.15 – Common end station component options**

Item	Feature	Status	References	Support
ESC-4	Does the end station component support the operation of the credit-based shaper?	O	5.10.1 a)	Yes [] No []
ESC-5	Does the end station component support talker end system behaviors?	O	5.10.1 b)	Yes [] No []
ESC-6	Does the end station component support listener end system behaviors?	O	5.10.1 c)	Yes [] No []

6451

6452 **A.7.4 Conformance class A end station component options**

6453 The form in Table A.16 is used to indicate options for end station components conforming to
6454 conformance class A.

6455 **Table A.16 – Conformance class A end station component options**

Item	Feature	Status	References	Support
CCA-ESC-1	Does the end station component support any of the common end station component options?	O	5.10.2 a)	Yes [] No [] N/A []

CCA-ESC-2	Does the end station component support more than 2 PTP Instances?	O	5.10.2 b)	Yes [] No [] N/A []
CCA-ESC-3	State the number of PTP Instances supported by the end-station component.	CCA-ESC-2:M	5.10.2 b)	Number ____
CCA-ESC-4	Does the end station component support enhancements for scheduled traffic for data rates 10 Mb/s, 2,5 Gb/s, 5 Gb/s, or 10 Gb/s?	O	5.10.2 c)	Yes [] No [] N/A []
CCA-ESC-5	Does the end station component support requirements for frame preemption for data rates 10 Mb/s, 2,5 Gb/s, 5 Gb/s, or 10 Gb/s?	O	5.10.2 d)	Yes [] No [] N/A []

6456

6457 **A.7.5 Conformance class B end station component options**

6458 The form in Table A.17 is used to indicate options for end station components conforming to
6459 conformance class B.

6460 **Table A.17 – Conformance class B end station component options**

Item	Feature	Status	References	Support
CCB-ESC-1	Does the end station component support any of the common end station component options?	O	5.10.3 a)	Yes [] No [] N/A []
CCB-ESC-2	Does the end station component support one or more PTP Instances?	O	5.10.3 b)	Yes [] No [] N/A []
CCB-ESC-3	State the number of PTP Instances supported by the end-station component.	CCB-ESC-2:M	5.10.3 b)	Number ____
CCB-ESC-4	Does the end station component support enhancements for scheduled traffic?	O	5.10.3 c)	Yes [] No [] N/A []
CCB-ESC-5	Does the end station component support requirements for frame preemption?	O	5.10.3 d)	Yes [] No [] N/A []

6461

Annex B (informative)

Representative Configuration Domain

6462
6463
6464
6465

6466 The following quantities are representative of what can be supported in a single Configuration
6467 Domain.

- 6468 • IA-stations: 1 024.
- 6469 • Network diameter: 64.
- 6470 • Streams per IA-controller for IA-controller to IA-device (C2D) communication:
 - 6471 – 512 Talker and ≥ 512 Listener streams, and
 - 6472 – 1 024 Talker and $\geq 1\,024$ Listener streams in case of seamless redundancy.
- 6473 • Streams per IA-controller for IA-controller to IA-controller (C2C) communication:
 - 6474 – 64 Talker and ≥ 64 Listener streams, and
 - 6475 – 128 Talker and ≥ 128 Listener streams in case of seamless redundancy.
- 6476 • Streams per IA-device for IA-device-to-IA-device (D2D) communication:
 - 6477 – 2 Talker and 2 Listener streams, and
 - 6478 – 4 Talker and 4 Listener streams in case of seamless redundancy.
- 6479 • Example calculation of data flow quantities for eight PLCs – without seamless redundancy:
 - 6480 – $8 \times 512 \times 2 = 8\,192$ streams for C2D communication,
 - 6481 – $8 \times 64 \times 2 = 1\,024$ streams for C2C communication, and
 - 6482 – $(8\,192 + 1\,024) \times 2\,000 = 18\,432\,000$ octets data for all streams.

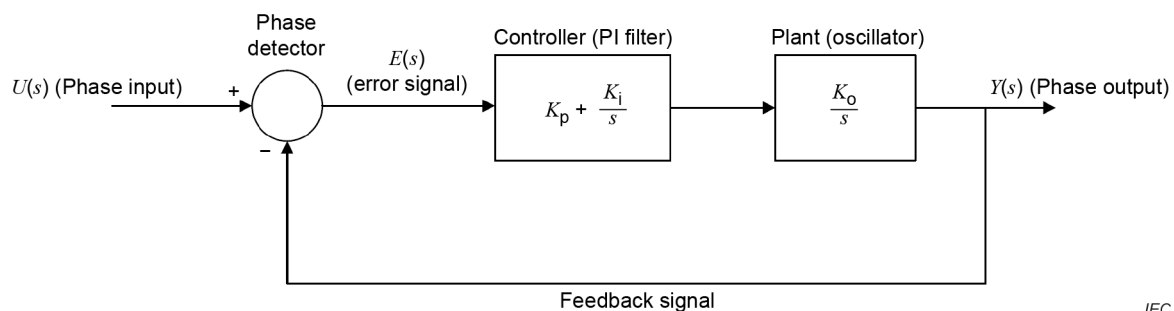
Annex C (informative)

Description of clock control system

C.1 Clock control system introduction

Annex C provides an introductory discussion of a basic clock control system. For more detailed information, see the Bibliography References ITU-T Recommendation G.8260 and ITU-T Series G Supplement 65.

Figure C.1 shows a basic control system model that uses a proportional plus integral (PI) controller. This is meant to be reference model, i.e. it is not meant to specify an implementation. Requirements for the clock control system can be expressed using parameters (e.g. 3 dB bandwidth, gain peaking, frequency response) that are based on this reference model. Any implementation whose parameters are within the requirements is considered to be acceptable. For example, the model of Figure C.1 is expressed in the analog domain (i.e. s-domain), and is shown to be second order. An actual implementation can be digital, and can be higher order, as long as it meets the respective requirements.



IEC

Figure C.1 – Reference model for clock control system

In Figure C.1, the plant, i.e. the entity being controlled, represents the clock oscillator. It is desired that the phase output, $y(t)$, of the oscillator follows the phase input, $u(t)$, as closely as possible (the signals are shown in the frequency domain (i.e. as Laplace Transforms) in Figure C.1; however, they can equivalently be expressed in the time domain, with t representing time). The parameter K_o is the oscillator gain; the oscillator frequency is equal to the oscillator input multiplied by K_o . In some implementations the input signal to the oscillator is a voltage, and the oscillator is referred to as a voltage-controlled oscillator (VCO). However, other implementations are possible, e.g. digital implementations, where the oscillator is a digital controlled oscillator (DCO). Since the input to the oscillator depends on the implementation, it is not labeled in Figure C.1.

The control system of Figure C.1 uses negative feedback to enable the phase output to follow the phase input. The phase detector computes the difference between the input and output signals to produce the error signal $e(t)$. The error signal is then filtered by the PI filter to produce the input to the oscillator. The filter is referred to as a PI filter because its output is the sum of the proportional gain, K_p , multiplied by the error signal and the integral gain, K_i , multiplied by the integral of the error signal. The gains K_o , K_p , and K_i must be chosen such that the performance of the control system is acceptable, i.e. the time-domain behavior of the output with respect to the input is acceptable. However, an alternative set of parameters, which are more convenient, can be defined in terms of K_o , K_p , and K_i ; this is done in Clause C.2.

C.2 Transfer function for control system

From the block diagram of Figure C.1, the input and output are related by:

$$Y(s) = \left(K_p + \frac{K_i}{s} \right) \left(\frac{K_o}{s} \right) (U(s) - Y(s)) \quad (\text{C.1})$$

where

$Y(s)$ is the phase output, expressed in the s-domain;

$U(s)$ is the phase input, in the s-domain;

K_p is the proportional gain;

K_i is the integral gain;

K_o is the oscillator gain.

or

$$Y(s) = \frac{\left(K_p + \frac{K_i}{s} \right) \left(\frac{K_o}{s} \right)}{1 + \left(K_p + \frac{K_i}{s} \right) \left(\frac{K_o}{s} \right)} U(s) \quad (\text{C.2})$$

This can be simplified by multiplying the numerator and denominator by s^2 to produce:

$$Y(s) = H(s) U(s) \quad (\text{C.3})$$

where the transfer function $H(s)$ is given by:

$$H(s) = \frac{K_p K_o s + K_i K_o}{s^2 + K_p K_o s + K_i K_o} \quad (\text{C.4})$$

In Formula (C.4), the parameter K_o does not appear independently of K_p and K_i ; rather, only the products $K_p K_o$ and $K_i K_o$ appear. The plant and PI filter could have been combined in the model of Figure C.1; this is consistent with the fact that the exact nature of the signal between the PI filter and plant is unimportant in this reference model. The units of $K_p K_o$ are (time)⁻¹ and the units of $K_i K_o$ are (time)⁻². The frequency units are the same as the units of s , e.g. if s has units rad/s, then $K_p K_o$ has units rad/s and $K_i K_o$ has units (rad/s)². The integration operation in the plant results in the transfer function being dimensionless, which is consistent with the fact that the input and output of the control system both have units of phase.

6542 The transfer function can be expressed in an equivalent form by defining the undamped natural
6543 frequency, ω_n , and damping ratio, ζ :

$$H(s) = \frac{2\zeta\omega_n s + \omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (\text{C.5})$$

6544

6545 where

6546 ζ is the damping ratio;6547 ω_n is the undamped natural frequency expressed in rad/s.6548 And, ζ and ω_n are given by:

$$\omega_n = \sqrt{K_i K_o} \quad (\text{C.6})$$

$$\zeta = \frac{K_p K_o}{2\sqrt{K_i K_o}} = \frac{K_p}{2} \sqrt{\frac{K_o}{K_i}} \quad (\text{C.7})$$

6549

6550 In the Formula (C.7), the first form shows explicitly that ζ depends only on the products $K_p K_o$
6551 and $K_i K_o$.

6552 C.3 Frequency response for control system

6553 The frequency response is obtained by setting $s = j\omega$ in Formula (C.5) and taking the absolute
6554 value (here j rather than i is used for $\sqrt{-1}$ to avoid confusion with other uses of i), where ω is
6555 the frequency in rad/s. The result is:

$$|H(j\omega)| = \left| \frac{2\zeta\omega_n \omega j + \omega_n^2}{-\omega^2 + \omega_n^2 + 2\zeta\omega_n \omega j} \right| = \left(\frac{4\zeta^2 \omega_n^2 \omega^2 + \omega_n^4}{(\omega_n^2 - \omega^2)^2 + 4\zeta^2 \omega_n^2 \omega^2} \right)^{1/2} \quad (\text{C.8})$$

6556

6557 Dividing the numerator and denominator of Formula (C.8) by ω_n^4 and defining the dimensionless
6558 frequency $x = \omega/\omega_n$ produces:

$$|H(j\omega)| = \left(\frac{4\zeta^2 x^2 + 1}{(1 - x^2)^2 + 4\zeta^2 x^2} \right)^{1/2} \quad (\text{C.9})$$

6559

where

ζ is the damping ratio;

ω is the frequency expressed in rad/s;

$x = \omega/\omega_n$.

Figure C.2 contains plots of frequency response (Formula (C.9)) versus dimensionless frequency x , on a log-log scale, for damping ratio ζ equal to 0,3, 0,5, 0,707, 1,0, 2,0, 3,0, 4,0, and 5,0. It is seen that the frequency response is very close to 1 for values of dimensionless frequency much less than 1 (i.e. for $\omega \ll \omega_n$). The frequency response increases as the frequency approaches the undamped natural frequency (i.e. as dimensionless frequency approaches 1) and reaches a peak for dimensionless frequency slightly less than 1. The frequency response then decreases, eventually having a slope (i.e. roll-off) of 20 dB/decade (i.e. frequency response decreases by a factor of 10 for every factor of 10 increase in x for $x \gg 1$). Figure C.3 shows the detail of frequency response for x in the range 0,1 to 10.

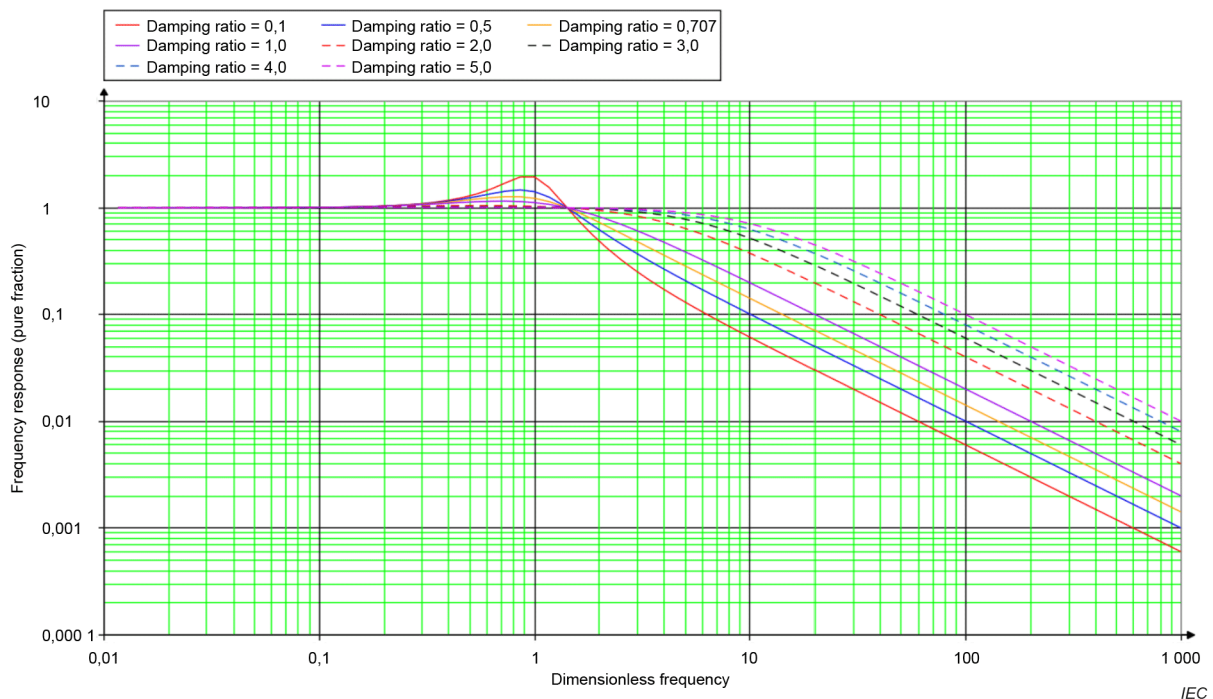


Figure C.2 – Frequency response for the control system of Figure C.1

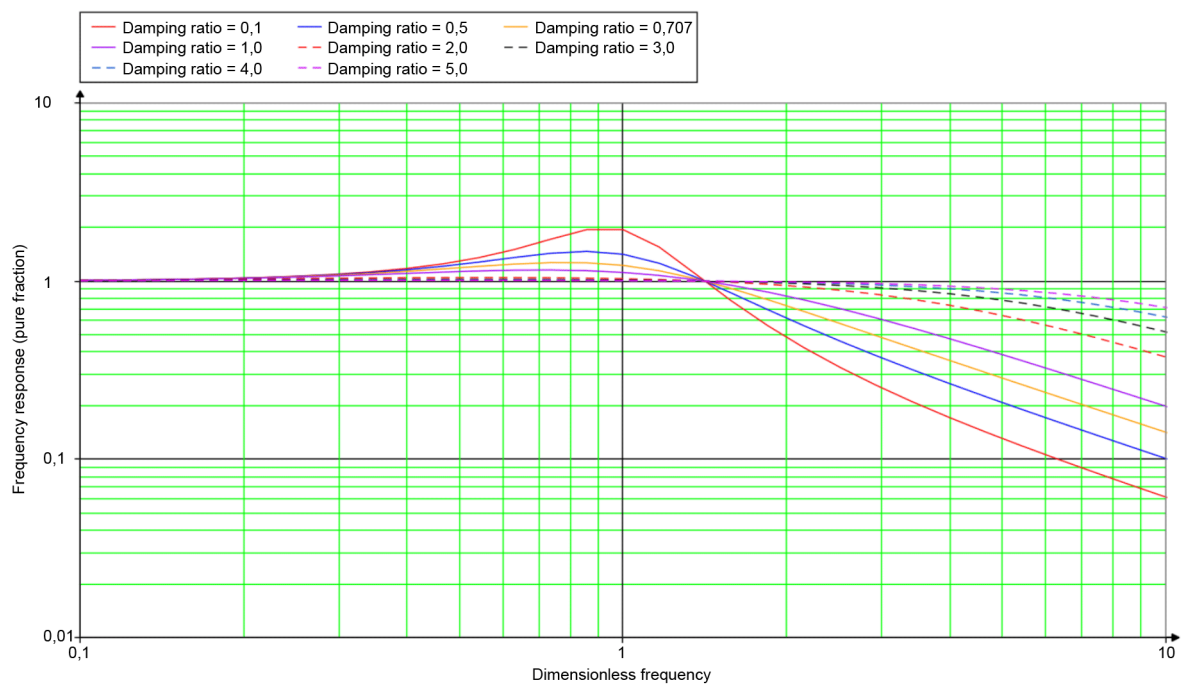


Figure C.3 – Detail of frequency response for the control system of Figure C.1 for dimensionless frequency in the range 0,1 to 10

In addition to undamped natural frequency ω_n and damping ratio ζ , the parameters 3 dB bandwidth and gain peaking are often used when specifying clock performance. The 3 dB bandwidth is defined as the value of frequency for which the frequency response is equal to -3 dB. Since dB is given by 10 multiplied by the logarithm to base 10 of the power ratio, which is 20 multiplied by the logarithm to base 10 of the amplitude ratio, -3 dB corresponds to the value $10^{-3/20}$. The 3 dB bandwidth can be computed by setting Formula (C.9) equal to $10^{-3/20}$ and solving for x in terms of ζ . This is equivalent to setting the quantity in parentheses (i.e. inside the square root) in Formula (C.9) equal to $10^{-3/10}$ and solving for x . Now, $10^{-3/10}$ is approximately equal to 0,501 2, i.e. it is very close to $\frac{1}{2}$. Then the 3dB bandwidth can be obtained by solving the following formula for x in terms of ζ :

$$\frac{4\zeta^2 x^2 + 1}{(1-x^2)^2 + 4\zeta^2 x^2} = \frac{1}{2} \quad (\text{C.10})$$

or

$$x^4 - 2(2\zeta^2 + 1)x^2 - 1 = 0 \quad (\text{C.11})$$

The result is:

$$x = \left[2\zeta^2 + 1 + \sqrt{(2\zeta^2 + 1)^2 + 1} \right]^{1/2} \quad (\text{C.12})$$

6592 or

$$\omega_{3\text{dB}} = \omega_n \left[2\zeta^2 + 1 + \sqrt{(2\zeta^2 + 1)^2 + 1} \right]^{1/2} \quad (\text{C.13})$$

6593

6594 The gain peaking is the maximum value of the frequency response, in dB. It is computed by
 6595 differentiating Formula (C.9) with respect to x , setting the result to zero, solving for x , and then
 6596 substituting this value of x into Formula (C.9) to obtain the maximum. The result is:

$$H_p = \left[1 - 2\alpha - 2\alpha^2 + 2\alpha(2\alpha + \alpha^2)^{1/2} \right]^{-1/2} \quad (\text{C.14})$$

6597

6598 where α is related to damping ratio by:

$$\alpha = \frac{1}{4\zeta^2} \quad (\text{C.15})$$

6599

6600 and H_p is the gain peaking expressed as a pure fraction. The gain peaking in dB is equal to
 6601 $20 \cdot \log_{10} H_p$. In some cases, it is necessary to compute damping ratio from gain peaking. The
 6602 result for this is:

$$\alpha = \frac{(1-q)(1+\sqrt{1-q})}{2q} \quad (\text{C.16})$$

6603

6604 where

$$q = \frac{1}{H_p^2} \quad (\text{C.17})$$

6605

6606 Damping ratio is obtained from α using Formula (C.15).

6607 If 3 dB bandwidth and gain peaking are given, damping ratio can be obtained using Formulas
 6608 (C.15), (C.16) and (C.17). Undamped natural frequency can then be obtained using Formula
 6609 (C.13).

6610 Figure C.4 shows gain peaking, expressed as a pure fraction, as a function of damping ratio.
 6611 Figure C.5 shows gain peaking in dB as a function of damping ratio.

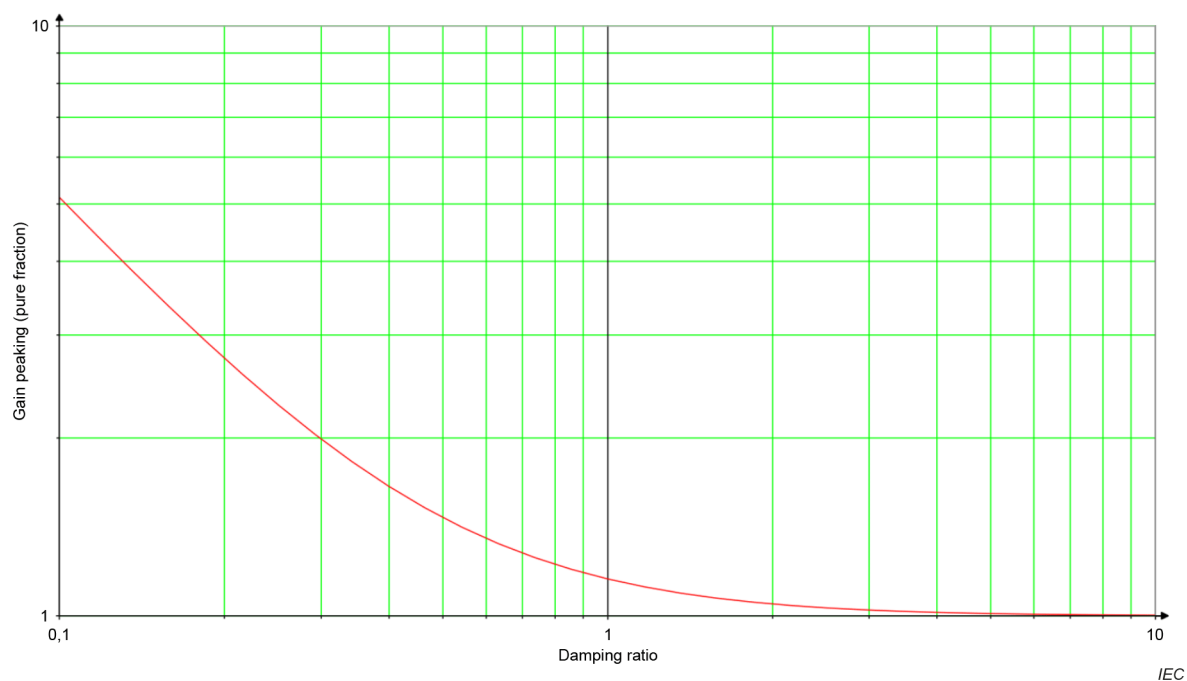


Figure C.4 – Gain peaking (pure fraction) as a function of damping ratio

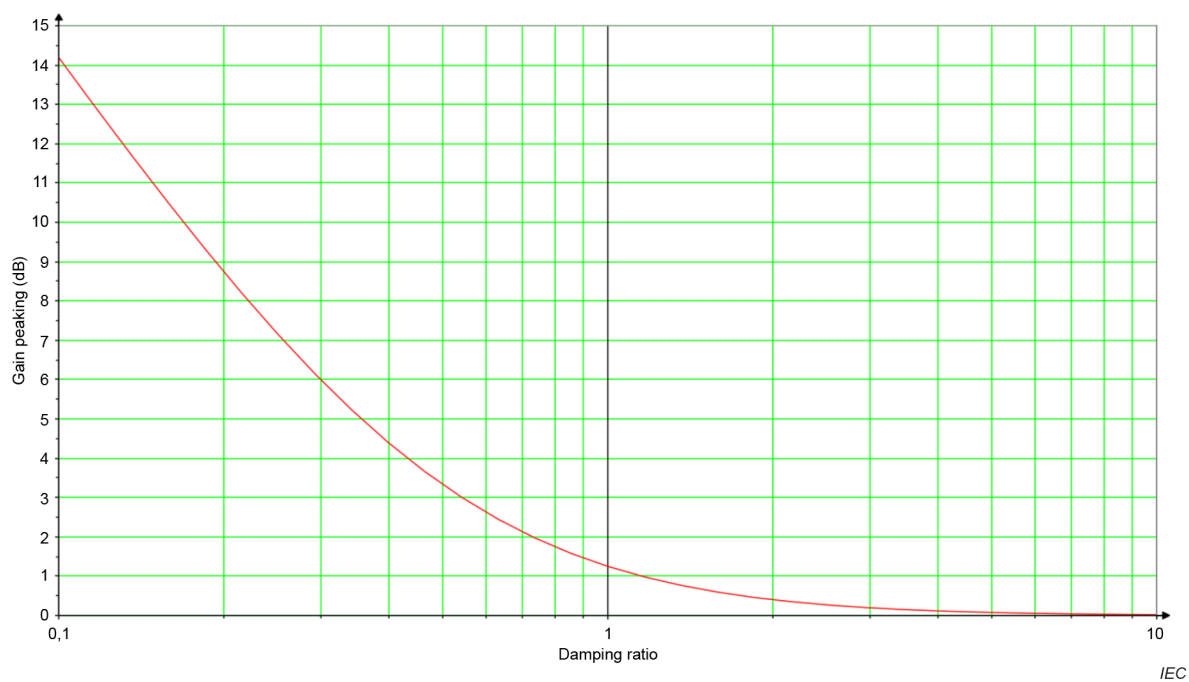


Figure C.5 – Gain peaking in dB as a function of damping ratio

The performance of the clock control system can be described using the frequency response as follows:

- a) Maximum 3 dB bandwidth in Hz,
- b) Maximum gain peaking in dB, and
- c) Frequency response plot (mask) corresponding to (a) and (b) that is not to be exceeded.

6621 C.4 Example

6622 Consider a clock control system with $K_p K_o = 4,23 \text{ rad/s}$ and $K_i K_o = 9,62 (\text{rad/s})^2$. The undamped
6623 natural frequency and damping ratio are:

$$\omega_n = \sqrt{K_i K_o} = \sqrt{9,62 (\text{rad/s})^2} = 3,10 \text{ rad/s} \quad (\text{C.18})$$

$$\zeta = \frac{K_p K_o}{2\sqrt{K_i K_o}} = \frac{4,23 \text{ rad/s}}{2\sqrt{9,62 (\text{rad/s})^2}} = 0,682 \quad (\text{C.19})$$

6624

6625 The gain peaking is obtained from:

$$\alpha = \frac{1}{4(0,682)^2} = 0,537 \quad (\text{C.20})$$

$$\begin{aligned} H_p (\text{purefraction}) &= \left[1 - 2(0,537) - 2(0,537)^2 + 2(0,537)\sqrt{2(0,537) + (0,537)^2} \right]^{-\frac{1}{2}} \\ &= 1,288 \text{ 03.} \end{aligned} \quad (\text{C.21})$$

$$H_p (\text{dB}) = 20 \log_{10}(1,29) \text{ dB} = 2,2 \text{ dB} \quad (\text{C.22})$$

6626

6627 The 3dB bandwidth is:

$$f_{3\text{dB}} (\text{Hz}) = \frac{\omega_n}{2\pi} \left[1 + 2\zeta^2 + \sqrt{(1 + 2\zeta^2)^2 + 1} \right]^{1/2} \quad (\text{C.23})$$

$$\begin{aligned} &= \frac{3,10}{2\pi} \left[1 + 2(0,682)^2 + \sqrt{(1 + 2(0,682)^2)^2 + 1} \right]^{1/2} . \quad = 1,0 \text{ Hz} \end{aligned}$$

6629

6630 The frequency response is shown in Figure C.6.

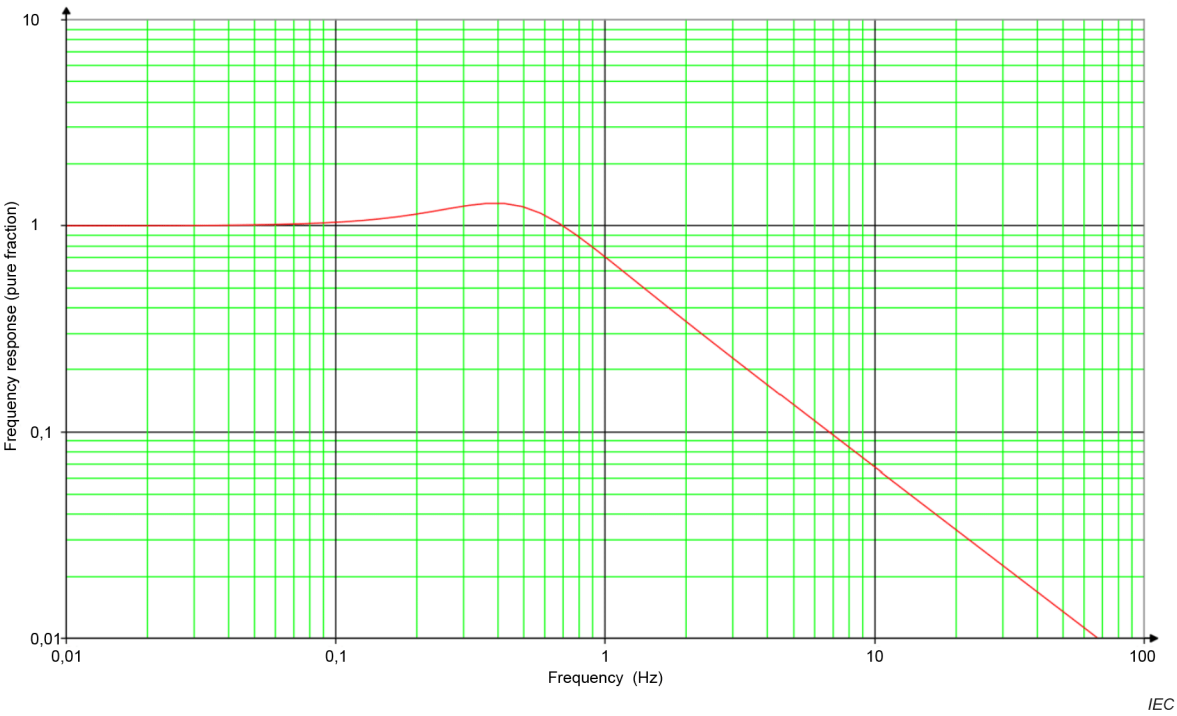


Figure C.6 – Example frequency response

Annex D (informative)

Time synchronization annex

D.1 Overview

Annex D describes how a network of compliant devices can achieve a time synchronization accuracy, at the application level, of $\pm 1 \mu\text{s}$, relative to the Clock Source at the Grandmaster, over 100 network hops. To achieve this, it allocates the overall error budget of 1 000 ns as described in Table D.1.

Table D.1 – Time synchronization error budget

Network Aspect	Error Type	Network-Level Error Budget (ns)
All PTP Instances	Constant time error	200
	Dynamic time error	600
All PTP links	Constant time error	200
	Dynamic time error	

A chain of 1 Grandmaster PTP Instance, 99 PTP Relay Instances and 1 PTP End Instance (100 network hops) that all comply with the normative requirements of 6.2.2, 6.2.3, 6.2.4, and 6.2.5 generates a network-level Time Error at or below the Error Budget for All PTP Instances.

Clause D.2 describes the principles of operation this document assumes.

Clause D.3 provides additional information on specific normative requirements.

The principles of operation include the use of crystal oscillators (XOs) as opposed to more accurate, stable, and costly options such as temperature-compensated crystal oscillators (TCXOs).

Clause D.4 describes a potential approach to testing the normative requirements. It is not a test specification, but rather a high-level overview of one potential approach that can be adopted by a full test specification.

The use of XOs means that some of the normative requirements are difficult or impossible to meet without employing algorithms that track Neighbor Rate Ratio (NRR) drift and Rate Ratio drift and compensate for consequent errors in calculating Rate Ratio and Correction Field.

Clause D.5 provides examples of algorithms that can be used for this purpose, and which have been shown to enable compliance with the normative requirements.

Implementations that employ TCXOs or other more accurate, stable oscillators can still find some of the normative requirements difficult or impossible to meet without employing algorithms to track and compensate for errors due to clock drift. This is because other PTP Instances that use XOs can still cause the implementation to experience Neighbor Rate Ratio drift, Rate Ratio drift or both.

There is no normative requirement to use the algorithms described in Clause D.5; an implementation can employ alternative algorithms provided the normative requirements are met. Clause D.5 describes the potential risks of deploying a network whose instances employ a mix of different algorithms. It is the responsibility of implementers to mitigate the risks and ensure alternative algorithms deliver the desired network-level performance.

This document does not include normative requirements for PTP Links. Annex D.2.4 describes PTP Link characteristics that influence achieving 1 μ s time synchronization accuracy. It includes some examples using common PTP Link characteristics.

This document's normative requirements regarding instance-level error generation are intended to ensure not just an overall level of dTE generation at each node, but also the performance of drift tracking and error compensation algorithms and the amount of dTE generation due to timestamp error versus clock drift. The algorithms are employed to mitigate errors due to clock drift but cannot mitigate timestamp errors.

D.2 Principles of operation

D.2.1 General

Achieving ± 1 μ s time synchronization accuracy across 100 network hops involves managing the accumulation of errors in the preciseOriginTimestamp plus correctionField and the Rate Ratio as they are passed, via Sync or Follow_Up messages, down the chain of PTP Instances and are then used by the PTP End Instance to keep its ClockTarget in line with the ClockSource at the Grandmaster PTP Instance. The majority of significant errors can ultimately be traced back to one of three sources: timestamp error, clock drift, or path delay asymmetry. The selection of PTP protocol parameters often involves trading off one source of error against the other. This document requires specific PTP protocol configurations, and assumes the use of mechanisms (algorithms), that reduce dTE due to timestamp error but would also – without additional measures – increase dTE due to clock drift to the point where the latter exceeds the allocated error budget. However, this document also assumes additional measures to minimize some sources of dTE due to clock drift, and mechanisms and to track and compensate for errors from other sources to a sufficient degree that the error budget is not exceeded.

The specific protocol configurations and other measures, along with their intended effects, are described in Table D.2.

Table D.2 – Protocol configurations & other measures to achieve dTE budget

Configuration or measure	Description and intended effect(s)
Sync Interval 125 ms	Effects: 1) Calibrates the balance between dTE from timestamp error vs error due to clock drift. Larger intervals lead to less timestamp error and more error due to clock drift. 2) Keeps below acceptable limits the impact of errors in Rate Ratio and Rate Ratio Drift estimation when keeping ClockTarget in line with ClockSource between arrival of Sync messages. Larger intervals increase the impact of any errors.
Drift_Tracking TLV – syncEgressTimestamp	Effect: Enables calculation of NRR using Sync message timestamps, which eliminates error due to NRR clock drift that would otherwise occur between calculation of NRR using Pdelay_Resp messages and use during Sync message processing (i.e. calculation of Rate Ratio and output Correction Field values).

Configuration or measure	Description and intended effect(s)
NRR Smoothing	Description: Algorithm to use timestamps from multiple past Sync messages when calculating NRR. Effect: Reduces the amount of error in the estimate of NRR due to timestamp error while increasing the amount of error due to clock drift.
NRR Drift Tracking & Compensation	Description: Algorithm to use timestamps from multiple past Sync messages to estimate NRR drift rate and then apply compensation to correct for consequent errors in NRR calculation. Effect: Mitigates the effect of errors due to clock drift when calculating and using the estimated NRR.
Drift_Tracking TLV – rateRatioDrift	Description: Carries estimate of Rate Ratio drift rate from one node to the next. Effect: Allows each node to estimate its own Rate Ratio drift rate by combining the incoming Rate Ratio drift rate with the local estimate of NRR drift rate.
RR Drift Compensation	Description: Algorithm that uses the estimate of RR drift rate to compensate for that drift, adjusting the estimated RR over time according to the drift rate. Effect: For PTP Relay Instances, minimizes errors in the Correction Field caused by Rate Ratio drift. For PTP End Instances, a similar approach can reduce errors in keeping ClockTarget in line with ClockSource between arrival of Sync messages, but is outside the scope of this document.
Pdelay Interval Consistency	Description: This document requires tighter control of the interval between Pdelay messages generated at the Grandmaster PTP Instance than the defaults in IEEE Std 802.1AS-2020. Effect: This document requires the use of Sync messages to calculate NRR (see above). However, when a sufficient number of Sync messages are not available, for example on startup or after a reconfiguration, Pdelay_Resp messages can be used instead. In such cases, errors due to clock drift at Relay Instances have a tendency to cancel out. A clock drift that generates a positive error in NRR measurement on receipt of a Pdelay_Resp message generates a negative error in NRR measurement at the next node. The degree of cancellation depends on the consistency of the intervals over which NRR is measured at neighboring nodes. Tighter control of the Pdelay Interval increases the consistency of the measurement interval and thus decreases the amount of error.
Mean Residence Time	Description: This document defines a mean Residence Time requirement, which is significantly lower than the default maximum Residence Time in IEEE Std 802.1AS-2020. Effect: The amount of error in the Correction Field at the PTP End Instance due to clock drift is proportional to the cumulative meanLinkDelay and residenceTime experienced by a Sync message during transit from the Grandmaster PTP Instance to the PTP End Instance. Specifying a lower mean residenceTime reduces this source of error.

D.2.2 Grandmaster PTP Instance implementation

Depending on implementation, a Grandmaster PTP Instance can:

- a) contain a single oscillator used for both Local Clock and Clock Source,
- b) contain separate oscillators for Local Clock and Clock Source, or
- c) contain only an oscillator for Local Clock and accept an external input for Clock Source.

In some cases, a Grandmaster PTP Instance can support more than one mode of operation and transition between them depending on changes in network configuration (see Splitting, Joining and Aligning Time Domains).

In the first case, the rateRatio transmitted by the Grandmaster PTP Instance is one (i.e. 0 ppm) the rateRatioDrift is 0 ppm/s, reflecting the fact there is no difference between the Local Clock and Clock Source frequencies.

In the second and third cases there can be differences between the Local Clock and Clock Source frequencies. Any differences are reflected in the rateRatio and rateRatioDrift fields transmitted by the Grandmaster PTP Instance. This means that the Grandmaster PTP Instance tracks rateRatio over time in order to calculate rateRatioDrift, similarly to PTP Relay Instances and PTP End Instances. The exact implementation can vary.

D.2.3 Splitting, joining and aligning time domains

Modular machines or production cells can allow the splitting and combining of machines if this is required by the production process. When separate, the ClockSources of two machines run separately, each with its own time domain. If both ClockSources are traceable to the same PTP timescale, the difference between the ClockSources can be relatively small. If traceable to different timescales, especially if one or both are ARB timescales, there can be a very large difference between the ClockSources.

When two machines are joined, the first machine's time domain remains unaffected, and it can continue operation without disruption. There are two typical approaches to how the second machine behaves. In the first case, at a time of the end user's choosing, the second machine's time domain ceases to exist, with its PTP Instances becoming part of the first machine's time domain. In the second case, the second machine's time domain is gradually aligned with the first machine's time domain such that control loop cycles are coordinated. In the first case the second machine's time domain is unaffected, and it can continue production even if the machines are accidentally connected, until the end user chooses to join the time domains. In the second case the second machine can continue production while its time domain is being aligned.

D.2.3.1 Joining machines with single time domain

In the first case, where the second machine's time domain ceases to exist, a discontinuity in timing for the second machine's PTP Instances can occur, as they switch to use the first machine's Grandmaster. Some implementations implement measures to limit such timing discontinuities, but these measures are outside the scope of this document. Typically, in this case, the second machine is not operational while it is joined to the first. It resumes operation once its PTP Instances have synchronized with the first machine's Grandmaster.

D.2.3.2 Joining machines with multiple coordinated time domains

In the second case, where the second machine's time domain is gradually aligned with the first machine's time domain, this typically requires both machines to be implementing the same control loop cycle time. The goal is that, once coordinated, each control loop cycle of the first machine is aligned with the start of a control loop cycle of the second machine, even though the two machines maintain separate time domains and there can be a large time difference between their Clock Sources.

6745 In this case, after being joined together, the first machine effectively drives the second
6746 machine's Clock Source faster or slower, during an alignment period, until coordination is
6747 achieved. During the alignment period, this drive from the first machine can result in the second
6748 machine's Clock Source temporarily exceeding the usual normative requirement on range of
6749 fractional frequency offset relative to the nominal frequency of ± 50 ppm. The usual normative
6750 requirement on range of rate of change of fractional frequency offset of ± 1 ppm/s, applicable
6751 when split (i.e. independent) or coordinated (i.e. joined and stable, after the alignment period),
6752 may also be temporarily exceeded. However, if the value stays within the range ± 3 ppm/s, the
6753 network-level performance of 1 μ s time synchronization accuracy can be maintained. For this
6754 reason, this document specifies a separate normative requirement for temporary, externally
6755 driven, rate of rate of change of fractional frequency offset.

6756 Since the second machine experiences no time discontinuities and the network-level
6757 performance is maintained the second machine can, if desired, continue operation during the
6758 alignment period.

6759 Once coordinated, the first machine continues to drive the Clock Source of the second machine
6760 to maintain coordination. In this stable, coordinated mode of operation the normal range of
6761 ± 1 ppm/s is not exceeded.

6762 The mechanism by which the first machine drives the Clock Source of the second machine is
6763 not addressed in this document.

6764 **D.2.3.3 Splitting machines**

6765 In the first case, where the second machine's time domain ceased to exist while joined to the
6766 first, splitting machines means that the second machine must create its own time domain again.
6767 The second machine's Clock Source typically starts at the PTP Grandmaster Instance's last,
6768 best estimate of the first machine's Clock Source. The goal is for no discontinuities in time sync
6769 to occur; however, depending on implementation, it can take some time to before the time
6770 synchronization accuracy of all the second machine's PTP Instances relative to its Grandmaster
6771 can be relied upon. For this reason, it is possible the second machine is not operational during
6772 the split. Hot Standby can be employed to mitigate this transition time, but the details of how to
6773 do so are out of scope for this document.

6774 In the second case, where the second machine maintains its time domain while joined to the
6775 first, splitting machines means that the first machine ceases driving the second machine's Clock
6776 Source to maintain coordination of control loop cycle times. Without this drive, the two time
6777 domains can drift relative to each other resulting in loss of coordination. Time synchronization
6778 performance within the second machine is maintained during the split and the second machine
6779 can, if desired, continue operation throughout the process.

6780 **D.2.4 PTP link characteristics**

6781 A vast majority of time synchronization error due to PTP link characteristics is caused by
6782 asymmetrical path delay in one direction versus the other. The mechanism to measure path
6783 delay assumes the link is symmetrical and cannot detect asymmetry, thus asymmetry causes
6784 an error. The potential maximum asymmetry and thus error typically scales linearly with physical
6785 path length.

6786 The error budget due to PTP link characteristics for an entire network is 200 ns. In any specific
6787 network this budget can be allocated as required with some links allocated a higher budget
6788 (typically longer length) than others.

6789 A typical specified maximum delay skew for Category 6 Ethernet cables is 50 ns per 100 m.
6790 Some Ethernet MAU types such as 100BASE-TX use different pairs for transmit and receive. If
6791 such cables are used, a maximum total cable length between Clock Source and Clock Target
6792 with 99 PTP Relay Instances between them (i.e. 100 network hops) is 400 m. Extending the

6793 cable length beyond 400 m without jeopardizing network-level performance would require the
6794 use of cables with less delay skew or asymmetry compensation for delay skew.

6795 It is possible for the delay skew in one section of cable to cancel all or part of a delay skew in
6796 the opposite direction from prior section but, depending on how cables are manufactured and
6797 deployed, it is feasible for the delay skews of every cable segment between a Grandmaster
6798 PTP Instance and a PTP End Instance to be additive.

6799 **D.3 Notes on normative requirements**

6800 **D.3.1 Oscillator requirements**

6801 Clock drift at the Grandmaster PTP Instance causes greater dTE than the same amount of clock
6802 drift at a PTP Relay Instance or the PTP End Instance. This document therefore requires tighter
6803 limits on maximum fractional frequency offset for an oscillator at the Grandmaster PTP Instance
6804 than at other instances.

6805 This document does not place requirements on operational temperature range or other
6806 environmental factors. The required oscillator behavior is delivered for the operational
6807 conditions across which a device claims it is compliant. These conditions typically include
6808 temperature range but can also include rate of change of ambient temperature, supply voltage
6809 stability, amount of vibration and others.

6810 **D.3.2 Timestamp Granularity Error**

6811 Timestamp Granularity Error (TSGE) is the error in timestamping each incoming and outgoing
6812 message due to the maximum timestamp resolution of which an implementation is capable. It
6813 is typically directly related to an implementation's clock rate.

6814 For example: a clock rate of 125 MHz typically results in a maximum resolution of 8 ns.
6815 Depending on implementation the consequent TSGE range can be –8 ns to 0 ns, 0 ns to 8 ns,
6816 or anything in between. In some implementations, offsets are applied to ensure the average
6817 TSGE is 0 ns with, assuming uniform error distribution, a range of –4 ns to +4 ns

6818 Similarly, a clock rate of 500 MHz results in a maximum resolution of 2 ns; a consequent TSGE
6819 range between –2 ns to 0 ns and 0 ns to 2 ns; and, if a suitable offset is applied to ensure a
6820 TSGE average of 0 ns, a range of –1 ns to +1 ns.

6821 A minimum resolution of 8 ns, i.e. minimum clock rate of 125 MHz is assumed. It is further
6822 assumed that TSGE for the sum of the preciseOriginTimestamp and followUpCorrectionField at
6823 the Grandmaster PTP Instance (see IEEE Std 802.1AS-2020 as amended by IEEE Std
6824 802.1ASdm-2024, 10.2.9.2.1) has an average of 0 ns and that the TSGE averages for other
6825 timestamps are stable and consistent across all a PTP Instance's ports. No assumption
6826 regarding the value of the TSGE average for these other timestamps is necessary as they are
6827 always used to measure intervals such that any stable, consistent offset cancels out.

6828 **D.3.3 Dynamic Timestamp Error**

6829 Dynamic Timestamp Error (DTSE) is the, effectively random, error in timestamping each
6830 incoming and outgoing event message due to an implementation's inherent inaccuracies,
6831 excluding TSGE. It is assumed to vary between a minimum of –6 ns and a maximum of + 6 ns
6832 with an average of 0 ns. Lower levels of DTSE are better.

6833 If an implementation timestamps an incoming or outgoing message at a point other than as
6834 specified in IEEE Std 802.3-2022 as amended by IEEE Std 802.3cx-2023 and IEEE Std
6835 802.3de-2022, any variability in delay between that point and the delay point specified in 802.3
6836 translates to DTSE. Some common implementations were not designed to limit this variability.
6837 If care is not taken to avoid implementations with high variability, the assumed DTSE range is

6838 easily exceeded. Such implementations can find some of the normative requirements difficult
6839 or impossible to meet.

6840 D.3.4 Grandmaster PTP Instance error generation

6841 Table 12 sets normative requirements for error generation at a Grandmaster PTP Instance that
6842 ensure the relevant fields in the Sync and Follow_Up messages it transmits are sufficiently
6843 accurate to deliver the network-level performance. Table D.3 describes how the normative
6844 requirements align with major sources of error.

6845 **Table D.3 – Main sources of error affecting normative requirements for Grandmaster**
6846 **PTP Instance**

Item	Normative requirement	Main sources of error
1	preciseOriginTimestamp + correctionField vs Direct measurement of Working Clock at Grandmaster (acting as a Clock Source)	Timestamp error relative to Clock Source plus accuracy measuring any internal delay between generation of the preciseOriginTimestamp and Sync message transmission.
2	rateRatio vs Direct measurement of Rate Ratio of Clock Source vs Local Clock	Accuracy of internal mechanism to measure Rate Ratio of Clock Source vs. Local Clock, potentially including algorithms that track RateRatioDrift and modify Rate Ratio accordingly ^a
3	syncEgressTimestamp vs Direct measurement of Local Clock	Timestamp error relative to Local Clock
4	(responseOriginTimestamp – requestReceiptTimestamp) vs Direct measurement of Local Clock interval between reception of Pdelay_Req and transmission of Pdelay_Resp	Timestamp error relative to Local Clock
^a Only applicable if Clock Source and Local Clock are not locked to the same frequency by the implementation. If they are locked, then rateRatio is 0 ppm and rateRatioDrift is 0 ppm/s.		

6847

6848 D.3.5 PTP Relay Instance error generation

6849 Table 13 sets normative requirements for error generation at a PTP Relay Instance that ensure
6850 the relevant fields in the Sync and Follow_Up messages it transmits as part of Sync processing
6851 are sufficiently accurate to deliver the network-level time sync performance. The requirements
6852 include the ability to mitigate errors in rateRatio and rateRatio drift that would otherwise occur
6853 due to clock drift at the current PTP Relay Instance, an adjacent PTP Relay Instance, or the
6854 Grandmaster PTP Instance. Table D.4 describes how the normative requirements align with
6855 major sources of error.

6856 **Table D.4 – Main sources of error affecting normative requirements for PTP Relay**
6857 **Instance**

Item	Normative requirement	Clock drifts	Main sources of error
1	preciseOriginTimestamp + correctionField vs Direct measurement of Clock Source at Grandmaster PTP Instance	None	Timestamp errors relative to Local Clock when measuring Residence Time, i.e. Sync message ingress and egress. Accuracy of meanLinkDelay measurement. Errors in Rate Ratio used when translating Residence Time measured in terms of Local Clock to Residence time in terms of Clock Source, although these are typically orders of magnitude smaller than those from Timestamp Errors.

Item	Normative requirement	Clock drifts	Main sources of error
2	rateRatio vs Direct measurement of Rate Ratio of Clock Source vs Local Clock	None	Timestamp error affecting measurement of NRR when there is no NRR Drift. The effect should be low. This normative requirement is a baseline for the next two requirements.
3		Clock Source (RR Drift)	Accuracy of measurement of NRR when there is no NRR Drift (as above). Accuracy of calculation of rateRatio, including algorithms for RR Drift tracking & error compensation.
4		Clock Source and Local Clock at previous PTP Instance (RR Drift & NRR drift)	Accuracy of measurement of NRR when there is NRR Drift, including algorithms for NRR Drift tracking & error compensation Accuracy of calculation of rateRatio, including algorithms for RR Drift tracking & error compensation. Combined with test 3 this effectively requires a level of performance regarding NRR Drift tracking & error compensation, whether the source of the NRR drift is the Local Clock of the current PTP Instance or the previous PTP Instance.
5	rateRatioDrift vs Direct measurement of Rate Ratio Drift of Clock Source vs Local Clock	None	Timestamp error affecting measurement of NRR Drift when there is no NRR Drift. The effect should be low. This normative requirement is a baseline for the next two requirements.
6		Clock Source (RR Drift)	Accuracy of measurement of NRR Drift when there is no NRR Drift (as above). Accuracy of calculation of rateRatioDrift, including algorithms for RR Drift tracking & error compensation.
7		Clock Source and Local Clock at previous PTP Instance (RR Drift & NRR drift)	Accuracy of measurement of NRR Drift when there is NRR Drift, including algorithms for NRR Drift tracking & error compensation. Accuracy of calculation of rateRatioDrift, including algorithms for RR Drift tracking & error compensation. Combined with test 6 this effectively requires a level of performance regarding NRR Drift tracking & error compensation, whether the source of the NRR drift is the Local Clock of the current PTP Instance or the previous PTP Instance.
8	syncEgressTimestamp vs Direct measurement of Local Clock	None	Timestamp error relative to Local Clock
9	(responseOriginTimestamp – requestReceiptTimestamp) vs Direct measurement of Local Clock interval between reception of Pdelay_Req and transmission of Pdelay_Resp	None	Timestamp error relative to Local Clock
10	Mean Link Delay	None	Timestamp error affecting measurement of meanLinkDelay. Each Pdelay measurement can be subject to relatively large errors from both the local and neighboring node. This normative requirement ensures a suitable algorithm

Item	Normative requirement	Clock drifts	Main sources of error
			mitigates the impact of any such single Pdelay measurement.

6858

6859 D.3.6 PTP End Instance error generation

6860 Table 14 sets normative requirements for error generation at a PTP End Instance that ensure
 6861 the ClockTarget it generates from incoming Sync and Follow_Up messages is sufficiently
 6862 accurate to deliver the network-level time sync performance. Table D.5 describes how the
 6863 normative requirements align with major sources of error.

6864 **Table D.5 – Main sources of error affecting normative requirements for PTP End**
 6865 **Instance**

Item	Normative requirement	Clock drifts	Main sources of error
1	ClockTarget vs ClockSource	None	Timestamp error affecting measurement of NRR Drift when there is no NRR Drift. The effect should be low. This normative requirement is a baseline for the next two tests.
2		Clock Source (RR Drift)	Accuracy of measurement of NRR Drift when there is no NRR Drift (as above). Accuracy of calculation of rateRatioDrift, including algorithms for RR Drift tracking & error compensation.
3		Clock Source and Local Clock at previous PTP Instance (RR Drift & NRR drift)	Accuracy of measurement of NRR Drift when there is NRR Drift, including algorithms for NRR Drift tracking & error compensation. Accuracy of calculation of rateRatioDrift, including algorithms for RR Drift tracking & error compensation. Combined with test 2 this effectively requires a level of performance regarding NRR Drift tracking & error compensation, whether the source of the NRR drift is the Local Clock of the current PTP Instance or the previous PTP Instance.
4	Mean Link Delay	None	Timestamp error affecting measurement of meanLinkDelay. Each Pdelay measurement can be subject to relatively large errors from both the local and neighboring node. This normative requirement ensures a suitable algorithm mitigates the impact of any such single Pdelay measurement.

6866

6867 D.3.7 Explanation for the asymmetric normative requirements for the allowable range 6868 of dTE in Table 14, rows 2 and 3

6869 The PTP End Instance requirements for dTE in Table 14, rows 2 and 3, are asymmetric, i.e.,
 6870 the allowable range in both cases is -145 ns to +15 ns. These tests are performed with the
 6871 fractional frequency offset of the Working Clock (acting as the ClockSource) at the Grandmaster
 6872 PTP Instance increasing at the rate of 1 ppm/s. dTE is measured at the output of the PTP End
 6873 Instance clock control system after steady-state is reached, i.e., after any transients due to
 6874 initiating the 1 ppm/s frequency drift rate have decayed. The clock control system is a second-
 6875 order filter, with bandwidth and gain peaking requirements given in Table 11. Table 11 also
 6876 indicates that the filter has a minimum roll-off of 20 dB/decade. The worst-case dTE response
 6877 can be analyzed using the second-order filter model described in Annex C. The transfer function
 6878 for this filter is given by Eq. (C.5). The example in C.4 shows that the damping ratio, ζ , for a

6879 filter whose gain peaking is 2.2 dB, i.e., the maximum allowed in Table 11, is 0.682 (see Eq.
6880 (C.19)). The allowable range for filter 3dB bandwidth in Table 11.

6881 is 0.9 Hz – 1.0 Hz. The example in C.4 indicates that the undamped natural frequency, ω_n
6882 corresponding to the maximum bandwidth of 1.0 Hz is 3.10 rad/s (see Eq. (C.18)).

6883 It is well-known that if a frequency drift is applied to a linear, second-order filter with 20
6884 dB/decade roll-off, the steady-state output has a time offset. This means that if the input to the
6885 PTP End Instance from the ClockSource (see Figure D.5) has a frequency drift of 1 ppm/s, the
6886 measured dTE output of the ClockTarget has a steady-state offset. Any errors due to other
6887 sources (e.g., timestamp granularity, dynamic timestamp error, clock noise) are in addition to
6888 this. If these other errors are symmetric about zero, the addition of the steady-state offset
6889 results in an asymmetric range of dTE.

6890 The requirement in Table 14 is on the Working Clock (acting as ClockTarget) at the PTP End
6891 Instance minus the Working Clock (acting as ClockSource) at the Grandmaster. This difference,
6892 which is the error between the input and output and is denoted by $E(s)$ in the s-domain, is given
6893 by:

$$\begin{aligned} E(s) &= Y(s) - U(s) \\ &= H(s)U(s) - U(s) \\ &= (H(s) - 1)U(s) \\ &= H_e(s)U(s) \end{aligned} \quad (D.1)$$

6894

6895 where $Y(s)$ and $U(s)$ are the Laplace transforms of the output and input, respectively, $H(s)$ is the
6896 transfer function between the input and output and is given by Eq. (C.5), and $H_e(s)$ is the transfer
6897 function between the input and error. Using Eq. (C.5), the error transfer function is given by:

$$H_e(s) = -\frac{s^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (D.2)$$

6898 The phase drift corresponding to an $A = 1$ ppm/s = 1000 ns/s² frequency drift is $0.5At^2$, where t
6899 is the time in seconds and the phase drift is in ns if A is in ns/s². The Laplace transform of this
6900 waveform is $U(s) = A/s^3$. The steady-state value of the filter output, $y_{\text{steady-state}}$, due to this drift
6901 is obtained from the final value theorem:

$$y_{\text{steady-state}} = \lim_{s \rightarrow 0} sH(s)U(s) = \lim_{s \rightarrow 0} s \cdot \left(-\frac{s^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \right) \cdot \frac{A}{s^3} = -\frac{A}{\omega_n^2} \quad (D.3)$$

6902 Since the 3dB bandwidth is proportional to the undamped natural frequency (see Eq. (C.23)),
6903 Eq. (D.32) indicates that the worst case, i.e., largest steady-state error occurs for the minimum
6904 3dB bandwidth, i.e., 0.9 Hz in Table 11. In the example of C.4, the undamped natural frequency
6905 of 3.10 rad/s and damping ratio 0,682 correspond to a 3dB bandwidth of 1,0 Hz. Then, due to
6906 the direct proportionality between 3dB bandwidth and undamped natural frequency (for the
6907 same damping ratio), a 3dB bandwidth of 0.9 Hz corresponds to an undamped natural frequency
6908 of:

$$\omega_n = \left(\frac{0.9 \text{ Hz}}{1.0 \text{ Hz}} \right) (3.10 \text{ rad/s}) = 2.79 \text{ rad/s} \quad (D.4)$$

6909 Then, using Eq. (D.32), the steady-state error corresponding to a frequency drift of 1 ppm/s is:

$$\gamma_{\text{steady-state}} = -\frac{1000 \text{ ns/s}}{(2.79 \text{ rad/s})^2} = -128 \text{ ns} \quad (\text{D.5})$$

6910 Then, assuming the error without the filter is 15 ns as given in Table 14, row 1, the lower end
6911 of the range of dTE when the filter is present is:

$$\text{dTE} = -128 \text{ ns} - 15 \text{ ns} = -143 \text{ ns} \approx -145 \text{ ns} \quad (\text{D.6})$$

6912 **D.4 Approach to testing normative requirements**

6913 **D.4.1 General**

6914 This document does not specify tests to ensure conformance with the requirements. However,
6915 it is important that the requirements are, in principle, testable. Clause D.4 describes, at a high
6916 level, approaches a test specification can take to testing conformance with some of the
6917 requirements related to time synchronization.

6918 It is assumed that test equipment can precisely measure the output of the ClockSource (at a
6919 Grandmaster PTP Instance), ClockTarget (at a PTP End Instance) and Local Clock (at any PTP
6920 Instance) to ensure conformance with frequency offset and frequency offset drift requirements.
6921 This might be via a Pulse per Second (PPS) plus Time-of-Day information or another
6922 mechanism.

6923 It is also assumed that test equipment can generate sequences of PTP messages with precise
6924 timing and content (for testing PTP Relay Instances and PTP End Instances) and receive, log,
6925 and process sequences of PTP messages with precise timing measurement (e.g. of message
6926 arrival).

6927 **D.4.2 Testing Grandmaster PTP Instance**

6928 Figure D.1 illustrates an approach to testing the three normative requirements discussed in
6929 D.3.4.

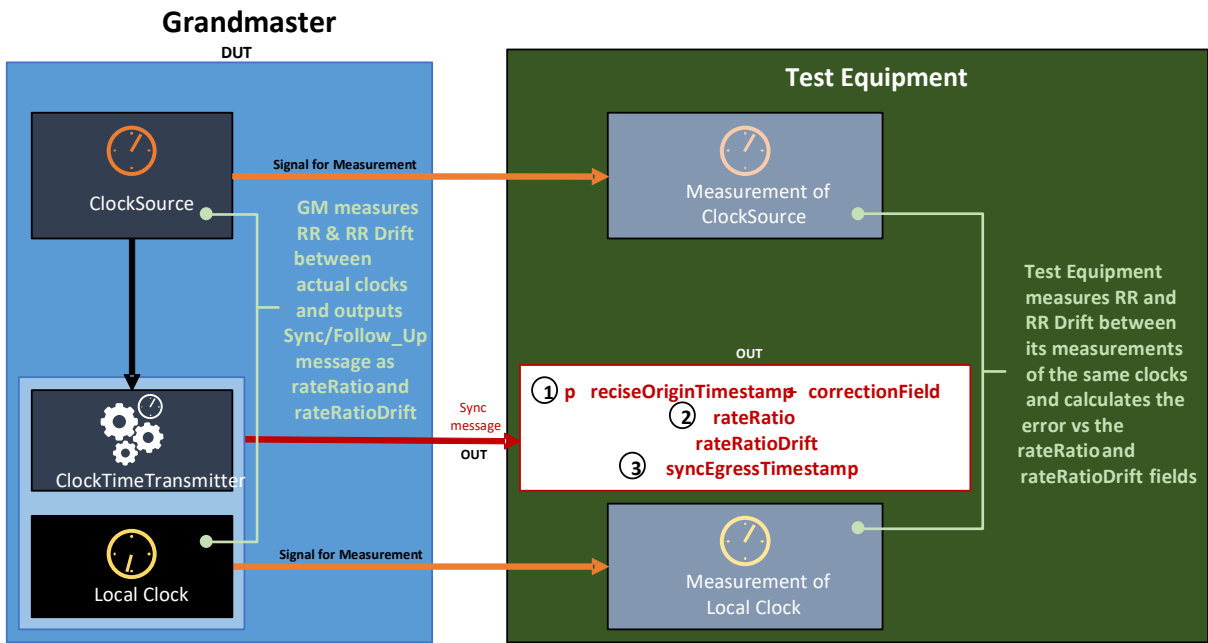


Figure D.1 – Approach to testing normative requirements for Grandmaster PTP Instance

The test equipment can calculate the time the Sync message is output at the DUT by subtracting the link delay from the measured arrival time at the test equipment.

For test 1, the test equipment compares the value of the preciseOriginTimestamp + correctionField against its measurement of the ClockSource.

For test 2, the test equipment compares the value in the rateRatio field with its calculation of the equivalent value based on its measurement of the ClockSource.

For test 3, the test equipment compares the value of the syncEgressTimestamp against its measurement of the Local Clock.

D.4.3 Testing PTP Relay Instance

Figure D.2 illustrates an approach to testing normative requirements 1, 2, 5 and 8 discussed in D.3.5.

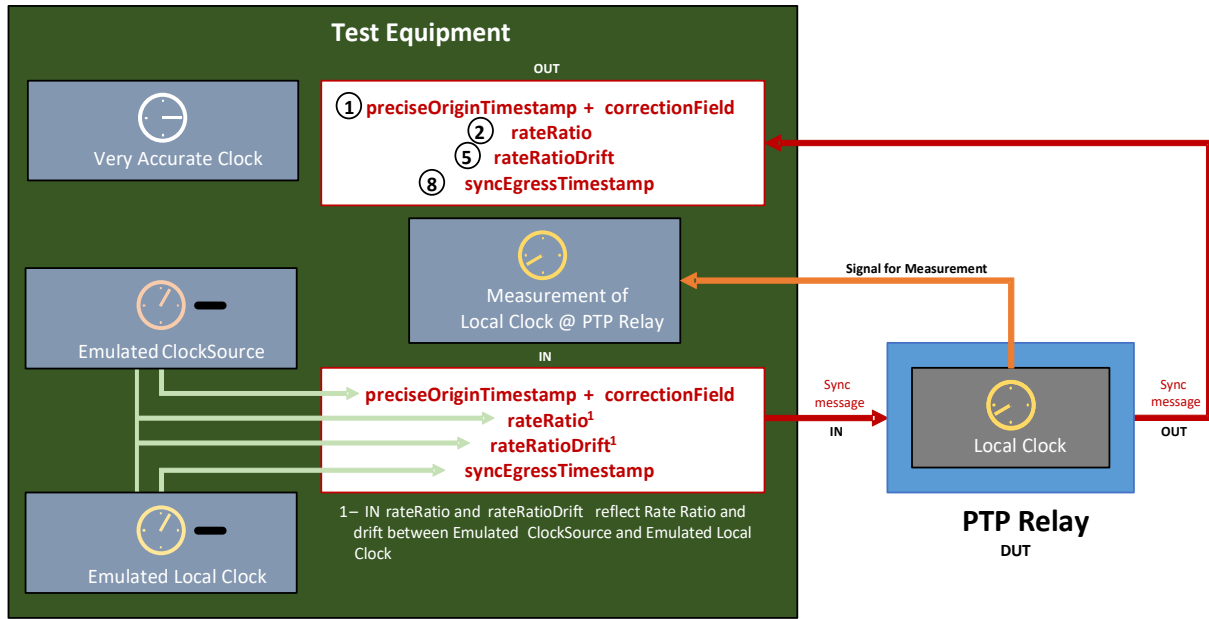


Figure D.2 – Approach to testing normative requirements for PTP Relay Instance - 1

The test equipment can compare the DUT's output Sync message to the expected result given the measurement of the Local Clock and the timing of the input PTP timing message transmission and output PTP timing message reception.

For these four tests, the Emulated ClockSource and Emulated Local Clock are stable and in sync. In practice, both can be equal to the test equipment's Very Accurate Clock. In the input Follow_Up information TLV, rateRatio is 0 ppm, and in the input Drift_Tracking TLV rateRatioDrift is 0 ppm/s. If the Local Clock of the PTP Relay Instance is also stable, it measures NRR of 0 ppm and NRR Drift of 0 ppm/s.

The test equipment can calculate the time the output Sync message is output at the DUT by subtracting the link delay from the measured arrival time at the test equipment.

For test 1, the test equipment can compare the increase in the value of the correctionField to the measured meanLinkDelay (from the test equipment to the DUT) plus residenceTime. The test equipment accounts for the additional delay between the PTP Relay Instance's transmission of the input PTP timing message and its reception by the test equipment.

For tests 2 and 5, the test equipment can compare the rateRatio and rateRatioDrift fields in the output PTP timing message with the equivalent calculated values between the measured Local Clock and the Emulated ClockSource.

For test 8, the test equipment can compare syncEgressTimestamp value in the output PTP timing message with its measurement of the Local Clock.

Figure D.3 illustrates an approach to testing normative requirements 3 and 6 discussed in D.3.5.

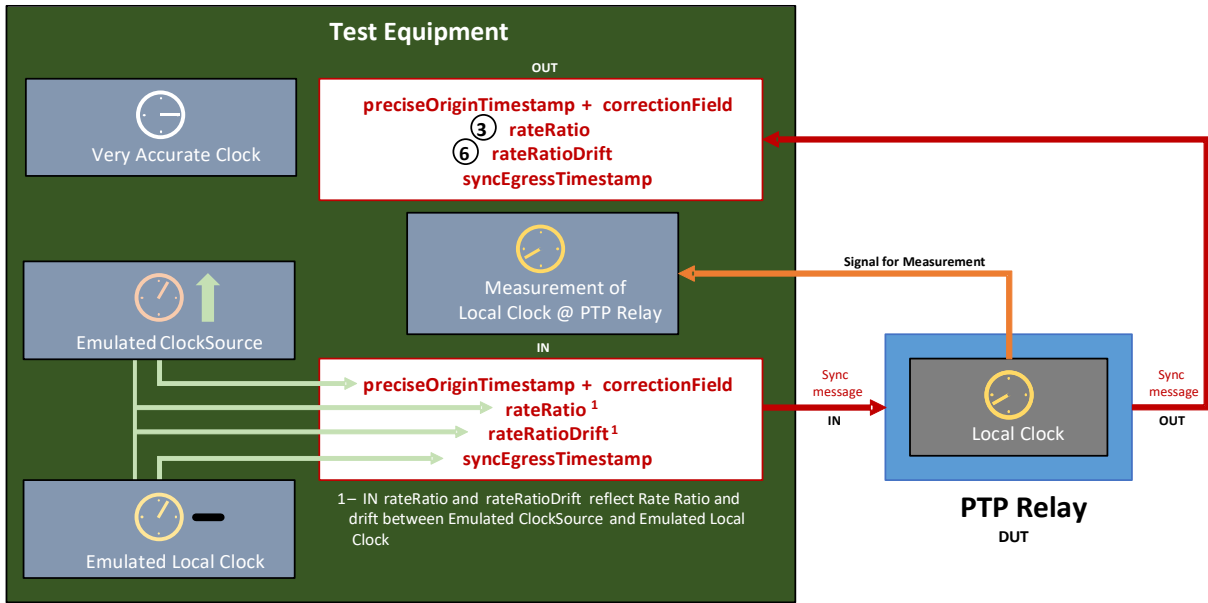


Figure D.3 – Approach to testing normative requirements for PTP Relay Instance – 2

For these two tests, the fractional frequency offset of the Emulated ClockSource is increasing at a defined ppm/s rate relative to the Very Accurate Clock. The Emulated Local Clock is stable; in practice, it can be equal to the test equipment's Very Accurate Clock. In the output Follow_Up information TLV, the rateRatio field increases over time, and in the output Drift_Tracking TLV, the rateRatioDrift field maintains a matching positive value. If the Local Clock of the PTP Relay Instance is also stable, it measures a NRR of 0 ppm and a NRR Drift of 0 ppm/s.

For tests 3 and 6, the test equipment can compare the rateRatio and rateRatioDrift fields in the output Follow_Up information TLV and Drift_Tracking TLV respectively with the equivalent calculated values between the measured Local Clock and the Emulated ClockSource.

Figure D.4 illustrates an approach to testing normative requirements 4 and 7 discussed in D.3.5.

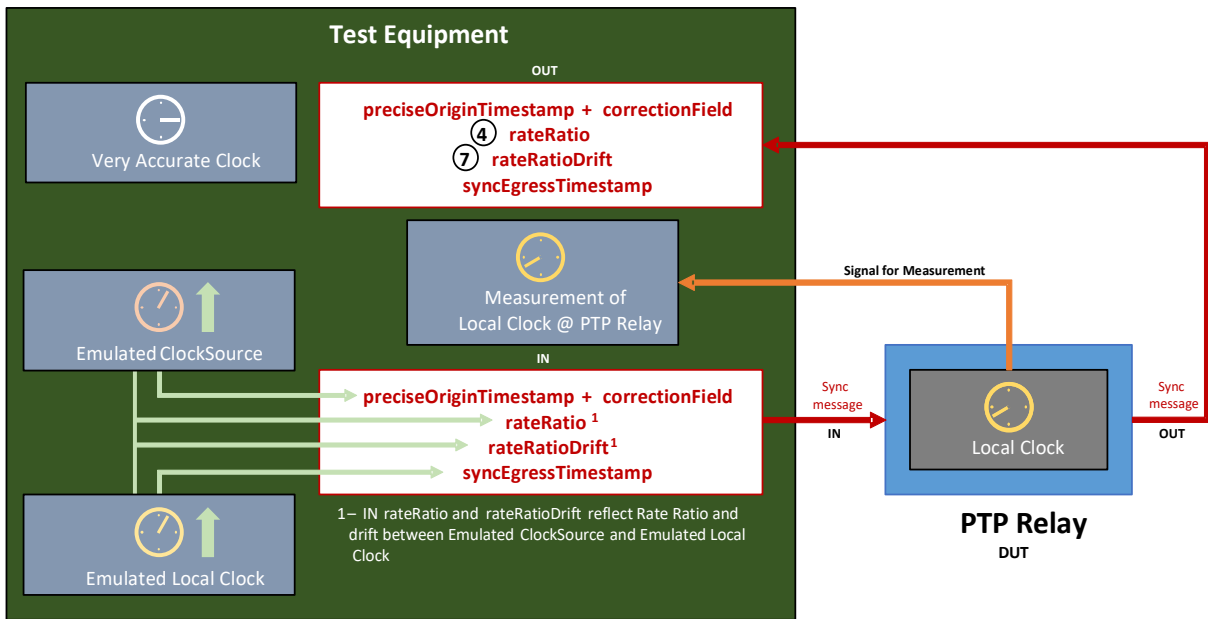


Figure D.4 – Approach to testing normative requirements for PTP Relay Instance – 3

For these two tests, the fractional frequency offsets of the Emulated ClockSource and the Emulated Local Clock are equal and increasing at a defined ppm/s rate relative to the Very Accurate Clock. In the output Follow_Up information TLV, the rateRatio field is 0 ppm, and in the output Drift_Tracking TLV the rateRatioDrift field is 0 ppm/s. If the Local Clock of the PTP Relay Instance is stable, the NRR it measures increases over time and the NRR Drift it measures maintains a matching positive value.

For tests 4 and 7, the test equipment can compare the rateRatio and rateRatioDrift fields in the output Follow_Up information TLV and Drift_Tracking TLV respectively with the equivalent calculated values between the measured Local Clock and the Emulated ClockSource.

D.4.4 Testing PTP End Instance

Figure D.5 illustrates an approach to testing the three normative requirements discussed in D.3.6.

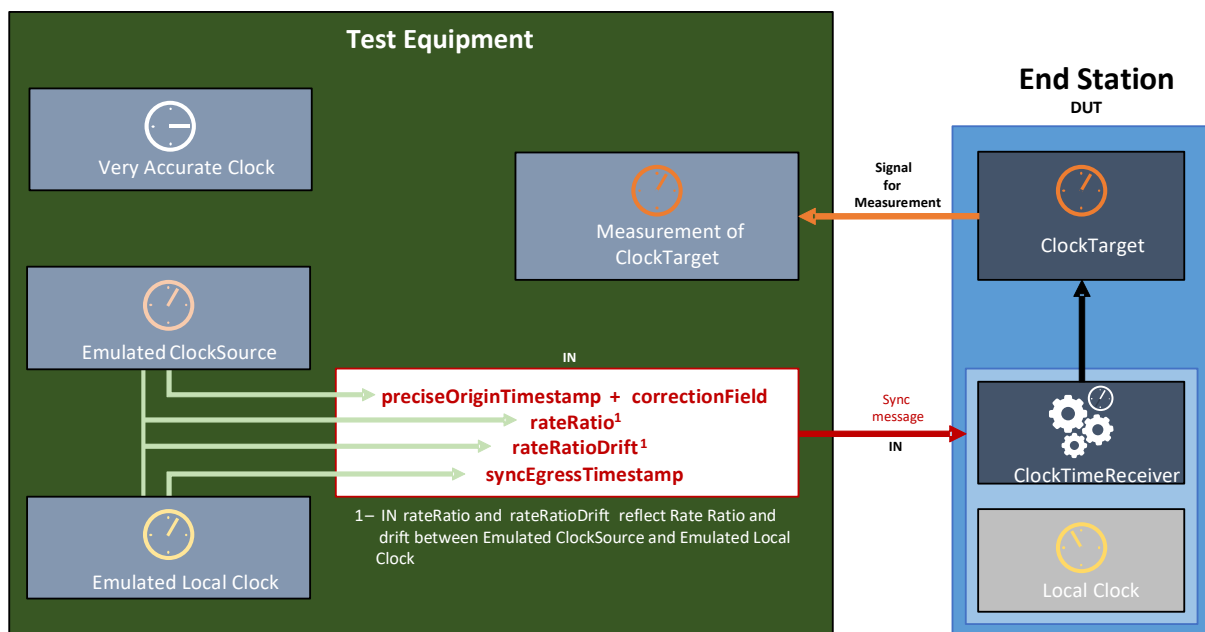


Figure D.5 – Approach to testing normative requirements for PTP End Instance

The test equipment can compare its measurement of the DUT's ClockTarget to the Emulated ClockSource. It accounts for the additional delay between its transmission of the input Sync message and the reception of the message by the DUT.

For test 1, the Emulated ClockSource and Emulated Local Clock are stable and in sync. In practice, both can be equal to the test equipment's Very Accurate Clock. In the input Follow_Up information TLV, rateRatio is 0 ppm, and in the input Drift_Tracking TLV, rateRatioDrift is 0 ppm/s. If the Local Clock of the PTP End Instance is also stable, it measures NRR of 0 ppm and NRR Drift of 0 ppm/s.

For test 2, the fractional frequency offset of the Emulated ClockSource is increasing at a defined ppm/s rate relative to the Very Accurate Clock. The Emulated Local Clock is stable; in practice, it can be equal to the test equipment's Very Accurate Clock. In the output Follow_Up information TLV, the rateRatio field increases over time, and the output Drift_Tracking TLV, the rateRatioDrift field maintains a matching positive value. If the Local Clock of the PTP Relay Instance is also stable, it measures NRR of 0 ppm and NRR Drift of 0 ppm/s.

7008 For test 3, the fractional frequency offsets of the Emulated ClockSource and the Emulated Local
7009 Clock are equal and increasing at a defined ppm/s rate relative to the Very Accurate Clock. In
7010 the output Follow_Up information TLV, the rateRatio field is 0 ppm, and in the output output
7011 Drift_Tracking TLV, the rateRatioDrift field is 0 ppm/s. If the Local Clock of the PTP Relay
7012 Instance is stable, the NRR it measures increases over time and the NRR Drift it measures
7013 maintains a matching positive value.

7014 **D.4.5 Testing meanLinkDelay Accuracy**

7015 D.3.5 and D.3.6 both discuss a normative requirement to test the accuracy of meanLinkDelay
7016 measurements for PTP Relay Instances and PTP End Instances respectively. This can be
7017 achieved by using a cable of known length and asymmetry to connect the test equipment to the
7018 DUT. The test equipment then responds to the DUT's Pdelay_Req messages with Pdelay_Resp
7019 messages, allowing the DUT to estimate the path delay.

7020 The DUT's estimate of the path delay can be accessed via the meanLinkDelay global variable
7021 (see IEEE 802.1AS-2020, Clause 14.8.8).

7022 The error in a single path delay measurement via a Pdelay_Req / Pdelay_Resp exchange can
7023 be large relative to the normative requirement, but an estimate with significantly less error can
7024 be generated by considering more than one measurement, hence the "mean" in
7025 "meanLinkDelay". Two consequences of this are noteworthy.

7026 Firstly, since errors can be introduced by Timestamp Errors at both the DUT and a neighboring
7027 node, and the DUT is responsible for mitigating both, the test equipment simulates typical errors
7028 on its timestamps for receiving Pdelay_Req and transmitting Pdelay_Resp messages to provide
7029 a complete test.

7030 Secondly, since the meanLinkDelay measurement can take some time to stabilize based on
7031 multiple Pdelay_Req / Pdelay_Resp exchanges, leaving sufficient time for these multiple
7032 exchanges before assessing meanLinkDelay accuracy results in a more accurate test.

7033 **D.5 Example algorithms**

7034 **D.5.1 General**

7035 This document does not place normative requirements on the use of specific algorithms.
7036 However, the normative requirements assume the use of algorithms to reduce the effect of
7037 errors in meanLinkDelay and to track clock drift and compensate for consequent errors. PTP
7038 Instances that do not implement algorithms can find it difficult or impossible to meet the
7039 normative requirements.

7040 D.5 provides examples of algorithms that can be used for:

- 7041 • tracking NRR drift;
- 7042 • correcting for errors in measured NRR (mNRR) due to NRR drift;
- 7043 • calculating RR drift;
- 7044 • correcting for errors in measured RR (mRR) due to RR drift;
- 7045 • reducing the effect of errors in meanLinkDelay.

7046 **D.5.2 Algorithm for tracking NRR drift**

7047 For measured NRR, measured RR, and meanLinkDelay, an example for how startup behavior
7048 can be handled is provided.

7049 NRR Drift Tracking and Error Correction is carried out for each network hop, i.e. at every node
7050 other than the Grandmaster. It is based on pairs of timestamps with each pair associated with
7051 a Sync message transmitted from the previous node ($n-1$) to the current node (n).

- 7052 • t_{s1outP} – Timestamp of the Sync message egress from the previous node ($n-1$), timestamped
7053 by that node's Local Clock. Unit: ns.
- 7054 • t_{s2in} – Timestamp of the Sync message ingress to the current node (n), timestamped by that
7055 node's Local Clock. Unit: ns.

7056 All timestamps are affected by Timestamp Errors.

7057 The algorithm uses information from the 32 most recent Sync messages. However, a node only
7058 keeps track of the 9 most recent pairs of timestamps from the most recent (x) to the 9th most
7059 recent ($x-8$) Sync message. The algorithm generates one measurement of NRR using the prior
7060 2 s of Sync message data (on average, based on a nominal Sync Interval of 125 ms), and a
7061 second measure based on the 2 s of Sync message data prior to that. It then uses the difference
7062 in the two measurements over the interval between the effective measurement points to
7063 calculate the NRR drift rate.

7064 On arrival of a new Sync message (x), or Follow_Up in the case of two-step time transport, a
7065 node executes a NRR calculation:

$$NRRcalc(x) = \left(\frac{t_{s1outP}(x) - t_{s1outP}(x-8)}{t_{s2in}(x) - t_{s2in}(x-8)} - 1 \right) \times 10^6 \quad (D.7)$$

7066

7067 where

7068 $NRRcalc$ is the calculated Neighbor Rate Ratio, expressed in ppm;

7069 x is the most recent Sync message;

7070 t_{s1outP} is the timestamp of the Sync message egress from the previous node ($n-1$),
7071 timestamped by that node's Local Clock, expressed in ns;

7072 t_{s2in} is the Timestamp of the Sync message ingress to the current node (n), timestamped
7073 by that node's Local Clock, expressed in ns.

7074 with an associated effective measurement point:

$$NRRcalcT(x) = \frac{t_{s2in}(x) + t_{s2in}(x-8)}{2} \quad (D.8)$$

7075

7076 where

7077 $NRRcalcT$ is the effective measurement point, expressed in ns;

7078 t_{s1outP} is the timestamp of the Sync message egress from the previous node ($n-1$),
7079 timestamped by that node's Local Clock, expressed in ns;

7080 t_{s2in} is the Timestamp of the Sync message ingress to the current node (n), timestamped
7081 by that node's Local Clock, expressed in ns.

7082 A node keeps track of the 24 most recent NRR calculations and effective measurement points,
7083 from the most recent (x) to the 24th most recent ($x-23$).

7084 After of a new most-recent NRR calculation, a node calculates an NRR drift rate:

$$NRR_{averageA} = \sum_{i=x-7}^x \frac{mNRR_{calc}(i)}{8} \quad (D.9)$$

7085

7086 where

7087 $NRR_{averageA}$ is the average of the 8 most recent Neighbor Rate Ratio calculations, expressed
7088 in ppm.

$$NRR_{averageB} = \sum_{i=x-23}^{x-16} \frac{mNRR_{calc}(i)}{8} \quad (D.10)$$

7089

7090 where

7091 $NRR_{averageB}$ is the average of the 8 least recent Neighbor Rate Ratio calculations, expressed
7092 in ppm.

$$NRR_{driftInterval} = \sum_{i=x-7}^x \frac{mNRR_{calcT}(i)}{8} - \sum_{i=x-23}^{x-16} \frac{mNRR_{calcT}(i)}{8} \quad (D.11)$$

7093

7094 where

7095 $NRR_{driftInterval}$ is the period across which Neighbor Rate Ratio drift is measured, expressed
7096 in ns.

$$NRR_{driftRate}(n) = \left(\frac{NRR_{averageA} - NRR_{averageB}}{NRR_{driftInterval}} \right) \times 10^9 \quad (D.12)$$

7097

7098 where

7099 $NRR_{driftRate}(n)$ is the the NRR drift rate for the current Node n , expressed in ppm/s.

7100 **D.5.3 Algorithm to compensate for errors in measured NRR due to clock drift**

7101 **D.5.3.1 General**

7102 The algorithm to measure NRR uses data from the previous 1 s of Sync message data,
7103 combined with the NRR drift estimate from the previous step. This smaller amount of data (vs
7104 that used for either of the NRR measurements in the previous step) is employed as it improves
7105 responsiveness to sudden changes in NRR drift with minimal loss of accuracy.

7106 On arrival of a Sync message (x), or Follow_Up in the case of two-step time transport, a node
7107 executes a NRR calculation:

$$mNRR_{calc}(x) = \left(\frac{t_{s1outP}(x) - t_{s1outP}(x-4)}{t_{s2in}(x) - t_{s2in}(x-4)} - 1 \right) \times 10^6 \quad (D.13)$$

7108

7109 with an associated effective measurement point:

$$mNRRcalcT(x) = \frac{t_{s2in}(x) + t_{s2in}(x-4)}{2} \quad (D.14)$$

7110

7111 A node keeps track of the 4 most recent mNRR calculations and effective measurement points,
7112 from the most recent (x) to the 4th most recent ($x-3$). The mNRR calculations use information
7113 from the 5 most recent Sync messages, but the node is already keeping track of information
7114 from the 9 most recent Sync messages for the NRR drift tracking algorithm.

7115 The node then calculates an error corrected measured NRR value.

7116 For $i = x$ to $(x-3)$

$$mNRRcorrected(i) = mNRRcalc(i) + \left(NRRdriftRate(n) \times \frac{(t_{s2in}(x) - mNRRcalcT(i))}{10^9} \right) \quad (D.15)$$

7117

7118 where

7119 $mNRRcorrected(i)$ is the error-corrected measured NRR value at message i , expressed in ppm.

$$mNRR(n) = \sum_{i=x-3}^x \frac{mNRRcorrected(i)}{4} \quad (D.16)$$

7120

7121 where

7122 $mNRR$ is the error-corrected measured NRR value, expressed in ppm.

7123 The result is a measured NRR value, error-corrected to the time when the most recent Sync
7124 message was received.

7125 **D.5.3.2 Measured NRR algorithm – Startup behavior**

7126 NRR is used when calculating meanLinkDelay and output Sync/Follow_Up message fields. The
7127 first NRR drift calculation is only be available after receipt of 32 Sync/Follow_Up messages, i.e.
7128 after approximately 4 seconds of operation given the 125 ms Sync Interval. During this time
7129 meanLinkDelay and output Sync/Follow_Up messages fields have not been calculated, so an
7130 alternative must be used, even if it cannot deliver the same assurances regarding network-level
7131 performance.

7132 If measured NRR from Sync/Follow_Up message information is unavailable but equivalent
7133 information from Pdelay_Resp messages is available, it may be substituted for Sync/Follow_Up
7134 message information. However, measuring NRR using Pdelay_Resp messages is vulnerable to
7135 additional error due to clock drift between the time NRR is measured, on receipt of the latest
7136 Pdelay_Resp message, and use of the measurement during Sync message processing. This is
7137 the reason using Sync/Follow_Up message information is preferable. It also means that a switch
7138 to using Sync/Follow_Up message information as soon as possible is desirable. It is technically
7139 possible to calculate a NRR using a combination of Pdelay_Resp and Sync messages but this
7140 can be risky due to the potential for very short intervals between messages and resulting high
7141 error due to timestamp errors, so it not recommended.

7142 It is the responsibility of implementers to decide whether and when to use Pdelay_Resp
 7143 message information and when to switch to using Sync/Follow_Up message information. The
 7144 normative requirements in this document can be achieved after 32 Sync/Follow_Up messages
 7145 have been received and assume the use of the algorithms in D.5.2 and D.5.3, or more effective
 7146 algorithms. Implementations that continue to use Pdelay_Resp message information to
 7147 calculate NRR after 32 messages have been received can find some of the normative
 7148 requirements difficult or impossible to meet.

7149 The following describes potential startup behavior applicable to either Sync/Follow_Up or
 7150 Pdelay_Resp message information.

- 7151 a) At least two messages must be received before calculating a NRR value.
- 7152 b) Prior to two messages being received, $NRR = 1$ (i.e. 0 ppm) should be used.
- 7153 c) Once two messages have been received, NRR should be calculated using the formula:

$$2^{\text{nd}} \text{ message: } mNRR = \left(\left(\frac{t_3(x) - t_3(x-1)}{t_4(x) - t_4(x-1)} \right) - 1 \right) \times 10^6 \quad (\text{D.17})$$

7154
 7155 where

7156 $t_3(i)$ is the timestamp of the i^{th} Pdelay_Resp message on egress from the previous node ($n-1$),
 7157 timestamped by that node's Local Clock, expressed in ns;

7158 $t_4(i)$ is the timestamp of the Pdelay_Resp message on ingress to the current node (n),
 7159 timestamped by that node's Local Clock, expressed in ns.

- 7160 d) When three to four messages have been received, NRR should be calculated using the
 7161 following formula:

$$3^{\text{rd}} \text{ message: } mNRR = \left(\left(\frac{t_{1\text{outP}}(x) - t_{1\text{outP}}(x-2)}{t_{2\text{in}}(x) - t_{2\text{in}}(x-2)} \right) - 1 \right) \times 10^6 \quad (\text{D.18})$$

$$4^{\text{th}} \text{ message: } mNRR = \left(\left(\frac{t_{1\text{outP}}(x) - t_{1\text{outP}}(x-3)}{t_{2\text{in}}(x) - t_{2\text{in}}(x-3)} \right) - 1 \right) \times 10^6 \quad (\text{D.19})$$

7162

- 7163 e) On arrival of the 5th Sync/Follow_Up message the first mNRRcalc and mNRRcalcT
 7164 calculations can take place and should be used for NRR:

$$mNRRcalc(x) = \left(\frac{t_{s1\text{outP}}(x) - t_{s1\text{outP}}(x-4)}{t_{s2\text{in}}(x) - t_{s2\text{in}}(x-4)} - 1 \right) \times 10^6 \quad (\text{D.20})$$

$$mNRRcalcT(x) = \frac{t_{s2\text{in}}(x) + t_{s2\text{in}}(x-4)}{2} \quad (\text{D.21})$$

$$5^{\text{th}} \text{ Sync message: } mNRR = mNRRcalc(x) \quad (\text{D.22})$$

7165

7166 f) As the 6th, 7th and 8th messages arrive an average can be taken and used for NRR, so:

$$6^{\text{th}} \text{ message: } mNRR = \sum_{i=x-1}^x \frac{mNRR_{calc}(i)}{2} \quad (\text{D.23})$$

$$7^{\text{th}} \text{ message: } mNRR = \sum_{i=x-2}^x \frac{mNRR_{calc}(i)}{3} \quad (\text{D.24})$$

$$8^{\text{th}} \text{ message: } mNRR = \sum_{i=x-3}^x \frac{mNRR_{calc}(i)}{4} \quad (\text{D.25})$$

7167 g) For the 9th to the 31st message, the Formula (D.19) can be used.

7168 h) Once the 32nd message arrives, the regular formulas with NRR drift tracking and error
7169 correction can be used.

7170 **D.5.4 Algorithm for tracking RR Drift**

7171 A Sync or Follow_Up message carries the rateRatio field, which informs each node of the
7172 previous node's estimate of its (the previous node's) Rate Ratio. This document also requires
7173 support for the Drift_Tracking TLV, which carries the rateRatioDrift field and informs each node
7174 of the previous node's estimate of its (the previous node's) Rate Ratio Drift.

7175 If the implementation of the Grandmaster PTP Instance means the ClockSource and Local Clock
7176 (at the Grandmaster PTP Instance) are linked such that the two are always operating at the
7177 same frequency, the rateRatio field received by the first node (Node 1) is always 0 ppm and the
7178 rateRatioDrift field is always 0 ppm/s. Thus, at Node 1, RR equals NRR, RR Drift equals NRR
7179 Drift, and therefore D.5.3 and D.5.2 describe how to calculate RR and RR Drift at Node 1.

7180 If the implementation of the Grandmaster PTP Instance means the ClockSource and Local Clock
7181 (at the Grandmaster PTP Instance) can operate at different frequencies, the implementation
7182 populates the rateRatio and rateRatioDrift field with values reflecting those differences.

7183 In either case all PTP Instances, other than the Grandmaster PTP Instance, calculate an
7184 estimate of the local Rate Ratio Drift when the latest Sync/Follow_Up Message is received,
7185 based on the received rateRatioDrift field and the local measure of NRR Drift. The Rate Ratio
7186 Drift Rate from the previous node is in ppm/s relative to the timebase of its Local Clock (i.e. the
7187 "s" in "ppm/s"). For highest precision, this can be converted to the timebase of the current
7188 node's Local Clock.

$$rateRatioDrift(n) = \frac{rateRatioDrift(n-1)}{\left(1 + \frac{mNRR(n)}{10^6}\right)} + NRRdriftRate(n) \quad (\text{D.26})$$

7189

7190 where

7191 *rateRatioDrift* is the Rate Ratio drift rate, expressed in ppm/s.

7192 However, given that adding ppm/s already lacks the precision of multiplying actual ratios, this
7193 simplification delivers similarly accurate results.

$$rateRatioDrift(n) = rateRatioDrift(n-1) + NRRdriftRate(n) \quad (D.27)$$

7194

7195 D.5.5 Algorithm to compensate for errors in measured RR due to clock drift

7196 On receipt of a Sync or Follow_Up message, all PTP Relay Instances estimate a measured RR
 7197 ($mRR(n)$) based on the received rateRatio field ($mRR(n-1)$) and the local measure of NRR
 7198 ($mNRR(n)$). An $mRR(n)$ value is used to translate the sum of meanLinkDelay and residenceTime
 7199 from Local Clock timebase into Grandmaster timebase. An $mRR(n)$ value is also passed in the
 7200 transmitted Sync or Follow_Up message's rateRatio field to the next node. Errors in these
 7201 estimates due to clock drift can be reduced by taking account of RR Drift. Since the optimal
 7202 point in time for each estimate is different, the amount of applicable RR Drift is different, and
 7203 hence the estimates are different.

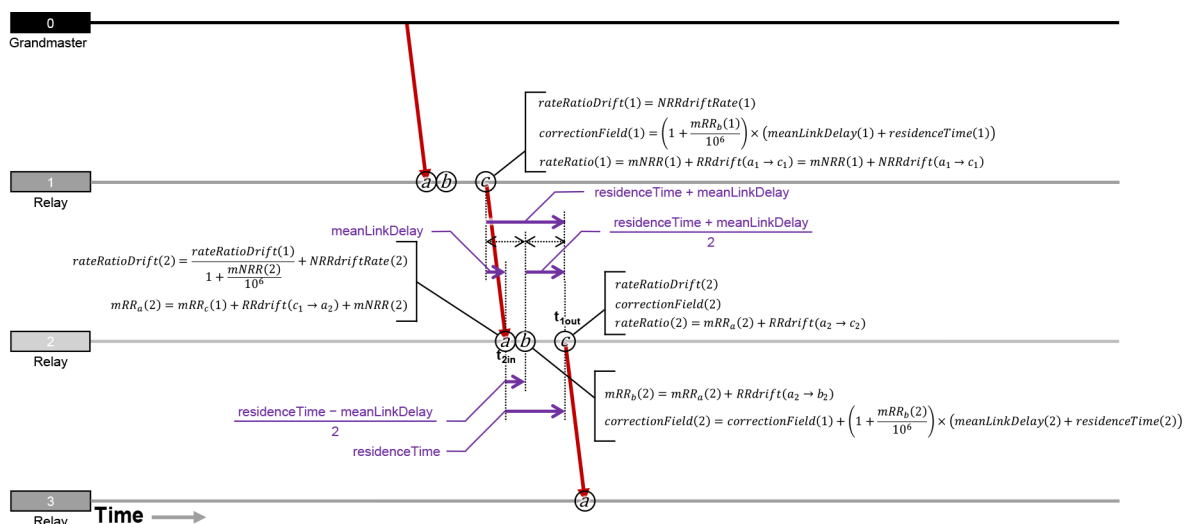
7204 (For discussion of how different Grandmaster PTP Implementations affect the behavior of a
 7205 PTP Relay Instance at Node 1 – or not – see D.5.4.)

7206 A PTP End Instance is similar in that it estimates $mRR(n)$ on receipt of a Sync message,
 7207 subsequently uses an $mRR(n)$ value, and errors in the latter due to clock drift can be mitigated
 7208 by taking account of RR Drift. However, unlike a PTP Relay Instance, the $mRR(n)$ value is used
 7209 to keep the ClockTarget in line with the ClockSource and transmitting a rateRatio field to a
 7210 subsequent node is unnecessary.

7211 For a PTP Relay Instance there are three points in time of interest:

- 7212 • Point a: receipt of the Sync Message by the current node (Node n),
- 7213 • Point b: mid-point between transmission of the Sync message by the previous node (Node
 7214 $n-1$) and transmission of the consequent Sync message by the current node (Node n), and
- 7215 • Point c: transmission of the Sync Message by the current node (Node n).

7216 Figure D.6 illustrates these points and the associated calculations.



7217

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7218 **Figure D.6 – RR Drift tracking and error compensation calculations –**
 7219 **PTP Relay Instance**

7220 The estimate of RR when the Sync message arrives can be calculated as follows:

$$mRR_a(n) = rateRatio(n-1) + RRdrift(c_{n-1} \rightarrow a_n) + mNRR(n). \quad (D.28)$$

$$7221 \quad = rateRatio(n-1) + \left(rateRatioDrift(n-1) \times \left(1 + \frac{mNRR(n)}{10^6} \right) \times meanLinkDelay(n) \right) + mNRR(n)$$

7222 where

7223 mRR_a is the estimate of RR when the Sync message arrives at node n , expressed in ppm;

7224 $RRdrift(c_{n-1} \rightarrow a_n)$ is the amount $rateRatio(n-1)$ drifts between transmission of the Sync
7225 message at Node $n-1$ and reception at Node n , expressed in ppm/s.

7226 $RRdrift(c_{n-1} \rightarrow a_n)$ is equivalent to $rateRatioDrift(n-1)$ multiplied by meanLinkDelay but,
7227 since meanLinkDelay is measured in terms of Node n 's Local Clock and rateRatioDrift is in
7228 terms of Node $n-1$'s Local Clock the former should be multiplied by the NRR at Node n for the
7229 highest accuracy.

7230 However, given that adding ppm/s already lacks the precision of multiplying actual ratios, the
7231 following simplification delivers similarly accurate results:

$$mRR_a(n) = rateRatio(n-1) + (rateRatioDrift(n-1) \times meanLinkDelay(n)) + mNRR(n) \quad (D.29)$$

7232

7233 Once the time when Node n transmits the consequent Sync message is known, the
7234 correctionField value can be calculated.

$$mRR_b(n) = mRR_a(n) + RRdrift_n(a \rightarrow b) \quad (D.30)$$

$$7235 \quad = mRR_a(n) + \left(rateRatioDrift(n) \times \frac{residenceTime(n) - meanLinkDelay(n)}{2} \right)$$

7236 where

7237 $mRR_b(n)$ is the estimate of RR when Node n transmits the consequent Sync message,
7238 expressed in ppm.

7239 The correctionField is calculated as follows:

$$correctionField(n) = correctionField(n-1) + \left(1 + \frac{mRR_b(n)}{10^6} \right) \quad (D.31)$$

$$\times (meanLinkDelay(n) + residenceTime(n))$$

7240

7241 where

7242 $correctionField(n)$ is the value of the correction field transmitted by node n , expressed in
7243 ns.

7244

7245 The rateRatio field is calculated as follows:

$$7246 \quad \begin{aligned} rateRatio(n) &= mRR_a(n) + RRdrift_n(a \rightarrow c) \\ &= mRR_a(n) + (RRdriftRate(n) \times residenceTime(n)) \end{aligned} \quad (D.32)$$

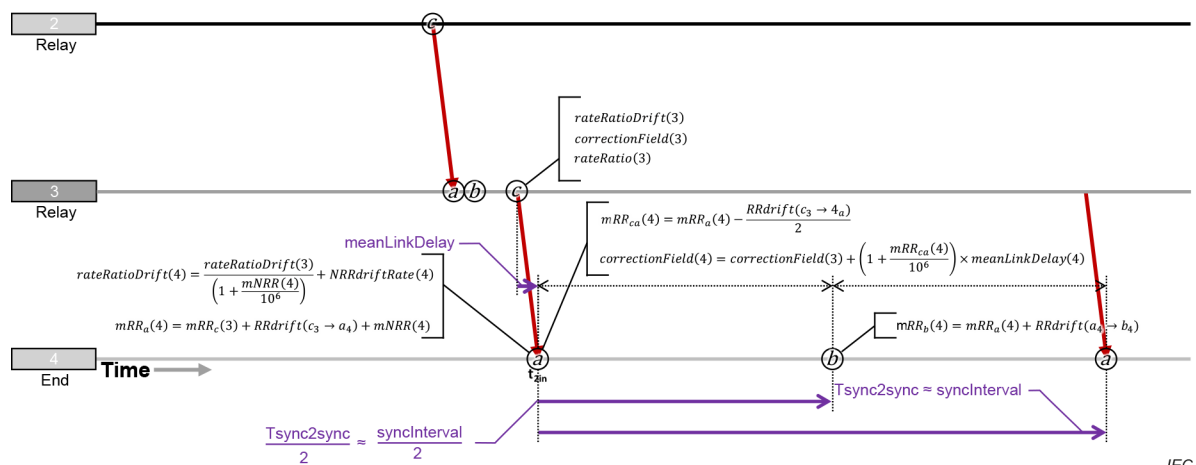
7247

7248 where

7249 $rateRatio(n)$ is the value of the rateRatio field transmitted by node n , expressed in ppm.

7250 D.5.6 Algorithm to compensate for errors in measured RR due to clock drift at PTP 7251 End Instance

7252 Figure D.7 illustrates a possible approach to applying similar RR drift tracking and error
7253 compensation at a PTP End Instance.



7254

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7255 **Figure D.7 – RR Drift tracking and error compensation calculations –**
7256 **PTP End Instance**

7257 The initial calculations for rateRatioDrift and mRR_a are exactly the same as for a PTP Relay
7258 Instance. Instead of using RR to translate meanLinkDelay + residenceTime from the Local Clock
7259 timebase to the Grandmaster timebase – as is done at a PTP Relay Instance – a PTP End
7260 Instance uses mRR to translate meanLinkDelay to the Grandmaster timebase (there is no
7261 residenceTime at an End Instance), adding the result to the incoming correctionField to obtain
7262 an estimate of the ClockSource at the time the Sync message arrives, then uses mRR to keep
7263 its ClockTarget in line with the ClockSource until arrival of the next Sync message. The optimal
7264 mRR value for translating meanLinkDelay is halfway between meanLinkDelay's transmission
7265 (at c_{n-1} , i.e. point C at the previous node) and reception (at a_n , i.e. point A at the current node);
7266 in the formulas below, this value is referred to as mRR_{ca} .

$$\begin{aligned}
mRR_{ca}(n) &= mRR_a(n) - \frac{RRdrift(c_{n-1} \rightarrow a_n)}{2} \\
&= mRR_a(n) - \left(rateRatioDrift(n) \times \frac{meanLinkDelay(n)}{2} \right)
\end{aligned} \tag{D.33}$$

7268

7269 where

7270 $mRR_{ca}(n)$ is the estimate of RR calculated based upon one-half of the meanLinkDelay,
 7271 expressed in ppm.

7272 The correction field at the PTP End Instance is given by:

$$correctionField(n) = correctionField(n-1) + mRR_{ca}(n) \times meanlinkDelay(n) \tag{D.34}$$

7273

7274 The optimal value of mRR for keeping the ClockTarget in line with the Clock Source is mRR_b ,
 7275 where Point B is halfway between the most recently received Sync message and the next Sync
 7276 message.

7277 Of course, the exact interval until the next Sync message's arrival (Tsync2sync in Figure D.7)
 7278 can't be known before it happens, but the Rate Ratio value is required as soon as possible after
 7279 arrival of the most recent Sync message. The solution is to use the nominal value of the interval,
 7280 i.e. syncInterval, which is 125 ms.

7281

$$mRR_b(n) = mRR_a(n) + RRdrift_n(a \rightarrow b)$$

7282

$$\begin{aligned}
&= mRR_a(n) + \left(rateRatioDrift(n) \times \frac{syncInterval}{2} \right) \\
&= mRR_a(n) + (rateRatioDrift(n) \times 0,0625)
\end{aligned} \tag{D.35}$$

7283

7284 where

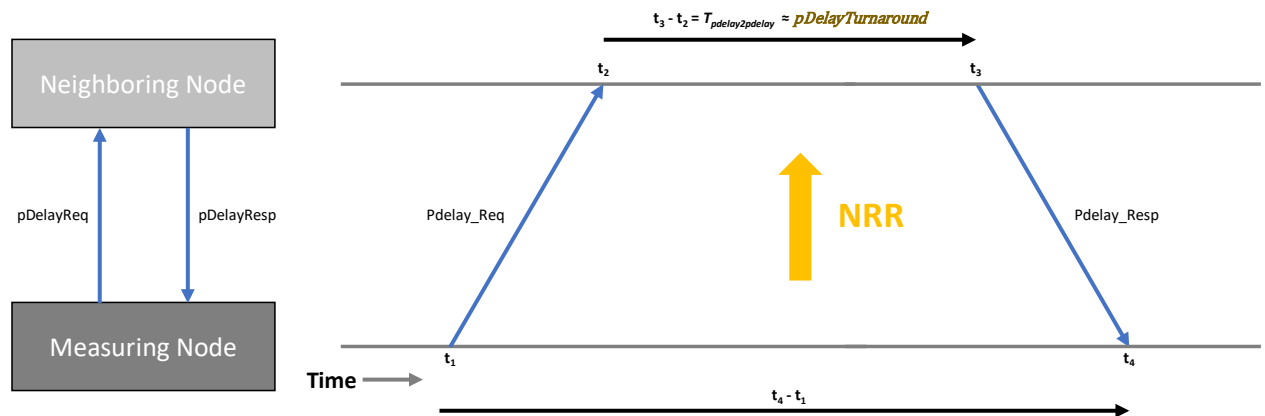
7285 *syncInterval* is the nominal value of the interval between sync messages, expressed in ms.

7286 It is also possible to use more complex algorithms that repeatedly or continuously adjust the
 7287 mRR value between Sync messages, but such an approach is not addressed in this document.

7288 **D.5.7 Mean Link Delay averaging**

7289 The actual Path Delay from one node to the next – for a wired connection – is very stable and
 7290 errors measuring it due to Timestamp Error average to zero. Thus, taking a long average or
 7291 applying a low-pass filter with a low bandwidth is an effective way to reduce error in
 7292 meanLinkDelay. Care is taken during system startup or after any other initialization of the
 7293 algorithm, to quickly converge on a stable value.

7294 The basic Pdelay calculation, used by the Common Mean Link Delay service, remains the same.
 7295 Figure D.8 illustrates it.



$$mPathDelay = \left(\frac{(t_4 - t_1) - \frac{(t_3 - t_2)}{NRR}}{2} \right)$$

7297 **Figure D.8 – Signals and timestamps to measure path delay**

7298 Following each Pdelay_Req – Pdelay_Resp exchange, the measured path delay (mPathDelay)
 7299 is calculated.

7300 For the x^{th} message after initialization

$$mPathDelay(x) = \frac{(t_4 - t_1) - \frac{(t_3 - t_2)}{NRR}}{2} \quad (\text{D.36})$$

7301
 7302 where

7303 $mPathDelay$ is the measured path delay between the measuring node and the neighboring
 7304 node expressed in ns;

7305 t_1 is a measurement point as defined in Figure D.8, expressed in ns;

7306 t_2 is a measurement point as defined in Figure D.8, expressed in ns;

7307 t_3 is a measurement point as defined in Figure D.8, expressed in ns;

7308 t_4 is a measurement point as defined in Figure D.8, expressed in ns.

7309 The meanLinkDelay is then updated via an IIR (Infinite Impulse Response) filter. For the first
 7310 measurement, the filter is initialized:

$$meanLinkDelay(x) = mPathDelay(x) \quad (\text{D.37})$$

7311

7312 For the next few minutes after initialization (when $x < 1\,000$) the filter is in startup mode. It then
 7313 transitions to steady-state mode.

7314 If $x < 1\,000$ then $f = x$ else $f = 1\,000$

$$meanLinkDelay(x) = \frac{(meanLinkDelay(x-1) \times (f-1)) + mPathDelay(x)}{f} \quad (D.38)$$

7315 For example:

$$meanLinkDelay(100) = \frac{(meanLinkDelay(99) \times 99) + pDelay(x)}{100} \quad (D.39)$$

7316

$$meanLinkDelay(5\,836) = \frac{(meanLinkDelay(5\,835) \times 999) + pDelay(x)}{1\,000} \quad (D.40)$$

7317

7318 It is possible to automatically reinitialize the algorithm if an *mPathDelay* value, or series of
7319 values, deviates too much from the *meanLinkDelay*, but the details are outside the scope of this
7320 document.

7321 The behavior of timestamp error means that, for shorter actual link delays, *mPathDelay* can be
7322 a negative value. It can seem tempting to reject negative values, since a negative delay is
7323 impossible. However, at a device level, including negative values of *mPathDelay* in the input to
7324 the IRR filter results in a more accurate filter output, i.e. *meanLinkDelay* value; values lower
7325 than the actual delay, even when negative, are balanced by values higher than the actual delay.

7326 Similarly, at a network level, using negative values of *meanLinkDelay*, i.e. the output of the IIR
7327 filter, results in a more accurate *correctionField* calculation at the PTP End Instance when there
7328 are many networking hops between it and the Grandmaster PTP Instance.

Annex E
(normative)

60802 device info TLV definition

E.1 60802 device info TLV

E.1.1 General

In the TLV definition that follows, any fields that are labelled as “Reserved” are transmitted as zero and ignored on receipt.

The 60802 device info TLV is a mandatory TLV that indicates a device’s configuration capabilities.

Figure E.1 shows the 60802 device info TLV.

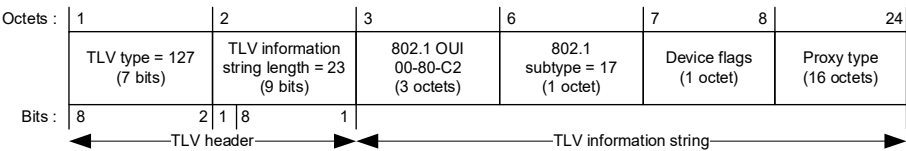


Figure E.1 – 60802 device info TLV

E.1.2 Device flags

The Device flags field shall contain a bit map indicating the device configuration capabilities as defined in Table E.1.

Table E.1 – Device flags

Bit	Value/meaning	Function
0	1 = Proxy required 0 = no Proxy required	Proxy required
1	1 = supported 0 = not supported	NETCONF configuration supported
2-7	(set to zero)	Reserved for future standardization

E.1.3 Proxy type

The proxy type field is conditionally mandatory if the device flags indicate that the device requires a proxy. It indicates the proxy type that is required to configure the device. The format is a human readable string, that is specific for a specific proxy and its associated devices.

Annex F (informative)

MAC and PHY delay

F.1 Accumulated delays in industrial automation

F.1.1 General

MAC and PHY delays affect the performance of industrial communication. These delays impact forwarding delay, the accuracy of time-triggered transmission, and synchronization accuracy. Factors contributing to these delays include:

- the clock system between the MAC application, the MAC, the Media Independent Interface (MII), and the PHY,
- the processing of the frame, and
- the transmission of bits from the MAC application over MAC, MII and PHY to the MDI.

These delays consist of two components:

- the propagation delay, and
- the propagation delay variance (often called jitter).

F.1.2 Forwarding delay

Linear topologies, as commonly used in industrial communication, result in cumulative MAC and PHY delays. With 64 hops, there are 130 MAC and PHY delays (64 hops x 2 + 2x end stations) in end-to-end transmission.

For example, a frame sent from the end station at the beginning of a linear topology (64 Hops) to an end station at the end of the line is delayed by 26 μ s / 65 μ s / 130 μ s solely due to a PHY delay of 200 ns / 500 ns / 1 μ s. This means that the lower the PHY delay, the shorter the achievable update interval and thus, the roundtrip delay.

F.1.3 Time-triggered transmission

The reference point for time-triggered transmission is at the MDI. However, the frames are handed over from the transmission control to the MAC in a time-controlled manner. To transmit a frame at the correct time at the reference point, the MAC and PHY delay are compensated by the transmission control.

For example, the magnitude of the propagation delay reduces the computation time available for the application to calculate new process values, as the frames are handed over from the transmission control to the MAC earlier so they can be sent at the correct time at the reference point. Additionally, the frames are also received later by the application, which further reduces the computation time (see 4.1).

The magnitude of the propagation delay variance affects the safety margin of the windows in TAS and is a consideration when configuring PSFP and scheduled traffic. Additionally, it increases the required TAS window size in linear topologies to ensure that all planned frames can pass through their assigned TAS windows.

F.1.4 Synchronization

The reference point for synchronization is also at the MDI. The actual point at which the transmitted or received synchronization frames are timestamped depends on the implementation and can be in the PHY, on the MII, or behind the MAC. Depending on the

7392 implementation, part or all the MAC and PHY delay is compensated to determine the time at
7393 which the synchronization frame was at the reference point.

7394 For example, during synchronization, the mean link delay and residence time are determined.
7395 The propagation delay variance may be different for the transmission of pdelay frames and
7396 synchronization frames. This results in an uncompensated delay error for the sync frames,
7397 which accumulates in linear topologies.

7398 **F.1.5 Recommended delays**

7399 Ethernet hardware used for industrial communication typically considers the requirements for
7400 MAC and MII propagation delay and propagation delay variance, minimizing these values during
7401 development. PHYs, if not integrated, are appropriately developed or selected.

7402 Table F.1 shows the recommended values for PHYs in industrial communication.

7403 **Table F.1 – Recommended PHY delay**

MAU Type	Recommended Propagation delay	Recommended Propagation delay variance
10BASE-T1L	maximum < 5 μ s minimum < 500 ns	maximum < 16 ns minimum < 4 ns
100BASE-TX, 100BASE-FX, 1000BASE-SX, and 1000BASE-T	maximum < 1 μ s minimum < 500 ns	maximum < 16 ns minimum < 4 ns

7404

7405

7406

7407

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